# Tax Heterogeneity, Duration Risk, and the Term Structure: An Application to the Municipal Bond Puzzle*

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**Abstract**

At short maturities, tax-exempt municipal yields closely match top-bracket households' after-tax yields on comparable taxable bonds. At thirty years, municipal yields exceed this household benchmark by 118 basis points, implying a marginal tax rate below the top household rate. I explain this longstanding municipal bond puzzle with a term-structure model jointly pricing taxable and tax-exempt bonds. Heterogeneous taxes and balance sheets endogenously segment ownership across markets and maturities under short-sale constraints. High-tax households value the exemption most and hold shorter municipal bonds. Some lower-tax investors have long-duration liabilities that make rate risk cheaper to bear, so they hold long municipal bonds despite smaller exemption benefits. Because they also hold taxable bonds, relative long-term yields reflect their lower taxes. Without targeting municipal yields, the calibrated model explains 83 percent of the puzzle. Cross-state differences in home-state exemptions and a discontinuity in a bank-only tax benefit support the model's predictions. Household taxes are almost fully reflected in short yields but less in long yields, while bank tax benefits mainly affect long yields.

**Keywords:** Term Structure, Investor Heterogeneity, Taxes, Endogenous Segmentation, Asset Ownership, Municipal Bonds
**JEL Codes:** G11, G12, G22, H24, H71, H74

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# 1 Introduction

The $4 trillion municipal bond market finances U.S. state and local governments, and the bonds' interest is generally exempt from federal income tax. The exemption is worth most to top-bracket households. The literature's standard benchmark therefore predicts that municipal yields equal top-bracket households' after-tax yields on comparable taxable bonds. The benchmark fits short maturities, but long municipal yields substantially exceed it, implying marginal tax rates that decline with maturity. This pattern, known as the municipal bond puzzle, has been documented since the 1970s but remains unresolved. The household benchmark uses a single tax rate, but ownership data reveal substantial tax heterogeneity among municipal bondholders. Top-bracket households hold only about half the market, while lower-tax households, banks and insurers hold substantial amounts despite their smaller exemption benefits. Motivated by these observations, I ask what drives the participation of investors with heterogeneous exemption benefits in the municipal bond market, and to what extent the resulting investor composition can explain the municipal bond puzzle.

To answer these questions, I develop an equilibrium term-structure model in which investors with heterogeneous taxes and balance sheets jointly choose taxable and tax-exempt bonds across maturities under short-sale constraints. Their portfolio choices endogenously segment ownership and determine both yield curves. Tax differences draw high-tax investors toward municipal bonds, and lower-tax investors toward taxable bonds, but long-duration liabilities let some lower-tax investors bear interest-rate risk more cheaply than high-tax households. This advantage grows with maturity and draws them into long municipal bonds despite their smaller exemption benefits. These lower-tax investors also hold taxable bonds, so relative long-term yields reflect their lower tax rates and municipal yields exceed the household benchmark. Calibrated to U.S. taxes, balance sheets and holdings, without targeting municipal yields, the model explains over 80% of the long-maturity yield gap in the municipal bond puzzle. Empirical tests exploit cross-state differences in home-state tax exemptions and a discontinuity in a bank-only tax benefit. Household taxes are almost fully priced in short municipal yields but less in long ones, while banks' benefit is priced mainly in long yields, as the model predicts. Counterfactual tax reforms raise borrowing costs in both municipal and taxable markets through investor portfolio substitution. Tax revenue relative to added borrowing costs depends on how holdings shift across investors with different tax rates.

Most municipal bonds pay federally tax-exempt interest, a benefit intended to lower public borrowing costs. How much the exemption saves public borrowers depends on which investors hold their bonds and the yields they require. Top-bracket households face the highest ordinary income tax rates and therefore benefit most from the exemption. Their tax advantage and limits on municipal short selling motivate the literature's household benchmark, under which municipal yields equal their after-tax yields on comparable taxable bonds (Schwert, 2017; Fleckenstein and Longstaff, 2025). The benchmark fits short maturities, but long municipal yields substantially exceed it. This municipal bond puzzle has been documented since the 1970s (Fortune, 1973). In my 2000–2020 sample, thirty-year municipal yields exceed top-bracket households' after-tax yields on comparable taxable bonds by 118 basis points. Equivalently, relative yields imply a marginal

tax rate of about 38% at two years, close to the top household rate, but only 21% at thirty. The thirty-year municipal yield is therefore about one third above the benchmark. Applied to municipal debt outstanding at each maturity, the excess over the benchmark amounts to about $13 billion of interest a year in 2000–2017. Adjustments for default risk, liquidity and capital-gains taxation explain only a small part of this gap. Despite its name, the puzzle concerns the relative pricing of municipal and taxable bonds, rather than only municipal bonds. Much of the relevant tax heterogeneity arises on the taxable side, where investors pay different tax rates and therefore value the exemption differently. The ownership data reveal substantial tax heterogeneity among municipal bondholders. Top-bracket households hold roughly half the market, while lower-tax households, banks and insurers hold much of the remainder despite their smaller exemption benefits.

Motivated by this tax heterogeneity among municipal bondholders, I develop a general model in which multiple investor groups with different tax treatment and liability duration choose portfolios jointly across markets and maturities, subject to short-sale constraints. This general framework underlies my calibration and can be applied to other bond markets where tax treatment differs across investors and assets. I then illustrate the mechanism analytically with two investors and two bond markets. A high-tax household and a lower-tax investor choose portfolios across taxable bonds and tax-exempt municipal bonds of different maturities. They have the same risk aversion, but the lower-tax investor has a larger liability hedge. Taxes sort the household into municipal bonds and the lower-tax investor into taxable bonds, because the exemption is worth more to the household. Bearing rate risk, however, is cheaper for the lower-tax investor, because its liabilities lose value when rates rise, offsetting losses on its bonds. Insurers with long-term obligations and lower-tax households with mortgage debt are examples. This duration benefit grows with maturity, because longer bonds carry more rate risk, whereas the difference in exemption benefits is the same at every maturity. Beyond a threshold maturity, the duration benefit outweighs this difference. In this two-investor equilibrium, the household therefore holds shorter municipal bonds, and the lower-tax investor holds longer municipal bonds together with risky taxable bonds.

This sorting raises municipal yields above the household benchmark through three channels, which I call exemption benefit, duration cost and rate loading. First, lower-tax investors hold long bonds in both markets. Because they pay less tax on taxable interest, they accept a smaller yield reduction for the municipal exemption than high-tax households do. Long municipal yields therefore lie closer to taxable yields and exceed the household benchmark. Second, the household alone holds shorter risky municipal bonds and bears rate risk at a higher cost than the lower-tax investor holding comparable taxable bonds. The household's higher required compensation raises municipal yields above the benchmark even at these maturities. Third, because taxes scale returns, they also scale risk. Under the household benchmark, the municipal yield is the taxable yield multiplied by the fraction of taxable interest the household keeps, so its sensitivity to interest rates is scaled by the same fraction. Lower-tax investors keep a larger fraction, so the long municipal yields they price move more with interest rates than the benchmark predicts. Holders require compensation for that additional rate risk, and household portfolio substitution carries the effect

to shorter municipal bonds. All three channels vanish at the riskless shortest maturity, where the household's tax advantage determines relative yields.

Other motives can also lead lower-tax investors to hold long municipal bonds and generate the same qualitative pricing pattern. Lower trading costs or better default diversification can do so when the advantage grows with maturity and applies equally to taxable and tax-exempt bonds (Section 4.7). I focus on liability hedging because investors' widely differing liability durations provide an economically important and measurable source of differences in the cost of bearing interest-rate risk. Observed balance sheets and maturity holdings discipline these differences in the quantitative model.

The quantitative application asks how much of the puzzle my model can explain given observed taxes and portfolios. It includes high- and lower-tax households in each state, banks, property-and-casualty insurers, life insurers and tax-neutral investors trading across state municipal markets and a taxable market. With many investor groups and bond-specific risk, holdings overlap across maturities, so segmentation is partial, unlike the sharp sorting of the two-investor illustration. I measure tax treatment, bond supplies and investor balance-sheet sizes directly from data and estimate rate persistence and volatility from historical short rates. Observed ownership and allocations mainly discipline investors' unobserved exposure to interest rates, while taxable yields help discipline risk aversion. Municipal yields and empirical test coefficients are not targeted in the calibration. The model accounts for 97 of the 118-basis-point puzzle at thirty years, and measured default, liquidity and capital-gains differentials explain most of the remainder.

I evaluate the model against the household benchmark using three empirical tests, none targeted in the calibration. The model predicts that variation in top-bracket households' exemption benefits is fully reflected in implied taxes at the shortest maturity and less strongly at longer maturities. Changes in lower-tax investors' exemption benefits have no effect at the shortest maturity and matter more at longer maturities. First, most states exempt residents' interest on their own municipal bonds from state income tax, so top-bracket households' exemption benefits vary with state tax rates. I compare bonds issued in states with different tax rates, separately by maturity bin. The household benchmark predicts that state tax rates are reflected one-for-one in implied state taxes at every maturity. Consistent with my model, they are reflected almost one-for-one at short maturities but progressively less at longer maturities. Second, I compare yields on designated bank-qualified and nonqualified municipal bonds. Bank-qualified bonds give banks an additional tax benefit that other investors do not receive. Bonds' eligibility for this designation depends on whether an issuer's annual tax-exempt issuance exceeds a cap. To identify how much of the bank-specific benefit is reflected in yields, I compare bonds from issuers just below and above the cap in a regression-discontinuity design, using below-cap eligibility as an instrument for actual designation. Because the benefit reaches only banks, the benchmark predicts that designation leaves yields unchanged at every maturity. In the data, designation barely lowers short municipal yields, but its yield reduction grows with maturity, so banks' tax benefit is priced mainly at long maturities, as the model predicts. Third, I ask how much municipal yields move when treasury yields move. The

benchmark makes municipal yields a fixed fraction of treasury yields, so this sensitivity should be the same at every maturity. I measure it by comparing each bond's yield change between consecutive trades with the change in the treasury yield of the same maturity between the same trade dates. The sensitivity rises with maturity, as the rate-loading channel predicts.

Two counterfactual reforms show how removing different tax benefits changes financing costs and tax receipts. The first removes a state's exemption for residents' interest on its own municipal bonds, as in Illinois, Iowa and Wisconsin, which generally tax that interest. Residents keep the federal exemption but lose the tax incentive to prefer home-state bonds, so high-tax households largely switch to other states' municipal bonds of similar maturity. The interest on these replacement bonds remains in the state's tax base, and the state collects tax at its full rate on residents' municipal interest. Local issuers' yields, however, rise by less than the full state tax rate would imply, especially at long maturities, where yields also reflect institutions and nonresidents whose tax treatment does not change. This matches the partial pass-through of state taxes at long maturities in the first empirical test. Accounting for resident ownership and portfolio adjustment, added state receipts exceed local issuers' added financing costs. Across the seven state experiments, own-state costs average \$0.68 to \$0.88 per dollar of added receipts over 2000–2017. Portfolio substitution also raises borrowing costs for other states' municipal issuers and taxable issuers.

The second reform also removes the federal exemption, taxing municipal interest like taxable interest for every investor. Taxable municipal issuance has a precedent in the 2009–2010 Build America Bonds program, which subsidized issuers' interest payments. Unlike under home-state repeal, high-tax investors can no longer keep an exemption by switching to other states' municipal bonds. Without the municipal exemption, high-tax investors reduce their long-bond holdings and shift toward shorter bonds and cash, which pay less interest and so generate less tax revenue. Tax-neutral investors take over more long bonds. Their increased rate exposure raises the compensation they require in both markets, while the reform imposes no additional tax on their interest. Thirty-year yields rise by 171 basis points in the municipal market and by 23 in the taxable market. In equilibrium, financing costs rise by more than tax receipts. Averaged over 2000–2017, the reform raises annual financing costs by \$46.3 billion and annual tax receipts by \$38.7 billion. Municipal and taxable issuers together incur \$1.20 in added costs per dollar of added federal and state receipts, including \$0.13 borne by taxable issuers. The excess becomes higher after-tax interest income, distributed unevenly across investors. Lower-tax households, life insurers and tax-neutral investors gain on average, while high-tax households, banks and property-and-casualty insurers, which used to benefit more from the exemption, lose.

The framework is not only about municipal bonds. It explains relative bond pricing across taxable and tax-exempt markets, with investor portfolio substitution transmitting tax changes in one market to yields in the other. I apply it to municipal and taxable bonds because municipal tax rules provide clean variation to test the mechanism, and the market presents a longstanding pricing puzzle. Its applications extend to other settings where tax treatment differs across investors and markets, including state-tax differences between treasury and corporate bonds,

tax-favored accounts and cross-border withholding. More broadly, heterogeneous taxes, together with balance-sheet differences and limits to arbitrage, shape endogenous market segmentation and are important for understanding bond yields and the effects of tax policy.

Section 2 reviews related work, and Section 3 presents the tax, ownership and yield evidence. Section 4 develops the model. Section 5 calibrates it and accounts for the municipal puzzle, and Section 6 tests its empirical predictions. Section 7 studies tax policy, and Section 8 concludes. Formal results, proofs and supporting evidence are in the appendix.

## 2 Related Literature

The paper contributes to research on how taxes affect portfolio choice and asset prices. Theoretical work studies how tax differences shape portfolios and equilibrium prices (Miller, 1977; Auerbach and King, 1983; Dybvig and Ross, 1986; Dammon and Green, 1987), and Sialm (2009) provides evidence that taxes are reflected in equity valuations. I study how these differences shape the bond yield curve. Under short-sale constraints, investors choose jointly across markets and maturities. Their taxes and costs of bearing rate risk determine who holds each bond and whose tax treatment its yield reflects.

Research on market segmentation links the yield curve to bond supply, investors' maturity preferences and their capacity to bear risk (Vayanos and Vila, 2021; Greenwood and Vayanos, 2014; Guibaud et al., 2013). Portfolio adjustment also transmits supply shocks across asset classes and currencies (Greenwood et al., 2018, 2023).[1] I explain how taxes and investor balance sheets generate partial segmentation across markets and maturities. Participation responds to taxes and balance-sheet exposures, so tax changes alter both yield curves through joint portfolio choice.

The model's explanation of participation connects to research on institutional balance sheets and asset pricing. Intermediaries' capital and exposure to interest-rate movements affect the risk premia they require (He and Krishnamurthy, 2013; Haddad and Sraer, 2020). Kekre et al. (2024) show how losses on arbitrageurs' bond portfolios can amplify the effect of monetary tightening on long yields. Insurers' long-term liabilities also shape their demand for long bonds and their capacity to bear credit risk (Domanski et al., 2017; Li, 2026).[2] I show how a balance-sheet advantage in bearing rate risk can offset a smaller exemption benefit, explaining why some lower-tax investors hold long tax-exempt bonds and how their holdings affect equilibrium yields.

The municipal bond puzzle has motivated a longstanding literature on relative bond yields. Early explanations emphasized differences in investors' tax rates (Fortune, 1973) and a tax arbitrage available only to commercial banks, which before 1986 could deduct most of the interest on funds used to carry municipal bonds. In these accounts, bank demand set short municipal yields

[1] Constraints on arbitrage can limit this adjustment (Gromb and Vayanos, 2018; Siriwardane et al., 2025). Related empirical work measures investors' asset demand and household rebalancing (Koijen and Yogo, 2019; Gabaix et al., 2026).

[2] Municipal offering spreads reflect banks' rate exposure and funds' liquidity needs (Pati, 2023). Liability structures also help explain investors' responses to credit-market downturns (Coppola, 2025).

and household demand set long ones (Mussa and Kormendi, 1979; Hendershott and Koch, 1980; Kidwell and Koch, 1983; Skelton, 1983). The Tax Reform Act of 1986 removed this deduction for most new purchases and reduced banks' municipal holdings (Hines and Horton, 2024), yet long municipal yields remained high relative to after-tax taxable yields (Green, 1993). Later work studied default risk, liquidity, callability and the taxation of discount bonds (Ang et al., 2010; Schwert, 2017; Brown, 2011).[3] The maturity pattern survives careful adjustment for these features: in default-free pre-refunded bonds (Chalmers, 1998), in swap rates free of bond-specific credit and liquidity risk (Longstaff, 2011) and in option-free bonds adjusted for credit and liquidity (Fleckenstein and Longstaff, 2025). On a common household after-tax scale, my bond-side adjustments leave a large long-maturity gap. My model explains most of this remaining gap through endogenous participation by investors with different taxes and costs of bearing rate risk.

Empirical research on municipal tax policy links state tax benefits to local ownership, financing conditions and risk sharing across states (Hendershott and Kidwell, 1978; Pirinsky and Wang, 2011; Babina et al., 2021; Moussawi, 2022, 2024). Changes in banks' tax benefits likewise affect municipal investment and access to financing (Hines and Horton, 2024; Dagostino, 2025; Benson and Marks, 2024).[4] I test how household and institutional tax effects vary with maturity. Household state-tax pass-through declines, while the yield reduction associated with bank qualification rises. This opposite pattern tests the participation mechanism against a single household tax rate applied across the curve.

Fiscal analyses of the municipal tax exemption examine its revenue cost and benefits to borrowers. Poterba and Ramírez Verdugo (2011) show that estimates of forgone revenue depend on how households would reinvest without the exemption. Garrett et al. (2023) show how auction competition affects the borrowing-cost savings from tax benefits. I evaluate reforms by running counterfactuals that incorporate the equilibrium impact of investor reallocation, capturing changes in tax receipts and financing costs.

## 3 Taxes, Ownership and the Municipal Puzzle

In this section, I describe the data and the tax benefits available to different investors, then show that municipal ownership extends well beyond high-tax households. I measure the municipal puzzle and examine whether bond characteristics or the choice of taxable benchmark explain its maturity pattern.

[3] Related pricing, tax-policy and maturity-clientele analyses include Fama (1977), Livingston (1982), Poterba (1986) and Brown and Sirmans (2011). Trading-cost evidence informs my liquidity adjustment (Harris and Piwowar, 2006; Green et al., 2007; Bagley et al., 2023). For surveys, Cestau et al. (2019) and Spreen and Gerrish (2022) summarize the long stream of literature on the municipal bond puzzle. Appendix C.4 compares the model with Green's benchmark.

[4] Municipal fund flows and fragility also affect yields and financing (Li et al., 2024; Adelino et al., 2026).

### 3.1 Data

I combine data on bond issuance, investor holdings and taxes. Mergent provides issuance records, ratings, bond characteristics and call schedules for municipal bonds and corporate bonds. Insurers' Schedule D filings with the National Association of Insurance Commissioners (NAIC) and a pension holdings panel collected through Freedom of Information Act requests allow me to examine institutional portfolios at the level of individual bond holdings. I use the Financial Accounts to measure aggregate holdings, the Survey of Consumer Finances to measure household balance sheets, and IRS Statistics of Income to document household tax-exempt interest income across income brackets.

The main yield comparison covers 2000–2020. Its taxable benchmark is the A-or-above High Quality Market (HQM) corporate zero-coupon curve from FRED. Federal Reserve treasury curves (Gürkaynak et al., 2007) provide the reference for the interest-rate response test. I measure trading and liquidity using municipal transactions from the Municipal Securities Rulemaking Board (MSRB) and corporate transactions from the Trade Reporting and Compliance Engine (TRACE) Enhanced database. Markit provides municipal and corporate credit default swap (CDS) quotes, and the Tax Foundation provides state tax schedules. Appendix D.6 details the sources and their coverage.

### 3.2 Municipal issuance

Table 1: Municipal issuance: descriptive statistics (sample means).

| Variable | Full sample | Tax-exempt, non-AMT | |
|---|---|---|---|
| | | AA and above (score $\leq 3$) | A or above (score $\leq 7$) |
| Offering yield (%) | 3.11 | 2.82 | 2.96 |
| Corporate benchmark yield (%) | 4.47 | 4.33 | 4.41 |
| Maturity at issuance (years) | 10.15 | 10.76 | 10.86 |
| Offering price (per $100 par) | 102.88 | 104.68 | 103.85 |
| Issue rating (1 = AAA, 3 = AA, 7 = A−) | 3.80 | 2.18 | 3.55 |
| Implied tax rate $1 - y^M/y^C$ | 0.29 | 0.35 | 0.33 |
| Share callable | 0.46 | 0.51 | 0.51 |
| Share tax-exempt, non-AMT | 0.92 | 1.00 | 1.00 |
| Sample size (CUSIPs) | 3,014,420 | 984,210 | 1,990,791 |

*Notes.* The table reports bond-level means for issues of 2000–2020 before the call and issuance-premium adjustments of Section 3.6, with one observation per municipal issue (CUSIP). The full sample contains all issues. The AA-and-above and A-or-above columns restrict to federally tax-exempt, non-AMT issues (not subject to the alternative minimum tax) with average agency score $\leq 3$ and $\leq 7$, the latter matching the A-or-above corporate credit band. The implied-tax row averages the bond-level ratio $1 - y^M/y^C$, winsorized at the first and ninety-ninth percentiles, so it need not equal one minus the ratio of the displayed yield means. Each row uses non-missing observations, so the full-sample rating mean uses the 2,292,092 rated issues. The corporate benchmark is the corporate zero yield for the issue month and maturity.

Table 1 describes municipal bonds issued during 2000–2020, with one observation per bond identifier (CUSIP). The full sample contains more than three million issues, with an average maturity of about ten years. Most pay federally exempt interest. Taxable municipal bonds account

for 5.8% of issues, bonds subject to the alternative minimum tax (AMT) account for 1.9%, and the remaining issues are tax-exempt, non-AMT bonds. Call provisions allow issuers to redeem bonds before maturity and are common in the data. Callable bonds account for 46% of all issues and 51% of the high-grade, tax-exempt samples, making callability important for comparing yields.

The table also reports two high-grade samples of tax-exempt, non-AMT bonds. The broader sample includes A-or-above issues (average agency score $\leq 7$ on the twenty-two-notch scale) and matches the credit band of the corporate benchmark used for the main yield curves. The state-tax and interest-rate tests of Section 6 use the AA-and-above sample ($\leq 3$) to minimize the impact of default risk.

### 3.3 Bond taxation and the representative-tax benchmark

For a taxable household, interest on taxable bonds is ordinary income, while interest on tax-exempt municipal bonds is federally exempt. Interest includes both coupon payments and annual original-issue-discount accrual. Institutions generally do not receive the full exemption benefit from holding municipal bonds. For example, property-and-casualty (P&C) insurers pay an effective federal tax of 5.25% on municipal interest, both before and after the 2017 Tax Cuts and Jobs Act (TCJA). Let $\tau_i^C$ and $\tau_i^M$ denote investor $i$'s effective tax rates on taxable and municipal interest. I define the relative after-tax share as the fraction of taxable interest retained divided by the fraction of municipal interest retained:

$$\xi_i \equiv \frac{1-\tau_i^C}{1-\tau_i^M}. \tag{1}$$

Investor $i$'s exemption benefit is $1-\xi_i = (\tau_i^C - \tau_i^M)/(1-\tau_i^M)$. A smaller $\xi_i$ means a larger benefit. When municipal interest is fully exempt, $\xi_i = 1-\tau_i^C$, so the benefit equals the tax rate on taxable interest. Appendix D explains the underlying tax rules.

The representative-tax benchmark uses the top-bracket household's tax treatment at every maturity. Write $y_T^M$ and $y_T^C$ for municipal and taxable yields at maturity $T$, and $H$ for this household. Under the benchmark, the two bonds offer the household the same after-tax yield, $y_T^M = \xi_H y_T^C$. I call $\xi_H y_T^C$ the household benchmark and $1-\xi_H$ the household wedge. The benchmark's yield-implied tax is therefore $1 - y_T^M/y_T^C = 1-\xi_H$. With fully exempt municipal interest, it equals $\tau_H^C$. Investors with different $\xi_i$ cannot all be indifferent at this yield ratio. Their participation in the municipal bond market is the central question of the model.

### 3.4 Investor taxes and ownership

The exemption benefit of municipal interest varies substantially across investors in the roughly $4 trillion municipal market. Table 2 summarizes these differences before the 2017 TCJA. For households and insurers, the net municipal bond exemption benefit is $1-\xi_i$ in (1). Households in higher tax brackets receive a larger benefit because they pay more tax on taxable interest. Exemption from home-state income tax adds to the benefit on in-state municipal bonds. Appendix D gives the detailed tax treatment of each investor group and the changes after the 2017 Act.

Table 2: Investor heterogeneity in the municipal market before the 2017 Tax Cuts and Jobs Act

| Investor type | Federal treatment | State treatment | Net exemption benefit |
|---|---|---|---|
| High-tax household | Interest exempt | Exempt on in-state bonds only | 0.35–0.43 |
| Middle-tax household | Interest exempt | Exempt on in-state bonds only | 0.25–0.33 |
| Low-tax household | Interest exempt | Exempt on in-state bonds only | 0.10–0.15 |
| P&C insurer | Exempt, less proration | Premium taxes, no distinction | 0.31 |
| Life insurer | Exempt, less proration | Premium taxes, no distinction | 0.21 |
| Bank | Exempt, financing cost disallowed | Varies by state | 0.15–0.33 |
| Other (pensions, etc.) | Varies | Varies | $< 0.05$ |

*Notes.* For households and insurers, the net exemption benefit is $1 - \xi_i$, from statutory rates and proration over 2000–2017. For banks it also deducts the financing-tax cost described below. Appendix Table 35 gives the values after the 2017 Act. Household values are federal marginal rates in three groups (35% and above, 25% to 33%, below 25%), and the top value includes the 3.8% net investment income tax from 2013. The quantitative model pools middle- and low-tax households into one lower-tax group, whose benefit with home-state taxes is 0.21 to 0.27 across states in 2015–2017, against 0.43 to 0.50 for the high-tax group (Table 3). Proration cuts a P&C insurer's losses-incurred deduction by 15% of its exempt interest and a life insurer's reserve deduction by the policyholders' share. Section 265(b) disallows the share of a bank's interest expense allocable to exempt holdings, of which 80% stays deductible for bank-qualified bonds. The bank value, net of that disallowance, is the break-even implied tax, equal to the bank's tax rate less the ratio of its annual financing-tax cost per dollar of municipal holdings to the taxable yield. That cost is 1.02% in 2000–2002 and 0.11% in 2015–2017, and the bank range covers the six three-year periods of 2000–2017 and maturities of two to thirty years. States generally tax insurers through premium taxes, which do not distinguish municipal from taxable interest, and banks through franchise or income taxes.

Insurers receive smaller benefits than high-tax households because exempt interest reduces their deductions. The reduction is less severe for P&C insurers than for life insurers, giving P&C insurers the larger benefit. Banks generally lose the interest deduction allocated to municipal holdings, with partial relief for bank-qualified bonds, so the table reports their benefit net of this financing-tax cost.[5] Tax-exempt institutions such as qualified pension funds and endowment funds already receive taxable interest free of current income tax and gain no additional benefit from the municipal exemption. Public pension funds in the sample hold almost no municipal bonds (Appendix C.8).

Figure 1 shows that municipal ownership spans these tax groups. Panel (a) reports aggregate holdings. Households and municipal funds hold the largest amounts, while banks and P&C insurers also hold substantial portfolios despite smaller exemption benefits. P&C insurers allocate about 40% of their bond portfolios to tax-exempt municipals and maintain a larger municipal portfolio share than life insurers throughout the sample (Appendix C.8). Panel (b) documents household tax-exempt interest income across income brackets. Assigning municipal-fund holdings to households and allocating household holdings using tax-exempt interest shares puts roughly half the market with high-tax households.[6] The remaining holdings by investors with smaller exemption benefits motivate a model in which investors with different tax treatment choose municipal and taxable bonds jointly. The quantitative model includes household groups by state of residence,

[5] The quantitative model charges the bank financing-tax cost separately. Table 3 therefore reports the bank's gross exemption benefit $1 - \xi_i$ before that cost.

[6] I use IRS Statistics of Income shares of tax-exempt interest on taxable returns, including fund distributions, to approximate household holdings by income group in 2010–2017. Adjusted gross income of at least $200,000 proxies for the high-tax group.

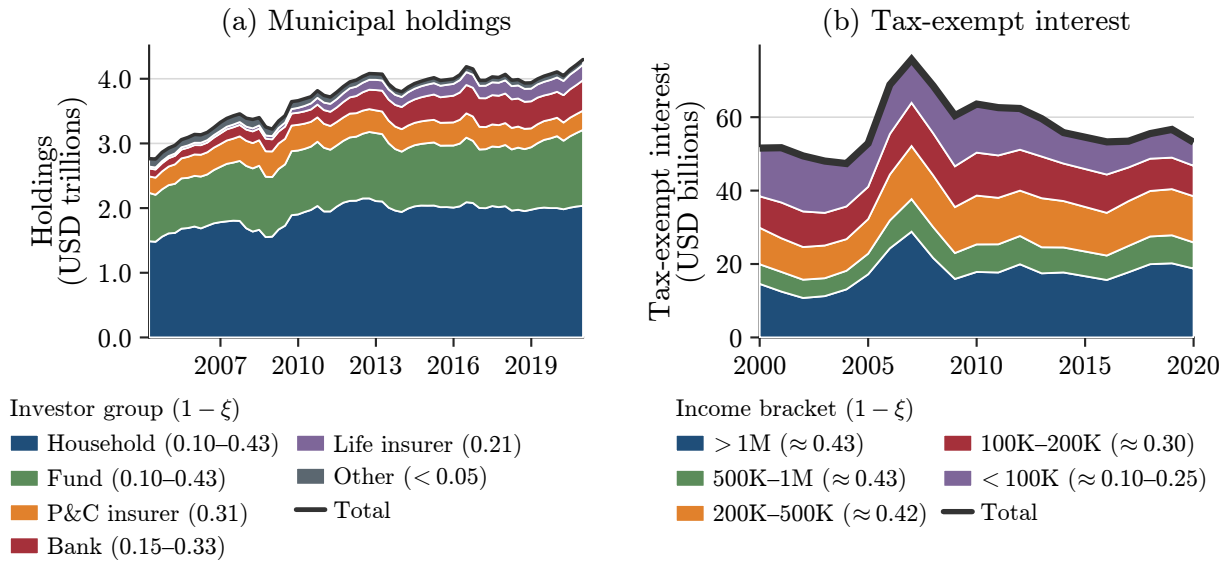



Figure 1: Municipal holdings and tax-exempt interest by investor group. Panel (a): direct holdings of taxable and tax-exempt municipal securities, 2005–2021 (USD trillions, Financial Accounts, with households including nonprofits). Panel (b): tax-exempt interest, an income flow, on taxable individual returns by adjusted-gross-income bracket, 2000–2020 (USD billions, IRS Statistics of Income). 2015 lacks comparable bracket detail and is omitted. Legend values in parentheses are net exemption benefits: in panel (a), the values before the 2017 Act from Table 2, with funds assigned their household shareholders' range, and in panel (b), approximate values by bracket.

insurers, banks and a residual group and uses this observed ownership to discipline the calibration (Section 5.1).

### 3.5 Limits to arbitrage

Tax rules and trading frictions limit the trades that could exploit yield differences or equalize investors' costs of bearing rate risk. Interest and short-sale payments incurred to carry exempt bonds generally lose their deduction. Consider a household that shorts a long-maturity corporate bond to hedge its municipal holdings. With the deduction denied, an equal-principal position earns coupon carry equal to the pretax municipal-minus-corporate yield difference. This carry is negative along the sample-average par curves and about $-95$ basis points a year at thirty years. Municipal shorts are scarce and costly because lenders owe tax on the payments that replace exempt coupons. Off-exchange swaps face eligibility restrictions, while futures losses can face deduction limits and deferral. The model summarizes these tax asymmetries and trading frictions through nonnegative bond positions and cash balances, ruling out short sales and borrowing. Removing the swap frictions for households leaves most of the gap. In the quantitative model for 2015–2017, giving every household fee-free swap access lowers the thirty-year gap by only 9.1% (Appendix D.5.4). Appendix D.5 examines the tax treatment, carry and residual risk of these trades.

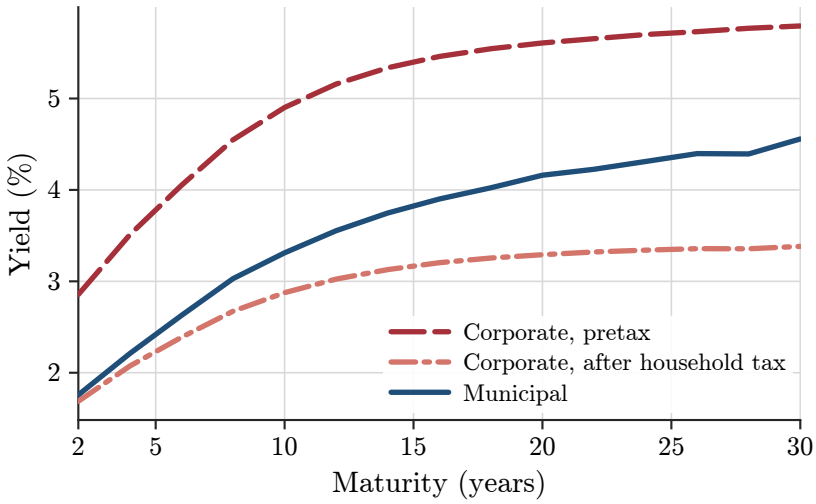



Figure 2: The municipal puzzle: A-or-above zero-coupon yields, 2000–2020. Municipal yields remove callability and the issuance premium. Averaging follows Appendix C.1.1. Appendix C.10.1 gives the AA-and-above counterpart.

### 3.6 The municipal puzzle

The municipal puzzle is a maturity pattern. Short municipal yields are close to the household benchmark, while long municipal yields substantially exceed it. To measure this pattern, I use the seasoned, option-free A-or-above corporate zero-coupon curve as the taxable comparison. High-grade corporate bonds are closer to municipals in credit and liquidity than treasuries. Treasury yields also contain a premium for safety and liquidity (Krishnamurthy and Vissing-Jorgensen, 2012), which would enter the measured gap if treasuries supplied the benchmark. Studies using treasury yields (Chalmers, 1998) or swap rates (Longstaff, 2011) as the taxable benchmark also find the municipal puzzle.

I construct municipal zero-coupon yields in the same credit band from offering prices and contractual cash flows. Issuance data provide broad coverage and a common measurement date. Offering yields also avoid the direct tax effects of capital gains and market discounts arising from prior secondary-market trades. I remove callability and the issuance premium to make municipal yields comparable with the seasoned, option-free corporate curve. The issuance premium is the extra yield at issuance relative to early secondary-market purchases of the same bond.[7] Let $y_t^C(T)$ denote the taxable yield and $y_{s,t}^M(T)$ the municipal yield after these two adjustments, where $T$ is maturity, $s$ is the issuer's state and $t$ is the issue date. The municipal puzzle is the gap between $y_{s,t}^M(T)$ and the household benchmark $\xi_{H,s,t}\, y_t^C(T)$, where $\xi_{H,s,t}$ is the relative after-tax share of a top-bracket household resident in state $s$ at date $t$, combining federal and home-state income taxes.

[7]The market-discount tax applies to later secondary purchasers. Its expected effect on resale values can also affect offering prices. The capital-gains adjustment includes this effect as well as taxes on realized gains.

The yield-implied tax expresses the same comparison in tax-rate units:

$$ImpliedTax_{s,t}(T) \equiv 1 - \frac{y^M_{s,t}(T)}{y^C_t(T)}. \tag{2}$$

The two measures satisfy

$$\underbrace{y^M_{s,t}(T) - \xi_{H,s,t}\, y^C_t(T)}_{\text{municipal yield gap}} = \underbrace{\left[(1-\xi_{H,s,t}) - ImpliedTax_{s,t}(T)\right]}_{\text{shortfall from the household wedge}}\, y^C_t(T). \tag{3}$$

For positive taxable yields, a positive gap means an implied tax below the household wedge $1-\xi_{H,s,t}$. In Figure 2, the municipal yield is near the household benchmark at the short end but about 118 basis points above it at thirty years. The implied tax falls from about 38% to 21% while the household wedge, one minus the ratio of averaged after-tax to pretax corporate yields, averages 41.3% and is nearly flat across maturities (Figure 7(d)).

The declining implied-tax pattern also appears within issuer-matched tax-exempt and taxable municipals (Section 3.8) and in the AA-and-above and unlimited-tax general-obligation samples (Appendices C.10.1 and C.3).

### 3.7 Comparability and bond-side adjustments

Studies such as Fleckenstein and Longstaff (2025) use secondary-market prices of option-free bonds to examine the municipal puzzle. To match that comparison, the puzzle already reflects two comparability adjustments, which remove callability and the issuance premium. I then assess three bond-side adjustments, for capital-gains taxation, liquidity and default risk, as explanations of the puzzle. I value the call option with a rate model fitted to yield differences between callable and noncallable bonds, measure the issuance premium from early trades of the same bond, and measure the bond-side differences from resale behavior under actual tax rules, matched trading costs and CDS spreads. Each bond-side adjustment measures the difference in compensation between municipal and corporate bonds, expressed in the household's after-tax yield units. Compensation common to both markets therefore cancels. Appendix C.1 explains the adjustments, and Appendix Table 14 reports their contributions by maturity. Appendix C.4 discusses the tax timing of taxable bonds in Green (1993).

At thirty years, the call and issuance-premium adjustments reduce the 160-basis-point spread over the household benchmark to the 118-basis-point municipal puzzle. The default, liquidity and capital-gains differentials explain 40% of the puzzle at ten years but only 12% at thirty. Subtracting them leaves a residual before the model of 103 basis points at thirty years for comparison with the model in Section 5.3. At two years, their sum exceeds the small puzzle, leaving −7 basis points, consistent with the convenience premium discussed in Fleckenstein and Longstaff (2025). The adjustments leave the central pattern intact. The residual before the model widens with maturity.

### 3.8 Issuer-matched evidence

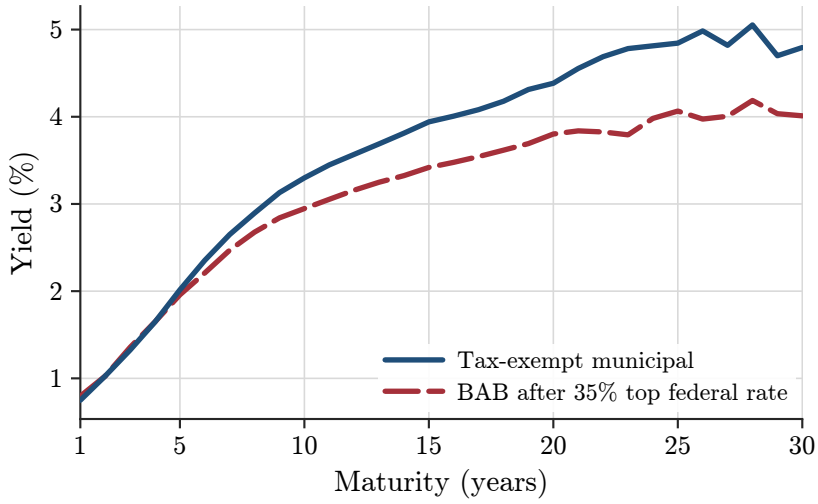



Figure 3: Tax-exempt and Build America Bond (BAB) offering yields by maturity, AA-and-above issues, April 2009–December 2010. BAB yields are multiplied by 0.65, one minus the 35% top federal rate in 2009–2010. Appendix Table 21 reports regressions with and without call-status matching. Appendix C.2 describes the sample.

Because the puzzle compares municipal with corporate bonds, its maturity pattern could come from differences between the two sets of borrowers, such as credit risk or liquidity, which the bond-side adjustments of Section 3.7 address. A cleaner test compares taxable and tax-exempt bonds sold by the same municipal issuer, which share its default risk and are matched on maturity, issue quarter and, in one specification, call status. Their main difference is tax treatment, so a yield gap that still widens with maturity shows that the pattern is not explained by issuer characteristics such as default risk. The Build America Bonds (BAB) program of 2009–2010 makes this comparison possible. It let municipal issuers sell taxable bonds, with a federal subsidy of 35% of their interest. Because the top federal household rate was also 35%, I multiply BAB yields by 0.65 to put them on the top-bracket household's federal after-tax scale and leave tax-exempt offering yields unchanged. Under the household benchmark, the two yields are then equal at every maturity. I estimate how their difference varies with maturity among bonds from the same issuer:

$$y_b \;=\; \alpha_1\, EXMP_b \;+\; \beta\,(EXMP_b \times n_b) \;+\; \varphi_C\, y^{C}_{n_b,t_b} \;+\; \lambda_{j(b)\times n_b\times q_b} \;+\; \varepsilon_b, \tag{4}$$

where $y_b$ is the offering yield for a tax-exempt bond and 0.65 times the offering yield for a BAB. The indicator $EXMP_b$ equals one for a tax-exempt bond and zero for a BAB, and $n_b$ is maturity in years. The fixed effect $\lambda_{j(b)\times n_b\times q_b}$ compares bonds from issuer $j(b)$ at the same maturity and in the same issue quarter $q_b$. The corporate yield $y^{C}_{n_b,t_b}$, measured at the bond's maturity and issue month $t_b$, controls for market yield movements within that quarter. The coefficient $\beta$ measures how the tax-exempt yield minus the household after-tax BAB yield changes with maturity. The household benchmark implies $\beta = 0$, and a positive $\beta$ is the municipal puzzle within a single issuer.

The estimated gradient $\beta$ is 3.4 basis points per year of maturity in the AA-and-above sample and 3.3 in the A-or-above sample, and 3.1 and 3.0 with call matching. With five-year maturity bins, the coefficients rise monotonically. In the AA-and-above sample with call matching, the $[25, 30]$-year coefficient is 63.9 basis points above the $[1, 5)$-year bin (Appendix Table 21). Figure 3 shows the same widening gap in mean yields. Long tax-exempt yields therefore exceed the household benchmark even against taxable bonds of the same issuer, so the puzzle does not come from differences between municipal and corporate borrowers.

# 4 Model

Motivated by the ownership structure of municipal bonds, I develop a term-structure model in which investors differ in tax treatment and balance-sheet exposure to interest rates. They choose bonds jointly across markets and maturities, subject to short-sale constraints. The model explains why investors with smaller exemption benefits hold tax-exempt bonds and how their participation changes yields relative to the household benchmark.

Section 4.1 presents the general model for multiple investor groups and markets. Sections 4.2–4.6 develop a two-investor, two-market analytical illustration, with proofs in Appendix A. Section 5 returns to the general model for calibration.

## 4.1 The general model

Time is discrete. Each investor solves an after-tax mean–variance portfolio choice problem across all assets and maturities, subject to short-sale constraints.

*Interest rates.* A single rate state $z_t$ drives the short rate $r$, the pretax return on cash, which is subject to a zero lower bound:

$$z_{t+1} = \rho z_t + \sigma_r \varepsilon_{t+1}, \qquad r(z_t) = \max\{\bar{r} + z_t,\, 0\}, \qquad 0 < \rho < 1. \tag{5}$$

Here $\varepsilon_{t+1}$ is standard normal, $\rho$ measures persistence, $\sigma_r$ the size of rate shocks and $\bar{r}$ the mean rate before the floor. Taxable cash earns this known log gross return $r(z_t)$ over the next period.

*Bonds and returns.* Bonds are zero-coupon and grouped into markets $a$. Tax rates differ by investor and market. For a given investor, all maturities in a market receive the same tax treatment. A bond of market $a$ with $T$ periods to maturity has price $P_T^a(z)$, pays one unit at maturity and becomes a $(T-1)$-period bond next period. Cash is the one-period taxable asset. Bond dollar supplies $Q_T^a$ are fixed.

An investor who buys a $T$-period bond today and sells it next period earns

$$\text{gross holding return} = 1 + R_{T,t+1}^a = \frac{P_{T-1}^a(z_{t+1})}{P_T^a(z_t)}, \qquad P_0^a = 1.$$

The price ratio is the resale value per dollar invested. It carries rate risk because next period's price depends on next period's rate state.

With one period to maturity, the payment is known, so the bond is fully riskless. A zero-coupon bond's yield is $y_T^a = -\log P_T^a/T$. Investors optimize over the one-period holding horizon.

As in the literature, I approximate the net holding return for portfolio choice as[8]

$$R_{T,t+1}^a \approx \mathcal{R}_{T,t+1}^a = \mu_T^a(z_t) + u_{T,t+1}^a. \tag{6}$$

Here $\mu_T^a(z_t) = \log \mathbb{E}_t[1 + R_{T,t+1}^a]$ is the mean of the approximated return used in the portfolio problem. The shock $u_{T,t+1}^a$ is the unexpected component of the log holding return, with conditional mean zero. Thus $\mu$ determines expected portfolio income, while $u$ determines risk. The approximation becomes exact in the continuous-time limit.

*Investors.* Investor $i$ has funding $W_i$ that can be invested in any bonds or cash, risk aversion $\gamma_i$ and a liability hedge $\ell_i$ from assets and liabilities beyond its bond portfolio. A positive $\ell_i$ raises net wealth when rates rise, as does a long-term liability whose value falls. A negative $\ell_i$ adds to bond losses when rates rise. Investor $i$ keeps the after-tax share $\xi_i^a = 1 - \tau_i^a$ of returns in market $a$, where $\tau_i^a$ is its tax rate on bonds in market $a$. Cash is taxed like taxable bonds, which form market $C$, so its after-tax share is $\xi_i^C$.

The investor chooses dollar positions $x_{i,T}^a \geq 0$ across all markets $a$ and all maturities $T$, keeps the rest of its funding in cash, $x_i^0 = W_i - \sum_{a,T} x_{i,T}^a \geq 0$, and maximizes expected next-period wealth less a penalty for its variance. With pretax returns measured by $\mathcal{R}$, next-period after-tax wealth is $\widehat{W}_i'$, and the problem is

$$\begin{aligned} \max_{x_i \geq 0,\, x_i^0 \geq 0} \quad & \mathbb{E}_t[\widehat{W}_i'] - \frac{\gamma_i}{2} \operatorname{Var}_t(\widehat{W}_i'), \\ \text{subject to} \quad & \widehat{W}_i' = W_i + \xi_i^C r(z_t)\, x_i^0 + \sum_{a,T} \xi_i^a x_{i,T}^a \mathcal{R}_{T,t+1}^a + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{7}$$

Next-period wealth adds after-tax cash income, after-tax bond returns and the change in outside wealth to current funding. Investors choose portfolios each period, taking prices as given and conditioning on today's rate state. Nonnegative positions rule out short sales and borrowing (Section 3.5).

*Required returns.* The portfolio problem gives the Karush–Kuhn–Tucker conditions

$$\underbrace{\xi_i^a \mu_T^a - \xi_i^C r}_{\text{after-tax return over cash}} \leq \underbrace{\gamma_i \xi_i^a \operatorname{Cov}_z\!\left(u_{T,t+1}^a, \widehat{W}_i'\right)}_{\text{cost of the risk the bond adds}} + \underbrace{\kappa_i,}_{\text{shorting constraint on cash}} \tag{8}$$

with equality whenever $x_{i,T}^a > 0$. The covariance measures the risk the bond adds to total wealth, conditional on today's state. The multiplier $\kappa_i \geq 0$ on the shorting constraint on cash is zero

[8] I follow the discrete-time return approximation in Greenwood et al. (2024), Section 2.1. See also Greenwood et al. (2023), Section III.A, and Hanson and Stein (2015), footnote 18. Appendix A.1.1 gives the precise return convention and wealth equation.

when the investor holds cash and can invest another dollar without borrowing. A held bond must compensate for both its added risk and, when this constraint binds, the dollar it displaces from other bonds. If a bond offers less than an investor's required return in equilibrium, that investor holds none. Differences in taxes and balance sheets therefore allow the same bond to suit one investor but not another in equilibrium.

*Equilibrium.* Prices clear every market at every rate state:

$$\sum_i m_i x^a_{i,T}(z) = Q^a_T, \tag{9}$$

where $m_i$ is the mass of type $i$, converting its positions and funding into aggregate holdings $m_i x^a_{i,T}$ and funding $m_i W_i$. In the calibration, one unit of mass represents one dollar of baseline portfolio assets (debt securities and cash). Thus $m_i$ measures a type's portfolio size, and $W_i$ is the share available for the bonds and cash included in the model. The rate process, supplies, taxes, funding, risk aversion and liability hedges are exogenous inputs. Equilibrium prices imply holding returns for which investors satisfy (8) and all markets clear. Who holds each bond is determined by these conditions. Appendix Table 6 collects the notation.

### 4.2 Analytical two-investor, two-market illustration

To isolate the mechanism, I use a special case of the general model with two investors, two markets and rate risk alone. A high-tax household $H$ and a lower-tax investor $I$ choose municipal bonds $M$, taxable bonds $C$ and cash. Municipal maturities begin at $T = 1$, while risky taxable bonds begin at $T = 2$. Maturity $T$ counts periods, which last two years in the quantitative application. The two determinants of required returns are the investor's tax treatment and the cost of adding risk to its portfolio.

The relative after-tax share of Section 3.3 summarizes the tax comparison:

$$\xi_i = \frac{\xi^C_i}{\xi^M_i} = \frac{1-\tau^C_i}{1-\tau^M_i}, \qquad \xi_H < \xi_I.$$

The household's smaller $\xi_H$ means a larger exemption benefit, $1 - \xi_H$. For portfolio choice, $\xi_i$ is a sufficient statistic for an investor's tax treatment, because dividing its after-tax payoffs by $1 - \tau^M_i$ keeps municipal returns in full and multiplies taxable returns and cash income by $\xi_i$, without changing its optimal portfolio.[9]

In the general model, prices are nonlinear because the rate floor and the shorting constraint on cash can bind, and participation can change as rates move. Future nonlinearities also affect today's prices through resale values. For tractability, the illustration computes prices as if the same investors held each bond at every future rate state, while allowing their positions to vary. I call this fixed pattern of positive holdings a participation branch. The illustration also removes the

[9] With payoffs divided by $\xi^M_i$, risk aversion $\gamma_i \xi^M_i$ and liability hedge $\ell_i / \xi^M_i$, the objective is $1/\xi^M_i$ times the original, up to a portfolio-independent constant. Below, $\gamma_i$ and $\ell_i$ denote these normalized values.

rate floor, setting $r = \bar{r} + z$, and the shorting constraint on cash, so $\kappa_i = 0$. Each log price is then a constant plus a constant multiple of the rate state, $\log P_T^a(z) = -A_T^a z - C_T^a$, where $A_T^a$ is the bond's rate loading defined below. The quantitative model in Section 5 retains the rate floor and the shorting constraint on cash and allows participation to change with rates. It therefore accounts for changes in who holds a bond over its remaining life and how these affect its rate sensitivity (Appendix equation (45)).

Define a bond's rate loading as $A_T^a \equiv -\partial \log P_T^a / \partial z$, which measures its proportional price decline when the rate state rises and is the model's counterpart of duration. It is an equilibrium outcome, as Section 4.5 shows. Rate shocks affect the holding return through next period's resale price, when the bond has $T-1$ periods remaining. Define the investor's net rate exposure as

$$\widehat{D}_i \equiv -\frac{\partial \widehat{W}_i'}{\partial z_{t+1}} = \sum_T x_{i,T}^M A_{T-1}^M + \xi_i \sum_T x_{i,T}^C A_{T-1}^C - \ell_i. \tag{10}$$

The derivative measures how much next-period wealth falls when the rate state rises, holding today's portfolio fixed. Liabilities whose value also falls offset losses on bonds, so a larger liability hedge $\ell_i$ lowers this net rate exposure.

The investor's cost per unit of additional rate exposure is $\gamma_i \sigma_r^2 \widehat{D}_i$. The portfolio first-order conditions are

$$\begin{aligned} \mu_T^M - \xi_i r &\le \gamma_i \sigma_r^2 \widehat{D}_i \, A_{T-1}^M, \\ \mu_T^C - r &\le \gamma_i \sigma_r^2 \widehat{D}_i \, A_{T-1}^C, \end{aligned} \tag{11}$$

with equality for a positive position. The municipal condition compares $\mu_T^M$ with after-tax cash, $\xi_i r$. The taxable condition, already divided by $\xi_i$, is on a pretax scale and compares $\mu_T^C$ with $r$. In both lines the right side is the return the investor requires for the rate risk added by the bond. The quantitative model uses common risk aversion in these normalized payoff units, so its investors differ in tax treatment $\xi_i$ and net rate exposure $\widehat{D}_i$, which depends on portfolio choice and the liability hedge. The analytical benchmark allows $\gamma_i$ to differ.

These conditions separate the two forces behind participation. The household's larger exemption benefit lowers its required municipal return. A lower unit cost of rate risk also lowers the required return, increasingly so for a bond with a larger loading.

### 4.3 Participation across markets and maturities

In the illustration, the cash alternative alone determines participation at the riskless maturity. With $A_0^M = 0$, a municipal holder requires $\mu_1^M = \xi_i r$. The household accepts the lowest return because it keeps the smallest share of taxable interest. With a positive short rate and a slack shorting constraint on cash, it holds the municipal supply and

$$y_1^M = \mu_1^M = \xi_H r, \qquad y_1^C = r. \tag{12}$$

At the riskless short end the municipal yield equals the household benchmark exactly (Appendix Proposition A.2).

At a risky maturity, compare the returns the two investors require on the same municipal bond. The lower-tax investor requires less when its duration benefit exceeds the difference in exemption benefits:

$$\underbrace{\sigma_r^2 A_{T-1}^M(\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)}_{\text{duration benefit}} > \underbrace{(\xi_I - \xi_H)r}_{\text{difference in exemption benefits}}. \tag{13}$$

All quantities are evaluated at equilibrium. A positive $\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$ means that the lower-tax investor requires less return per unit of rate exposure. With the same $\gamma_i$, this happens when its liability hedge $\ell_I$ is large enough to give it a lower equilibrium $\widehat{D}_I$. Where the duration benefit is smaller than the difference in exemption benefits, the household requires less return and holds the bond. Where it is larger, the lower-tax investor requires less and holds the bond. Both can hold a maturity at which the two terms are equal (Appendix Theorem A.1).

In the illustration, the municipal rate loading $A_{T-1}^M$ is endogenous and increases with maturity (Section 4.5). At a given equilibrium, the difference in unit costs of rate risk is common across maturities. The duration benefit therefore grows with maturity, while $(\xi_I - \xi_H)r$ stays fixed. Lower-tax participation in municipal bonds can begin at the long end, from a threshold maturity $T^*$. For taxable bonds, bond returns and cash income receive the same tax treatment, so there is no difference in exemption benefits to overcome. The investor with lower $\gamma_i \widehat{D}_i$, the lower-tax investor in this illustration, requires less compensation for rate risk at every risky maturity and holds risky taxable bonds.

Figure 4 shows sharp sorting when rate risk is the only risk. The lower-tax investor holds all risky taxable bonds and the long municipal maturities, while the household holds the short municipal maturities. Adding a small bond-specific shock allows joint holdings at more than one maturity. In the otherwise identical illustration of Appendix A.2.9, both investors hold maturities 17 to 30, while municipal yields change by at most 0.33 basis points. Bond-specific risk thus matters for who holds each bond but has little effect on prices. The quantitative model includes this risk and multiple investor types, which generate overlapping holdings and partial segmentation.

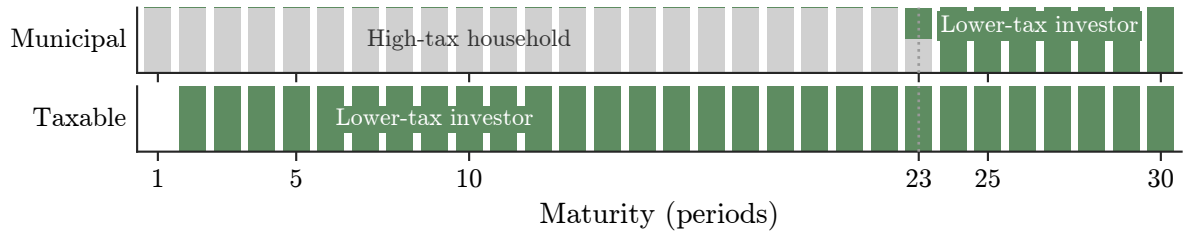



Figure 4: Participation in the two-investor illustration without bond-specific risk. Bars show supply shares at the mean rate state. Both investors hold municipal maturity $T^* = 23$ (dotted line). The high-tax household holds shorter municipal maturities, and the lower-tax investor holds longer municipal maturities and risky taxable bonds. The one-period taxable claim is cash. Appendix A.2.7 gives the parameters and checks.

### 4.4 Three channels of the one-period return gap

Using this allocation, I now compare returns on municipal and taxable bonds of the same maturity. Define the one-period return gap relative to the household benchmark as

$$g_T = \mu_T^M - \xi_H \mu_T^C. \tag{14}$$

The portfolio conditions separate the gap into three channels. Section 4.5 then translates holding returns into yields.

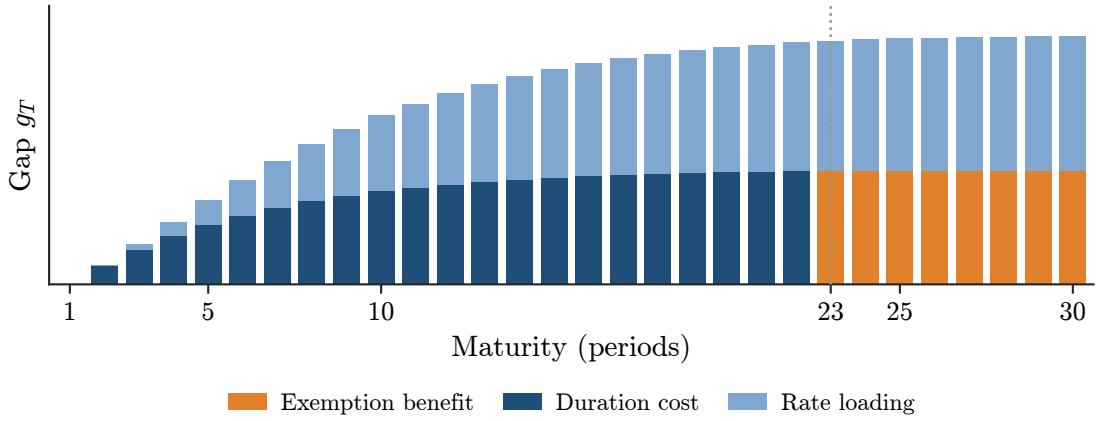



Figure 5: One-period return gap by channel. The allocation is that of Figure 4. Bars decompose $g_T$ into the exemption-benefit, duration-cost and rate-loading terms of (15). The vertical scale is omitted because levels depend on the illustrative parameters.

**Proposition 1 (Three channels of the one-period return gap).** In the two-investor affine allocation just described, the lower-tax investor alone holds risky taxable bonds. For municipal bonds, use the household's condition ($i = H$) below $T^*$ and the lower-tax investor's condition ($i = I$) from $T^*$ onward. Subtracting $\xi_H$ times the taxable holder's condition from the municipal holder's condition in (11) gives

$$\begin{aligned} g_T = \; & \underbrace{(\xi_i - \xi_H) r}_{\text{exemption benefit}} + \underbrace{\sigma_r^2 A_{T-1}^M (\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I)}_{\text{duration cost}} \\ & + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I (A_{T-1}^M - \xi_H A_{T-1}^C)}_{\text{rate loading}}. \end{aligned} \tag{15}$$

Each term compares one element of the municipal holder's condition with the corresponding element of $\xi_H$ times the taxable holder's condition. The exemption-benefit term compares the municipal holder's gain from tax exemption with the household's. The duration-cost term compares the two holders' costs per unit of rate exposure. The rate-loading term compares the municipal loading with the taxable loading scaled by $\xi_H$. Appendix A.2.4 states and proves the decomposition.

*Exemption benefit.* Below $T^*$, the household holds the municipal bond. The benchmark already uses its tax treatment, so this term is zero. From $T^*$ onward, the lower-tax investor holds both mu-

nicipal and taxable bonds. Because it pays less tax on taxable interest, tax exemption saves it less than it saves the high-tax household. It therefore requires a higher municipal return than the household benchmark predicts, and the difference in exemption benefits contributes $(\xi_I - \xi_H)r$ to the gap.

*Duration cost.* At risky maturities below $T^*$, the household alone holds municipal bonds, while the lower-tax investor holds taxable bonds. Rate risk is more costly for the household, so it needs more return to bear the same rate exposure. This difference contributes $\sigma_r^2 A_{T-1}^M(\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)$ to the gap, even though the lower-tax investor holds no municipal bonds at these maturities. It is the duration benefit in (13), expressed as the household's higher risk cost. From $T^*$ onward, the lower-tax investor holds both bonds and prices them with the same unit cost of rate risk, so this term is zero.

*Rate loading.* When taxes scale returns, they also scale risk. The household benchmark scales taxable yields and rate exposure by $\xi_H$, predicting $A_{T-1}^M = \xi_H A_{T-1}^C$. In the illustration, lower-tax participation raises municipal rate loadings relative to taxable loadings, $A_{T-1}^M \geq \xi_H A_{T-1}^C$. The extra loading contributes positively to the gap when the lower-tax investor's net rate exposure is positive, as in the illustration. The contribution equals the extra loading times $\gamma_I \sigma_r^2 \widehat{D}_I$, the lower-tax investor's unit cost of rate risk. The lower-tax investor holds the taxable bond at every risky maturity, so this term applies on both sides of $T^*$. I defer the derivation of the loading gap $A_{T-1}^M - \xi_H A_{T-1}^C$ to Section 4.5.

All three channels vanish at the riskless maturity, and $g_1 = 0$. Since $A_0^M = A_0^C = 0$, the duration-cost and rate-loading terms vanish. The exemption-benefit term is also zero because the household holds the riskless municipal bond. Figure 5 shows how the channels combine across maturities.

### 4.5 Rate loadings and the yield gap

Under the household benchmark, $y_T^M = \xi_H y_T^C$ at every rate state, so the same fraction scales yield sensitivity and price exposure to rates: $A_T^M = \xi_H A_T^C$. This is the sense in which taxes scale risk as well as returns. Lower-tax investors keep a larger fraction of taxable interest, so the cash-return component of their required municipal return rises more with rates. Portfolio reallocation also changes risk compensation in both markets. I first derive the resulting loading gap $A_{T-1}^M - \xi_H A_{T-1}^C$ of Proposition 1, then show how the one-period return gaps $g_T$ accumulate into the yield gap $y_T^M - \xi_H y_T^C$.

*Loadings accumulate return responses.* Because $\mu_T^a(z)$ is the log expected gross holding return, today's log price equals the log expected resale value less the required holding return:

$$\log P_T^a(z) = -\mu_T^a(z) + \log \mathbb{E}_z\left[P_{T-1}^a(z')\right].$$

Let $s_T^a = \partial \mu_T^a / \partial z$ denote the response of the required holding return to the rate state. With affine log prices and $\mathbb{E}_z[z'] = \rho z$ for next-period state $z'$, the log expected resale value has loading $\rho A_{T-1}^a$. Taking the negative state derivative gives

$$A_T^a = \rho A_{T-1}^a + s_T^a, \qquad A_0^a = 0. \tag{16}$$

A bond's loading carries forward the loading of the bond it becomes next period, weighted by rate persistence, and adds the response of its current required return. Subtracting $\xi_H$ times the taxable recursion from the municipal recursion gives

$$A_T^M - \xi_H A_T^C = \rho\big(A_{T-1}^M - \xi_H A_{T-1}^C\big) + \big(s_T^M - \xi_H s_T^C\big).$$

The loading gap therefore accumulates the response gaps $s_T^M - \xi_H s_T^C$. Under the household benchmark, municipal required returns are $\xi_H$ times taxable required returns, so $s_T^M = \xi_H s_T^C$ and the loading gap is zero. A loading gap arises when municipal required returns respond to rates by more than $\xi_H$ times taxable required returns.

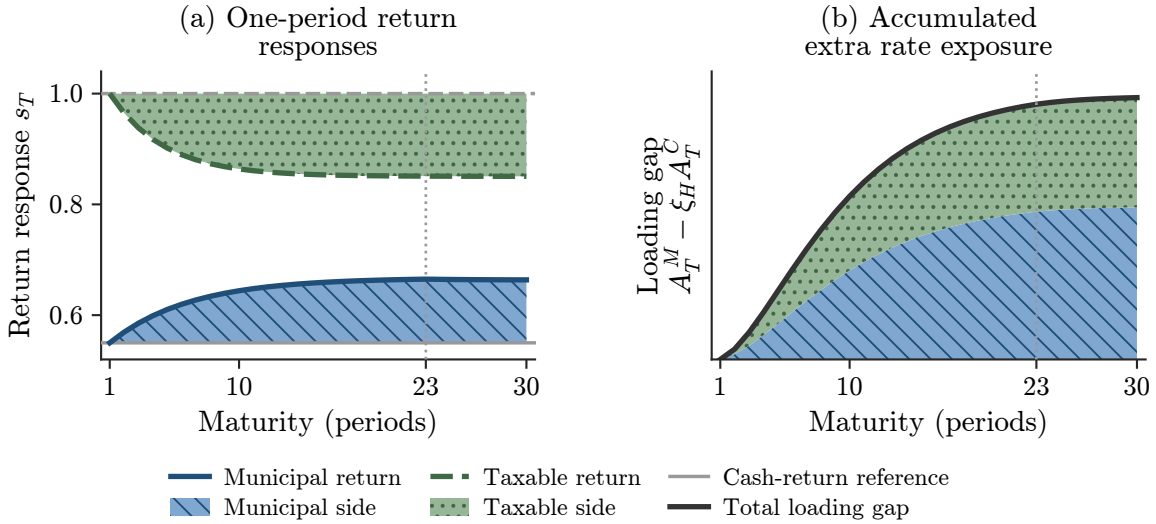



Figure 6: Rate responses in the illustration of Figure 4. Panel (a) shows municipal and taxable holding-return responses. Gray lines show the household's after-tax cash response, $\xi_H$, and the pretax cash response, 1. Blue shading combines the municipal tax mix and portfolio response into $s_T^M - \xi_H$. Green shading shows $1 - s_T^C$. Panel (b) accumulates $s_T^M - \xi_H$ and $\xi_H(1 - s_T^C)$ into the loading gap. The dotted line marks the shared municipal maturity $T^* = 23$.

*How the return gap responds to rates.* The response gap is the response of the return gap $g_T = \mu_T^M - \xi_H \mu_T^C$ of Section 4.4 to the rate state, $s_T^M - \xi_H s_T^C = \partial g_T / \partial z$. With the participation pattern fixed, the loadings $A_{T-1}^a$ do not depend on the rate state, and the short rate moves one for one with it while $r > 0$. Differentiating (15), with $i$ the municipal holder as in Proposition 1, gives

$$s_T^M - \xi_H s_T^C = \underbrace{\xi_i - \xi_H}_{\text{exemption benefit}} + \underbrace{\sigma_r^2\Big(\gamma_i A_{T-1}^M \frac{\partial \widehat{D}_i}{\partial z} - \xi_H \gamma_I A_{T-1}^C \frac{\partial \widehat{D}_I}{\partial z}\Big)}_{\text{portfolio reallocation}}. \tag{17}$$

The return gap thus responds to rates in two ways. First, the exemption benefit $(\xi_i - \xi_H)r$ grows with the short rate. Below $T^*$, the household holds the municipal bond and this term is zero. From $T^*$ onward, the lower-tax investor holds it. The cash-return component of its required municipal return

then rises with rates by $\xi_I$, rather than the household's smaller $\xi_H$. Second, $\partial \widehat{D}_i/\partial z$ measures how investor $i$'s net rate exposure changes as rates rise. Portfolio reallocation changes these exposures and hence the risk compensation each investor requires on every risky bond it holds.

The reallocation works as follows. As rates rise, the difference in exemption benefits $(\xi_I - \xi_H)r$ widens, so municipal bonds become less attractive to the lower-tax investor. It sells municipal bonds of the threshold maturity $T^*$ to the household, so the household's net exposure rises and the lower-tax investor's falls. As a result, $\partial \widehat{D}_H/\partial z > 0 > \partial \widehat{D}_I/\partial z$. Below $T^*$, both changes raise the response gap. The household requires more compensation on all its risky municipal bonds, including those it holds alone, and the lower-tax investor requires less on its taxable bonds. From $T^*$ onward, the lower-tax investor holds both bonds, so its falling exposure lowers its required returns in both markets when rates rise. Intuitively, the household bears more rate risk when rates rise and therefore requires more compensation, which raises the rate loadings of the bonds it holds. The opposite holds for the lower-tax investor.

**Proposition 2 (Participation and rate responses).** On an affine participation branch of the two-investor model with $r > 0$, suppose the household alone holds municipal maturities below $T^*$, both investors hold $T^*$, and the lower-tax investor alone holds longer municipal maturities and the risky taxable bonds, as in Figure 4. The municipal holders' tax mix equals $\xi_H$ when the household holds the bond alone, $\xi_I$ when the lower-tax investor holds it alone, and $\xi^* = \omega_H \xi_H + \omega_I \xi_I$ under joint holdings. Here $\omega_i$ is each investor's share of total risk tolerance.[10] Differentiating (11) and clearing (9) gives

$$
\begin{aligned}
s_T^M &= \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M \frac{\partial \widehat{D}_H}{\partial z}, & \text{household alone,} \\ \xi^*, & \text{both investors,} \\ \xi_I + \gamma_I \sigma_r^2 A_{T-1}^M \frac{\partial \widehat{D}_I}{\partial z}, & \text{lower-tax investor alone,} \end{cases} \\
s_T^C &= 1 + \gamma_I \sigma_r^2 A_{T-1}^C \frac{\partial \widehat{D}_I}{\partial z}.
\end{aligned}
\tag{18}
$$

Figure 6(a) groups the same response gap by market:

$$
s_T^M - \xi_H s_T^C = \underbrace{s_T^M - \xi_H}_{\text{municipal side}} + \underbrace{\xi_H (1 - s_T^C)}_{\text{taxable side}}.
$$

The municipal side measures how much more the municipal required return responds than the household's after-tax cash return, and the taxable side how much less the taxable required return responds than the pretax cash return. Both sides are positive at every risky maturity in the illustration. Accumulated through (16), they give a positive loading gap, $A_T^M > \xi_H A_T^C$ for $T \geq 2$ (Figure 6(b)). Municipal bonds therefore carry more rate exposure than the household benchmark implies, and with the lower-tax investor's positive net exposure, the rate-loading channel raises $g_T$. The appendix proves the response formulas and the positive loading gap under condition (S)

[10] The weights are $\omega_i = (m_i/\gamma_i)/(m_H/\gamma_H + m_I/\gamma_I)$.

in Appendix equation (72), which keeps the lower-tax investor's falling risk compensation from reversing the sign of the loading gap (Proposition 2).

*From return gaps to yield gaps.* A bond with $T$ periods to maturity earns a sequence of one-period returns as it ages: the return on a $T$-period bond today, on a $(T-1)$-period bond next period, and so on until it matures. Its yield averages these expected returns, less a convexity term (Appendix equation (54)). Applying this relation to both markets gives

$$y_T^M - \xi_H y_T^C = \frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t[g_{T-j}(z_{t+j})] - \text{convexity difference}. \tag{19}$$

The left side is the municipal yield above the household benchmark, the model counterpart of the measured puzzle. The right side averages the one-period gaps the bond earns over its remaining life, so the yield gap inherits the three channels of Proposition 1. A long municipal bond earns the exemption-benefit and rate-loading terms while its remaining maturity is at least $T^*$, and the duration-cost and rate-loading terms after it falls below $T^*$. Since $g_1 = 0$ and the one-period gaps rise with maturity in the illustration (Figure 5), longer bonds average larger gaps. The convexity difference is small, so the equilibrium municipal yield exceeds the household benchmark at every maturity beyond one period, by more at longer maturities.

### 4.6 Empirical predictions

The illustration motivates three empirical tests. The representative-tax benchmark predicts full pass-through of household tax changes at every maturity and no response to tax benefits specific to lower-tax institutions. In the illustration, household taxes determine the riskless short municipal yield, while longer yields also reflect lower-tax participants' tax treatment and portfolio responses. P1 uses tax variation specific to households to test whether household tax pass-through declines with maturity. P2 uses tax variation specific to banks to test whether lower-tax institutions' benefits affect longer yields while leaving the riskless short yield unchanged. P3 tests the associated prediction that municipal rate sensitivity rises with maturity relative to the taxable benchmark.

The outcome of interest is the yield-implied tax $\tau_T^{Y,\text{impl}} = 1 - y_T^M/y_T^C$ from (2). Investor $i$'s pass-through measures how much this equilibrium implied tax changes when investor $i$'s exemption benefit changes:

$$\mathcal{P}_i^Y(T) = \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad i \in \{H, I\}. \tag{20}$$

Here, pass-through of one means full transmission and zero means no effect.

By (12), the implied tax at the riskless shortest maturity is $1 - \xi_H$. Therefore

$$y_1^M = \xi_H y_1^C \quad \Longrightarrow \quad \mathcal{P}_H^Y(1) = 1, \qquad \mathcal{P}_I^Y(1) = 0. \tag{21}$$

Only the household's benefit enters this pricing relation.

At longer maturities, lower-tax participation changes whose exemption benefit affects returns. A larger benefit makes municipal bonds more attractive to its recipient and changes portfolios in both markets. The household's benefit therefore no longer has the exclusive role it has at the riskless short end. An institution's tax treatment affects returns where it holds bonds, while portfolio substitution transmits the effect to other risky maturities.

Pass-through is an equilibrium derivative. Changing one investor's after-tax share re-solves portfolios, loadings and both yield curves, holding non-tax primitives fixed. At the long end, the lower-tax investor holds both municipal and taxable bonds, so the long-end municipal-to-taxable yield ratio equals its after-tax share $\xi_I$, up to a relative convexity term whose tax derivatives are small (Appendix Propositions A.6(iii) and A.8). Long-end pass-through is therefore below 1 for the household and above 0 for the lower-tax investor:

$$\mathcal{P}_H^Y(\infty) < 1, \qquad \mathcal{P}_I^Y(\infty) > 0. \tag{22}$$

Between the two ends, household pass-through falls and the lower-tax investor's rises with maturity in the illustration (Appendix A.2.7). Appendix A.2.8 gives the counterpart when bond-specific risk extends joint holdings across long maturities.

**Household taxes (P1).** A home-state exemption gives resident households an additional benefit on their state's bonds. That benefit is fully reflected in implied state taxes at the riskless short end and less strongly at longer maturities in the quantitative model. The representative-tax benchmark predicts full pass-through at every maturity.

**Institutional taxes (P2).** A benefit specific to a lower-tax institution has no effect at the riskless short end and matters more at longer maturities. Bank qualification tests this prediction. The measure is the share of the bank-only benefit reflected in yields (Section 6.2). The representative-tax benchmark gives these benefits no role.

**Interest-rate responses (P3).** Section 4.5 shows why municipal bonds carry more rate exposure than the household benchmark predicts. Since $y_T^a = -\log P_T^a/T$, yield sensitivity is $A_T^a/T$. Comparing bonds at the same maturity cancels this division by $T$. For the illustration's affine branch and its long-end continuation, this gives

$$\frac{\partial y_T^M/\partial z}{\partial y_T^C/\partial z} = \frac{A_T^M}{A_T^C}, \qquad \frac{A_1^M}{A_1^C} = \xi_H, \qquad \lim_{T\to\infty} \frac{A_T^M}{A_T^C} = \xi_I > \xi_H.$$

At long maturities, the lower-tax investor alone holds both municipal and taxable bonds. Its changing rate exposure affects both municipal and taxable bond loading recursions through (18). This common adjustment cancels in the limiting ratio, leaving $\xi_I$ (Appendix Proposition A.6). It still affects each bond's loading and return response. On the illustrated branch, the ratio rises with maturity from $\xi_H$ toward $\xi_I$. P3 tests whether municipal yield sensitivity relative to same-maturity taxable bonds increases with maturity. The representative-tax benchmark predicts the constant ratio $\xi_H$.

### 4.7 Other motives for lower-tax participation

The illustration uses liability hedging to explain why lower-tax investors hold long municipal bonds. A lower-tax investor can also hold long municipal bonds at lower cost if it trades less often or more cheaply than households, or diversifies default risk across markets. When these advantages grow with maturity, they can outweigh the maturity-invariant difference in exemption benefits and sustain participation in long municipal bonds.

The pricing mechanism extends to these motives when the relevant advantage is neutral across taxable and tax-exempt bonds. Neutrality means that the advantage itself creates no return gap relative to the household benchmark when the household holds both bonds. Lower-tax participation then brings the participant's smaller exemption benefit into the return comparison. Appendix A.3 states the result, and Appendix A.3.4 gives solved trading-cost and default-diversification examples. A preference for municipal bonds as such, including warm glow from financing local projects, lowers the return an investor requires on them. Such a preference alone does not explain yields above the household benchmark.

I use liability hedging in the quantitative application because rate risk is common to both markets and balance sheets provide a measurable source of differences in investors' risk-bearing costs. Observed differences in maturity holdings help discipline the liability hedges.

## 5 Quantitative Results

I now ask how much of the municipal puzzle the model of Section 4 explains when taxes, balance sheets and bond supplies take their measured values. The quantitative model keeps what the analytical benchmark set aside: many investor types, separate state municipal markets, the rate floor, the shorting constraint on cash and participation that changes with rates. Section 5.1 describes its markets and investors, Section 5.2 explains what is measured and what is estimated, and Section 5.3 compares its yield curves with the data and accounts for the puzzle.

### 5.1 The quantitative model

The model has 45 municipal markets and one taxable market. Each state with a positive personal income tax and the District of Columbia has its own municipal market. The seven states with no state income tax are combined into one market because investors treat their bonds alike in the model. The taxable market pools treasury and corporate supply. A model period is two years, giving fifteen maturities from two to thirty years in each municipal market. Cash and the two-year municipal claim are riskless.

Six investor groups hold these bonds: high-tax households, lower-tax households, P&C insurers, life insurers, banks and a tax-neutral residual group, which consists mainly of foreign investors, pensions, governments and nonprofits. They differ in three ways that matter for the mechanism of Section 4. The first is tax treatment, summarized by the exemption benefit $1 - \xi_i$, which follows from tax law (Table 3). High-tax households face a federal marginal rate of at least 35%.

Table 3: Investor groups in the quantitative model, 2015–2017

| Group | Types | Types differ by | Gross exemption benefit $1-\xi_i$ | Portfolio assets $m_g$ |
|---|---|---|---|---|
| High-tax households | $7 \times 45 = 315$ | Outside exposure, home state | 0.43–0.50 | 6.97 |
| Lower-tax households | $7 \times 45 = 315$ | Outside exposure, home state | 0.21–0.27 | 8.09 |
| P&C insurers | 2 | Outside exposure | 0.31 | 1.11 |
| Life insurers | 3 | Outside exposure | 0.21 | 3.05 |
| Banks | 1 | None | 0.35 | 4.83 |
| Residual group | 7 | Outside exposure | 0 | 26.47 |
| Total | 643 | | | 50.52 |

*Notes.* Each household group has seven outside-exposure types in each of the 45 markets and in the one-market model used to estimate the coefficients (Section 5.2). Estimated coefficients map each household type's measured outside assets and debt into exposure. Institutional exposures are estimated directly (Appendix Table 9). The gross exemption benefit $1-\xi_i$, with $\xi_i = (1-\tau_i^C)/(1-\tau_i^M)$, follows from tax law. Households multiply the federal and home-state after-tax shares of each SCF tax unit, with the range running across states, and institutions face the 35% corporate rate, with proration for insurers. Banks also pay a financing-tax cost of 0.11% a year on municipal holdings, which lowers their net benefit below this gross value (Table 2). Portfolio assets $m_g$, in trillions of dollars, are all of a group's debt securities and cash in the Financial Accounts, and they scale the group's risk-bearing capacity.

Households also differ by home state, because a resident's own state's bonds are exempt from state tax while other states' bonds and taxable bonds are not.[11] Insurers and banks face the corporate rate. Proration taxes part of insurers' municipal interest, and banks also pay a financing-tax cost on municipal holdings. The second is size. An investor's portfolio assets $m_i$ are its debt securities and cash in the Financial Accounts, including holdings outside the model's markets, and they scale its risk-bearing capacity. A measured fraction $W_i$ is available for the investments in the model: municipal bonds, corporate bonds, treasury notes and bonds, and cash. Its funding is therefore $m_i W_i$ (Appendix B.1). Positions $x^a_{i,T}$ are measured per dollar of portfolio assets, so aggregate bond holdings are $m_i x^a_{i,T}$.

The third difference is outside rate exposure, which measures how the rest of an investor's balance sheet changes in value when rates move. Outside assets that lose value when rates rise add to the losses on bonds, while liabilities that lose value offset them. Outside rate exposure is the negative of the liability hedge $\ell_i$ of Section 4. As there, an investor whose outside exposure offsets bond losses bears rate risk more cheaply and can hold long bonds even when its exemption benefit is small. Within each group, types differ in outside rate exposure. Each household group has seven types in each of the 45 home markets, formed by sorting Survey of Consumer Finances (SCF) households on outside assets and debt relative to their bond and deposit holdings. P&C insurers have two types and life insurers three, which differ in exposure and in their shares of the sector's assets. Banks form a single type. The residual group has seven types whose exposures approximate a normal distribution. Appendix B.1 defines every type.

[11] Treasury interest is exempt from state income tax, while interest on corporate and other taxable bonds is not. The model taxes all interest in its taxable market at federal and state rates, as in the corporate comparison curve of Section 3.6.

The quantitative model keeps the structure of Section 4. Investors solve (7) with market-specific after-tax shares, and prices satisfy the covariance-based portfolio conditions (8) and clearing (9) in every market and rate state. Banks' measured financing-tax cost is subtracted from their municipal returns (Appendix B.1). The model retains the rate floor in (5) and allows the shorting constraint on cash to bind and participation to change across rate states. I solve $P_T^a(z)$ numerically through the general resale relation in Appendix equation (40), integrating over next-period states. Returns, portfolio risks, holdings and both yield curves are determined jointly.

The model of Section 4.1 has rate risk only. The quantitative model adds a small independent bond-specific return shock to every risky bond, which adds its own variance to the bond's log return, as in the general economy of Appendix A.1 (equation (32)). The shock, a realistic source of return risk, makes bonds imperfect substitutes and lets several investor types hold the same maturities. In the two-investor model, it extends joint holdings from one maturity to a block of maturities while changing municipal yields by at most 0.33 basis points (Appendix A.2.8). It vanishes at the riskless claim, and changing its variance by 20% moves the thirty-year gap by less than 0.1 basis points (Appendix B.5).

Measured inputs change over time, so I solve the model separately for six three-year periods, 2000–2002 through 2015–2017, each centered on an SCF wave. Each period has its own tax rules, portfolio assets, bond supplies, household balance sheets and observed rate states. Financial Accounts stocks give each sector's portfolio assets and holdings and the taxable supply. Mergent gives corporate maturity profiles and municipal supply by maturity and state. The SCF supplies household balance sheets and the income and filing status needed to apply statutory taxes. Historical short rates determine rate persistence and volatility, while two-year corporate yields set the rate level and observed states. Appendix B.1 details the construction and sources. I also solve the model for a seventh period, 2018–2020, with its own measured inputs and the tax rules of the 2017 TCJA. This period is not used in estimation. It keeps the coefficients estimated on the first six periods, and Section 5.3 includes it when comparing model and observed yields.

### 5.2 Calibration and fit

Everything the data measure directly is fixed before estimation: investor tax treatment, each group's portfolio assets and funding, bond quantities and maturity profiles, household outside assets and debt, and the short-rate process. These inputs differ across the six periods. The remaining seventeen coefficients cannot be measured directly and are common to all six periods and all states. One is the common risk aversion. Two set the size and maturity profile of the bond-specific shock. The other fourteen set outside rate exposures, which balance-sheet amounts alone cannot measure: three translate a household type's measured outside assets and debt into exposure, eight set the exposures and asset shares of the insurer types, one sets bank exposure, and two set the mean and dispersion of residual-group exposure.

I estimate the coefficients by fitting twenty-two targets. Nine are ownership stocks: municipal holdings of the two household groups, P&C insurers, life insurers and banks, and taxable holdings

of households and the same three institutional sectors. Four are the mean maturities of insurers' municipal and taxable portfolios. Six describe the composition of insurer portfolios. Two, one for each insurer sector, measure how unevenly municipal bonds are spread across insurers. The other four are the fractions of each insurer sector's municipal and taxable holdings that mature beyond twenty years. The last three are the taxable yields at ten, twenty and thirty years, measured as term spreads over the two-year yield.

Ownership provides the main information about outside exposure. Given a group's tax treatment and portfolio assets, how much it holds in each market and at which maturities reveals how much rate risk it is willing to bear. Insurer portfolio composition disciplines the differences among insurer types. The taxable term spreads and long-bond allocations discipline risk aversion and bond-specific risk. Each coefficient affects every target through equilibrium prices and portfolios, so the coefficients are estimated jointly. I evaluate the targets at each period's observed rate states, average them across the six periods and minimize their weighted squared deviations from the data.[12]

Calibration on the 45-market model is not computationally tractable, so I estimate the coefficients in a one-market version with a single national municipal market. In that market, each household type's municipal after-tax share averages its home-state shares, weighted by each state's municipal supply. I then restore the 45 municipal markets and their measured tax differences without re-estimating anything, since no coefficient is state-specific.[13] The one-market and 45-market models give similar relative municipal yields. Measured against each model's own taxable curve, their thirty-year gaps average 102.6 and 102.4 basis points across the six periods and differ by at most 2.5 basis points in any period, while their thirty-year taxable yields differ by at most 0.52 basis points. The gap is also insensitive to the individual coefficients. Changing one group of coefficients at a time, for example scaling an exposure by 20%, moves the one-market gap by at most 4.1 basis points (Appendix Table 11). The exception is the common risk aversion. It scales the compensation for rate risk in both markets, so it moves the taxable and municipal curves nearly in proportion. Because the mechanism works through the tax rate embedded in yields, the gap in basis points moves with the level of taxable yields. Lowering risk aversion by 20% lowers the thirty-year gap by 13.6 basis points, mostly because it lowers the thirty-year taxable yield by 70 basis points. The gap per unit of taxable yield falls only from 0.170 to 0.166. This ratio, the shortfall of the implied tax below the household wedge, is what the comparison with data in Section 5.3 uses. The same change also leaves taxable term spreads about 20% below their targets, so the data rule it out.[14]

Municipal yields, the municipal puzzle and empirical pass-through coefficients are not targeted. Ownership disciplines outside rate exposure, which combines with observed taxes to determine municipal yields in equilibrium. The relative municipal yield curve and cross-state pass-through

[12] Each averaged deviation is scaled by 10% of its observed normalizer. Maturity targets and the fractions held beyond twenty years include model-portfolio weights. Appendix equation (113) gives the criterion, and Appendix Tables 9 and 10 report the coefficients and the observed and fitted target values.

[13] Fourteen of the twenty-two targets are matched to within 10% of their observed normalizers. The best small move away from the estimates improves the fit by less than 0.001% and changes the average thirty-year gap by 0.004 basis points. Appendices B.3 and B.4 report the estimation procedure and fit.

[14] It also raises the estimation criterion from 36.3 to 55.0 (Appendix Table 11).

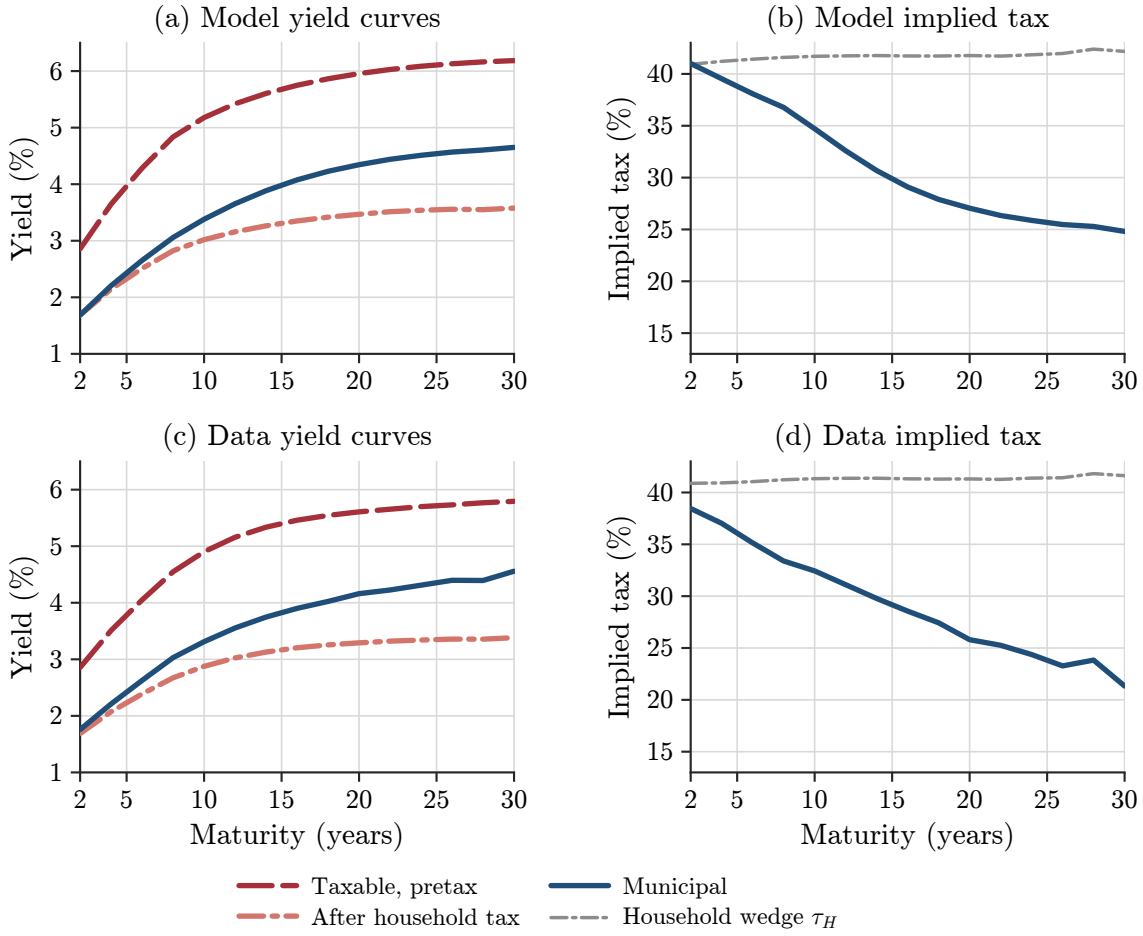



Figure 7: Model (top) and adjusted data (bottom), 2000–2020. Panels (a) and (c) plot the pretax taxable yield, the taxable yield after household tax (the household benchmark) and the municipal yield. Panels (b) and (d) plot the implied tax $1 - y_T^M / y_T^C$ and the household wedge, denoted $\tau_H$ in the legend. Corresponding panels share axis scales. The taxable curve is the model's pooled taxable market (top) and the observed corporate benchmark (bottom). The municipal data curve removes callability (Appendix C.1.2) and the issuance premium, and panel (d) uses this same adjusted curve. The model evaluates the estimated coefficients at observed issue-quarter rate states in the six calibration periods and the 2018–2020 period. Both rows use the empirical bond, rating and quarter weights and nominal semiannual yields. The household benchmark multiplies each observation's taxable yield by the high-tax household's relative after-tax share for that bond's state and date before averaging. Implied taxes and the household wedge are ratios of averaged yields.

therefore test the mechanism (Sections 5.3 and 6.1).[15] In the headline comparison of Section 5.3, model and data share the same taxable curve and weights. I average model and observed yields with the weights of the 2000–2020 bond sample and apply the model's implied tax at each maturity to the observed corporate yield.

### 5.3 Model yields and the municipal puzzle

Figure 7 sets the calibrated model (top) against the data (bottom). The left panels plot three yield curves: the pretax taxable yield, the household benchmark, which is that yield after the high-tax

[15] Appendix B describes the solution method, and Appendix B.7.2 reports accuracy.

household's taxes, and the municipal yield. The right panels convert these curves into the implied tax $1 - y_T^M / y_T^C$ and compare it with the household wedge $\tau_H$. In both rows, the municipal yield is close to the household benchmark at two years and rises above it with maturity, so the implied tax starts near the household wedge and falls well below it at long maturities. In nominal semiannual yields, the model's implied tax is 41.0% at two years, 34.7% at ten and 24.8% at thirty, a decline of 16.2 percentage points. In the adjusted data, it is 38.5%, 32.4% and 21.3%, a decline of 17.1 points. The household wedge stays between 40.9% and 42.4% in both rows. The model's thirty-year municipal and taxable yields are 4.651% and 6.185%, against 4.558% and 5.794% in the adjusted data.

The accounting below asks how much of the municipal puzzle the model explains, alongside the measured capital-gains, liquidity and default differentials. Because the model's thirty-year taxable yield is 39 basis points above the observed one, its gap in basis points is not directly comparable with the puzzle. A higher taxable yield enlarges the gap even when the implied tax is unchanged. I therefore express all contributions on the same observed taxable yield curve.

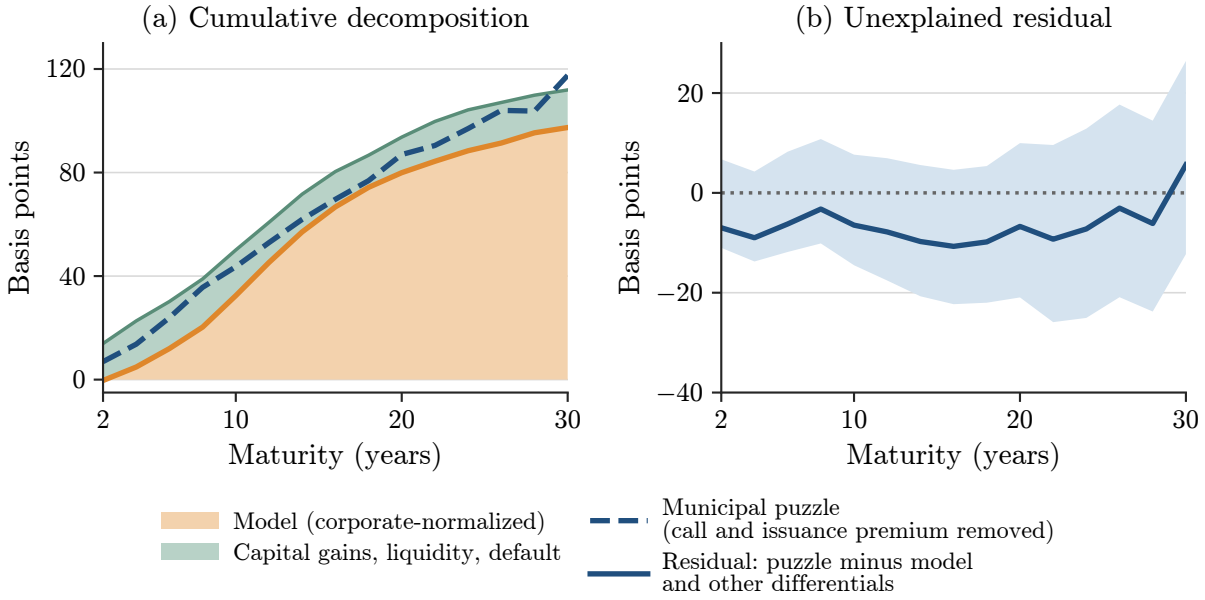



Figure 8: Cumulative decomposition of the municipal puzzle, 2000–2020, as in (24). The puzzle is the observed spread after removing callability and the issuance premium. Panel (a) stacks the model gap $\widetilde{\mathcal{G}}_T$ of (23) and the capital-gains, liquidity and default differentials against that puzzle. Panel (b) shows Residual($T$), the puzzle less their sum. Shading in panel (b) spans the ranges of the five empirical adjustments (Appendix C.1) and excludes model uncertainty. Weights and yield conventions match Figure 7.

At each maturity, I keep the model's municipal-to-taxable yield ratio and multiply it by the observed taxable yield. I then subtract the household benchmark. The resulting gap, denoted $\widetilde{\mathcal{G}}_T$, measures the model's contribution on the data's taxable curve:

$$\begin{aligned} \text{Normalized municipal yield} &= \frac{\text{Model municipal yield}}{\text{Model taxable yield}} \times \text{Observed taxable yield}, \\ \widetilde{\mathcal{G}}_T &= \text{Normalized municipal yield} - \text{Household benchmark}. \end{aligned} \tag{23}$$

Table 4: Accounting for the municipal puzzle. Removing callability and the issuance premium from the raw spread over the household benchmark gives the puzzle. The model contribution applies its implied tax to observed corporate yields, as in equation (23).

| | | Comparability | | | Accounting for the puzzle | | | |
|---|---|---|---|---|---|---|---|---|
| Maturity (years) | Raw spread | Call | Issuance premium | Puzzle | Model | Model share | Bond-side adjustments | Unexplained |
| 2 | 9.9 | 0.1 | 2.9 | 6.9 | −0.3 | | 14.3 | −7.0 |
| 6 | 27.9 | 0.7 | 3.3 | 23.9 | 12.0 | 50% | 18.2 | −6.2 |
| 10 | 52.2 | 4.6 | 4.0 | 43.6 | 32.4 | 74% | 17.6 | −6.5 |
| 14 | 87.0 | 17.9 | 7.2 | 61.9 | 57.1 | 92% | 14.6 | −9.7 |
| 18 | 110.3 | 23.9 | 9.6 | 76.9 | 74.3 | 97% | 12.4 | −9.8 |
| 22 | 128.8 | 27.0 | 11.4 | 90.5 | 84.3 | 93% | 15.4 | −9.3 |
| 26 | 145.0 | 28.5 | 12.5 | 104.0 | 91.4 | 88% | 15.7 | −3.1 |
| 30 | 159.8 | 29.5 | 12.8 | 117.5 | 97.4 | 83% | 14.5 | 5.6 |

*Notes.* Basis points, except maturity and shares. The accounting is puzzle = model + bond-side + unexplained, as in (24). The model column uses the corporate normalization, and the bond-side adjustments sum the capital-gains, liquidity and default differentials. The model share divides its contribution by the puzzle, using unrounded values, and is omitted where the puzzle is below 20 bp. Appendix C.1.7 reports the unnormalized gaps. Model and data both use the 2000–2020 sample, the averaging of Appendix C.1.1 and the nominal semiannual convention. The separately measured adjustments are not jointly estimated. Rows show every second model maturity. Figure 8 shows all fifteen.

I call this the corporate normalization. It preserves the model's implied tax and changes only the taxable yield on which the gap is measured. The solved equilibrium and the curves in Figure 7 are unchanged. Appendix C.1.7 gives the averaging and normalization details.

The accounting has three steps at each maturity $T$:

$$\begin{aligned} \text{Spread}(T) - \underbrace{\big[\text{Call}(T) + \text{Prem}(T)\big]}_{\text{comparability adjustments}} &= \underbrace{\text{Puzzle}(T)}_{\text{municipal puzzle}}, \\ \text{Puzzle}(T) - \underbrace{\big[\text{CG}(T) + \text{Liq}(T) + \text{Def}(T)\big]}_{\text{bond-side adjustments}} &= \underbrace{\text{Puzzle}^{\text{adj}}(T)}_{\text{residual before the model}}, \\ \text{Puzzle}^{\text{adj}}(T) - \underbrace{\widetilde{\mathcal{G}}_T}_{\text{model}} &= \underbrace{\text{Residual}(T)}_{\text{unexplained}}. \end{aligned} \tag{24}$$

The first line removes the comparability adjustments of Section 3.7, callability and the issuance premium, from the raw spread and gives the municipal puzzle. The second line subtracts the capital-gains, liquidity and default differentials, CG, Liq and Def. These features affect municipal yields but are absent from the model, so $\text{Puzzle}^{\text{adj}}(T)$ is the part of the puzzle left for the model to explain. The third line subtracts the model gap and leaves the unexplained residual. The household benchmark cancels in this last difference, so the residual equals the observed corporate yield times the difference between the model's implied tax and the implied tax of the fully adjusted data. A negative residual therefore means that the model's implied tax is below the data's and the model gap exceeds $\text{Puzzle}^{\text{adj}}(T)$. Appendix C.1 explains how each of the five adjustments is measured, in the order callability, issuance premium, capital gains, liquidity and default (Appendices C.1.2–C.1.6).

At thirty years (Figure 8 and Table 4), the model accounts for 97 basis points of the 118-basis-point puzzle, or 83%. The bond-side adjustments account for 15 basis points, or 12%, leaving about 6 basis points unexplained. The model's contribution rises with maturity, and its share of the puzzle is 74% at ten years, 97% at eighteen and 83% at thirty. The unexplained residual is negative at every maturity except thirty years. From twelve to twenty-four years, the model gap exceeds $\text{Puzzle}^{\text{adj}}(T)$ by 7 to 11 basis points, and the largest absolute residual is 11 basis points at sixteen years. A negative residual is consistent with the convenience-premium interpretation in Fleckenstein and Longstaff (2025).

# 6 Empirical Tests

I test the model's participation mechanism using variation in household taxes, institutional tax benefits and interest rates (Section 4.6). None of the estimates is a calibration target. P1 uses differences in residents' state income-tax rates. P2 uses bank qualification. These designs compare household and institutional tax effects across maturities. P3 compares changes in municipal yields with changes in treasury yields at the same remaining maturity.

The P1 and P3 tests use the AA-and-above municipal sample (agency score three or better) to limit credit differences that could affect both yields and trading. The curve accounting of Sections 3 and 5 uses the wider A-or-above sample (score seven or better), which matches the corporate benchmark's credit band, and Appendix C.10.2 repeats both tests on that sample, where the same maturity ordering holds. The bank-qualified design uses all new issues with rating-by-year fixed effects. Throughout, $n_b$ denotes a bond's remaining maturity in years.

## 6.1 P1. State-tax pass-through

P1 tests whether residents' home-state exemption is reflected less strongly in long municipal yields than in short yields. I compare bonds of similar maturity across states with different top income-tax rates, controlling for bond and state characteristics. Higher state rates make the home-state exemption more valuable to resident households. To measure how much of that benefit is reflected in yields, I infer a state tax rate from municipal yields relative to comparable taxable yields, accounting for federal taxation. Under the household benchmark, a one-percentage-point increase in the statutory rate raises this implied state tax by one percentage point at every maturity. The model predicts a smaller response at longer maturities.

State rates move resident households' exemption benefits, which institutions largely do not receive. Pensions and tax-deferred accounts owe no current state tax on interest, and insurer premium taxes generally do not distinguish municipal from taxable interest at the state level. Under its stated conditions, Appendix Proposition A.8 gives household pass-through of one at the riskless short end and below one in the long-end limit, and Appendix Proposition A.18 gives the same short-end result for state taxes and the part of state-tax pass-through that comes from the holders' tax mix.

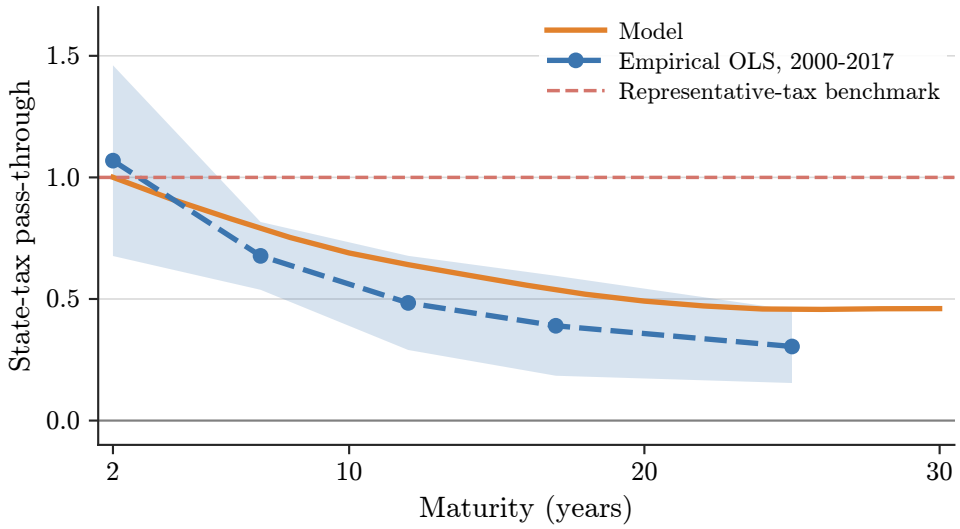



Figure 9: State-tax pass-through by maturity. Blue: OLS estimates for 2000–2017 call-adjusted zeros, with state-clustered 95% intervals using $t(46)$. The regression is unweighted and includes the controls in (26). Orange: the model's cross-state slope at the mean rate state $z = 0$, using its own federal after-tax share and nominal semiannual yields, averaged equally across the six pre-2018 periods. Both use (25). The red dashed line is the representative-tax benchmark.

The yield ratio $y^M/y^C$ contains both federal and state taxes. Under the pre-2018 federal deduction convention, the after-tax share of taxable interest is $\xi_s^{\text{st}}\xi_t^{\text{fed}}$, where $\xi_s^{\text{st}} = 1 - \tau_s^{\text{st}}$ is the state share and $\xi_t^{\text{fed}}$ the federal share, while home-state municipal interest is exempt at both levels. Household indifference under the representative-tax benchmark gives $y_{s,t}^M = \xi_t^{\text{fed}}\xi_s^{\text{st}}\, y_t^C$. Dividing out the federal share defines the implied state tax

$$\mathit{ImpliedStateTax}_{s,t}(T) \;\equiv\; 1 - \frac{y_{s,t}^M(T)/y_t^C(T)}{\xi_t^{\text{fed}}}. \tag{25}$$

It equals the statutory rate $\tau_s^{\text{st}}$ under the benchmark, giving pass-through of one. I set $\xi_t^{\text{fed}}$ to one minus the dated top federal ordinary-income rate, excluding the net investment income tax: 0.604 in 2000, 0.609 in 2001, 0.614 in 2002, 0.65 in 2003–2012 and 0.604 in 2013–2017. The sample ends before the TCJA capped state and local tax deductions at \$10,000 from 2018, and Appendix C.6 reports the later period under its tax mapping.

I estimate pass-through separately in five maturity bins ($[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$ and $[20,30]$ years):

$$\begin{aligned}\mathit{ImpliedStateTax}_{b,t} = &\sum_{g=1}^{5}\beta_g\big(\mathit{StateTax}_{s(b),t}\times\mathbf{1}\{n_b\in g\}\big) + \sum_{g=1}^{5}\delta_g'\big(Z_{b,t}\times\mathbf{1}\{n_b\in g\}\big)\\ &+\lambda' V_b + \alpha_{t\times n\times \mathit{callable}} + \varepsilon_{b,t},\end{aligned} \tag{26}$$

where $\mathit{StateTax}_{s,t}$ is the state's top statutory rate and the outcome (25) uses call-adjusted zero-coupon yields. The coefficient $\beta_g$ is the change in the implied state tax per percentage point of

statutory rate in maturity bin $g$, and the benchmark gives $\beta_g = 1$ in every bin. The controls $Z_{b,t}$, also interacted with the bins, are the rounded average rating and six lagged state variables: par-weighted outstanding maturity, the maturity of the preceding year's issuance, outstanding par relative to personal income, log income, log preceding-year issuance and the outstanding share in the bond's maturity bin. $V_b$ contains bond-specific characteristics, and $\alpha_{t\times n\times callable}$ denotes issue-year–month–maturity–callability fixed effects.

The empirical sample excludes Illinois, Iowa and Wisconsin, which generally do not exempt residents' interest on in-state municipal bonds, and Oklahoma, whose exemption depends on the issuer and issue date. It retains bonds recorded as exempt from own-state tax and bonds issued in states with a zero rate. The 2000–2017 sample has 681,838 observations with complete controls. Standard errors are clustered by state, allowing dependence among a state's bonds across years and maturities. Cross-sectional variation in state tax rates identifies pass-through at each maturity. Because current tax policy could respond to current municipal yields, I also report an instrumental-variables (IV) comparison that instruments the current statutory rate with its 1990 counterpart, following the policy-history approach of Babina et al. (2021). The IV interpretation requires that, conditional on the controls, historical rates affect current yields only through current tax rates. The results are robust to the choice of historical year, and Appendix C.6 reports other vintages with the same controls and sample.

Figure 9 compares the ordinary least squares (OLS) estimates of (26) with the cross-state slope in the full 45-market model. For the model, I compute the implied state tax (25) with its high-tax household's federal share $\xi^{\text{fed}}$ and nominal semiannual yields. At each maturity, I regress it on each state's top rate, averaged within the three-year calibration period, weighting the 43 income-tax states and the District of Columbia equally, and I average the slopes across the six pre-2018 periods.

Estimated pass-through declines across all five bins, from 1.069 below five years to 0.305 beyond twenty (Appendix Table 22). A state top rate ten percentage points higher raises the implied state tax by about 10.7 points at the short end and 3.0 at the long end. The short-to-long decline is 0.764 (s.e. 0.235, $p = 0.002$). Unit pass-through cannot be rejected in the shortest bin and is strongly rejected in the longest, and every bin coefficient is positive and significant at one percent. The IV estimates also decline, from 1.088 to 0.229.

The model's untargeted profile declines from 1 at two years to 0.69 at ten and 0.46 at thirty. It lies within the 95% confidence intervals of the four bins below twenty years and just above the interval of the longest bin. The empirical decline survives alternative yield constructions, credit controls, weights, instruments, state fixed effects and single-state omissions (Appendices C.5 and C.6).

### 6.2 P2. Bank qualification

P2 tests how much of a tax benefit available only to banks is reflected in municipal yields. I compare offering yields on bank-qualified (BQ) and nonqualified municipal bonds, separately by maturity. BQ designation gives banks an additional tax deduction without changing the bond's promised

payments or other investors' tax treatment. The yield reduction attributable to designation measures how much of this benefit reaches municipal issuers through lower borrowing costs.

Banks ordinarily cannot deduct interest expense allocated to tax-exempt municipal holdings (Section 265(b) of the Internal Revenue Code). For BQ bonds, Sections 265(b)(3) and 291 restore 80% of that deduction. A municipal issuer can designate its bonds as bank-qualified if its reasonably anticipated tax-exempt issuance for the year does not exceed $10 million ($30 million in 2009–2010 under the American Recovery and Reinvestment Act).

I identify the yield effect by comparing bonds from issuers just below and above the annual issuance cap. Below-cap status raises the probability of designation without determining it, so the design is a fuzzy regression discontinuity. Designation averages 0.79 below the cap and 0.21 above it across the estimation window. Measured annual issuance need not equal the issuance anticipated at each sale or counted under the statute, so some designated bonds appear above the measured cap. I use below-cap status as an instrument for actual designation. Within each maturity bin, the estimated premium is the yield increase on crossing above the cap divided by the corresponding fall in the designation rate.

To measure the share of the benefit reflected in yields, I calculate the expected annual tax saving on each bond. Tax rules allocate a bank's interest expense across its assets in proportion to tax basis, so each dollar of municipal holdings carries the same annual financing cost $c$, whatever the bond's maturity. Designation makes 80% of that cost deductible and saves $X = 0.8\,\tau_B c$ a year per dollar of designated holdings, where $\tau_B$ is the federal bank tax rate. In any year, the benefit is therefore the same for short and long bonds. Measured from Call Report interest expense, $X$ moves with banks' financing costs. It is about 65 basis points in 2005–2008 and 9 to 14 from 2011 to 2019 (Appendix C.7.2). A bond's benefit, however, accrues over its whole life, so the benefit relevant for its yield is $\bar{X}_b$, the expected annual benefit over bond $b$'s life, which evaluates $X$ at expected future financing costs (Appendix C.7.2). Its median is 45 basis points.

Three hypotheses differ in how much of the bank benefit is reflected in the designation premium $\varpi_k = y_k^N - y_k^Q$, the nonqualified minus the bank-qualified municipal yield at maturity $k$. Under the representative-tax benchmark, households hold both bonds at the same yield, $y_k^N = y_k^Q = (1-\tau_H)y_k^C$, where $\tau_H = 1 - \xi_H$ is the household wedge. A bank then prefers a taxable bond, which pays $(1-\tau_B)y_k^C$ after tax, to a designated bond, which pays the household benchmark yield less the tax value of the disallowed fifth of its financing cost, whenever

$$(1-\tau_B)\,y_k^C > (1-\tau_H)\,y_k^C - 0.2\,\tau_B c \quad \Longleftrightarrow \quad \tau_H\,y_k^C > \tau_B\,y_k^C - 0.2\,\tau_B c,$$

that is, whenever the household's exemption benefit exceeds the bank's net exemption benefit on a designated bond. This holds when household rates are at least as high as banks', as they are for top-bracket households throughout the sample period. Banks then hold neither municipal bond, and $\varpi_k = 0$. Under a bank-only benchmark, banks' indifference to taxable bonds gives full capitalization. The premium equals the benefit at every maturity. In general, holding outside investment opportunities and other bond characteristics fixed, the premium is the benefit times the

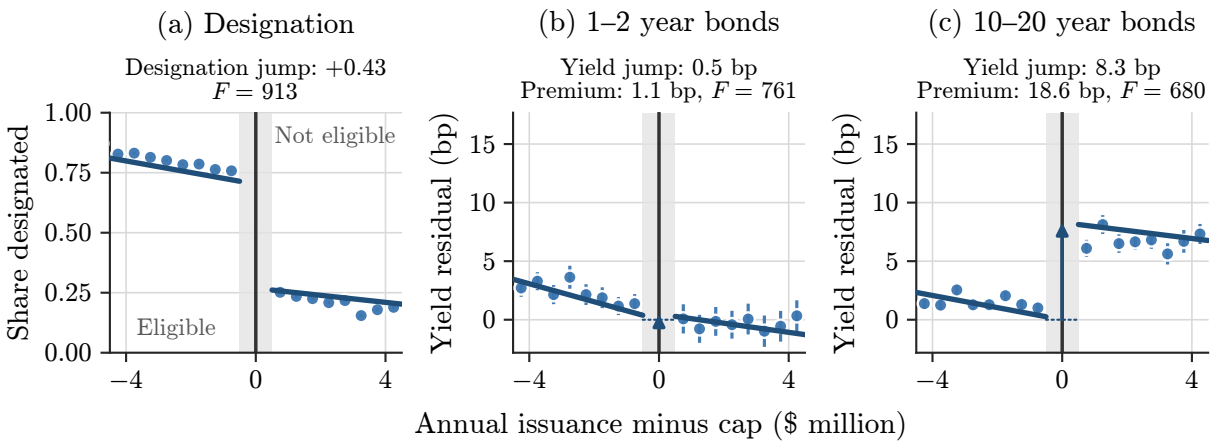



Figure 10: The bank-qualified discontinuity. The horizontal axis is an issuer's measured annual tax-exempt issuance minus that year's cap. Designation is more frequent to the left of zero. Dots are means in $0.5 million bins with 95% intervals: the share of bonds designated in panel (a), and offering-yield residuals after the fixed effects and log par of (28) for 1–2 and 10–20 year bonds in panels (b) and (c). Lines are the estimated local linear fits, which exclude the shaded donut. Yield fits are shifted to zero on the below-cap side at the cap, and the arrow marks the yield jump. Each premium divides the yield jump by its bin's designation jump. $F$ is the first-stage Wald statistic with issuer-clustered standard errors. Appendix Figure 24 shows the other maturities.

bank's capitalization share $\theta_k$, the average bank share of the total demand response to the bond's yield (Appendix C.7):

$$\varpi_k = \theta_k X, \qquad \underbrace{\theta_k = 0}_{\text{representative-tax}}, \quad \underbrace{\theta_k = 1}_{\text{bank-only}}, \qquad \underbrace{\theta_{\text{short}} \simeq 0 < \theta_{\text{long}}}_{\text{model}}. \tag{27}$$

To estimate the capitalization share, I repeat the designation comparison with each bond's yield divided by its expected benefit $\bar{X}_b$. The estimated yield reduction in these benefit units measures $\theta_k$: zero under the representative-tax benchmark, one under the bank-only benchmark and rising with maturity in the model. The capitalization share is the bank counterpart of $\mathcal{P}_I^Y(T)$ in Section 4.6. Household after-tax cash returns anchor short yields and banks hold few short bonds, so $\theta_k$ is near zero at short maturities. In the calibrated model, banks hold intermediate maturities, where their duration benefit offsets their smaller exemption benefit (Section 4.3), and yields there reflect more of the bank benefit.

The variable $R_{jt}$ is issuer $j$'s annual tax-exempt issuance minus that year's cap, in millions of dollars. The indicator $E_{jt} = \mathbf{1}\{R_{jt} \leq 0\}$ marks below-cap issuer-years, and $B_b$ indicates whether bond $b$ is bank-qualified. The sample covers tax-exempt new issues from 2000 to early 2021 with $0.5 \leq |R_{jt}| \leq 4$. The half-million exclusion around the cap, the donut, removes managed bunching and issuer-years exactly at the cap. With bins $g$ of maturity at issuance (1–2, 2–5, 5–10, 10–20 and more than 20 years), I estimate

$$y_b = \sum_g \beta_g B_b \mathbf{1}\{n_b \in g\} + Z_b'\delta + u_b, \tag{28}$$

by two-stage least squares (2SLS), instrumenting each designation interaction $B_b\mathbf{1}\{n_b \in g\}$ with the eligibility interaction $E_{jt}\mathbf{1}\{n_b \in g\}$ of the same bin. The first stage is the local designation jump conditional on controls, 0.42 to 0.45 across bins (Appendix Table 28). Here $Z_b$ contains local linear controls in $R_{jt}$ on each side of the cap, log tranche par, and fixed effects for maturity, state, offering month, rating, security type, use of proceeds, coupon and callability, with the interactions listed in Appendix C.7.2. The outcome is the offering yield in basis points, or that yield divided by $\bar{X}_b$, so $-\beta_g$ is the premium in basis points or, in expected-benefit units, an estimate of the capitalization share $\theta_k$. Standard errors are clustered by issuer.

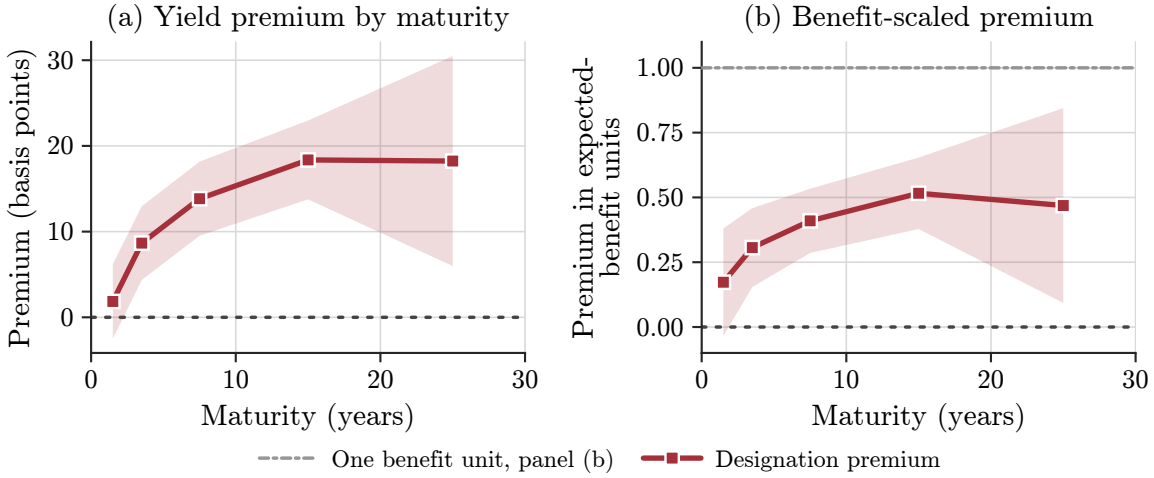



Figure 11: The bank-only benefit and municipal yields. Panel (a): 2SLS yield premium in basis points. Panel (b): the premium in expected-benefit units, with each yield divided by the expected annual benefit $\bar{X}_b$ and the same controls. Financing costs are municipal-holdings-weighted. Expected rates use the Kim–Wright expected-short-rate component, capped at ten years (Appendix C.7). Shading is the issuer-clustered 95% interval. The dashed line marks one benefit unit.

Identification requires that, given these controls, crossing the cap moves offering yields only through designation. Callability and the number of maturities per issue are balanced at the cap. A placebo cutoff at \$5 million shows essentially no designation effect, and its 10–20 year yield jump exceeds the 1–2 year jump by only 0.4 basis points, against 7.8 at the actual cap (Appendix C.7).

Figure 10 illustrates the discontinuity with separate local fits for each maturity bin. The 1–2 year yield barely moves at the cap, by 0.5 basis points, while the 10–20 year yield jumps by 8.3, implying designation premia of 1.1 and 18.6 basis points after dividing by the designation jumps. The main estimates come from the joint 2SLS regression (28), which estimates all five bins together. Figure 11 reports these estimates (Appendix Table 28). The premium is 1.8 basis points at 1–2 years, indistinguishable from zero, and 8.7, 13.8, 18.4 and 18.2 at 2–5, 5–10, 10–20 and more than 20 years. The 10–20 year premium exceeds the 1–2 year premium by 16.5 basis points (s.e. 2.7). As a share of the expected benefit, the premium rises from 0.17 at 1–2 years to 0.52 at 10–20 years. The bank-only benefit is barely priced in short municipal yields, and about half of it is priced at 10–20 years.

The positive premia at 5–20 years reject the representative-tax benchmark, with $t$ statistics of 6 to 8, and the rise in the share, 0.34 (s.e. 0.11), rejects full capitalization, which implies a share of one at every maturity. The gradient appears in every benefit regime (terciles of years ranked by $X$) and survives alternative donuts and bandwidths, credit-enhancement controls, a general-obligation sample, issuer effects, year clustering and the binned estimator of Dagostino (2025) (Appendix C.7).

The rising bank profile is the reverse of household pass-through in P1, which falls from 1.07 to 0.30 (Figure 9), as the model predicts for institutional and household tax benefits.

### 6.3 P3. Rate responses by maturity

P3 asks whether municipal yields respond more to a given treasury yield change at longer maturities. The representative-tax benchmark makes municipal zero yields a constant fraction $\xi_H$ of treasury zero yields, so it predicts the same relative response at every maturity. The model predicts a relative response that rises with maturity, from $\xi_H$ at the riskless short end toward the lower-tax investor's share $\xi_I$ at the long end (Section 4.6). That section derives the pattern relative to the model's jointly priced taxable market $C$. The test measures rate movements with treasury yields, and for this reference the analytical illustration also gives a relative response that rises with maturity.[16] The observed coupon-yield response is about 0.35 at one year, below the model's zero-coupon ratio $\xi_H$. The test concerns how this relative response varies with maturity.

I compare consecutive trades in the same municipal bond, using its yield to maturity and the maturity-matched treasury par yield. With $\Delta y^{M}_{b,t}$ the change in its yield between the two trades and $\Delta y^{\text{Tsy}}_{n_b,t}$ the change over the same interval in the treasury yield matched to its remaining maturity $n_b$, I estimate

$$\Delta y^{M}_{b,t} = \beta_0 \, \Delta y^{\text{Tsy}}_{n_b,t} + \beta_1 \big( \Delta y^{\text{Tsy}}_{n_b,t} \times n_b \big) + \beta_2 \, n_b + \varepsilon_{b,t}. \tag{29}$$

The response to a treasury yield change is $\beta_0 + \beta_1 n_b$. A positive $\beta_1$ means that the same treasury yield change is associated with a larger municipal yield change at longer maturities. Comparing trades within a bond removes its time-invariant characteristics. The sample covers January 2005 to August 2020 and includes noncallable, federally tax-exempt municipal bonds rated AA and above. Trades are at most 90 days apart, and the treasury yield must move by at least 10 basis points in either direction to limit transaction noise. The treasury comparison follows the bond's remaining maturity at each trade, and standard errors are clustered by the second trade's date. Appendix C.9 details the construction.

The estimates are $\hat{\beta}_0 = 0.346$ and $\hat{\beta}_1 = 0.007$ (Appendix Table 31). A 100-basis-point rise in the treasury yield is associated with a municipal yield increase of about 35 basis points at one year and 42 at ten years.

[16] In the two-investor illustration, municipal sensitivity relative to a taxable reference whose one-period return responds one for one to the rate state increases over the thirty illustrated maturities. Appendix Proposition A.12(ii) establishes the maturity ordering for that reference when bond-specific risk extends joint holdings across long maturities.

All thirty combinations of alternative treasury-move cutoffs and trade windows give a positive maturity interaction, from 0.0047 to 0.0104, significant at five percent (Appendix Table 32). On the wider rating sample, the maturity pattern holds and the gradient is larger, 0.013 against 0.007 (Appendix Table 34).

# 7 Policy Counterfactuals

Tax reforms that change investors' exemption benefits change who holds municipal bonds. The Tax Reform Act of 1986 generally removed banks' interest deduction for new municipal purchases and was followed by a sharp decline in their holdings (Appendix D.5.1). The 2017 TCJA again changed banks' and insurers' relative exemption benefits. I study two reforms with existing policy precedents. The first removes a state's exemption for residents' interest on its own municipal bonds, which Illinois, Iowa and Wisconsin already generally tax. The second taxes municipal interest like taxable interest for every investor, as the 2009–2010 Build America Bonds program did for its taxable municipal bonds while subsidizing issuers' interest.

The two reforms differ in whether investors can retain the federal benefit by switching municipal markets. Home-state repeal leaves that benefit in place, so high-tax households mainly switch between municipal markets while their replacement investments generate state-taxable interest. Equal treatment removes the federal benefit as well. Investors who previously benefited most shorten their bond portfolios, while tax-neutral investors expand their long-bond holdings. The financing and revenue effects depend on both the interest these investors earn and the compensation they require for bearing rate risk.

## 7.1 Design and fiscal accounting

I solve each reform in each of the six three-year periods of 2000–2017 and report calendar-year-weighted averages. Each period uses the pre-2018 tax regime and period-specific investor and bond-market inputs (Section 5.2). I change the relevant tax rules and re-solve holdings and both yield curves, keeping bond supplies, investors' total funding, risk preferences and outside exposures at their dated baseline values.[17]

Annual financing costs equal debt outstanding at each maturity times its yield, summed across maturities. Their increase is the replacement-financing cost, the additional annual cost of financing the same debt at the new yields.[18] Each group's pretax interest on bonds and taxable cash divides into tax receipts and after-tax interest income, with households taxed by residence and institutions

[17] Tax changes are applied in each investor's baseline payoff units, preserving preferences and liability hedges. Yield changes average each period's 36 monthly rate states. Dollar amounts, California's implied state taxes and Figure 12(a) use the mean rate state, $z = 0$. Ratios for 2000–2017 divide average dollar financing costs by average dollar receipts.

[18] The calculation uses continuously compounded yields to construct coupon-equivalent flows. Existing contractual payments and original-issue-discount schedules remain unchanged. The taxable market pools treasury and corporate debt. Other liability flows, issuer deductions, refinancing timing and fiscal feedback are outside this interest-income accounting (Appendix D.4).

under their own tax treatment. Because aggregate funding, bond supplies and the short rate are fixed, total cash interest does not change. Every additional dollar issuers pay therefore goes either to governments or to investors. Summing across both markets and all governments gives

$$\Delta(\text{Financing costs}) = \Delta(\text{Tax receipts}) + \Delta(\text{After-tax interest income}). \tag{30}$$

I report financing costs per dollar of added receipts. For home-state repeal, I compare both own-state and all-issuer costs with the repealing state's added revenue, excluding changes in other governments' receipts. For equal treatment, I compare total municipal and taxable issuer costs with the combined increase in federal and state revenue. This ratio exceeds one when investors' after-tax interest income rises. Issuers then pay more than governments collect, and investors receive the difference.[19]

### 7.2 The home-state exemption

Most income-tax states exempt residents' interest on their own municipal bonds and tax interest on other states' bonds, which encourages residents to hold local debt. Illinois, Iowa and Wisconsin generally tax interest on their own bonds as well.[20]

Panel (a) of Figure 12 gradually removes California households' exemption on local municipal interest in the 2015–2017 period, keeping statutory state and federal tax rates and household composition fixed. The horizontal axis measures the remaining tax benefit of a local bond relative to an out-of-state bond for the high-tax household. The implied state tax is the percentage by which California's yield lies below the pooled municipal yield of the states without an income tax.[21] At the riskless two-year maturity, the implied state tax moves one for one with this benefit. Removing the exemption entirely lowers the implied state tax by 13.3 percentage points at two years but only 5.0 percentage points at thirty years. As in P1, the household's tax benefit is fully reflected in short yields and only partly in long yields, where lower-tax residents and institutions also hold the bonds.

Panel (b) removes residents' state exemption on home-state municipal bonds, separately in California, New York, New Jersey, Massachusetts, Ohio, Pennsylvania and Minnesota, seven states with high supply-weighted tax benefits. Residents then pay the same state tax on home-state as on out-of-state municipal interest, while the tax treatment of institutions, nonresidents, cash

[19] These ratios compare financing and interest-income flows. Welfare would also depend on changes in risk bearing, investment opportunities and public spending.

[20] Each of these three states exempts particular issues. The Supreme Court upheld the home-state preference in *Department of Revenue of Kentucky v. Davis* (2008). Home-state preferences also changed within the sample. Oklahoma exempts qualifying local obligations issued after July 1, 2001, and Utah has taxed out-of-state municipal interest on bonds acquired after January 1, 2003, except from states that do not tax Utah's bonds. Since 2018, the federal state-and-local-tax deduction cap has also increased the exemption's value for residents whose deductions it limits.

[21] The implied state tax is $1 - y^M_{\mathrm{CA},T}/y^0_T$, where $y^0_T$ is the pooled no-income-tax-state municipal yield. It is the model counterpart of the P1 measure in (25), with $y^0_T$ in place of $\xi^{\mathrm{fed}}_t y^C_T$. Both yields are solved at rate state $z = 0$. If a fraction $f$ of local interest becomes state-taxable, the high-tax household's relative home-state tax benefit is $(1 - f)\tau^{\mathrm{st}}_H/(1 - f\tau^{\mathrm{st}}_H)$. This compares its after-tax shares on local and out-of-state interest and equals its state rate when the local exemption is full.

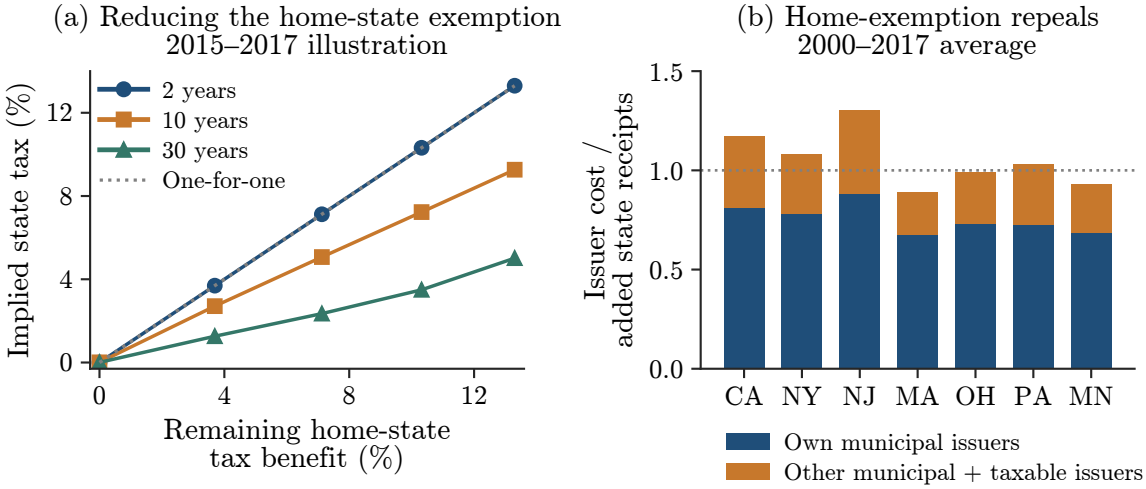



Figure 12: Home-state exemptions, municipal yields and financing costs. Panel (a) makes 0, 25, 50, 75 and 100% of local municipal interest state-taxable for both California household groups in the 2015–2017 period, keeping all statutory tax rates fixed. It plots $1 - y^M_{\text{CA},T}/y^0_T$ at two, ten and thirty years against the high-tax household's remaining relative home-state tax benefit. The comparison yield $y^0_T$ pools the seven states without an income tax. Markers are solved equilibria at $z = 0$, and the dotted line is one-for-one. Panel (b) removes each listed state's home-state exemption at fixed tax rates in each of the six periods of 2000–2017. It divides average replacement-financing costs for own-state municipal issuers and other municipal and pooled taxable issuers by average added state receipts from residents' bond and taxable cash interest at $z = 0$.

and taxable bonds is unchanged. Added state receipts exceed local issuers' added financing costs, which rise by \$0.68–\$0.88 per receipt dollar, below one in all seven six-period averages (Appendix Table 38). Including other municipal and taxable issuers raises the ratio to \$0.89–\$1.30, above one in California, New York, New Jersey and Pennsylvania. Both ratios use the same receipts, so their difference, \$0.21–\$0.42 per receipt dollar, is the cost transmitted to other markets. Other states' municipal issuers bear \$0.08–\$0.11 and taxable issuers \$0.11–\$0.31.

The mechanism is clearest for one dollar of state $s$'s maturity-$T$ debt held by a resident. As the comparison yield, take $y^0_T$, the pooled municipal yield of the states without an income tax, whose bonds carry no home-state benefit for any investor. The implied state tax $\tau^{\text{impl}}_{s,T} = 1 - y^M_{s,T}/y^0_T$ then measures how much of the home-state exemption state $s$'s yield reflects. Repeal has two effects on this dollar. The implied state tax falls, so with $y^0_T$ fixed the issuer's yield rises by $-y^0_T\,\Delta\tau^{\text{impl}}_{s,T}$. The state collects its full rate $\tau^{\text{st}}_s$ on the resident's interest. For a dollar earning $y^0_T$, the annual comparison is

$$\underbrace{\Delta y^M_{s,T} = -y^0_T\,\Delta\tau^{\text{impl}}_{s,T}}_{\text{added cost to the issuer}} \qquad \text{against} \qquad \underbrace{\tau^{\text{st}}_s\, y^0_T}_{\text{tax on the formerly exempt dollar}}.$$

Under the representative-tax benchmark, residents determine relative yields at every maturity, so repeal lowers the implied state tax by the full rate, $-\Delta\tau^{\text{impl}}_{s,T} = \tau^{\text{st}}_s$, and the two sides are equal. In

the model, this equality holds only at the riskless short end. At long maturities, lower-tax residents and investors who do not receive the home-state exemption also hold the bonds, so long yields reflect only part of the high-tax household's benefit, consistent with the partial pass-through documented in P1. Repeal therefore raises long yields by less than the benefit it removes. Across the six periods, California's thirty-year implied state tax falls by 3.8 percentage points, against a mean high-tax household rate of 10.8%, so at thirty years the issuer's added cost is about a third of the tax a high-tax resident pays on the same dollar. Aggregate costs and receipts also depend on resident ownership. Higher yields apply to all of the state's debt, while the state taxes only its residents' interest.

Repeal removes only the state exemption. Residents keep the federal exemption by buying other states' municipal bonds, so high-tax households mainly switch from home-state to other states' bonds, with modest changes in the maturities they hold. Across the seven experiments, residents' aggregate risky-bond share changes by less than two percentage points (Appendix Table 37). This switching does not erode the state's tax base, because after repeal the state taxes residents' interest on home-state bonds, other states' bonds and taxable bonds alike. What matters for receipts is the gross yield on residents' new holdings. Reinvestment at the same-maturity comparison yield $y_T^0$ leaves the receipts in the comparison above unchanged. Reallocation in fact adds to receipts, mainly through substitution into higher-yielding taxable bonds (Appendix D.4.3), and in equilibrium the added receipts exceed the increase in local issuers' financing costs.

Reallocation also spreads financing costs to other issuers because investors choose jointly across municipal and taxable markets. In the California experiment, other states' municipal issuers bear an additional \$0.11 and taxable issuers \$0.25 per dollar of added California receipts. These spillovers raise the cost from \$0.81 for California's own issuers to \$1.17 for all issuers. The state's receipts can therefore exceed its own borrowers' costs even when financing costs across all markets exceed those receipts.

### 7.3 Equal municipal and taxable treatment

The second reform taxes each investor's municipal interest as it taxes that investor's taxable interest. It removes federal and state exemption benefits for every investor group.[22] The Build America Bonds program of 2009–2010 tested a version of it. Issuers could sell taxable bonds and receive a federal subsidy of 35% of their interest. That program changed individual issues while the rest of the market stayed exempt. Equal treatment changes all municipal debt at once, so it also changes who holds long bonds and the yields they require.

I implement the reform by moving each municipal after-tax share 99.9% of the way to its taxable share and removing the same share of banks' financing-tax cost, so that banks' municipal interest is taxed like their taxable interest. With exact equality, the riskless municipal bond and cash would be perfect substitutes, leaving investors' split between them indeterminate. The remaining 0.1% pins down that split and keeps yields and the allocation of bonds across maturities close to those of full removal (Appendix B.7.4). The tax treatment of cash and taxable interest does not change.

[22] The reform is broader than the investor-specific changes in the 2017 TCJA.

Equal treatment raises financing costs by more than it raises tax receipts. All amounts below are average annual changes across the six periods of 2000–2017. Per dollar of added federal and state receipts, municipal issuers bear about $1.06 in added financing costs and taxable issuers bear another $0.13. Total costs rise by about $1.20, leaving roughly $0.20 per receipt dollar as additional after-tax interest income for investors (Table 5 and Figure 13(a)). In annual dollars, municipal issuers pay an additional $41.2 billion and taxable issuers $5.1 billion, for a total of $46.3 billion. Governments collect $38.7 billion and investors retain the remaining $7.6 billion after tax, by (30). Appendix Table 39 reports each period separately.

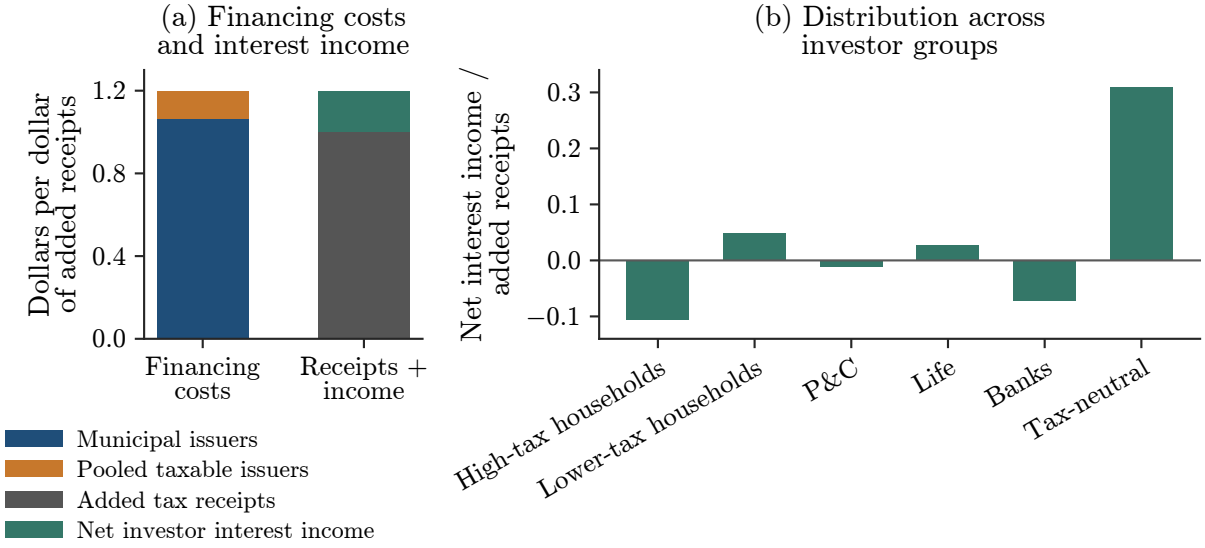



Figure 13: Average annual flows after removing 99.9% of each tax distinction in the six periods of 2000–2017. Panel (a) reports identity (30), dividing average financing costs and net investor interest income by average added federal and state receipts. Panel (b) allocates the income change across investor groups. Receipts and income include taxable cash interest. Each period uses continuous yields and equilibrium holdings at $z = 0$. Taxable issuers pool treasury and corporate quantities.

Unlike home-state repeal, equal treatment leaves investors no municipal market in which they can retain the federal exemption. High-tax households, banks and P&C insurers shorten the average maturity of their risky bonds by 1.8 to 3.7 years, and high-tax households also increase their riskless holdings (Appendix Table 37). Shorter bonds and cash generate less interest than long bonds, limiting the revenue collected from the affected investors. Tax-neutral investors' share of bonds beyond ten years rises from 39% to 57%, while lower-tax households' and life insurers' shares decline slightly to 27% and 14%. Long-bond interest thus shifts toward the tax-neutral group, on whom the reform imposes no additional tax, limiting receipts.

At the same time, the tax-neutral group's larger long-bond holdings increase its rate exposure and the returns it requires in both markets. Municipal yields rise nearly to taxable yields, increasing by 170.6 basis points at thirty years, while thirty-year taxable yields rise by 23.1 basis points. The transfer of rate risk therefore raises borrowing costs in both markets.

Table 5: Average annual fiscal effects of equal municipal and taxable treatment, 2000–2017

| Δ(Financing costs) | | | = | Δ(Tax receipts) + Δ(After-tax interest income) | | |
|---|---|---|---|---|---|---|
| | $bn/ year | Per receipt dollar | | | $bn/ year | Per receipt dollar |
| Municipal issuers | 41.16 | 1.063 | | Federal receipts | 34.29 | 0.886 |
| Pooled taxable issuers | 5.15 | 0.133 | | Residence-state receipts | 4.42 | 0.114 |
| | | | | Total receipts | 38.71 | 1.000 |
| | | | + | Net investor interest income | 7.59 | 0.196 |
| Total financing costs | 46.30 | 1.196 | = | Total receipts and income | 46.30 | 1.196 |

*Notes.* Calendar-year-weighted changes across the six three-year periods, each relative to its own baseline, with 99.9% of the tax distinction removed. Dollar amounts use dated-period quantities without inflation adjustment and continuous yields at $z = 0$. Per-receipt-dollar columns divide each average dollar change by average total receipts. Receipts and income include taxable cash interest, and net investor interest income is the change in after-tax interest income. Before rounding, total issuer costs equal receipts plus net investor interest income.

The $7.6 billion net increase in annual after-tax interest income is distributed unevenly (Figure 13(b)). The tax-neutral group gains $12.0 billion a year, lower-tax households gain $1.9 billion and life insurers gain $1.1 billion. High-tax households lose $4.1 billion, banks lose $2.8 billion and P&C insurers lose $0.5 billion. Investors who benefited more from the exemption lose income, while those less affected by the exemption gain income.

A direct subsidy, as under Build America Bonds, can return the added receipts to municipal issuers. Returning all $38.7 billion of added annual receipts still leaves municipal issuers paying $2.4 billion more per year and taxable issuers paying $5.1 billion more.

## 8 Conclusion

I develop an equilibrium term-structure model in which heterogeneous investor taxes endogenously segment bond markets and maturities. Investors choose taxable and tax-exempt bonds jointly across maturities, weighing exemption benefits against the cost of bearing rate risk, which depends on their balance sheets. Under short-sale constraints, these choices determine who holds each bond and therefore whose taxes its yield reflects. Portfolio substitution links markets and maturities, so an investor's tax treatment can affect even bonds it never holds.

The municipal application shows why one representative investor's taxes cannot explain an entire yield curve. Top-bracket households value the exemption most, yet investors with smaller exemption benefits hold substantial municipal portfolios. The model links this ownership to relative yields. Household taxes anchor the riskless short end, while longer yields also reflect lower-tax investors' tax treatment and their lower cost of bearing rate risk. The puzzle is therefore one of relative pricing, due to a benchmark that ignores the heterogeneity of investors, not a mispricing of municipal bonds. Overlapping investors price the taxable and tax-exempt markets jointly, and the allocation of rate risk among them shapes both yield curves. Without targeting municipal

yields, the calibrated model accounts for most of the puzzle, and the empirical tests confirm that household tax effects weaken with maturity while those of lower-tax investors strengthen.

These results matter beyond the municipal bond puzzle. Under the U.S. tax code, investors face very different tax treatment of the same asset, and the code changes frequently. For bond pricing, a model that ignores heterogeneity in investors' taxes and balance sheets misses the resulting partial segmentation and misprices bonds. For tax policy, the incidence of a tax change extends beyond the market and the investors it targets. Because investors reallocate across markets and maturities, a change in the taxation of one market shifts financing costs across issuers in different markets and redistributes after-tax income among investors. Evaluations that ignore investor heterogeneity and this reallocation misstate the effects of such policies. The same mechanism operates in any setting where investors with different tax treatments choose portfolios under limits to arbitrage, including the state-tax difference between treasury and corporate bonds, tax-favored accounts and cross-border withholding. Wherever taxes differ across investors, who holds a bond shapes its yield, making tax heterogeneity a first-order determinant of the term structure.

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# A The model: formal results and proofs

This appendix states and proves the model results used in Section 4. Appendix A.1 states portfolio choice, market clearing and the price recursion for the general economy. It also shows when the riskless municipal yield equals the household benchmark. Appendix A.2 derives participation and pricing in the two-investor affine model, then verifies the illustrations with and without bond-specific risk. Appendices A.3–A.6 consider other participation motives, multiple states and investor groups, and other bond markets.

Unless stated otherwise, the analytical solutions allow borrowing through the cash account. Wherever cash stays nonnegative, they also solve the problem with borrowing ruled out. Appendix B calibrates the quantitative model and solves it with the shorting constraint on cash. Appendix C constructs the empirical adjustments and tests, and Appendix D supplies the tax rules, institutional evidence, fiscal accounting and data sources. Table 6 collects the main model notation.

Table 6: Notation for the model and the proofs

| Symbol | Meaning |
|---|---|
| $z$, $r(z) = \max\{\bar{r} + z, 0\}$ | Latent rate state and the one-period log gross cash return |
| $\tau_i^a$, $\xi_i^a$, $\xi_i = \xi_i^C / \xi_i^M$ | Investor $i$'s tax rate and after-tax share in market $a$. In the two-market benchmark, $\xi_i$ is its relative after-tax share and $1 - \xi_i$ its exemption benefit |
| $\xi_H < \xi_I$, $\xi^*$ | Benchmark relative after-tax shares of the high-tax household and lower-tax investor, and their tax mix $\omega_H \xi_H + \omega_I \xi_I$ with risk-tolerance shares $\omega_i$ |
| $p_T^a(z)$, $A_T^a = -\partial_z p_T^a$, $y_T^a = -p_T^a / T$ | Log price of the maturity-$T$ bond in market $a$, its rate loading (the model's duration), and its yield, with $A_0^a = 0$ |
| $C_T^a(z) = -p_T^a(z) - A_T^a(z) z$ | Level term, the constant intercept in the affine model |
| $\mu_T^a(z) = \log \mathbb{E}_z[1 + R_T^a]$, $s_T^a = \partial_z \mu_T^a$ | Mean approximated return, equal to its holders' required return in the portfolio conditions, and its response to the state |
| $R_{T,t+1}^a$, $\mathcal{R}_{T,t+1}^a = \mu_T^a + u_{T,t+1}^a$ | Simple holding-period return and the return used in portfolio choice, with mean $\mu_T^a$ and centered log-return innovation $u_{T,t+1}^a$ |
| $\sigma_{\eta,T}^a$, $v_T^a = (\sigma_{\eta,T}^a)^2$ | Standard deviation of the bond-specific log-return shock and its variance, zero for the one-period claim |
| $\nu_{i,T}^a$ | Slack in the portfolio condition: required return minus expected return, zero for a positive holding |
| $x_{i,T}^a$, $Q_T^a$, $m_i$, $\gamma_i$ | Holdings per unit of mass, total supply, investor mass, risk aversion. In the calibration, $m_i$ measures baseline portfolio assets |
| $W_i$, $W_i'$, $\widehat{W}_i'$, $\kappa_i$ | Funding per unit of mass, next-period net wealth, its discrete-time approximation, and the multiplier on the shorting constraint on cash |
| $\widehat{D}_i$, $\ell_i$ | Net rate exposure and the signed liability hedge. Appendix B.1 uses the outside rate exposure $h_i = -\ell_i$ |
| $\widehat{\Lambda}$ | Difference in risk-weighted net rate exposures, $\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$, at the equilibrium allocation |
| $\omega_{i,T}^a$, $\mathcal{U}_T^a$ | Risk-tolerance share of a holder and the set of holders of the maturity-$T$ bond in market $a$ |
| $g_T = \mu_T^M - \xi_H \mu_T^C$ | One-period return gap, the municipal return less the high-tax household's after-tax taxable return |
| $\tau_T^{Y,\text{impl}} = 1 - y_T^M / y_T^C$ | Model yield-implied tax, the counterpart of *ImpliedTax* in (2) |

## A.1 The general economy

This subsection sets out the general economy, which adds bond-specific return risk and trading exclusions to the model of Section 4.1. As in that model, investors with different tax rates and liability hedges choose jointly across bond markets and maturities. Their portfolio conditions (the Karush–Kuhn–Tucker, or KKT, first-order conditions with the nonnegativity constraints) give the return each investor requires on each bond. Market clearing determines who holds it, and a price recursion turns one-period returns into prices and yields. The quantitative model of Section 5 solves this economy with several investor groups, state municipal markets, shorting constraints on cash that can bind and the rate floor.

### A.1.1 Assets and portfolio choice

**The rate state.** Time is discrete and indexed by $t$. One aggregate rate state $z_t$ follows a stationary first-order autoregression, AR(1), and the short rate, the pretax log gross return on cash, cannot fall below zero:

$$z_{t+1} \;=\; \rho z_t \;+\; \sigma_r \varepsilon_{t+1}, \qquad r_t = r(z_t) := \max\{\bar{r} + z_t, 0\}, \qquad \rho \in (0,1), \quad \varepsilon_{t+1} \sim \mathcal{N}(0,1). \tag{31}$$

Here $\rho$ measures the persistence of the rate state and $\sigma_r$ the size of its shocks. The parameter $\bar{r}$ is the cash rate before the floor at the stationary mean $z = 0$. The floor applies to the cash return and leaves the state $z_{t+1}$ Gaussian. It makes resale prices, the next-period prices of bonds, nonlinear in the state (Appendix A.1.2).

**Bonds and holding returns.** Let $\mathcal{A}$ be a finite set of bond markets, indexed by $a$ and distinguished by asset class or issuer. In the two-investor affine model, $\mathcal{A}$ contains a taxable market $C$ and a municipal market $M$. The quantitative model has 45 municipal markets, pooling the seven states without an income tax. Each market contains zero-coupon bonds indexed by their remaining maturity $T \in \{1, \dots, T_{\max}\}$, the number of periods until the bond pays one unit, so $P^a_0 = 1$. The one-period claim of the taxable market $C$ is the cash account, with log gross return $r(z)$, so risky taxable bonds begin at $T = 2$. A bond with $T$ periods remaining has price $P^a_T(z_t)$ and holding-period return

$$1 + R^a_{T,t+1} \;=\; \frac{P^a_{T-1}(z_{t+1})}{P^a_T(z_t)} \exp\!\big(\sigma^a_{\eta,T}\, \eta^a_{T,t+1}\big). \tag{32}$$

The standard normal shocks $\eta^a_{T,t+1}$ are independent of each other and of $\varepsilon_{t+1}$. They add bond-specific log-return variance $v^a_T := (\sigma^a_{\eta,T})^2$, and setting $v^a_T = 0$ gives the main-text return. Every market has $\sigma^a_{\eta,1} = 0$, so a one-period claim is fully riskless. Bond-specific risk represents residual variation in holding returns, such as an issue-specific change in trading demand. It makes longer bonds imperfect substitutes and allows joint holdings at more than one maturity, without adding a state variable that affects future prices.

**Investors.** Investors belong to finitely many types, indexed by $i$. Type $i$ has mass $m_i > 0$, risk aversion $\gamma_i > 0$, nonnegative funding $W_i$ and liability hedge $\ell_i$ against the rate shock from assets and liabilities beyond its bonds. It keeps the absolute after-tax share $\xi^a_i = 1 - \tau^a_i > 0$ of returns in market $a$, where $\tau^a_i$ is its tax rate on those returns. Investor $i$ is eligible for a bond unless a predetermined trading exclusion fixes its holding at zero, and the portfolio conditions below apply to eligible investor–bond pairs. Because after-tax shares differ across investors and markets, the same structure covers municipal and taxable bonds, state taxes on treasury and corporate interest, tax-favored accounts and cross-border withholding (Appendix A.6).

**Holdings and the return approximation.** Investor $i$ holds bonds $x^a_{i,T,t}$ and cash $x^0_{i,t} = W_i - \sum_{a,T} x^a_{i,T,t}$, in dollars per unit of mass. Aggregate holdings are $m_i x^a_{i,T,t}$, in the same units as supply. With simple holding-period returns $R^a_{T,t+1}$, exact next-period net wealth is

$$W'_i = W_i + \xi^C_i x^0_i \big(e^{r(z_t)} - 1\big) + \sum_{a,T} \xi^a_i x^a_{i,T} R^a_{T,t+1} + \ell_i \sigma_r \varepsilon_{t+1}.$$

Taking logs of (32) gives the change in log price plus the bond-specific shock. Following Greenwood et al. (2024), I approximate the simple bond return by this log return plus a Jensen adjustment, and $e^r - 1$ by $r$ for cash. The approximated bond return is

$$\mathcal{R}^a_{T,t+1} = \log(1 + R^a_{T,t+1}) + \underbrace{\log \mathbb{E}_t[1 + R^a_{T,t+1}] - \mathbb{E}_t[\log(1 + R^a_{T,t+1})]}_{\text{Jensen adjustment}}.$$

The Jensen adjustment is known at time $t$, so it shifts the mean but leaves log-return variances and covariances unchanged. As in Section 4.1, write $\mathcal{R}^a_{T,t+1} = \mu^a_T(z_t) + u^a_{T,t+1}$. Taking the conditional expectation gives the expected return, Jensen adjustment included, $\mathbb{E}_t[\mathcal{R}^a_{T,t+1}] = \mu^a_T(z_t) = \log \mathbb{E}_t[1 + R^a_{T,t+1}]$. Subtracting this mean cancels the adjustment and leaves the centered log-return innovation

$$u^a_{T,t+1} = \mathcal{R}^a_{T,t+1} - \mu^a_T(z_t) = \log(1 + R^a_{T,t+1}) - \mathbb{E}_t[\log(1 + R^a_{T,t+1})].$$

In the affine model, log returns are conditionally Gaussian, so

$$\mu^a_T(z_t) = \mathbb{E}_t[\log(1 + R^a_{T,t+1})] + \tfrac{1}{2} \operatorname{Var}_t[\log(1 + R^a_{T,t+1})].$$

This mean and the centered innovation $u$ match equations (7)–(8) of Greenwood et al. (2024). The quantitative model uses $\mu = \log \mathbb{E}_t[1 + R]$, retaining the full Jensen adjustment even when log returns are not Gaussian.

**Exactly taxed simple returns.** The price recursion (42) below is exact for the chosen mean $\mu = \log \mathbb{E}_t[1+R]$. Using this mean in the linear wealth equation (33) below approximates simple returns, whose conditional mean is $e^{\mu} - 1$. Appendix D.3 compares the two conventions for a deterministic return. Section 5 estimates the coefficients in a one-market model with a single national municipal market. To measure the approximation's effect on the solved model, I re-solve that model with exactly taxed simple returns. I refit risk aversion to restore the targeted taxable yields and hold the other coefficients fixed. The thirty-year gap in that model rises from 102.6 to 107.7 basis points, so the approximation does not inflate the gap.

**The portfolio problem.** Substituting the approximated returns into exact wealth gives approximated next-period wealth $\widehat{W}_i'$. Each investor maximizes the mean of $\widehat{W}_i'$ less a penalty for its variance:[23]

$$\begin{aligned} \max_{x_i \geq 0,\, x_{i,t}^0 \geq 0} \quad & \mathbb{E}_t[\widehat{W}_i'] - \frac{\gamma_i}{2} \operatorname{Var}_t(\widehat{W}_i'), \\ \text{subject to} \quad & \widehat{W}_i' = W_i + \xi_i^C r(z_t) x_{i,t}^0 \\ & \qquad + \sum_{a,T} \xi_i^a x_{i,T,t}^a \mathcal{R}_{T,t+1}^a + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{33}$$

Positions and cash are nonnegative, which rules out short sales and borrowing (Section 3.5). Funding and masses are fixed across states. The liability hedge $\ell_i$ can take either sign.

Collecting the cash and bond terms in (33) gives each bond's expected after-tax excess return over cash,

$$\widetilde{\mu}_{i,T}^a(z_t) := \underbrace{\xi_i^a \mu_T^a(z_t)}_{\text{after-tax bond return}} - \underbrace{\xi_i^C r(z_t)}_{\text{after-tax cash return}}. \tag{34}$$

The bond and outside innovations have conditional mean zero, so expected approximated wealth is initial funding grown at the after-tax cash rate plus the portfolio's excess returns:

$$\mathbb{E}_t[\widehat{W}_i'] = W_i\big(1 + \xi_i^C r(z_t)\big) + \sum_{a,T} x_{i,T,t}^a \widetilde{\mu}_{i,T}^a(z_t).$$

The variance of wealth is that of the after-tax portfolio innovation, the random part of (33):

$$\widetilde{\Pi}_i(t+1) = \sum_{a \in \mathcal{A}} \sum_T \xi_i^a x_{i,T,t}^a u_{T,t+1}^a + \underbrace{\ell_i \sigma_r \varepsilon_{t+1}}_{\text{liability hedge}}. \tag{35}$$

Dropping the portfolio-independent term $W_i(1 + \xi_i^C r(z_t))$ and eliminating cash gives the equivalent reduced problem:

$$\max_{\substack{x_i \geq 0 \\ \sum_{a,T} x_{i,T,t}^a \leq W_i}} \underbrace{\sum_{a \in \mathcal{A}} \sum_T x_{i,T,t}^a \widetilde{\mu}_{i,T}^a(z_t)}_{\text{after-tax excess return}} - \underbrace{\frac{\gamma_i}{2} \operatorname{Var}_t\Big(\widetilde{\Pi}_i(t+1)\Big)}_{\text{whole-portfolio risk cost}}. \tag{36}$$

**The portfolio conditions.** Let $\kappa_i(z) \geq 0$ denote the multiplier on the shorting constraint on cash, $\sum_{a,T} x_{i,T}^a \leq W_i$. The objective in (36) is concave and its constraints are linear, so the following portfolio conditions are necessary and sufficient for optimality:

$$\begin{aligned} x_{i,T}^a(z) > 0 \quad &\Longrightarrow \quad \widetilde{\mu}_{i,T}^a(z) = \underbrace{\gamma_i \xi_i^a \operatorname{Cov}_z\Big(u_{T,t+1}^a, \widetilde{\Pi}_i(t+1)\Big)}_{\text{marginal portfolio-risk cost}} + \kappa_i(z), \\ x_{i,T}^a(z) = 0 \quad &\Longrightarrow \quad \widetilde{\mu}_{i,T}^a(z) \leq \gamma_i \xi_i^a \operatorname{Cov}_z\Big(u_{T,t+1}^a, \widetilde{\Pi}_i(t+1)\Big) + \kappa_i(z). \end{aligned} \tag{37}$$

The right side is the investor's required after-tax excess return. The multiplier satisfies complementary slackness, $\kappa_i(z)[W_i - \sum_{a,T} x_{i,T}^a(z)] = 0$, so it is positive only when the investor holds no cash. Buying more of one bond then means selling another, and $\kappa_i$ raises the required pretax return on each held bond by $\kappa_i/\xi_i^a$. Since $\widehat{W}_i' - \widetilde{\Pi}_i(t+1)$ is known at $t$, the covariance equals the main-text $\operatorname{Cov}_z(u_{T,t+1}^a, \widehat{W}_i')$ in (8).

Rate risk in the covariance comes from the next-period resale price. With log price $p_T^a := \log P_T^a$ and rate loading $A_T^a := -\partial_z p_T^a$, a bond with $T$ periods remaining has the rate-driven log-return innovation $-A_{T-1}^a(\rho z_t)\sigma_r \varepsilon_{t+1}$ to first order around the expected resale state. A one-period bond carries no rate risk, since $A_0^a = 0$, although its price can respond to today's state.

[23] Appendices C.1.4 and D.3 give the correction for taxing resale gains separately from interest.

**Market clearing and equilibrium.** Every bond other than cash clears at every rate state:

$$x^a_{i,T}(z) \geq 0, \qquad \sum_i m_i\, x^a_{i,T}(z) = Q^a_T, \tag{38}$$

where $Q^a_T$ is the fixed dollar supply. An equilibrium consists of log-price functions with $p^a_0 \equiv 0$, holdings and multipliers $\kappa_i(z) \geq 0$ such that, at every state, returns follow (32), every investor satisfies $\sum_{a,T} x^a_{i,T}(z) \leq W_i$, (37) and complementary slackness, and (38) holds.

**Riskless one-period claims.** Consider a riskless one-period claim in any market $a \neq C$. Its bond-specific variance $v^a_1 = (\sigma^a_{\eta,1})^2$ is zero and $p^a_0 = 0$, so its return has no covariance with portfolios or liabilities. Retaining $\kappa_i$ in (37), each eligible investor's portfolio condition becomes

$$\mu^a_1 \leq \frac{\xi^C_i r + \kappa_i}{\xi^a_i}, \qquad \text{with equality for a positive holding.}$$

With positive supply, at least one investor holds the claim, so its return equals the smallest of these bounds:

$$\mu^a_1(z) = \min_{i \text{ eligible}} \frac{\xi^C_i r(z) + \kappa_i(z)}{\xi^a_i}. \tag{39}$$

The price recursion (42) below then gives $p^a_1 = -\mu^a_1$, hence $y^a_1 = \mu^a_1$.

### A.1.2 Bond prices and required returns

A bond's price depends on the one-period returns it earns as it ages to maturity, and its rate loading accumulates their responses to the rate state. Section 4.5 uses both relations on an affine branch, and this subsection derives them for the general economy.

**The price recursion.** Taking expectations of the gross holding return (32), with next-period state $Z' = \rho z + \sigma_r \varepsilon'$, gives

$$P^a_T(z) = e^{-\mu^a_T(z) + v^a_T/2}\, \mathbb{E}_z\big[P^a_{T-1}(Z')\big], \qquad P^a_0 = 1. \tag{40}$$

The factor $e^{v^a_T/2}$ accounts for the mean of the bond-specific shock, and its yield effect is small (Appendix B.5).

With fixed supplies and funding, I seek an equilibrium log-price function for each market and maturity:

$$p^a_{T,t} = p^a_T(z_t), \qquad p^a_0(z) \equiv 0, \qquad a \in \mathcal{A}. \tag{41}$$

Writing (40) in logs, with $\mu^a_T$ the log expected gross return (32), gives

$$p^a_T(z) = \underbrace{\log \mathbb{E}_z\big[\exp\{p^a_{T-1}(Z')\}\big]}_{\text{log expected resale price}} + \underbrace{\tfrac{1}{2}(\sigma^a_{\eta,T})^2}_{\text{bond-specific payoff term}} - \underbrace{\mu^a_T(z)}_{\text{one-period return}}. \tag{42}$$

This recursion links adjacent maturities, while the portfolio conditions (37) link markets through investors' holdings. At every state, prices and portfolios jointly solve these equations and clearing (38). Appendix B gives the numerical implementation.

**Sources of nonlinearity.** The rate floor, changes in participation and binding shorting constraints on cash make prices nonlinear in the state. Resale carries these nonlinearities across maturities, because today's expected resale value averages future prices across states that differ in these respects.

**Prices and yields through one-period returns.** Iterating (42) from $p^a_0 = 0$ expresses the price through the one-period returns the bond earns as it ages through maturities $T, T-1, \ldots, 1$:

$$P^a_T(z_t) = e^{p^a_T(z_t)} = \mathbb{E}_t\Big[\exp\Big\{-\sum_{j=0}^{T-1}\Big(\mu^a_{T-j}(z_{t+j}) - \tfrac{1}{2}\big(\sigma^a_{\eta,T-j}\big)^2\Big)\Big\}\Big]. \tag{43}$$

Taking the negative log of (43) and dividing by $T$ gives the yield $y^a_T = -p^a_T/T$. In the affine model, (54) writes this yield as the average of expected one-period returns less a convexity term.

**Rate loadings and level terms.** In the general economy, I summarize the yield's dependence on the state by writing the log price in loading form,

$$p^a_T(z) = -A^a_T(z)\, z - C^a_T(z), \qquad A^a_T(z) := -\frac{\partial p^a_T(z)}{\partial z}, \tag{44}$$

so that $P_T^a = \exp\{-A_T^a z - C_T^a\}$ and, at differentiability points,

$$y_T^a(z) = \frac{A_T^a(z)z + C_T^a(z)}{T}, \qquad \frac{\partial y_T^a(z)}{\partial z} = \frac{A_T^a(z)}{T}.$$

The level term $C_T^a$ includes accumulated one-period returns and convexity, and the mean-state yield is $C_T^a(0)/T$. Outside the affine model, both $A_T^a$ and $C_T^a$ can vary with the state. The yield derivative above still holds because $y_T^a = -p_T^a/T$. The loading ratio $A_T^M/A_T^C$ is therefore the model counterpart of the relative response in P3.

**The loading recursion.** The loading recursion links $A_T^a$ to the one-period return response $s_T^a := \partial_z \mu_T^a$. Above the rate floor, cash has response one, and a riskless one-period claim in another market has response $\xi_i^C/\xi_i^a$ when, at all nearby states, investor $i$ attains the minimum in (39) and has $\kappa_i = 0$. At longer maturities the response also includes the change in the holders' risk premia. Suppose that $\mu_T^a$ is differentiable at a given state, that each resale price is the integral of its slope, which allows kinks, and that near this state the integrand $\exp\{p_{T-1}^a(\rho z + \sigma_r \varepsilon')\}$ and its derivative in $z$ are bounded by one function of $\varepsilon'$ with finite expectation. Differentiating under the expectation in (42) then gives

$$A_T^a(z) = \underbrace{s_T^a(z)}_{\text{current return response}} + \underbrace{\rho \frac{\mathbb{E}_z\left[\exp\{p_{T-1}^a(Z')\}\, A_{T-1}^a(Z')\right]}{\mathbb{E}_z\left[\exp\{p_{T-1}^a(Z')\}\right]}}_{\text{resale price response}}. \tag{45}$$

The fraction averages the next-period loading across next-period states, giving more weight to states where the resale bond is more valuable. Through this average, future changes in who holds the bond affect today's loading. In the affine model, (45) reduces to the loading recursion (53).

### A.1.3 Market clearing and the riskless short yield

Proposition A.1 shows that, given resale prices, clearing returns exist, and gives sufficient conditions for their uniqueness. Proposition A.2 gives conditions under which the riskless one-period municipal yield equals the household benchmark, $\xi_H r$.

**Supports and participation branches.** An allocation's support is the set of investor–bond pairs with positive holdings. A participation branch is a fixed pattern of which investors hold which bonds, that is, a support held fixed across rate states (Section 4.2). Affine results use floorless rates and a fixed participation branch.

**Portfolio conditions for all markets.** Index market–maturity coordinates other than cash by $j = (a, T)$ and write $v_j = (\sigma_{\eta,T}^a)^2$, $K_i = \mathrm{diag}(\xi_i^a)$ and the relative after-tax share $\xi_{ij} = \xi_i^C/\xi_i^a$, collected in the vector $\boldsymbol{\xi}_i$. With one municipal and one taxable market, as in Section 4.2, $\xi_{ij} = \xi_i$ at municipal coordinates and $\xi_{ij} = 1$ at taxable coordinates. With resale log price $p_{j,-} = p_{T-1}^a$, the exact covariance and the liability credit (the amount by which the liability hedge lowers the covariance of bond $j$'s return with investor $i$'s wealth) are

$$\Sigma_{jk}(z) = \mathrm{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j, \qquad c_{ij}^L(z) = -\mathrm{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r \varepsilon'). \tag{46}$$

With a slack shorting constraint on cash, dividing each eligible investor's first-order condition by the positive after-tax share of its market gives

$$\begin{gathered} \nu_{ij} := \xi_{ij} r + \gamma_i [\Sigma K_i x_i - c_i^L]_j - \mu_j \geq 0, \qquad x_{ij} \geq 0, \qquad x_{ij}\nu_{ij} = 0, \\ \sum_i m_i x_{ij} = Q_j. \end{gathered} \tag{47}$$

Here the slack $\nu_{ij}$ is the required minus the expected return. Concavity of (36) makes these conditions sufficient for investor optimality. With future resale prices held fixed, return covariances and liability credits are given, and current returns must adjust until investors' optimal holdings match bond supplies.

**Proposition A.1** (Market clearing with given resale prices)**.** *Fix the state and resale prices, with the shorting constraint on cash dropped, $\kappa_i \equiv 0$, finite nonnegative supplies, finite $\Sigma \succeq 0$, finite liability credits and at least one eligible investor for every positive-supply coordinate.*

*(i) The system of portfolio and clearing conditions (47) has a current-state solution. Holdings that also satisfy $\sum_{a,T} x_{i,T}^a \leq W_i$ for every $i$, one-period claims included, solve the problem with that constraint.*

*(ii) If, in addition, $\Sigma = V + \sigma_r^2 AA'$ with $V = \mathrm{diag}(v_j)$, and every risky coordinate $j = (a, T)$ has $v_j > 0$, supply $Q_j > 0$ and next-period loading $A_j = A_{T-1}^a > 0$ (the loading of the bond it becomes next period), the*

*current-state clearing returns and risky holdings are unique, even with endogenous support changes. Part (ii) covers resale log prices that are affine in the state, with bond-specific risk on every risky bond, taxable bonds included. A riskless one-period claim in positive supply has zero covariances and liability credits, so (39) with $\kappa_i = 0$ sets its return. Its allocation among investors tied at that minimum need not be unique. Neither part establishes the dynamic fixed point of resale prices.*

**Proof.** *(i) Existence.* Let $X = \{x \geq 0 : \sum_i m_i x_i = Q,\ x_{ij} = 0 \text{ for ineligible } (i,j)\}$ be the set of clearing allocations, with inner product $\langle x, y\rangle_m = \sum_i m_i x_i' y_i$. Positive masses, finite supplies and an eligible investor for each positive-supply coordinate make $X$ nonempty and bounded, and its linear constraints make it closed and convex. Let $\Upsilon_i(x_i) = \boldsymbol{\xi}_i r + \gamma_i(\Sigma K_i x_i - c_i^L)$ be investor $i$'s required return on each coordinate, and $\mathrm{Proj}_X(w)$ the point of $X$ nearest to $w$ in the distance defined by $\langle\cdot,\cdot\rangle_m$. By Brouwer's fixed-point theorem, a continuous map of a closed, bounded and convex set into itself has a fixed point. Applied to the continuous map $x \mapsto \mathrm{Proj}_X(x - \Upsilon(x))$ of $X$ into itself, it gives an $x \in X$ that is the point of $X$ nearest to $x - \Upsilon(x)$. Since $X$ contains the segment from $x$ to any $y \in X$, moving from $x$ toward $y$ cannot reduce the squared distance to $x - \Upsilon(x)$, so its derivative in that direction is nonnegative, which gives

$$\sum_i m_i \langle \Upsilon_i(x_i), y_i - x_i\rangle \geq 0 \qquad (y \in X).$$

Moving a small amount $\delta > 0$ of coordinate $j$ from a holder $k$ to another eligible investor $l$, $y_{kj} = x_{kj} - \delta/m_k$ and $y_{lj} = x_{lj} + \delta/m_l$, keeps $y$ in $X$ and gives $\delta(\Upsilon_{lj} - \Upsilon_{kj}) \geq 0$. Thus each holder minimizes $\Upsilon_{ij}$, and $\mu_j = \min_{i \text{ eligible}} \Upsilon_{ij}$ and $\nu_{ij} = \Upsilon_{ij} - \mu_j$ satisfy (47).

*(ii) Uniqueness.* Bonds are substitutes in demand, and own-return effects dominate once returns are weighted by loadings, so the average demand Jacobian $\overline{\mathcal{J}}$ defined below is invertible. Restrict to the risky coordinates. Since every $v_j > 0$, $\Sigma$ is positive definite, so each objective is strictly concave and demand $x_i(\mu)$ is unique and continuous. Where investor $i$ holds exactly the coordinates in a set $B_i$, its binding portfolio conditions are linear, so its demand is affine with derivative

$$\mathcal{J}_i = \gamma_i^{-1} \mathbf{E}_{B_i} K_{i,B_i}^{-1} \Sigma_{B_i B_i}^{-1} \mathbf{E}_{B_i}',$$

where the matrix $\mathbf{E}_B$ embeds the held coordinates. The Sherman–Morrison formula gives

$$\Sigma_{BB}^{-1} = V_B^{-1} - \frac{\sigma_r^2 V_B^{-1} A_B A_B' V_B^{-1}}{1 + \sigma_r^2 A_B' V_B^{-1} A_B}, \qquad \Sigma_{BB}^{-1} A_B = \frac{V_B^{-1} A_B}{1 + \sigma_r^2 A_B' V_B^{-1} A_B} \gg 0.$$

Off the diagonal, $V_B^{-1}$ is zero and the subtracted term is positive, so the off-diagonal entries of $\Sigma_{BB}^{-1}$ are negative, and $\Sigma_{BB}^{-1} A_B$ is positive. Where every $B_i$ is fixed, $\mathcal{J} = \sum_i m_i \mathcal{J}_i$ therefore has nonpositive off-diagonal entries and $\mathcal{J} A \geq 0$, strictly on held coordinates. Now suppose $\mu^0$ and $\mu^1$ both clear. There are finitely many holding patterns, and the returns at which each occurs are cut out by linear inequalities, so the segment between $\mu^0$ and $\mu^1$ splits into finitely many intervals on each of which aggregate demand is affine. Integrating the derivative of aggregate demand along the segment gives $0 = \overline{\mathcal{J}}(\mu^1 - \mu^0)$, with $\overline{\mathcal{J}} = \int_0^1 \mathcal{J}(\mu^0 + t(\mu^1 - \mu^0))\, dt$. Positive supply at $\mu^0$ and demand continuity give each coordinate a holder on an initial interval of the segment, where $(\mathcal{J}A)_j > 0$, so $(\overline{\mathcal{J}}A)_j > 0$ for every $j$. With nonpositive off-diagonal terms and positive row sums, $\overline{\mathcal{J}}\,\mathrm{diag}(A)$ has each diagonal term larger than the sum of the absolute values of the other terms in its row, so it is invertible, returns coincide, and unique demand gives identical risky holdings. □

**Regularity for derivatives.** Derivatives below are taken at regular states, at which the cash rate and price moments are continuously differentiable, an eligible investor that does not hold a risky bond requires more than the bond's expected return, and the Jacobian of the binding portfolio conditions and clearing is nonsingular. Proposition A.1(ii) gives conditions under which the demand Jacobian is nonsingular. At regular states, the implicit function theorem implies that the participation branch stays fixed at nearby states and that holdings and returns change smoothly with the rate state. Fixed supplies then imply, for the set $\mathcal{U}_j$ of each bond's holders,

$$\sum_{i \in \mathcal{U}_j} m_i \frac{\partial x_{ij}}{\partial z} = 0. \tag{48}$$

**The riskless short end.** Specializing the riskless-claim return (39) to one municipal and one taxable market gives the one-period yields (12) and tax pass-throughs (21).

**Proposition A.2** (Riskless one-period yields)**.** *With one municipal and one taxable market, any number of investors and normalized payoffs, suppose the household's shorting constraint on cash is slack, $\kappa_H = 0$, the one-period municipal claim has positive supply, and $r(z) > 0$. Suppose the high-tax household is eligible for the claim and has the minimal relative after-tax share among eligible investors, $\xi_H = \min_{i \text{ eligible}} \xi_i$. Then only investors with this minimal share can hold the claim, and*

$$y_1^M = \mu_1^M = \xi_H r(z), \qquad y_1^C = r(z), \qquad \tau_1^{Y,\text{impl}} = 1 - \xi_H. \tag{49}$$

*If the household is the unique minimizer and $\kappa_H = 0$ at nearby after-tax shares, then the pass-throughs (20) are $\mathcal{P}_H^Y(1) = 1$ and $\mathcal{P}_i^Y(1) = 0$ for every investor with $\xi_i > \xi_H$, which is (21). At a tie, the household pass-through requires perturbing all tied minimal after-tax shares together. At $r = 0$ the yield ratio is undefined.*

**Proof.** With $\xi_i^M = 1$ and $\xi_i^C = \xi_i$, the riskless-claim condition preceding (39) is $\mu_1^M \leq \xi_i r + \kappa_i$, with equality for positive holdings. The household's condition gives $\mu_1^M \leq \xi_H r$. Positive supply gives a holder $i$, and $\mu_1^M = \xi_i r + \kappa_i \geq \xi_H r$ because $\xi_i \geq \xi_H$ and $\kappa_i \geq 0$. Hence $\mu_1^M = \xi_H r$, and both inequalities hold with equality, so $\kappa_i = 0$ and, since $r > 0$, $\xi_i = \xi_H$. The recursion gives $p_1^M = -\mu_1^M$, so $y_1^M = \xi_H r$. Taxable cash has $p_1^C = -r$, so $y_1^C = r$ and $\tau_1^{Y,\text{impl}} = 1 - y_1^M / y_1^C = 1 - \xi_H$. Only the minimal after-tax share appears in this ratio, so differentiating it proves the tax statements. □

## A.2 The two-investor affine model

This subsection specializes the general economy to two investors and two jointly priced bond markets. It proves the participation and pricing results of Sections 4.3–4.6 in the setting used there, where risky bonds carry rate risk only, with no bond-specific return shock. Without bond-specific risk, investors sort by maturity whenever $r > 0$ and rate loadings rise with maturity. When the lower-tax investor bears rate risk more cheaply, the high-tax household holds the shorter municipal bonds, the lower-tax investor holds the longer ones, and at most one maturity is shared (Theorem A.1). Appendices A.2.1–A.2.7 work on the illustrated branch, the participation branch of the main-text illustration (Section 4.3 and Figure 4). On this branch, the household alone holds municipal maturities below the threshold maturity $T^*$, both investors hold $T^*$, and the lower-tax investor alone holds longer municipal bonds and all risky taxable bonds. Appendix A.2.8 then adds a small bond-specific shock, as in the quantitative model. The shock gives a branch with a jointly held block of maturities. Except at the long end, the pricing results carry over with the threshold maturity replaced by that block.

For the riskless one-period municipal claim, a slack shorting constraint on cash gives $\mu_1^M = \xi_H r$ whenever $r > 0$, and the household, which keeps the smallest share of taxable interest, holds the claim (Proposition A.2). At longer maturities, participation determines whose taxes and risk costs shape equilibrium returns.

### A.2.1 Assumptions and portfolio conditions

This part sets out the economy of Section 4.2, the simplifications that make prices affine and the net rate exposure behind the portfolio conditions (11).

**Investors, assets and units.** The high-tax household $H$ and the lower-tax investor $I$ choose among municipal maturities $T = 1, \ldots, T_{\max}$, taxable maturities $T = 2, \ldots, T_{\max}$ and cash, the one-period taxable claim. Bond supplies $Q_T^a$ are fixed. Risk aversions $\gamma_i$ can differ, and the relative after-tax share $\xi_i = \xi_i^C / \xi_i^M$ is that of Section 4.2. Payoffs are in units of after-tax municipal income, so taxable returns and cash income are multiplied by $\xi_i$, with $\gamma_i$ and $\ell_i$ normalized to match. All two-investor results assume $0 < \rho < 1$, $\sigma_r > 0$, $m_i, \gamma_i > 0$, $0 < \xi_H < \xi_I$ and positive traded supplies. Bond-specific variance $v_T^a = (\sigma_{\eta,T}^a)^2$ is zero through Appendix A.2.7. The general formulas below keep the bond-specific terms, which Appendix A.2.8 uses.

**Participation branches and three simplifications.** A participation branch, as in Section 4.2, specifies which investors hold each bond. I fix a branch, solve its portfolio conditions and market clearing, and check that no investor benefits from changing its holdings. Closed forms use three simplifications:

(i) the shorting constraint on cash is dropped;

(ii) the floored cash rate is replaced by the linear rule $r(z) = \bar{r} + z$; and

(iii) prices are computed as if the same participation branch applied at every state, which makes log prices affine, and the branch is verified at the states studied.

The quantitative model lets participation change across states and keeps the shorting constraint on cash and the rate floor.

On a fixed branch, holdings, one-period returns and log prices are affine in $z$, and the price coefficients solve a fixed-point problem whose feasibility and uniqueness are examined below. One-period returns take the form $\mu_T^{a,\text{aff}}(z) = \mu_T^{a,\text{aff}}(0) + s_T^{a,\text{aff}} z$, so $s_T^{a,\text{aff}}$ is the return's state response, and the rate loading $A_T^{a,\text{aff}}$ is the coefficient on $z$ in the negative log price. The superscript "aff", where shown, marks this branch.

**Net rate exposure.** Investor $i$'s net rate exposure is the after-tax exposure of its whole balance sheet to the next-period rate state:

$$\widehat{D}_i(z) := \underbrace{\sum_T x_{i,T}^M(z) A_{T-1}^M + \xi_i \sum_T x_{i,T}^C(z) A_{T-1}^C}_{\text{tax-weighted asset exposure}} - \underbrace{\ell_i}_{\text{liability hedge}}. \tag{50}$$

With today's positions fixed and wealth in units of after-tax municipal income, $\widehat{D}_i = -\partial \widehat{W}_i'/\partial z_{t+1}$, as in (10). It is dollar exposure before division by wealth. The loadings in (50) are next-period loadings $A_{T-1}^a$, because the rate shock moves the bonds' resale prices next period. The rate component of the wealth innovation is $-\widehat{D}_i \sigma_r \varepsilon_{t+1}$, and a larger liability hedge $\ell_i$ lowers $\widehat{D}_i$. A bond's rate-risk compensation equals its loading times the investor's unit cost of rate risk, $\gamma_i \sigma_r^2 \widehat{D}_i$, and the portfolio conditions are those of (11). Loadings are constant on an affine branch, so changes in $\widehat{D}_i$ across rate states reflect trading alone.

**Uniqueness and illustrations.** On the illustrated branch, each threshold maturity gives a unique response of net rate exposures to the rate state and hence unique loadings (Proposition 2). A linear level condition then pins the holding split at the threshold maturity. Proposition A.1 gives sufficient conditions for unique current-state clearing when every risky bond carries bond-specific risk. The main-text illustration shares only municipal maturity 23 (Appendix A.2.7). Adding small bond-specific risk gives overlapping holdings from maturity 17 (Appendix A.2.9). Both appendices report the parameters and check holdings and pricing.

### A.2.2 Affine price loadings and yields

A bond's rate loading and yield build up from its one-period returns. I derive the loading recursion (16) and the yield average behind (19) of Section 4.5, then construct and check affine solutions.

**Prices, loadings and responses.** Write $p_T^a = \log P_T^a$ for the log price and $y_T^a = -p_T^a/T$ for the yield. Under the affine assumptions above,

$$\log P_T^a(z) = -A_T^a z - C_T^a, \qquad y_T^a(z) = \frac{A_T^a z + C_T^a}{T}. \tag{51}$$

The coefficient $A_T^a$, the bond's rate loading, is the proportional fall in its price per unit rise in the rate state and the model's counterpart of duration. Thus $A_T^a/T$ is the yield's response to the rate state and $C_T^a/T$ is its level at the mean state. The response of the one-period return to the rate state is

$$s_T^a(z) := \frac{\partial \mu_T^a(z)}{\partial z}. \tag{52}$$

**The loading recursion.** Substituting affine prices into the price recursion (42) and matching its state and constant terms gives

$$A_T^a = \underbrace{\rho A_{T-1}^a}_{\text{inherited loading}} + \underbrace{s_T^a}_{\text{current response}}, \qquad C_T^a = C_{T-1}^a + \mu_T^a(0) - \tfrac{1}{2}\left[\sigma_r^2 \left(A_{T-1}^a\right)^2 + \left(\sigma_{\eta,T}^a\right)^2\right]. \tag{53}$$

The level term accumulates mean-state one-period returns less half-variance terms. Starting from $A_0^a = C_0^a = 0$, the loading accumulates the responses, weighted by rate persistence, as in (56) below. With a riskless one-period claim, $\sigma_{\eta,1}^a = 0$, the same recursion gives the yield as the average of the expected one-period returns along the bond's life less convexity:

$$y_T^a(z_t) = \underbrace{\frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t\left[\mu_{T-j}^a(z_{t+j})\right]}_{\text{one-period returns as the bond ages}} - \underbrace{\frac{1}{2T}\sum_{q=2}^{T}\left[\sigma_r^2\left(A_{q-1}^a\right)^2 + \left(\sigma_{\eta,q}^a\right)^2\right]}_{\text{convexity}}. \tag{54}$$

On the illustrated branch, the bond-specific terms are zero. Appendix A.2.8 makes them positive for risky municipal bonds.

**Constructing affine solutions.** Fix a participation branch, set the positions it excludes to zero, set $r(z) = \bar{r} + z$ and use affine resale prices at all states in $\mathbb{R}$. For affine candidates $p_T^a(z) = -A_T^a z - C_T^a$ and next-period state $Z' = \rho z + \sigma_r \varepsilon'$, the log resale price is normal, so the lognormal mean formula gives exactly

$$\log \mathbb{E}_z e^{p_{T-1}^a(Z')} = -\rho A_{T-1}^a z - C_{T-1}^a + \tfrac{1}{2}\sigma_r^2 (A_{T-1}^a)^2. \tag{55}$$

The current-state covariance $\Sigma = \sigma_r^2 AA' + \mathrm{diag}(v)$ and liability credit $c_i^L = \ell_i \sigma_r^2 A$ of (46) are then constant, where $A$ collects the next-period loadings and $v$ the bond-specific variances $v_j$. The binding portfolio conditions and clearing then form one linear system in holdings and returns across all maturities, with a right side affine in $z$. If this system has a unique solution, holdings and $\mu$ are affine in $z$, and the price recursion returns affine prices with updated loadings $\rho A_{T-1}^a + s_T^a$.

A fixed point of this update is a set of coefficients that it returns unchanged. At a fixed point, all binding portfolio conditions, clearing equations and exact resale equations hold at every state, and, with $A_0^a = 0$,

$$A_T^a = \rho A_{T-1}^a + s_T^a = \sum_{j=1}^{T} \rho^{T-j} s_j^a, \qquad s_T^a = \frac{\partial \mu_T^a}{\partial z}. \tag{56}$$

The candidate is feasible at a state if holdings are nonnegative, the slacks on bonds an investor does not hold are nonnegative and, for finite funding, $W_i - \sum_j x_{ij}(z) \geq 0$. These quantities are affine in $z$, so the states where all of them are nonnegative form an interval, possibly empty. Normal rate shocks leave any interval other than the whole real line with positive probability, so feasibility on an interval does not make the candidate an equilibrium at every state. For example, on any branch on which the household holds the one-period municipal claim, the lower-tax investor's slack on that claim, $(\xi_I - \xi_H)(\bar{r} + z)$, turns negative at negative rates.

### A.2.3 Participation across maturities

Theorem A.1 proves the sorting described in Section 4.3. Without bond-specific risk, the two investors sort by maturity in equilibrium, with the lower-tax investor's municipal holdings at the long end. As loadings rise with maturity, that investor's duration benefit grows, while the difference in exemption benefits it must overcome is flat.

**The participation comparison.** At an equilibrium allocation, the household's and the lower-tax investor's municipal conditions are

$$\underbrace{\mu_T^M - \xi_H r}_{\text{offered to the household}} \leq \underbrace{\gamma_H \sigma_r^2 A_{T-1}^M \widehat{D}_H}_{\text{household's marginal risk cost}}, \tag{57}$$

$$\underbrace{\mu_T^M - \xi_I r}_{\text{offered to the lower-tax investor}} \leq \underbrace{\gamma_I \sigma_r^2 A_{T-1}^M \widehat{D}_I}_{\text{lower-tax investor's marginal risk cost}}, \tag{58}$$

each with equality for a positive holding. Both face the same expected return $\mu_T^M$, so the lower-tax investor's required return is no higher than the household's exactly when

$$\underbrace{\sigma_r^2 \, A_{T-1}^M \, \widehat{\Lambda}}_{\text{duration benefit (grows with maturity)}} \geq \underbrace{(\xi_I - \xi_H)\, r}_{\text{difference in exemption benefits (flat in maturity)}}, \quad \widehat{\Lambda} := \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I. \tag{59}$$

This is the comparison (13) of Section 4.3. The duration benefit, the lower-tax investor's saving in rate-risk cost on the bond, is the bond's next-period loading times $\sigma_r^2 \widehat{\Lambda}$, the difference between the two investors' unit costs of rate risk at the equilibrium allocation. Net rate exposures are common to all of an investor's bonds, so $\widehat{\Lambda}$ is the same at every maturity. The loading rises with maturity in the illustration and, by the proof of Proposition A.6(i), on the illustrated branch under condition (S) in (72) when the feedback $b$ of Proposition 2 is below $\rho$. A positive $\widehat{\Lambda}$ means that the household's risk-weighted net rate exposure exceeds the lower-tax investor's, as when the lower-tax investor's outside liabilities hedge more of its bond risk. In the main-text illustration, $\widehat{\Lambda} \approx 4.50$ at the mean rate state.

**Theorem A.1** (Sorting without bond-specific risk)**.** *At any state of a two-investor affine solution without bond-specific risk, define*

$$\Psi_T^0 = (\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda}. \tag{60}$$

*The superscript marks zero bond-specific risk. The household alone holds municipal maturity $T$ if $\Psi^0_T > 0$, the lower-tax investor alone holds it if $\Psi^0_T < 0$, and both hold it only if $\Psi^0_T = 0$. Suppose $r > 0$, $\widehat{\Lambda} > 0$ and the next-period loadings $A^M_{T-1}$ increase strictly with maturity, and put*

$$T^*(z) \;=\; \min\Big\{T \geq 1 :\; A^M_{T-1} \geq \frac{(\xi_I - \xi_H)\, r}{\sigma_r^2\, \widehat{\Lambda}}\Big\}, \tag{61}$$

*with $\min \varnothing = +\infty$. Then the household alone holds every municipal maturity below $T^*$, the lower-tax investor alone holds every maturity above $T^*$, and both can hold $T^*$ only if $\Psi^0_{T^*} = 0$. In particular, the household holds exactly the municipal maturities up to some cutoff, the lower-tax investor holds exactly those from some cutoff on, and at most one maturity is held by both.*

**Proof.** Each investor's municipal condition is $\mu^M_T - \xi_i r \leq \gamma_i \sigma_r^2 A^M_{T-1} \widehat{D}_i$, with equality for a positive holding. If the lower-tax investor holds maturity $T$, subtracting its equality from the household's inequality gives $(\xi_I - \xi_H) r \leq \sigma_r^2 A^M_{T-1} \widehat{\Lambda}$, that is, $\Psi^0_T \leq 0$. If the household holds it, subtracting the lower-tax investor's inequality from the household's equality gives $\Psi^0_T \geq 0$. Positive supply means that someone holds every maturity. Hence $\Psi^0_T > 0$ excludes the lower-tax investor, $\Psi^0_T < 0$ excludes the household, and joint holdings require $\Psi^0_T = 0$.

With $\widehat{\Lambda} > 0$ and strictly increasing next-period loadings, $\Psi^0_T$ decreases strictly with maturity. By (61), it is positive below $T^*$, negative above $T^*$ and zero at most at $T^*$. Since $A^M_0 = 0$ and $r > 0$, $\Psi^0_1 > 0$, so $T^* \geq 2$ and the household holds the one-period claim. □

**What sets the threshold.** Taxes, balance sheets and loadings set the threshold $T^*$, and no investor is assigned a maturity habitat. At given loadings and $\widehat{\Lambda}$, a larger difference in exemption benefits weakly raises the threshold, and a larger $\widehat{\Lambda}$ weakly lowers it. In equilibrium, $\widehat{\Lambda}$ and the loadings are determined jointly with holdings, so Theorem A.1 characterizes participation within a given equilibrium.

**Joint holdings at the threshold maturity.** At a regular state, holdings and returns move smoothly with the rate state, and each investor either holds a risky bond or has a strictly positive slack in its portfolio condition for that bond (Appendix A.1.3). There, $\Psi^0_{T^*} = 0$ is equivalent to joint holdings, so the threshold $T^*$ is the first maturity the lower-tax investor holds. Without bond-specific risk, a maturity is shared only when the duration benefit equals the difference in exemption benefits, $\Psi^0_T = 0$. At that maturity, holdings are not fixed by clearing alone. The split between the investors adjusts their net rate exposures until the condition holds, and it absorbs the trading as rates move (Proposition 2). In the main-text illustration, this maturity is 23, and the lower-tax investor holds about 47% of it (Appendix A.2.7). Appendix A.2.8 shows how bond-specific risk extends joint holdings to a block of maturities.

### A.2.4 Return-gap decomposition and rate loadings

This part proves the results behind Sections 4.4 and 4.5. The one-period return gap splits into the exemption benefit, the duration cost and the rate loading, and the holders' tax mix and portfolio responses determine the loadings in the rate-loading channel.

**The three channels of the one-period return gap.** Subtracting $\xi_H$ times the taxable equality from a municipal holder's equality separates the one-period return gap $g_T = \mu^M_T - \xi_H \mu^C_T$ into three channels. Proposition 1 below restates (15) with its conditions and gives the exemption-benefit and duration-cost contributions in each holding case.

**Proposition 1** (Three channels of the one-period return gap)**.** *In the two-investor affine model without bond-specific risk, consider a state at which the lower-tax investor alone holds the risky taxable bonds. Let $i$ denote a holder of the maturity-$T$ municipal bond, with $i = I$ whenever the lower-tax investor holds it. Then*

$$g_T = \underbrace{(\xi_i - \xi_H) r}_{\text{exemption benefit}} + \underbrace{\sigma_r^2 A^M_{T-1}\big(\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I\big)}_{\text{duration cost}} + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I \big(A^M_{T-1} - \xi_H A^C_{T-1}\big)}_{\text{rate loading}}. \tag{62}$$

*At $T = 1$ the duration-cost and rate-loading contributions are zero, and so is the exemption-benefit contribution when $r > 0$. Where the household holds the bond alone, the exemption-benefit contribution is zero and the duration-cost contribution is at most $(\xi_I - \xi_H) r$. Where the lower-tax investor holds it, the exemption-benefit contribution is $(\xi_I - \xi_H) r$ and the duration-cost contribution is zero.*

**Proof.** Investor $i$'s municipal equality gives $\mu_T^M = \xi_i r + \gamma_i \sigma_r^2 A_{T-1}^M \widehat{D}_i$. The lower-tax investor's taxable equality, divided by its after-tax share, gives $\mu_T^C = r + \gamma_I \sigma_r^2 A_{T-1}^C \widehat{D}_I$, which also covers cash at $T = 1$ because $A_0^C = 0$. Subtracting $\xi_H$ times the second equality from the first, then adding and subtracting $\gamma_I \sigma_r^2 A_{T-1}^M \widehat{D}_I$, gives (62).

At $T = 1$, $A_0^a = 0$, so the duration-cost and rate-loading terms vanish. When $r > 0$, the household alone holds the claim (Proposition A.2), so $i = H$ and the exemption-benefit term also vanishes.

If the household alone holds maturity $T$, then $i = H$, and Theorem A.1 gives $\Psi_T^0 \geq 0$, that is, $\sigma_r^2 A_{T-1}^M \widehat{\Lambda} \leq (\xi_I - \xi_H) r$. The duration-cost contribution is $\sigma_r^2 A_{T-1}^M \widehat{\Lambda}$ and is therefore at most $(\xi_I - \xi_H) r$. If the lower-tax investor holds the bond, the identity with $i = I$ has a zero duration-cost contribution and an exemption-benefit contribution of $(\xi_I - \xi_H) r$. □

At the threshold maturity, $\Psi_T^0 = 0$, so either investor's equality gives the same total gap. Figure 5 plots this decomposition for the main-text illustration. At $T = 30$, the rate-loading term is 54% of the return gap, and the exemption-benefit term is the remainder. Figure 14 gives the counterpart with bond-specific risk.

**How the rate loading links maturities.** Iterating the loading recursion (53) from $A_0^a = 0$ gives

$$A_{T-1}^M - \xi_H A_{T-1}^C = \sum_{j=1}^{T-1} \rho^{T-1-j} \left( s_j^M - \xi_H s_j^C \right),$$

so the loading gap accumulates the response gaps $s_j^M - \xi_H s_j^C$ of the shorter maturities the bond passes through. For $T = 2$, the next-period loading gap $A_1^M - \xi_H A_1^C$ is zero, since the one-period responses are $s_1^M = \xi_H$ and $s_1^C = 1$ when $r > 0$.

**Tax mix and portfolio responses.** One-period returns respond to the rate state for two reasons. Each holder's after-tax return on cash rises by its after-tax share per unit of the rate state, and the holders' tax mix averages these shares weighted by risk tolerance (Proposition 2). Trading also changes how much net rate exposure each holder carries, and with it the compensation each requires. Aggregating the holders' equalities separates the two. Multiplying holder $i$'s municipal equality by $m_i/\gamma_i$, summing over the holders and differentiating with respect to $z$ gives

$$\left( \sum_{i \in \mathcal{U}_T^M} \frac{m_i}{\gamma_i} \right) s_T^M - \sum_{i \in \mathcal{U}_T^M} \frac{m_i \xi_i}{\gamma_i} = \sigma_r^2 A_{T-1}^M \sum_{i \in \mathcal{U}_T^M} m_i \frac{\partial \widehat{D}_i(z)}{\partial z}, \tag{63}$$

where the set $\mathcal{U}_T^M \subseteq \{H, I\}$ collects the investors who hold the maturity-$T$ municipal bond. With bond-specific risk, the same aggregation removes the bond-specific terms, because fixed supply implies $\sum_i m_i \, \partial x_{i,T}^M(z)/\partial z = 0$. Define the holders' risk-tolerance shares and aggregate exposure response as

$$\omega_{i,T}^M := \frac{m_i/\gamma_i}{\sum_{j \in \mathcal{U}_T^M} m_j/\gamma_j}, \qquad E_T^M := \frac{\sum_{i \in \mathcal{U}_T^M} m_i \, \partial \widehat{D}_i(z)/\partial z}{\sum_{i \in \mathcal{U}_T^M} m_i/\gamma_i}.$$

The weight $\omega_{i,T}^M$ is investor $i$'s share of the holders' risk tolerance, and $E_T^M$ is the response of their aggregate net rate exposure to the rate state (Section 4.5), divided by that total risk tolerance. The taxable counterpart $E_T^C$ is defined in the same way from the holders of the taxable bond. Dividing (63) by total risk tolerance gives Proposition A.3, the general form of the response formulas (18) in Section 4.5.

**Proposition A.3** (Holding-return response to the rate state)**.** *Consider a regular state of the two-investor affine model, at which price moments are differentiable, each investor either holds a risky bond or has strictly positive slack in its portfolio condition for that bond, and the binding portfolio conditions and clearing, linearized in holdings and returns, have a unique solution. The set of holders $\mathcal{U}_T^M \subseteq \{H, I\}$ therefore does not change near this state. Then*

$$s_T^M = \underbrace{\sum_{i \in \mathcal{U}_T^M} \omega_{i,T}^M \xi_i}_{\text{holders' tax mix}} + \underbrace{\sigma_r^2 A_{T-1}^M E_T^M}_{\text{portfolio response}} \tag{64}$$

*exactly. The tax-mix term depends on who holds maturity $T$. The portfolio response depends on how equilibrium trading changes those holders' exposure across their whole portfolios.*

**Proof of Proposition A.3.** Dividing (63) by the holders' total risk tolerance and using the definitions of $\omega_{i,T}^M$ and $E_T^M$ gives $s_T^M - \sum_{i\in\mathcal{U}_T^M} \omega_{i,T}^M \xi_i = \sigma_r^2 A_{T-1}^M E_T^M$, which is (64). With bond-specific risk, (63) holds because fixed supply cancels the derivative of the bond-specific terms by (48). □

**Exact aggregation with nonlinear prices.** Proposition A.4 extends the same aggregation to nonlinear prices by collecting the holders' tax treatment and covariance-based risk compensation before taking the state derivative. The notation $\Sigma$, $K_i$, $c_i^L$ and $\xi_{ij}$ is that of (46)–(47). In the normalized municipal–taxable economy, the entries $\xi_i^a$ of $K_i$ are one at municipal and $\xi_i$ at taxable coordinates. Appendix A.4 applies the proposition market by market in the multi-state model.

**Proposition A.4** (Required returns aggregated across a bond's holders)**.** *In the general economy of Appendix A.1, at a regular state with slack shorting constraints on cash for the holders, let* $\Gamma_j = \sum_{i\in\mathcal{U}_j} m_i/\gamma_i$ *be their total risk tolerance,* $\bar{\xi}_j = \Gamma_j^{-1} \sum_{i\in\mathcal{U}_j} m_i \xi_{ij}/\gamma_i$ *and* $\Omega_j = \Gamma_j^{-1} \sum_{i\in\mathcal{U}_j} m_i [\Sigma K_i x_i - c_i^L]_j$*. Here* $\bar{\xi}_j$ *is their tax mix and* $\Omega_j$ *their average risk compensation, weighted by risk tolerance. Then*

$$\begin{aligned} \mu_j &= \bar{\xi}_j r + \Omega_j, \\ s_j &= \bar{\xi}_j \, \partial_z r + \frac{1}{\Gamma_j} \sum_{i\in\mathcal{U}_j} m_i \left[ (\partial_z \Sigma) K_i x_i + \Sigma K_i \partial_z x_i - \partial_z c_i^L \right]_j . \end{aligned} \tag{65}$$

*In the normalized municipal–taxable economy, with* $\bar{\xi}_T^a = \bar{\xi}_j$ *and* $\Omega_T^a = \Omega_j$ *at* $j = (a, T)$*, above the rate floor the response gap is*

$$s_T^M - \xi_H s_T^C = (\bar{\xi}_T^M - \xi_H) + \partial_z \Omega_T^M - \xi_H \partial_z \Omega_T^C. \tag{66}$$

**Proof.** Weighting the holders' equalities $\mu_j - \xi_{ij} r = \gamma_i [\Sigma K_i x_i - c_i^L]_j$ by $m_i/\gamma_i$, summing and dividing by $\Gamma_j$ gives the first line. Near a regular state the holders do not change, so $\Gamma_j$ and $\bar{\xi}_j$ are constant, and differentiating gives the second line. In the two-market normalization, $\bar{\xi}_j = \bar{\xi}_T^M$ at municipal and $\bar{\xi}_j = 1$ at taxable coordinates. Above the rate floor $\partial_z r = 1$, and subtracting $\xi_H$ times the taxable response from the municipal response gives (66). □

When the shorting constraint on cash binds, each holder's equality also contains $\kappa_i/\xi_i^a$ by (37). The return identity in (65) then gains $\Gamma_j^{-1} \sum_{i\in\mathcal{U}_j} m_i \kappa_i / (\gamma_i \xi_i^a)$, and the response identity gains its state derivative.

**Responses in the affine model.** Substituting the affine covariance and liability credits into the binding portfolio conditions gives each bond's response

$$s_j = \bar{\xi}_j + \frac{\sigma_r^2 A_j}{\Gamma_j} \sum_{i\in\mathcal{U}_j} m_i \frac{\partial \widehat{D}_i}{\partial z} + \frac{v_j}{\Gamma_j} \sum_{i\in\mathcal{U}_j} m_i \xi_i^a \frac{\partial x_{ij}}{\partial z}, \tag{67}$$

where $A_j = A_{T-1}^a$ is the next-period loading and $\xi_i^a$ is the normalized after-tax share. The final term is zero without bond-specific risk. For municipal bonds, whose normalized share is one, (48) also cancels it when it is present. The tax-mix term is $\xi_H$ under household-only holdings, $\xi_I$ under lower-tax-only holdings and, under joint holdings, the tax mix

$$\xi^* := \omega_H \xi_H + \omega_I \xi_I, \qquad \omega_i := \frac{m_i/\gamma_i}{m_H/\gamma_H + m_I/\gamma_I},$$

where $\omega_i$ is investor $i$'s share of total risk tolerance. The lower-tax investor's after-tax share therefore appears in this term only at maturities it holds.

**Taxable counterpart.** Proposition 2 also uses the taxable version of (67). With $\xi_{ij} = 1$ at taxable coordinates, its tax term is one. Its bond-specific term vanishes in three cases: if $v_T^C = 0$, if one investor holds the taxable bond, because clearing then fixes its holding, or if all holders have equal after-tax shares, because these then factor out of (48). Otherwise, unweighted clearing does not cancel the term. With the lower-tax investor the sole holder of the taxable bond,

$$s_T^C = 1 + \gamma_I \sigma_r^2 A_{T-1}^C \partial_z \widehat{D}_I. \tag{68}$$

**Signing the responses.** Proposition 2 signs these responses on the illustrated branch, the case of Theorem A.1 in which one maturity is shared. A higher rate state increases the difference in exemption benefits. At the threshold maturity, the lower-tax investor sells municipal bonds to the household, whose net rate exposure therefore rises with rates, and so does its required compensation, even on shorter bonds it holds alone.

The lower-tax investor sheds net exposure, which lowers the response of the returns it requires on taxable and long municipal bonds. The proposition supplies the responses and signs in (18) and the positive loading gap behind (76). Figure 6(a) plots this case.

**Proposition 2** (Participation and rate responses)**.** *At a regular state with $r(z) > 0$ in the two-investor affine model without bond-specific risk, suppose the lower-tax investor alone holds risky taxable bonds, the household alone holds the municipal maturities below the threshold maturity $k = T^* \geq 2$, both investors hold maturity $k$, and the lower-tax investor alone holds the municipal maturities above $k$. Assume $A^M_{T-1}, A^C_{T-1} > 0$ for every $T \geq 2$, and put $d := \partial \widehat{D}_H/\partial z$. Then*

$$d = \frac{(\xi_I - \xi_H)m_I}{(\gamma_H m_I + \gamma_I m_H)\sigma_r^2 A^M_{k-1}} > 0, \qquad \frac{\partial \widehat{D}_I}{\partial z} = -\frac{m_H}{m_I} d. \tag{69}$$

*With $\xi^* = \omega_H \xi_H + \omega_I \xi_I$ and the feedback $b := \gamma_I \sigma_r^2 (m_H/m_I) d = \omega_H(\xi_I - \xi_H)/A^M_{k-1}$, the responses are*

$$s^M_T = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A^M_{T-1} d, & T < k, \\ \xi^*, & T = k, \\ \xi_I - b A^M_{T-1}, & T > k, \end{cases} \qquad s^C_T = 1 - b A^C_{T-1}. \tag{70}$$

*The exposure response $d$ is the unique positive solution of $d\, A^M_{k-1}(d) = (\xi_I - \xi_H)m_I/[(\gamma_H m_I + \gamma_I m_H)\sigma_r^2]$, where $A^M_{k-1}(d) = \xi_H \sum_{q=0}^{k-2} (\rho + \gamma_H \sigma_r^2 d)^q$ is the loading generated at household-held maturities. The response gap $s^M_T - \xi_H s^C_T$ is positive at every risky maturity $T \leq k$. If $b \leq \rho$, which condition (S) in (72) below ensures, then $A^M_T - \xi_H A^C_T > 0$ at every risky maturity $T \geq 2$, with equality at $T = 1$.*

**Proof.** Clearing fixes each sole holder's position, $Q^M_T/m_H$ for the household below $k$, $Q^M_T/m_I$ for the lower-tax investor above $k$ and $Q^C_T/m_I$ for its risky taxable bonds. Only the shared position at $k$ varies, so $\partial_z \widehat{D}_i = A^M_{k-1} \partial_z x^M_{i,k}$, and differentiating clearing at $k$ and multiplying by $A^M_{k-1}$ gives

$$m_H \partial_z \widehat{D}_H + m_I \partial_z \widehat{D}_I = 0. \tag{71}$$

Both investors' municipal equalities hold at $k$. With constant loadings and $\partial_z r = 1$, subtracting the lower-tax investor's differentiated equality from the household's gives $\xi_I - \xi_H = \sigma_r^2 A^M_{k-1}(\gamma_H \partial_z \widehat{D}_H - \gamma_I \partial_z \widehat{D}_I)$. Substituting (71) gives (69). The identity for $b$ follows from $\gamma_I m_H/(\gamma_H m_I + \gamma_I m_H) = \omega_H$.

A sole holder's differentiated equality gives $s^M_T = \xi_i + \gamma_i \sigma_r^2 A^M_{T-1} \partial_z \widehat{D}_i$, which is the first line below $k$ and, with (69), the third line above $k$. At $k$, weighting both equalities by risk tolerance as in (63) cancels the exposure responses by (71) and leaves $\xi^*$. The household's own equality at $k$ gives the same value, $\xi_H + \gamma_H \sigma_r^2 A^M_{k-1} d = \xi_H + \omega_I(\xi_I - \xi_H) = \xi^*$. Equation (68) gives the taxable formula.

Below $k$, substituting the household's response into (53) gives $A^M_T = (\rho + \gamma_H \sigma_r^2 d) A^M_{T-1} + \xi_H$, so $A^M_{k-1}(d) = \xi_H \sum_{q=0}^{k-2} (\rho + \gamma_H \sigma_r^2 d)^q$ is positive and nondecreasing in $d$. The product $d\, A^M_{k-1}(d)$ therefore rises strictly from zero without bound, and it equals the positive constant in the statement exactly once.

By (70), the response gap is

$$s^M_T - \xi_H s^C_T = \begin{cases} \sigma_r^2 d \left[ \gamma_H A^M_{T-1} + \xi_H \gamma_I \dfrac{m_H}{m_I} A^C_{T-1} \right] > 0, & 2 \leq T < k, \\ \omega_I(\xi_I - \xi_H) + \xi_H b A^C_{k-1} > 0, & T = k, \end{cases}$$

and it is zero at $T = 1$. Let $L_T = A^M_T - \xi_H A^C_T$. Then $L_1 = 0$ and $L_T = \rho L_{T-1} + (s^M_T - \xi_H s^C_T) > 0$ for $2 \leq T \leq k$. Above $k$, the responses give

$$L_T = (\rho - b) L_{T-1} + (\xi_I - \xi_H), \qquad T > k,$$

which is positive by induction when $b \leq \rho$. □

If no maturity satisfies $\Psi^0_T = 0$, clearing fixes every holding, both exposure responses are zero, and the municipal responses are $\xi_H$ below the threshold and $\xi_I$ from it on. The loading gap is then zero below the threshold and positive from it on.

**Condition (S).** The positive loading gap in Section 4.5 and the pricing results below use one restriction on taxes, risk-tolerance shares and rate persistence:

$$\text{(S)} \qquad \omega_H \frac{\xi_I - \xi_H}{\xi_H} \leq \rho, \tag{72}$$

written in masses and risk aversions as

$$\text{(S)} \qquad \frac{\gamma_I m_H(\xi_I - \xi_H)}{\xi_H(\gamma_H m_I + \gamma_I m_H)} \leq \rho. \tag{73}$$

On the illustrated branch, the threshold maturity's next-period loading is at least $A_1^M = \xi_H$, so (S) gives $b = \omega_H(\xi_I - \xi_H)/A_{k-1}^M \leq \rho$. The feedback from the lower-tax investor's falling exposure into its required returns is then too weak to reverse the sign of the loading gap or the rise of taxable loadings with maturity, since $A_T^C - A_{T-1}^C = (\rho - b)(A_{T-1}^C - A_{T-2}^C)$ and $A_1^C - A_0^C = 1$. In the main-text illustration, the left side of (S) is about 0.209, against $\rho = 0.8$.

**From responses to the rate-loading channel.** The response gap $s_T^M - \xi_H s_T^C = \partial_z g_T$ accumulates into the loading gap of the rate-loading channel in (62). Applying Proposition A.3 to municipal bonds and (67) to risky taxable bonds, which the lower-tax investor alone holds, splits it into tax-mix and portfolio parts:

$$s_T^M - \xi_H s_T^C = \underbrace{\left(\bar{\xi}_T^M - \xi_H\right)}_{\text{difference in tax mix}} + \underbrace{\sigma_r^2\left(A_{T-1}^M E_T^M - \xi_H A_{T-1}^C E_T^C\right)}_{\text{difference in portfolio responses}}, \tag{74}$$

where $\bar{\xi}_T^M := \sum_i \omega_{i,T}^M \xi_i$ is the holders' tax mix ($\xi_H$, $\xi^*$ or $\xi_I$). With one municipal holder, these are the exemption-benefit and portfolio-reallocation terms of (17). Below the threshold maturity, trading raises the household's municipal response and lowers the taxable response. At the threshold maturity, the holders' exposure responses cancel in aggregate, the tax mix adds $\xi^* - \xi_H$ and the taxable response falls by $bA_{k-1}^C$. There, (17) uses the lower-tax investor's condition, which splits $\xi^* - \xi_H$ into the exemption benefit $\xi_I - \xi_H$ and the municipal portfolio response $-\omega_H(\xi_I - \xi_H)$. Above it, the tax mix adds $\xi_I - \xi_H$, and the lower-tax investor's falling exposure lowers the municipal and taxable responses by $bA_{T-1}^M$ and $bA_{T-1}^C$. The response gap above the threshold is therefore $(\xi_I - \xi_H) - b(A_{T-1}^M - \xi_H A_{T-1}^C)$. All contributions vanish at the riskless one-period claim, and (66) gives the exact counterpart with nonlinear covariances. Figure 6 combines the municipal tax and portfolio terms and shows the taxable contribution separately.

The rate-loading channel at maturity $T$ uses the next-period loadings, which accumulate only the response gaps below $T$. This channel of Proposition 1 therefore incorporates portfolio responses from maturity three on and the change in tax mix from maturity $k+1$ on.

**Comparison with a treasury reference.** The model counterpart of the treasury comparison in Section 6.3 divides municipal loadings by $G_T := \sum_{j=1}^T \rho^{T-j} = A_T^{\text{Tsy}}$, the loading of a treasury reference whose one-period return responds one for one to the rate state. Since $A_T^M/G_T = \sum_{j=1}^T (\rho^{T-j}/G_T) s_j^M$, the relative response averages municipal return responses with positive weights and equals $\xi_H$ at the allocation of Proposition A.5 below. Its difference from $\xi_H$ reflects municipal holders' tax mix and portfolio responses, both of which depend on participation. Figure 6(b) measures loadings against the equilibrium taxable market $C$, so the lower-tax investor's portfolio response also adds the taxable side $\xi_H(1 - s_j^C)$ to the loading gap $A_T^M - \xi_H A_T^C$.

**State-independent holding benefits.** Suppose an investor's objective includes $\sum_{a,T} \vartheta_{i,T}^a x_{i,T}^a$, with each $\vartheta_{i,T}^a$ independent of the rate state, for example a trading-cost advantage (Section 4.7) or the calibration's fixed bank financing-tax cost. Its municipal condition for a held bond becomes

$$\mu_T^M - \xi_i r + \vartheta_{i,T}^M = \gamma_i \left[\sigma_r^2 A_{T-1}^M \widehat{D}_i + v_T^M x_{i,T}^M\right],$$

and the taxable condition similarly gains $\vartheta_{i,T}^C/\xi_i$. With covariance, financing and constraints fixed, differentiation removes state-independent benefits and costs, here and in (65). The response formulas and loading equations therefore hold on the same participation branch, so a given loading solution is unchanged. Such benefits and costs still change which maturities an investor holds and the levels of one-period returns, so they affect return intercepts, gap levels (Appendix A.3) and the rate states where the branch holds. State-dependent benefits add their derivatives. Fixed participation fees affect global participation without a per-unit term in the portfolio conditions.

**Household-only loadings.** The last result covers the allocation in which the household holds every municipal bond. In that allocation, the loading gap of Section 4.5 is zero and the municipal-to-taxable loading ratio $A_T^M/A_T^C$ of P3 equals $\xi_H$ at every maturity.

**Proposition A.5** (Household-only loadings)**.** *In the two-investor affine model, if the high-tax household alone holds every municipal bond and the lower-tax investor alone holds every taxable bond, then* $s_T^M = \xi_H$, $s_T^C = 1$ *and* $A_T^M = \xi_H A_T^C$ *at every maturity.*

**Proof.** Each bond has one holder, so clearing fixes every holding and both net rate exposures have zero derivatives. The portfolio and bond-specific terms of (67) then vanish, leaving $s_T^M = \xi_H$ and $s_T^C = 1$, and (56) gives $A_T^M = \xi_H \sum_{j=1}^{T} \rho^{T-j} = \xi_H A_T^C$. The identity concerns loadings, and risk premia in the level terms $C_T^M$ and $C_T^C$ can still move the yield ratio away from $\xi_H$. □

### A.2.5 Yield gaps and implied taxes

The three channels act on one-period returns, while the municipal puzzle concerns yields to maturity. This subsection carries the channels into yields, formalizing (19) of Section 4.5. Proposition A.6 then orders by maturity the loading ratio $A_T^M/A_T^C$, the relative response of P3 (Section 6.3). It also orders the yield-implied tax when municipal-to-taxable forward-rate ratios rise with maturity, and gives the long-end implied tax. These results underlie the long-end pass-throughs and prediction P3 of the main text.

**From one-period gaps to yield gaps.** Applying (54) to both markets turns the one-period gap $g_T$ of Proposition 1 into the yield gap:

$$y_T^M - \xi_H y_T^C = \underbrace{\frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t\big[g_{T-j}(z_{t+j})\big]}_{\text{average gap along the bond's life}} \underbrace{- \frac{1}{2T}\sum_{q=2}^{T}\Big[\sigma_r^2\Big(\big(A_{q-1}^M\big)^2 - \xi_H\big(A_{q-1}^C\big)^2\Big) + \big(\sigma_{\eta,q}^M\big)^2 - \xi_H\big(\sigma_{\eta,q}^C\big)^2\Big]}_{\text{convexity difference}}. \tag{75}$$

The last term of the average, the riskless one-period gap, is zero. The yield-implied tax is

$$\tau_T^{Y,\text{impl}} := 1 - \frac{y_T^M}{y_T^C}, \qquad \text{the model counterpart of } \textit{ImpliedTax} \text{ in (2).}$$

It equals the household wedge less the yield gap per unit of taxable yield, $\tau_T^{Y,\text{impl}} = (1-\xi_H) - (y_T^M - \xi_H y_T^C)/y_T^C$, as in the identity (3) of Section 3.6. A positive gap accumulated in (75) therefore pulls the implied tax below the household wedge. Proposition A.6(ii) gives a forward-rate condition under which the implied tax falls with maturity.

**The excess yield response.** The excess yield response is the loading gap divided by maturity, and the loading gap accumulates the response gaps in (74). These response gaps reflect portfolio responses below the threshold maturity and, from that maturity on, also the holders' tax mix, their after-tax shares averaged with risk-tolerance weights. Under condition (S), the loading gap is positive at every risky maturity (Proposition 2). Hence, for $T \geq 2$,

$$\underbrace{\frac{\partial y_T^M}{\partial z} - \xi_H \frac{\partial y_T^C}{\partial z}}_{\text{excess yield response}} = \frac{1}{T}\sum_{j=2}^{T} \rho^{T-j} \underbrace{\left(\frac{\partial \mu_j^M}{\partial z} - \xi_H \frac{\partial \mu_j^C}{\partial z}\right)}_{\text{response gap}} > 0, \tag{76}$$

in model-period units, and both sides are zero for the riskless one-period bond. Long municipal yields therefore move more with the short rate than the household benchmark implies. Through the household's exposure response, the change in its net rate exposure as rates rise (Section 4.5), the lower-tax investor's tax affects these responses even at maturities it does not hold.

**Forward rates.** Parts (ii) and (iii) of Proposition A.6 use forward rates. Define the forward rate implied by today's prices for the period from $t+T-1$ to $t+T$:

$$f_T^a(z) = \log \frac{P_{T-1}^a(z)}{P_T^a(z)}, \qquad y_T^a(z) = \frac{1}{T}\sum_{j=1}^{T} f_j^a(z). \tag{77}$$

**Proposition A.6** (Maturity ordering of the loading ratio and the yield-implied tax)**.** *In the two-investor affine model without bond-specific risk, consider the illustrated branch of Proposition 2. The high-tax household alone holds municipal bonds shorter than $k$, both investors hold maturity $k$, and the lower-tax investor alone holds longer municipal bonds and all risky taxable bonds. Assume condition (S) in (72).*

***(i) Rate sensitivity.*** *The municipal-to-taxable loading ratio is nondecreasing in maturity. It starts at the household's after-tax share. When the recursions are extended to infinite maturity with the household's exposure response $d$ held at its solved value, the ratio converges to the lower-tax investor's after-tax share:*

$$\frac{A_1^{M,\mathrm{aff}}}{A_1^{C,\mathrm{aff}}} = \xi_H \qquad \text{and} \qquad \lim_{T\to\infty} \frac{A_T^{M,\mathrm{aff}}}{A_T^{C,\mathrm{aff}}} = \xi_I. \tag{78}$$

***(ii) Yield levels.*** *Suppose also that taxable forward rates are positive and the municipal-to-taxable forward-rate ratio is nondecreasing in maturity at the stationary mean $z = 0$. Then the yield ratio $y_T^M/y_T^C$ is nondecreasing at $z = 0$ (Proposition A.7), so the yield-implied tax $\tau_T^{Y,\mathrm{impl,aff}}$ is nonincreasing.*

***(iii) Long end.*** *With the recursions extended to infinite maturity and the lower-tax investor's risk-premium coefficient $\pi_I = \gamma_I \sigma_r^2 \widehat{D}_I(0)$ held at its solved value, assume the limiting taxable forward is positive. Then*

$$\lim_{T\to\infty} \tau_T^{Y,\mathrm{impl,aff}} = 1 - \frac{\xi_I \bar{r} + A_\infty^M \pi_I - \frac{1}{2}\sigma_r^2 (A_\infty^M)^2}{\bar{r} + A_\infty^C \pi_I - \frac{1}{2}\sigma_r^2 (A_\infty^C)^2} = (1-\xi_I) - \frac{\frac{1}{2}\sigma_r^2 \xi_I (1-\xi_I)(A_\infty^C)^2}{y_\infty^C}, \tag{79}$$

*where $A_\infty^C = 1/(1-\rho+b)$, $A_\infty^M = \xi_I A_\infty^C$, the limiting taxable yield $y_\infty^C$ is the denominator of the first fraction, and $b$ is the feedback in (70).*

**Proof of Proposition A.6(i).** Put $\alpha = \rho - b$ and $\beta = \rho + \gamma_H \sigma_r^2 d$. Condition (S) and $A_{k-1}^M \geq \xi_H$ give $b \leq \rho$, as shown after (72). Together with $d > 0$ and $b > 0$ (Proposition 2), this gives $0 \leq \alpha < \rho < \beta$. By (70) and the loading recursion (53), the loadings satisfy

$$A_T^C = \alpha A_{T-1}^C + 1, \qquad A_T^M = \begin{cases} \beta A_{T-1}^M + \xi_H, & T \leq k, \\ \alpha A_{T-1}^M + \xi_I, & T > k. \end{cases}$$

The first municipal recursion also holds at $k$, because the household's equality there gives $s_k^M = \xi_H + \gamma_H \sigma_r^2 A_{k-1}^M d$, which equals $\xi^*$ (Proposition 2).

Up to the threshold maturity, $A_T^M/\xi_H = \sum_{j=0}^{T-1} \beta^j$ and $A_T^C = \sum_{j=0}^{T-1} \alpha^j$, so the ratio is $\xi_H$ at $T = 1$. Cross-multiplying shows that it does not fall from $T$ to $T+1 \leq k$ when $\beta^T \sum_{j=0}^{T-1} \alpha^j \geq \alpha^T \sum_{j=0}^{T-1} \beta^j$, which holds term by term because $\beta > \alpha \geq 0$ gives $\beta^T \alpha^j \geq \alpha^T \beta^j$ for $j < T$.

From the threshold maturity on, write $A_T^M/A_T^C = \xi_I + F_T/A_T^C$ with $F_T = A_T^M - \xi_I A_T^C$. By (69), $\beta - \alpha = (\xi_I - \xi_H)/A_{k-1}^M$, so for $T \leq k$

$$F_T = \alpha F_{T-1} + (\xi_I - \xi_H)\left(\frac{A_{T-1}^M}{A_{k-1}^M} - 1\right) \leq \alpha F_{T-1},$$

because household-only loadings rise with maturity. Since $F_1 = \xi_H - \xi_I < 0$ and $\alpha \geq 0$, induction on maturity gives $F_T \leq 0$ through $k$. For $T > k$, $F_T = \alpha F_{T-1}$, so from $k$ on $|F_T|$ is nonincreasing, while $A_T^C$ is nondecreasing because $A_T^C - A_{T-1}^C = \alpha^{T-1} \geq 0$. The ratio is therefore nondecreasing from $k$ on.

As $T \to \infty$, $A_T^C = \sum_{j=0}^{T-1} \alpha^j \to 1/(1-\alpha) = 1/(1-\rho+b) > 0$ and $F_T = \alpha^{T-k} F_k \to 0$, so the ratio converges to $\xi_I$. When $b < \rho$, $\alpha > 0$, so $A_T^C$ rises strictly, and so does $A_T^M$, the product of $A_T^C$ and a positive nondecreasing ratio. The illustrated branch therefore has the rising loadings that Theorem A.1 uses. □

**Proof of part (ii).** Part (ii) follows from the next result.

**Proposition A.7** (Yield ratios from forward-rate ratios)**.** *For any price functions, affine or nonlinear, let $f_T^a(z) = -p_T^a(z) + p_{T-1}^a(z)$. If, over the maturities considered, $f_T^C(z) > 0$ and $f_T^M(z)/f_T^C(z)$ is nondecreasing, then $y_T^M(z)/y_T^C(z)$ is nondecreasing and $\tau_T^{Y,\mathrm{impl}}(z)$ is nonincreasing.*

**Proof.** Summing $f_j^a = p_{j-1}^a - p_j^a$ over $j \leq T$ cancels the intermediate log prices, and $p_0^a = 0$, so $T y_T^a = -p_T^a = \sum_{j\leq T} f_j^a$. Hence $y_{T+1}^M/y_{T+1}^C$ averages $y_T^M/y_T^C$ and $f_{T+1}^M/f_{T+1}^C$ with positive weights $\sum_{j\leq T} f_j^C$ and $f_{T+1}^C$. The added forward-rate ratio is at least every earlier one, hence at least $y_T^M/y_T^C$, their average with weights $f_j^C > 0$, so the yield ratio does not fall. On a two-year grid, $T$ counts periods and annualization cancels. □

The forward rate compares adjacent-maturity prices at the same date, whereas the holding return uses next period's resale price. The affine recursions (53) link the two:

$$f_T^a(z) = \mu_T^a(z) - (1-\rho)A_{T-1}^a z - \tfrac{1}{2}\left[\sigma_r^2 (A_{T-1}^a)^2 + (\sigma_{\eta,T}^a)^2\right]. \tag{80}$$

The premise of Proposition A.7 concerns full forward rates, including convexity terms, so monotone loadings alone do not establish it. Section 5 checks the nonlinear model's maturity profiles numerically.

**Proof of part (iii).** Long-end yield levels reflect the long-end holders' taxes, risk premia and convexity. For bond $j = (a, T)$, put $\Gamma_T^a := \sum_{i \in \mathcal{U}_j} m_i/\gamma_i$ and

$$\bar{\xi}_T^a = \bar{\xi}_j = \frac{\sum_i (m_i/\gamma_i)\xi_{ij}}{\Gamma_T^a}, \quad \pi_T^a = \frac{\sigma_r^2}{\Gamma_T^a}\sum_i m_i \widehat{D}_i(0), \quad \iota_T^a = v_j\left[\frac{\sum_i m_i \xi_i^a x_{i,T}^a(0)}{\Gamma_T^a} - \tfrac{1}{2}\right],$$

with sums over holders. As in Section 4.2, $\xi_i^a$ is the after-tax share divided by the municipal one, so it is one for municipal bonds and $\xi_i$ for taxable bonds. Here $\pi_T^a$ is the holders' cost of rate risk per unit of loading, weighted by risk tolerance, and $\iota_T^a$ is their compensation for bond-specific risk net of its convexity term. Weighting the holders' portfolio equalities at $z = 0$ by risk tolerance and summing, as in Proposition A.4, gives $\mu_T^a(0) = \bar{\xi}_T^a \bar{r} + A_{T-1}^a \pi_T^a + \iota_T^a + \frac{1}{2}v_j$. The level recursion in (53) subtracts $\frac{1}{2}\sigma_r^2 (A_{T-1}^a)^2 + \frac{1}{2}v_j$, so

$$C_T^a - C_{T-1}^a = \bar{\xi}_T^a \bar{r} + A_{T-1}^a \pi_T^a - \tfrac{1}{2}\sigma_r^2 (A_{T-1}^a)^2 + \iota_T^a. \tag{81}$$

The increment $C_T^a - C_{T-1}^a$ is the forward rate at $z = 0$, so $C_T^a/T$ averages the first $T$ forward rates. If the displayed coefficients and loadings converge, these forward rates converge, and their average converges to the same limit. Since $y_T^a(z) = (A_T^a z + C_T^a)/T$ and $A_T^a z/T \to 0$ at fixed $z$,

$$y_T^a(z) \longrightarrow f_\infty^a = \bar{\xi}_\infty^a \bar{r} + A_\infty^a \pi_\infty^a - \tfrac{1}{2}\sigma_r^2 (A_\infty^a)^2 + \iota_\infty^a. \tag{82}$$

On the illustrated branch without bond-specific risk, the lower-tax investor alone holds both long bonds, so $\bar{\xi}_\infty^M = \xi_I$, $\bar{\xi}_\infty^C = 1$, $\pi_\infty^M = \pi_\infty^C = \pi_I$ and $\iota_\infty^a = 0$. The proof of part (i) gives $A_\infty^C = 1/(1-\rho+b)$ and $A_\infty^M = \xi_I A_\infty^C$. With $f_\infty^C > 0$, the implied-tax limit $1 - f_\infty^M/f_\infty^C$ is therefore the first form in (79). Since $A_\infty^M = \xi_I A_\infty^C$, the interest and rate-risk terms of the municipal yield are $\xi_I$ times those of the taxable yield, and

$$y_\infty^M - \xi_I y_\infty^C = \tfrac{1}{2}\sigma_r^2 \xi_I (1-\xi_I)(A_\infty^C)^2,$$

which, divided by $y_\infty^C$, gives the second form, the lower-tax investor's wedge $1-\xi_I$ less a relative convexity term.

### A.2.6 Tax pass-through and relative rate sensitivity

The mechanism predicts whose taxes are priced at each maturity. This subsection proves the pass-through contrasts and the rate-sensitivity ordering behind predictions P1–P3 of Section 4.6 on the illustrated branch. Appendix A.2.8 gives their counterparts when bond-specific risk extends joint holdings across long maturities.

**Two implied taxes and their pass-throughs.** P1 and P2 concern yield levels, while P3 concerns rate sensitivities. To keep these objects distinct, I pair the yield-implied tax of Appendix A.2.5 with a loading-implied tax:

$$\tau_T^{Y,\text{impl}} = 1 - \frac{y_T^M}{y_T^C}, \qquad \tau_T^A = 1 - \frac{A_T^M}{A_T^C}.$$

To measure pass-through, I vary one investor's after-tax share at a fixed rate state and re-solve the equilibrium. Holdings, risk costs and both yield curves adjust while non-tax primitives stay fixed. The pass-throughs of investor $i$'s exemption benefit $1-\xi_i$ into the two implied taxes are

$$\mathcal{P}_i^Y(T) := \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad \mathcal{P}_i^A(T) := \frac{d\tau_T^A}{d(1-\xi_i)}, \qquad i \in \{H, I\},$$

where an argument $\infty$ denotes the limit as $T \to \infty$. Equivalently, they are the derivatives of $y_T^M/y_T^C$ and $A_T^M/A_T^C$ with respect to $\xi_i$.

Under the representative-tax benchmark of Section 3.3, $y_T^M = \xi_H y_T^C$ and $A_T^M = \xi_H A_T^C$, so in both measures household pass-through is one and the lower-tax investor's is zero at every maturity. With heterogeneous taxes, the household alone holds the riskless short bond, whose return is $\xi_H r$. The lower-tax investor holds the long bonds, whose returns depend on its after-tax share directly and on the household's through the exposure response at the threshold maturity. For the empirical treasury comparison of Section 6.3, I also use a treasury

reference, a taxable bond whose one-period return moves one-for-one with the rate state, with loading $A_T^{\text{Tsy}} = G_T$ (Appendix A.2.4).

**Proposition A.8** (Tax pass-through at the short and long ends)**.** *Consider the illustrated branch of Proposition 2 without bond-specific risk, under condition (S) of (72), with a positive cash rate and a slack household shorting constraint on cash. With its recursions extended to infinite maturity, suppose long yields, loading ratios and their tax derivatives converge as maturity grows, uniformly for after-tax shares near their given values, with a positive long taxable yield.*

*(i) (Short end) Both implied taxes equal the household wedge, $\tau_1^{Y,\text{impl}} = \tau_1^A = 1 - \xi_H$, so $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.*

*(ii) (Long end, P1 and P2) The long-end loading-implied tax is $1 - \xi_I$, so*

$$\mathcal{P}_H^A(\infty) = 0, \qquad \mathcal{P}_I^A(\infty) = 1.$$

*Yield pass-throughs equal these values plus the derivatives with respect to $\xi_i$ of the relative convexity term $\frac{1}{2}\sigma_r^2\xi_I(1-\xi_I)(A_\infty^C)^2/y_\infty^C$ in (79), which are $O(1/\bar{r})$ when the branch's equations are solved at other values of $\bar{r}$ with participation held fixed. Whenever these derivatives are below one in absolute value,*

$$\mathcal{P}_H^Y(\infty) < 1, \qquad \mathcal{P}_I^Y(\infty) > 0.$$

*(iii) (Loading ratios, P3) The municipal-to-taxable zero-coupon loading ratio $A_T^M/A_T^C$ is nondecreasing in maturity, equals $\xi_H$ at the short end and converges to $\xi_I$.*

On the illustrated branch, the lower-tax investor alone holds the long bonds of both markets, so their returns are its required returns. Its exposure response appears in both loading recursions through the same feedback $b$, which cancels from the loading ratio. Long-end loading pass-throughs are therefore zero for the household and one for the lower-tax investor, and yield pass-throughs differ from them only through relative convexity. At finite maturities, the loading pass-throughs with exposure responses switched off average the derivatives of the holders' tax mix over household-only, shared and lower-tax-only maturities, weighted by rate persistence (Proposition A.9). The remainder reflects exposure responses. In both illustrations, household yield pass-through falls and lower-tax pass-through rises with maturity (Appendices A.2.7 and A.2.9). With more clienteles, each has its own threshold maturity and risk-tolerance share (Proposition 2 and Appendix A.5). Section 6 tests P1–P3.

**Proof of Proposition A.8.** *(i) The riskless short end.* The one-period claims have zero variance. With a slack household shorting constraint on cash, Proposition A.2 gives $y_1^M = \xi_H r$ and $y_1^C = r$ at every rate state with $r > 0$. The one-period loadings equal the responses $s_1^M = \xi_H$ and $s_1^C = 1$ by (53), so $y_1^M/y_1^C = A_1^M/A_1^C = \xi_H$. Differentiating the two ratios with respect to $\xi_H$ and $\xi_I$ gives $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.

*(ii) The long end.* By the convergence assumption, the limit of each pass-through equals the tax derivative of the long-end ratio. Proposition A.6 gives $A_\infty^M = \xi_I A_\infty^C$ on the branch, so the long-end loading ratio equals $\xi_I$ at every pair of after-tax shares at which the branch is feasible. Differentiating it gives $\mathcal{P}_H^A(\infty) = 0$ and $\mathcal{P}_I^A(\infty) = 1$.

For yields, part (iii) of the same proposition gives

$$\frac{y_\infty^M}{y_\infty^C} = \xi_I + \frac{K_\infty}{y_\infty^C}, \qquad K_\infty = \tfrac{1}{2}\sigma_r^2\xi_I(1-\xi_I)(A_\infty^C)^2,$$

so $\mathcal{P}_i^Y(\infty) = \mathcal{P}_i^A(\infty) + d(K_\infty/y_\infty^C)/d\xi_i$, with holdings, loadings and both yields adjusting. By Proposition 2, the exposure response $d$ solves $d\,A_{k-1}^M(d) = (\xi_I - \xi_H)m_I/[(\gamma_H m_I + \gamma_I m_H)\sigma_r^2]$, an equation without $\bar{r}$, so $d$ and the feedback $b$ do not depend on $\bar{r}$. Hence $K_\infty$ and its tax derivatives are independent of $\bar{r}$.

The mean rate affects $y_\infty^C$ directly and through $\pi_I$. At $z = 0$, both investors' municipal equalities hold at the threshold maturity, and subtracting one from the other gives $(\xi_I - \xi_H)\bar{r} = \sigma_r^2 A_{k-1}^M(\gamma_H\widehat{D}_H - \gamma_I\widehat{D}_I)$. The aggregate exposure $m_H\widehat{D}_H + m_I\widehat{D}_I = \sum_T A_{T-1}^M Q_T^M + \xi_I\sum_T A_{T-1}^C Q_T^C - m_H\ell_H - m_I\ell_I$ depends neither on how the threshold maturity is split nor on $\bar{r}$. Solving the two equations for $\widehat{D}_I$ gives $\pi_I = \gamma_I\sigma_r^2\widehat{D}_I = \pi_0 - b\bar{r}$, with $\pi_0 = \gamma_H\gamma_I\sigma_r^2(m_H\widehat{D}_H + m_I\widehat{D}_I)/(\gamma_H m_I + \gamma_I m_H)$ independent of $\bar{r}$. Since $1/A_\infty^C = 1 - \rho + b$,

$$y_\infty^C = \bar{r} + A_\infty^C(\pi_0 - b\bar{r}) - \tfrac{1}{2}\sigma_r^2(A_\infty^C)^2 = (1-\rho)A_\infty^C\bar{r} + O(1),$$

where the $O(1)$ term does not depend on $\bar{r}$, so $dy_\infty^C/d\xi_i = O(\bar{r})$. With $K_\infty$ and its tax derivatives independent of $\bar{r}$, differentiating the quotient gives $d(K_\infty/y_\infty^C)/d\xi_i = O(1/\bar{r})$ in the branch's equations. When both derivatives

are below one in absolute value, $\mathcal{P}_H^Y(\infty) < 1$ and $\mathcal{P}_I^Y(\infty) > 0$. In the illustration of Appendix A.2.7, these derivatives are 0.001 and −0.052.

*(iii) The loading ratios.* At a common maturity, dividing both price loadings by $T$ leaves their ratio unchanged. Proposition A.6(i) gives the ordering, the value $\xi_H$ at $T = 1$ and the limit $\xi_I$. □

**The tax-mix pass-through across maturities.** The next result gives the loading pass-through with exposure responses switched off at every maturity, the intermediate-maturity profile used above and in Appendix A.4.

**Proposition A.9** (Tax-mix pass-through)**.** *Switch off exposure-response terms in the affine pricing recursions, fixing each maturity's holders and risk-tolerance shares, and let $\bar{\xi}_j$ be the holders' tax mix at municipal maturity $j$. Put $G_T = \sum_{j=1}^{T} \rho^{T-j}$. The tax-mix pass-throughs $\mathcal{D}_i(T) := -\partial_{\xi_i} \tau_T^A$ are the loading pass-throughs $\mathcal{P}_i^A(T)$ of this economy, and*

$$\mathcal{D}_i(T) = \sum_{j=1}^{T} \frac{\rho^{T-j}}{G_T} \frac{\partial \bar{\xi}_j}{\partial \xi_i}. \tag{83}$$

*On the illustrated branch with threshold maturity $k$, put*

$$\Theta_S(T) = \frac{\sum_{j=1}^{\min(T,k-1)} \rho^{T-j}}{G_T}, \qquad \Theta_k(T) = \frac{\rho^{T-k}}{G_T} \mathbf{1}_{\{T \geq k\}}, \qquad \Theta_U(T) = 1 - \Theta_S(T) - \Theta_k(T).$$

*Then*

$$\mathcal{D}_H(T) = \Theta_S(T) + \omega_H \Theta_k(T), \qquad \mathcal{D}_I(T) = \omega_I \Theta_k(T) + \Theta_U(T), \tag{84}$$

*which equal $(1, 0)$ before participation and converge monotonically to $(0, 1)$. If both investors hold every municipal maturity from $T^*$ on (a jointly held block, Section 4.3), as with bond-specific risk in Appendix A.2.8, define $\Theta_S$ with $T^*$ in place of $k$ and put $\Theta_L(T) = 1 - \Theta_S(T)$. Then*

$$\mathcal{D}_H(T) = \Theta_S(T) + \omega_H \Theta_L(T), \qquad \mathcal{D}_I(T) = \omega_I \Theta_L(T), \tag{85}$$

*which equal $(1, 0)$ before participation and converge monotonically to $(\omega_H, \omega_I)$.*

**Proof.** With exposure responses switched off, $s_j^C = 1$ and $s_j^M = \bar{\xi}_j$, so the recursions give $A_T^C = G_T$ and $A_T^M = \sum_{j=1}^{T} \rho^{T-j} \bar{\xi}_j$. Differentiating $A_T^M / G_T$ gives (83). The derivative $\partial \bar{\xi}_j / \partial \xi_H$ is 1, $\omega_H$ and 0 at household-only, jointly held and lower-tax-only maturities, and $\partial \bar{\xi}_j / \partial \xi_I$ is 0, $\omega_I$ and 1, which gives (84) and (85). Before participation, $\Theta_S = 1$ and both profiles equal $(1, 0)$. For $T \geq k - 1$, or $T \geq T^* - 1$ on a jointly held block, the numerator of $\Theta_S$ is multiplied by $\rho$ each period and $G_{T+1} = \rho G_T + 1$, so $\Theta_S(T+1) = \Theta_S(T) \rho G_T / (\rho G_T + 1) < \Theta_S(T)$, and from $T = k$ on $\Theta_k$ satisfies the same recursion. The factor is below $\rho$ because $G_T < 1/(1-\rho)$, so both fall to zero. At $T = k$, $\mathcal{D}_H$ falls from 1 to $1 - \omega_I / G_k$. On the illustrated branch, $\mathcal{D}_H$ therefore falls to zero and $\mathcal{D}_I = 1 - \Theta_S - \omega_H \Theta_k$ rises to one. On the branch with a jointly held block, $\mathcal{D}_H = \omega_H + \omega_I \Theta_S$ falls to $\omega_H$ and $\mathcal{D}_I = \omega_I (1 - \Theta_S)$ rises to $\omega_I$. □

On the illustrated branch, the tax-mix pass-throughs average $(1, 0)$ at household-only maturities, $(\omega_H, \omega_I)$ at the threshold maturity and $(0, 1)$ at lower-tax-only maturities. The weights are the persistence-weighted shares of the bond's remaining life spent at each type of maturity.

### A.2.7 Illustration without bond-specific risk

The main-text figures use this rate-risk-only illustration. This subsection gives its parameters, checks the illustrated branch and the conditions of the results above, and verifies the values reported in the main text.

**Parameters and allocation.** Both investors have risk aversion $\gamma_H = \gamma_I = 5$, and their liability hedges differ, $\ell_H \approx 1.053$ and $\ell_I \approx 1.452$. Reported values are rounded, but calculations use unrounded inputs. The remaining parameters are $\rho = 0.8$, $\sigma_r = 0.025$, $(\xi_H, \xi_I) = (0.55, 0.78)$, equal unit masses, $\bar{r} = 0.04$, municipal supply 0.1 per maturity and taxable supply 0.03 at each $T \geq 2$. With common risk aversion, the lower-tax investor's balance-sheet advantage comes entirely from its greater liability hedge, which leaves it the smaller net exposure and the lower unit cost of rate risk. In the solved allocation, the household alone holds municipal maturities $T = 1, \ldots, 22$, both investors hold maturity $k = 23$, the threshold $T^*$ of Theorem A.1 and Figure 4, and the lower-tax investor alone holds $T = 24, \ldots, 30$. The lower-tax investor holds about 47% of maturity 23 and all risky taxable bonds. The analytical price recursions hold this participation branch fixed at every rate state, and positive holdings and slack conditions for bonds not held are checked near the mean rate state $z = 0$.

**Solution checks and return-gap channels.** The solved branch has $d \approx 11.24$, $\widehat{D}_H \approx 4.47$, $\widehat{D}_I \approx 3.57$ and $\rho - b \approx 0.765$. The index $\Psi_T^0$ of (60) is positive below maturity 23, zero at 23 and negative above it, as Theorem A.1 requires. The left side of condition (S) is about 0.209, below $\rho = 0.8$. The municipal yield exceeds the household benchmark at every maturity beyond one period, by at least 8.1 basis points, and the excess rises with maturity. At $z = 0$, taxable forwards are positive and the municipal-to-taxable forward-rate ratio rises monotonically from 0.55 to about 0.79 at $T = 30$, so the yield-implied tax falls with maturity (Proposition A.6(ii)). Both $s_T^M - \xi_H$ and $1 - s_T^C$ are positive at every risky maturity. Portfolio equalities for held bonds and market clearing hold to numerical precision, and conditions for bonds not held are strictly slack. Independent portfolio optimization reproduces the return responses (70), and direct finite sums reproduce the loading recursion. At $T = 30$, the return gap is about 201 basis points, comprising 92 of exemption benefit and 109 of rate loading. The duration-cost contribution is about 92 basis points at $T = 22$ and zero from $T = 23$, where the lower-tax investor's conditions are used for both bonds.

**Tax pass-through.** For P1 and P2, I differentiate the exposure fixed point, loading recursions, portfolio conditions and yield ratio with respect to one after-tax share at a time, fixing the remaining primitives. Re-solving after small tax changes confirms these derivatives, with the threshold maturity and portfolio feasibility preserved. Household yield pass-through falls from 1 at $T = 1$ to 0.581 at $T = 10$, 0.301 at 23 and 0.230 at 30. Lower-tax pass-through rises from 0 to 0.402, 0.668 and 0.736 at those maturities. Both profiles are monotone in the solved illustration.

**Long-end limit.** Extending the recursions to infinite maturity keeps the risk-premium coefficients of the solved finite-supply economy. Since the lower-tax investor's conditions bind in both markets, its exposure response appears in both loading recursions through $-b = \gamma_I \sigma_r^2 \partial \widehat{D}_I / \partial z$. Thus $A_\infty^C = 1/(1 - \rho + b)$ and $A_\infty^M = \xi_I/(1 - \rho + b)$. The feedback $b$ changes both loadings and cancels from their ratio. The individual return responses are $s_\infty^C = (1-\rho)/(1-\rho+b) = 0.851$ and $s_\infty^M = \xi_I s_\infty^C = 0.663$, so both responses include the portfolio adjustment. With $\pi_I = \gamma_I \sigma_r^2 \widehat{D}_I$, the limiting yields are

$$y_\infty^C = \bar{r} + \pi_I A_\infty^C - \tfrac{1}{2}\sigma_r^2 (A_\infty^C)^2, \qquad y_\infty^M = \xi_I \bar{r} + \pi_I A_\infty^M - \tfrac{1}{2}\sigma_r^2 (A_\infty^M)^2.$$

Re-solving the finite economy under the same tax perturbations and differentiating these limits gives $\mathcal{P}_H^Y(\infty) = 0.001$ and $\mathcal{P}_I^Y(\infty) = 0.948$, compared with the loading-implied limits of 0 and 1 in Proposition A.8(ii). For the treasury reference of P3, the ratio $A_T^M/G_T$ increases over the thirty illustrated maturities. Its infinite-maturity limit is $(1-\rho)A_\infty^M = s_\infty^M$, below $\xi_I$, because this reference lacks the portfolio response of the jointly priced taxable market $C$.

### A.2.8 Bond-specific risk and overlapping holdings

The quantitative model gives every risky bond a small bond-specific return shock. Here I add that shock to the two-investor affine model. The shock's one qualitative effect is that both investors can now hold a block of maturities, because the shock adds a risk cost that rises with each investor's holding of that particular bond. The pricing results of Appendices A.2.3–A.2.6 carry over except at the long end, with the threshold maturity replaced by the jointly held block (see the end of this subsection).

**Portfolio conditions with bond-specific risk.** Fix maturity $T$, and let $\mathcal{U}_T^M \subseteq \{H, I\}$ be the set of investors who hold that municipal bond. A holder's portfolio condition (37) becomes the main-text condition (11) with bond-specific risk added:

$$\underbrace{\mu_T^M - \xi_i r}_{\text{after-tax excess return offered to } i} = \gamma_i \left[ \underbrace{\sigma_r^2 A_{T-1}^M \widehat{D}_i}_{\text{whole-portfolio rate risk}} + \underbrace{(\sigma_{\eta,T}^M)^2 x_{i,T}^M}_{\text{bond-specific risk}} \right]. \tag{86}$$

On the branches studied here, the lower-tax investor alone holds the risky taxable bonds. Dividing its taxable condition by its after-tax share gives, for $T \geq 2$,

$$\mu_T^C - r = \gamma_I \left[ \sigma_r^2 A_{T-1}^C \widehat{D}_I + \xi_I (\sigma_{\eta,T}^C)^2 x_{I,T}^C \right], \tag{87}$$

and $\mu_1^C = r$ for cash.

**The return gap.** The subtraction in the proof of Proposition 1 now carries the bond-specific terms, which add bond-specific compensation to the three channels:

$$g_T = (\xi_i - \xi_H)r + \sigma_r^2 A_{T-1}^M\big(\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I\big) + \gamma_I \sigma_r^2 \widehat{D}_I\big(A_{T-1}^M - \xi_H A_{T-1}^C\big) + \underbrace{\gamma_i(\sigma_{\eta,T}^M)^2 x_{i,T}^M - \xi_H \gamma_I \xi_I (\sigma_{\eta,T}^C)^2 x_{I,T}^C}_{\text{bond-specific compensation}}. \tag{88}$$

Where the lower-tax investor holds the bond, the exemption-benefit and duration-cost contributions are $(\xi_I - \xi_H)r$ and zero, as in Proposition 1. Where the household holds it alone, the exemption-benefit contribution is zero and the index of (89) below satisfies $\Psi_T \geq 0$, so the duration-cost contribution is at most $(\xi_I - \xi_H)r - \gamma_H v_T^M Q_T^M / m_H$.

**Participation.** Bond-specific risk adds the bond-specific risk cost of the existing holder, the household, to the participation comparison. A new buyer bears no bond-specific risk cost at a zero position, so this cost adds to the lower-tax investor's participation incentive. The equilibrium index (89) makes this explicit.

**Theorem A.2** (Participation by the lower-tax investor in all maturities at or above a cutoff on an affine branch)**.** *At a regular state of a two-investor affine solution, at which holdings and returns move smoothly with the rate state and each investor either holds a risky bond or has a strictly positive slack in its portfolio condition for it (Appendix A.1.3), suppose $v_T^M > 0$ on the risky maturities considered. With the risk-weighted net-exposure difference $\widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$ at the equilibrium allocation, define*

$$\Psi_T = (\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} - \gamma_H v_T^M Q_T^M / m_H. \tag{89}$$

*Then $x_{I,T}^M > 0$ if and only if $\Psi_T < 0$. If $\widehat{\Lambda} > 0$ and the next-period loadings and $v_T^M Q_T^M$ are nondecreasing, the lower-tax investor holds exactly the municipal maturities from some cutoff on, possibly none. This identifies the cutoff within a given equilibrium and does not establish uniqueness across equilibria.*

**Proof.** If the lower-tax investor holds none of the bond, $x_{H,T}^M = Q_T^M / m_H$. Substituting the household's equality into the lower-tax investor's inequality gives $\Psi_T = \nu_{I,T} \geq 0$, the lower-tax investor's slack. If both investors hold the bond, subtracting their equalities and using clearing gives

$$\Psi_T = -v_T^M \left(\gamma_I + \gamma_H m_I / m_H\right) x_{I,T}^M < 0.$$

If the lower-tax investor alone holds the bond, its equality and the household's inequality imply $(\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} \leq -\gamma_I v_T^M Q_T^M / m_I < 0$, which also gives $\Psi_T < 0$. The three cases are exhaustive, so $x_{I,T}^M > 0$ if and only if $\Psi_T < 0$.

With $\widehat{\Lambda} > 0$ and nondecreasing next-period loadings and $v_T^M Q_T^M$, adjacent indices satisfy

$$\Psi_{T+1} - \Psi_T = -\sigma_r^2 (A_T^M - A_{T-1}^M)\widehat{\Lambda} - \frac{\gamma_H}{m_H}(v_{T+1}^M Q_{T+1}^M - v_T^M Q_T^M) \leq 0.$$

The index is therefore nonincreasing in maturity, so the maturities with a negative index, which the lower-tax investor holds, are all those from the first one on. That first maturity is the threshold maturity $T^*$. □

**Joint holdings.** At the equilibrium net rate exposures, define $N_T = (\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda}$, the difference in exemption benefits less the duration benefit. For $v_T := v_T^M > 0$, subtracting the two binding portfolio conditions and imposing clearing gives

$$x_{H,T}^M = \frac{\gamma_I Q_T^M + m_I N_T / v_T}{\gamma_H m_I + \gamma_I m_H}, \qquad x_{I,T}^M = \frac{\gamma_H Q_T^M - m_H N_T / v_T}{\gamma_H m_I + \gamma_I m_H}. \tag{90}$$

Both holdings are nonnegative exactly when $-\gamma_I Q_T^M / m_I \leq N_T / v_T \leq \gamma_H Q_T^M / m_H$, and strict inequalities give positive holdings for both investors. Each maturity's holdings depend on the net rate exposures, which are common to all maturities and must match the resulting portfolios. Two maturities $T < T'$ can both be jointly held only if $\sigma_r^2 \widehat{\Lambda}(A_{T'-1}^M - A_{T-1}^M) < v_T \gamma_H Q_T^M / m_H + v_{T'} \gamma_I Q_{T'}^M / m_I$, because both investors' holdings must be positive at both maturities. When the bond-specific variances are small relative to $\sigma_r^2 \widehat{\Lambda}$ times the loading differences, at most one maturity is jointly held, as in Theorem A.1.

**Responses on a jointly held block.** The next proposition is the counterpart of Proposition 2 on the branch that the illustration with bond-specific risk selects. On this branch, the household alone holds municipal bonds below $T^*$, and both investors hold them from $T^*$ on.

**Proposition A.10** (Responses with a jointly held block)**.** *At a regular state with $r(z) > 0$ in the two-investor affine model, suppose the lower-tax investor alone holds risky taxable bonds. Let $\varnothing \neq J \subseteq \{2, \dots, T_{\max}\}$ be the fixed set of municipal maturities held by both investors. The high-tax household alone holds every municipal maturity outside $J$, including the riskless one-period claim. Assume $A^M_{T-1}, A^C_{T-1} > 0$ for every $T \geq 2$ and $v^M_T > 0$ on $J$. Define*

$$P_J = \sum_{T \in J} \frac{A^M_{T-1}}{v^M_T}, \quad S_J = \sum_{T \in J} \frac{(A^M_{T-1})^2}{v^M_T}, \quad d := \frac{\partial \widehat{D}_H}{\partial z}.$$

*Then*

$$d = \frac{(\xi_I - \xi_H) m_I P_J}{(\gamma_H m_I + \gamma_I m_H)(1 + \sigma_r^2 S_J)} > 0, \qquad \frac{\partial \widehat{D}_I}{\partial z} = -\frac{m_H}{m_I} d. \tag{91}$$

*With $b = \gamma_I \sigma_r^2 (m_H/m_I) d$, the responses are*

$$s^M_T = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A^M_{T-1} d, & T \notin J, \\ \xi^*, & T \in J, \end{cases} \qquad s^C_T = 1 - b A^C_{T-1}. \tag{92}$$

*Hence $s^M_T - \xi_H s^C_T > 0$ and $A^M_T - \xi_H A^C_T > 0$ at every risky maturity $T \geq 2$, with equality in both comparisons at $T = 1$.*

**Proof.** Clearing fixes each sole holder's position, at $Q^M_T/m_H$ for the household at maturities outside $J$ and $Q^C_T/m_I$ for the lower-tax investor's risky taxable bonds, so only jointly held municipal positions vary with the rate state. With constant liabilities and fixed taxable holdings, differentiating clearing, weighting by municipal next-period loadings and summing gives

$$m_H \partial_z \widehat{D}_H + m_I \partial_z \widehat{D}_I = 0. \tag{93}$$

At each $T \in J$, subtracting the lower-tax investor's differentiated municipal equality from the household's gives

$$\xi_I - \xi_H = \sigma_r^2 A^M_{T-1} (\gamma_H \partial_z \widehat{D}_H - \gamma_I \partial_z \widehat{D}_I) + v^M_T (\gamma_H \partial_z x^M_{H,T} - \gamma_I \partial_z x^M_{I,T}).$$

With $k_0 = (\gamma_H m_I + \gamma_I m_H)/m_H$, clearing gives $\gamma_H \partial_z x^M_{H,T} - \gamma_I \partial_z x^M_{I,T} = -k_0 \partial_z x^M_{I,T}$, and (93) gives the same relation for net rate exposures. Substituting both and dividing by $-k_0 v^M_T$ gives $\partial_z x^M_{I,T} = -(\xi_I - \xi_H)/(k_0 v^M_T) - \sigma_r^2 A^M_{T-1} \partial_z \widehat{D}_I / v^M_T$. Since $\partial_z \widehat{D}_I = \sum_{T \in J} A^M_{T-1} \partial_z x^M_{I,T}$, multiplying by $A^M_{T-1}$ and summing over $J$ gives

$$(1 + \sigma_r^2 S_J) \partial_z \widehat{D}_I = -\frac{\xi_I - \xi_H}{k_0} P_J < 0.$$

Equation (93) then gives (91).

The return responses follow from (67). Outside $J$, the household's holding is fixed, so the bond-specific term vanishes and the first case of (92) remains. At jointly held maturities, weighting the holders' equalities by risk tolerance cancels the exposure and bond-specific derivatives by (93) and clearing and leaves $\xi^*$. Equation (68) gives the taxable response.

Equation (92) gives the response gap $\partial_z g_T = s^M_T - \xi_H s^C_T$ as

$$\partial_z g_T = \begin{cases} \sigma_r^2 d \left[ \gamma_H A^M_{T-1} + \xi_H \gamma_I \dfrac{m_H}{m_I} A^C_{T-1} \right], & T \notin J, \\ \omega_I (\xi_I - \xi_H) + \xi_H b A^C_{T-1}, & T \in J. \end{cases}$$

Since $d$, $b$ and the next-period loadings are positive, $\partial_z g_T > 0$ for every $T \geq 2$, while $\partial_z g_1 = 0$ because both next-period loadings vanish. Iterating (56), with $0 < \rho < 1$, yields

$$A^M_T - \xi_H A^C_T = \sum_{j=2}^{T} \rho^{T-j} \partial_z g_j > 0, \qquad T \geq 2.$$

□

The derivation of $d$ uses only that clearing fixes the positions outside $J$, so (91) also holds when the lower-tax investor alone holds the maturities above $J$, which then respond by $\xi_I - b A^M_{T-1}$. With $J = \{k\}$, the factor $P_J/(1 + \sigma_r^2 S_J) = A^M_{k-1}/(v^M_k + \sigma_r^2 (A^M_{k-1})^2)$ tends to $1/(\sigma_r^2 A^M_{k-1})$ as $v^M_k \to 0$, so (91) tends to the exposure response (69). Every response then converges to its value on the illustrated branch. Proposition A.10 needs no

condition (S), because without lower-tax-only municipal maturities the response gap is positive at every risky maturity.

**Uniqueness of the loading fixed point.** For given $d$, let $\mathcal{F}_J(d)$ be the right side of (91) at the municipal loadings that (92) and the loading recursion (56) generate, so the equilibrium $d$ is a fixed point of $\mathcal{F}_J$. If $\mathcal{F}_J$ maps $[0,\infty)$ into a bounded interval $[0, d_{\max}]$ and $\sup_{[0,d_{\max}]} |\mathcal{F}_J'| < 1$, every nonnegative fixed point lies in $[0, d_{\max}]$, which $\mathcal{F}_J$ maps into itself while shrinking distances by a factor below one. The contraction mapping theorem then gives exactly one fixed point, with its loading path on this branch. Holdings and non-holders' portfolio conditions must still be checked, and equilibria with other feasible threshold maturities remain possible. Appendix A.2.9 verifies both contraction conditions in the illustration, using the derivative

$$\mathcal{F}_J'(d) = c\frac{P_J'(1+\sigma_r^2 S_J) - \sigma_r^2 P_J S_J'}{(1+\sigma_r^2 S_J)^2}, \qquad c = \frac{(\xi_I - \xi_H)m_I}{\gamma_H m_I + \gamma_I m_H},$$

with the derivatives of $P_J, S_J$ obtained by differentiating the same loading recursions.

**Long-end pricing with a jointly held block.** For the jointly held block $J = \{T^*, \dots, T_{\max}\}$ of all maturities from the threshold maturity on, Proposition A.11 below uses condition (S) and the bound (M) on the threshold loading:

$$\textbf{(M)} \quad A^M_{T^*-1} < \frac{\xi^*}{1-\rho}. \tag{94}$$

The bound (M) places the threshold loading below its long-end value, so municipal loadings rise beyond the threshold maturity. The long-end taxable response $s^{C,\text{aff}}_\infty := \lim_{T\to\infty} s^{C,\text{aff}}_T$ lies in $(0,1)$, because the lower-tax investor's net rate exposure falls as the rate state rises.

The bound (M) holds at every solution of (91) on this branch. Put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$, so that (91) reads $\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 P_J/(1+\sigma_r^2 S_J)$. The recursions at household-only and jointly held maturities give $A^M_n = \xi_H \sum_{q=0}^{n-1}(\rho+\lambda)^q$ and $A^M_{n+l} = \rho^l A^M_n + (1-\rho^l)\xi^*/(1-\rho)$ for $l \geq 0$. Suppose $A^M_n \geq \xi^*/(1-\rho)$. Every next-period loading on $J$ is then at least $\xi^*/(1-\rho)$, so their weighted average satisfies $S_J/P_J \geq \xi^*/(1-\rho)$. Since $\sigma_r^2 P_J/(1+\sigma_r^2 S_J) < P_J/S_J$ and $\xi^* - \xi_H = \omega_I(\xi_I - \xi_H)$, this gives $\lambda < (1-\rho)(\xi^* - \xi_H)/\xi^* < 1-\rho$. Then $A^M_n < \xi_H/(1-\rho-\lambda) < \xi^*/(1-\rho)$, the second inequality by the bound on $\lambda$, a contradiction.

**Proposition A.11** (Loading ratio and long-end implied tax with a jointly held block)**.** *Consider the branch of Proposition A.10 with $J = \{T^*, \dots, T_{\max}\}$. The household alone holds municipal bonds shorter than $T^*$, both investors hold municipal bonds from $T^*$ to $T_{\max}$, and the lower-tax investor alone holds the risky taxable bonds. Assume condition (S).*

*(i) The municipal-to-taxable loading ratio is nondecreasing in maturity, starts at $\xi_H$ and, with the recursions extended to infinite maturity and $d$ held at its solved value, converges to $\xi^*/s^{C,\text{aff}}_\infty$. The ratio to the treasury reference, $A^M_T/G_T$, is also nondecreasing, starts at $\xi_H$ and converges to $\xi^* > \xi_H$.*

*(ii) With the recursions extended to infinite maturity, suppose the loadings, the risk-premium coefficients $\pi^a_T$ and the bond-specific compensation terms $\iota^a_T$ of (81) converge, and the limiting taxable forward is positive. Then*

$$\lim_{T\to\infty} \tau^{Y,\text{impl,aff}}_T = 1 - \frac{\xi^*\bar{r} + A^M_\infty \pi^M_\infty - \frac{1}{2}\sigma_r^2 (A^M_\infty)^2 + \iota^M_\infty}{\bar{r} + A^C_\infty \pi^C_\infty - \frac{1}{2}\sigma_r^2 (A^C_\infty)^2 + \iota^C_\infty}. \tag{95}$$

**Proof.** The loading ratio averages ratios of return responses, which rise with maturity. At household-only maturities, $A^M_T = (\rho + \gamma_H \sigma_r^2 d) A^M_{T-1} + \xi_H$, so $A^M_T$ is increasing from zero. At jointly held maturities, $A^M_T = \rho A^M_{T-1} + \xi^*$, whose increments have the sign of $\xi^* - (1-\rho)A^M_{T^*-1}$. The bound (M) makes them positive.

By (91), $b = \omega_H(\xi_I - \xi_H)\sigma_r^2 P_J/(1+\sigma_r^2 S_J) < \omega_H(\xi_I - \xi_H) P_J/S_J$. Rising municipal loadings give $A^M_{T-1} \geq \xi_H$ on $J$, hence $P_J/S_J \leq 1/\xi_H$, and (S) then gives $0 \leq b \leq \rho$. Therefore

$$A^C_T = \sum_{q=0}^{T-1} (\rho - b)^q, \qquad s^C_T = 1 - b A^C_{T-1} \downarrow \frac{1-\rho}{1-\rho+b} > 0.$$

The municipal response rises at household-only maturities, is constant at jointly held maturities and does not fall at $T^*$, where the lower-tax investor begins to participate. For the last claim, the response at $T^* - 1$, $\xi_H + \gamma_H \sigma_r^2 A^M_{T^*-2} d$, must not exceed $\xi^*$, which equals $\xi_H + \omega_I(\xi_I - \xi_H)$. Substituting (91) shows that the required inequality $\omega_I(\xi_I - \xi_H) \geq \gamma_H \sigma_r^2 A^M_{T^*-2} d$ is equivalent to $\sigma_r^2 A^M_{T^*-2} P_J \leq 1 + \sigma_r^2 S_J$. Municipal loadings rise strictly

under (M), so $A^M_{T-1} > A^M_{T^*-2}$ for every $T \in J$, which gives $\sigma_r^2 A^M_{T^*-2} P_J < \sigma_r^2 S_J$. Since $s^C_j$ is positive and nonincreasing, the ratio of return responses $s^M_j/s^C_j$ is nondecreasing in maturity.

For any positive $e_j$ and nondecreasing ratios $q_j = a_j/e_j$, the ratio weighted by rate persistence,

$$\bar{q}_T = \frac{\sum_{j\le T} \rho^{T-j} a_j}{\sum_{j\le T} \rho^{T-j} e_j}$$

is nondecreasing because, as a positive-weight average of $q_1, \dots, q_T$, it is at most $q_T \le q_{T+1}$, and

$$\bar{q}_{T+1} - \bar{q}_T = \frac{e_{T+1}(q_{T+1} - \bar{q}_T)}{\rho \sum_{j\le T} \rho^{T-j} e_j + e_{T+1}} \ge 0.$$

By (56), $A^M_T/A^C_T$ has this form. Applying the argument with $a_j = s^M_j$ and $e_j = s^C_j$ proves the loading-ratio ordering, starting from $\xi_H$. With $e_j = 1$, for which (M) suffices, the same argument proves that $A^M_T/G_T$ is nondecreasing.

Extending the same branch recursions to arbitrarily long maturities with fixed $d$ gives $A^M_\infty = \xi^*/(1-\rho)$, $A^C_\infty = 1/(1-\rho+b)$ and $G_\infty = 1/(1-\rho)$, hence the two limits in part (i). For part (ii), the level recursion (81) and the long-end yield (82) it implies apply with $\bar{\xi}^M_\infty = \xi^*$ at jointly held long maturities and $\bar{\xi}^C_\infty = 1$, which gives (95). □

**The long-end implied tax to first order.** Aggregating by risk tolerance over both investors for long municipal bonds and over the lower-tax investor alone for long taxable bonds gives

$$\pi^M_\infty - \pi^C_\infty = \omega_H \, \sigma_r^2 \, \widehat{\Lambda}, \qquad \widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I, \tag{96}$$

where $\widehat{D}_i = \widehat{D}_i(0)$ is the net rate exposure at the stationary mean. The approximation below treats the convexity and bond-specific terms as negligible. With a small feedback ($b \ll 1-\rho$) and a positive denominator, substituting $A^M_\infty = \xi^*/(1-\rho)$, $A^C_\infty \approx 1/(1-\rho)$ and (96) into (82) gives the first-order long-end limit

$$\tau^{Y,\text{impl,aff}}_\infty \approx \underbrace{(1-\xi^*)}_{\text{one minus tax mix}} - \underbrace{\frac{\xi^* \, \omega_H \, \sigma_r^2 \, \widehat{\Lambda}}{(1-\rho)\bar{r} + \pi^C_\infty}}_{\text{municipal term-premium excess}}. \tag{97}$$

With $\widehat{\Lambda} > 0$, this first-order limit lies below one minus the tax mix, $1-\xi^*$, and hence below the one-period implied tax $1-\xi_H$.

**Pass-through with a jointly held block.** On this branch the long-end loading ratio is $\xi^*/s^C_\infty$, and both after-tax shares affect it through the tax mix and the feedback $b$.

**Proposition A.12** (Tax pass-through with a jointly held block)**.** *Consider the branch of Proposition A.11, under condition (S), with a positive cash rate and a slack household shorting constraint on cash. With its recursions extended to infinite maturity, suppose long yields, loading ratios and their tax derivatives converge as maturity grows, uniformly for after-tax shares near their given values, with a positive long taxable yield. The short-end pass-throughs are those of Proposition A.8(i).*

*(i) (Long end, P1 and P2) Suppose the after-tax shares satisfy the tax-difference condition*

$$2\omega_H e + \omega_I e^2 \le 1, \qquad e := \frac{\xi_I - \xi_H}{\xi_H}.$$

*Then*

$$\mathcal{P}^A_H(\infty) < \omega_H, \qquad \mathcal{P}^A_I(\infty) > \omega_I.$$

*Yield pass-throughs add to these values the correction terms in (98) below, which vanish at rate $1/\bar{r}$ when the branch's equations are solved at other values of $\bar{r}$ with participation held fixed. Whenever the corrections are smaller in absolute value than the margins $\omega_H - \mathcal{P}^A_H(\infty)$ and $\mathcal{P}^A_I(\infty) - \omega_I$ of Lemma A.1,*

$$\mathcal{P}^Y_H(\infty) < \omega_H < 1, \qquad \mathcal{P}^Y_I(\infty) > \omega_I > 0.$$

*(ii) (Loading ratios, P3) The municipal-to-taxable zero-coupon loading ratio $A^M_T/A^C_T$ is nondecreasing in maturity, equals $\xi_H$ at the short end and converges to $\xi^*/s^C_\infty$, with $0 < s^C_\infty < 1$. For the treasury reference, $A^M_T/A^{\text{Tsy}}_T$ has the same ordering and the same value at the short end, and converges to $\xi^* > \xi_H$.*

With the tax mix alone, each investor's long-end loading pass-through on this branch equals its risk-tolerance share (Proposition A.9). The tax-difference condition holds whenever $\xi_I < \frac{3}{2}\xi_H$, regardless of risk-tolerance shares or the threshold maturity. Under it, portfolio adjustment lowers the household's pass-through and raises the lower-tax investor's, as Lemma A.1 proves with both portfolios and loadings adjusting.

**Proof of Proposition A.12.** *(i) The long end.* Let $d$ be the household's exposure response in (91) and $b = \gamma_I \sigma_r^2 (m_H/m_I) d$ the feedback into taxable responses, as in (92). The solved long responses and loadings satisfy

$$s_\infty^M = \xi^*, \qquad s_\infty^C = \frac{1-\rho}{1-\rho+b}, \qquad A_\infty^a = \frac{s_\infty^a}{1-\rho}.$$

Thus $R_\infty = A_\infty^M / A_\infty^C = \xi^* / s_\infty^C$. At the mean state, the solved yield formula (82) gives

$$\begin{aligned} y_\infty^M &= \xi^* \bar{r} + A_\infty^M \pi_\infty^M - \tfrac{1}{2}\sigma_r^2 (A_\infty^M)^2 + \iota_\infty^M, \\ y_\infty^C &= \bar{r} + A_\infty^C \pi_\infty^C - \tfrac{1}{2}\sigma_r^2 (A_\infty^C)^2 + \iota_\infty^C. \end{aligned}$$

The four terms are interest income, portfolio rate-risk compensation, convexity and bond-specific compensation. Dividing the two yields separates the loading ratio from the remaining yield-level difference:

$$\frac{y_\infty^M}{y_\infty^C} = R_\infty + \frac{y_\infty^M - R_\infty y_\infty^C}{y_\infty^C}.$$

All derivatives below are with respect to $\xi_i$, with holdings, loadings and both yields adjusting. The assumed convergence permits differentiation of the limiting ratio. The quotient rule gives

$$\begin{aligned} \mathcal{P}_i^Y(\infty) = \mathcal{P}_i^A(\infty) &+ \frac{1}{y_\infty^C} \frac{d(y_\infty^M - R_\infty y_\infty^C)}{d\xi_i} \\ &- \frac{y_\infty^M - R_\infty y_\infty^C}{(y_\infty^C)^2} \frac{dy_\infty^C}{d\xi_i}. \end{aligned} \tag{98}$$

The three terms are the responses of the loading ratio, the relative yield compensation and the taxable denominator. The first retains the endogenous taxable response:

$$\mathcal{P}_i^A(\infty) = \frac{\omega_i}{s_\infty^C} - \frac{\xi^*}{(s_\infty^C)^2} \frac{ds_\infty^C}{d\xi_i}.$$

Lemma A.1 below proves $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$ under the tax-difference condition, with uniform positive margins.

For the two correction terms, the calculation after Lemma A.1 shows that $y_\infty^M - R_\infty y_\infty^C$ and its tax derivatives are independent of $\bar{r}$. Also, $y_\infty^C = A_\infty^C(1-\rho)\bar{r} + O(1)$ and $dy_\infty^C/d\xi_i = O(\bar{r})$. Both correction terms in (98) therefore vanish at rate $1/\bar{r}$ in the branch's equations, while the lemma's margins do not depend on $\bar{r}$ and are uniform over the finitely many cutoffs. With other primitives fixed, the solution of the branch's equations is an equilibrium only for bounded $\bar{r}$, since some of the lower-tax investor's holdings $x_{I,T}^M$ in (90) turn negative as $\bar{r}$ grows. When the corrections are smaller in absolute value than these margins, $\mathcal{P}_H^Y(\infty) < \omega_H < 1$ and $\mathcal{P}_I^Y(\infty) > \omega_I > 0$. In the illustration of Appendix A.2.9, the corrections are 0.006 and $-0.054$, against margins above one half.

*(ii) The loading ratios.* At a common maturity, dividing both price loadings by $T$ leaves their ratio unchanged. Proposition A.11(i) gives the ordering, starting values and limits for the equilibrium taxable market $C$ and for the treasury reference, for which $A_T^{\text{Tsy}} = G_T$, $A_1^{\text{Tsy}} = 1$ and $A_\infty^{\text{Tsy}} = 1/(1-\rho)$. □

**Lemma A.1** (The loading response)**.** *Under the tax-difference condition in Proposition A.12, $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$. With other primitives fixed, both strict margins have positive lower bounds over all feasible threshold maturities.*

**Proof.** The proof differentiates the exposure fixed point and bounds its feedback into the loading ratio.

Put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$, the response of the household's unit cost of rate risk to the rate state. Equation (91), $R_\infty = \xi^*/s_\infty^C$ and $b = (\omega_H/\omega_I)\lambda$ give

$$\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 f, \qquad f = \frac{P_J}{1+\sigma_r^2 S_J}, \qquad R_\infty = \xi^* \left[1 + \frac{\omega_H \lambda}{\omega_I(1-\rho)}\right].$$

The sums $P_J$ and $S_J$ are those of Proposition A.10, and the ratio $f$ is unrelated to the forward rates $f_T^a$. With $G_n(p) = \sum_{q=0}^{n-1} p^q$, the loading recursion gives

$$A_n^M = \xi_H G_n(\rho+\lambda), \qquad A_{n+l}^M = \rho^l A_n^M + (1-\rho^l)A_\infty^M, \quad l \geq 0.$$

By (94), $A_n^M < A_\infty^M$, so loadings rise along the jointly held block. Thus $S_J/P_J \geq A_n^M$ and

$$\lambda G_n(\rho+\lambda) < \omega_I e, \qquad \lambda \frac{G_n'(\rho+\lambda)}{G_n(\rho+\lambda)} < \omega_I e,$$

where $e$ is the relative tax difference of Proposition A.12. The first bound combines $\lambda < \omega_I(\xi_I - \xi_H)P_J/S_J$ with $S_J/P_J \geq A_n^M = \xi_H G_n(\rho+\lambda)$. The second follows from $G_n' \leq G_n^2$, which holds coefficient by coefficient in $p$. The tax-difference condition implies $e < 1$, since at $e \geq 1$ its left side is at least $1 + \omega_H$. Hence $b = (\omega_H/\omega_I)\lambda < \omega_H e < 1$, so the taxable loadings converge.

To differentiate the loading sums at fixed $\lambda$, let $t_j = \rho^l A_n^M / A_{j-1}^M$ for $j = T^* + l$, the share of the next-period loading inherited from the threshold loading $A_n^M$, and

$$X = \bar{t}_P - \frac{2\sigma_r^2 S_J}{1+\sigma_r^2 S_J}\bar{t}_S.$$

Here $\bar{t}_P$ is the mean of $t_j$ with weights proportional to $A_{j-1}^M/v_j^M$, and $\bar{t}_S$ uses weights proportional to $(A_{j-1}^M)^2/v_j^M$. Since $0 \leq t_j \leq 1$, $-2 < X \leq 1$. Let $\partial_i$ denote differentiation with respect to $\xi_i$ holding $\lambda$ fixed. Differentiating $A_{j-1}^M = \rho^l A_n^M + (1-\rho^l)\xi^*/(1-\rho)$, in which $A_n^M$ is proportional to $\xi_H$ at fixed $\lambda$, gives $\partial_H \ln A_{j-1}^M = \omega_H/\xi^* + t_j(1/\xi_H - \omega_H/\xi^*)$ and $\partial_I \ln A_{j-1}^M = (1-t_j)\omega_I/\xi^*$. Averaging these with the weights of $\bar{t}_P$ and $\bar{t}_S$ gives

$$\partial_H \ln f = \left(\frac{1}{\xi_H} - \frac{\omega_H}{\xi^*}\right) X + \frac{\omega_H}{\xi^*}\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J},$$
$$\partial_I \ln f = \frac{\omega_I}{\xi^*}\left(\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J} - X\right).$$

The equilibrium response of $\lambda$ follows from taking logs of its fixed point $\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 f$ and differentiating with respect to $\xi_i$, using $\partial \ln f/\partial\lambda = (G_n'/G_n)X$, which comes from $\partial \ln A_{j-1}^M/\partial\lambda = t_j G_n'/G_n$. With $G_n$ and $G_n'$ evaluated at $\rho+\lambda$,

$$\frac{d\lambda}{d\xi_i} = \frac{\lambda\,\partial_i \ln[(\xi_I - \xi_H)f]}{D}, \qquad D = 1 - \lambda\frac{G_n'}{G_n}X > 1 - \omega_I e > 0.$$

The bound on $D$ uses $X \leq 1$ and $\lambda G_n'/G_n < \omega_I e$.

Substituting $d\lambda/d\xi_i$ into $dR_\infty/d\xi_i = \mathcal{P}_i^A(\infty)$ gives $\mathcal{P}_i^A(\infty) - \omega_i = \frac{\omega_H\lambda}{\omega_I(1-\rho)D}\left[\omega_i D + \xi^*\partial_i \ln\big((\xi_I - \xi_H)f\big)\right]$, whose bracket follows from the derivatives of $\ln f$ at fixed $\lambda$. In the household's bracket, $X$ has coefficient $\omega_I\xi_I/\xi_H - \omega_H\lambda G_n'/G_n > \omega_I(1+\omega_I e) > 0$. In the lower-tax investor's bracket, it has coefficient $-\omega_I(1+\lambda G_n'/G_n) < 0$. Using $X \leq 1$, $\lambda G_n'/G_n < \omega_I e$ and $\xi^*/(\xi_I - \xi_H) = 1/e + \omega_I$ gives

$$\mathcal{P}_H^A(\infty) - \omega_H \leq \frac{\omega_H\lambda}{\omega_I(1-\rho)D}\left(\frac{2\omega_H}{1+\sigma_r^2 S_J} - \frac{1}{e} + \omega_I e\right) < 0,$$
$$\mathcal{P}_I^A(\infty) - \omega_I > \frac{\omega_H\lambda}{\omega_I(1-\rho)D}(1/e - \omega_I^2 e) > 0.$$

Dividing the tax-difference condition by $e$ gives $2\omega_H + \omega_I e \leq 1/e$, so the household's parenthesized term is at most $-2\omega_H\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$, which is negative, and $1/e - \omega_I^2 e > 0$ because $e < 1$. Each investor therefore has a positive margin.

For each of the finitely many $n \in \{1, \ldots, T_{\max} - 1\}$, the values $x \geq 0$ with $xG_n(\rho+x) \leq \omega_I e$ form a closed bounded interval that depends only on primitives and, by the first bound, contains every equilibrium value of $\lambda$. On this interval the loading sums are positive and continuous in $x$, so $f$ and $\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$ have positive minima, while $D \geq 1 - \omega_I e$ keeps the tax derivatives bounded. At a fixed point, $\lambda \geq \omega_I(\xi_I - \xi_H)\sigma_r^2 \min f > 0$ and, since $X > -2$, $D < 1 + 2\omega_I e$. The displayed margins therefore have positive lower bounds over all feasible threshold maturities. □

**The relative yield compensation.** This calculation completes the proof of Proposition A.12(i). Subtract the two solved long yields, weighting the taxable yield by $R_\infty = A^M_\infty/A^C_\infty$. Equation (96) gives $\pi^M_\infty - \pi^C_\infty = \omega_H \sigma_r^2 \widehat{\Lambda}$, while $\xi^* - R_\infty = -bA^M_\infty$. Therefore

$$y^M_\infty - R_\infty y^C_\infty = A^M_\infty(\omega_H \sigma_r^2 \widehat{\Lambda} - b\bar{r}) + \tfrac{1}{2}\sigma_r^2 A^M_\infty (A^C_\infty - A^M_\infty) + \iota^M_\infty - R_\infty \iota^C_\infty.$$

The three terms are the difference in risk costs on the bonds each investor holds alone, relative convexity and bond-specific compensation. On the jointly held block, substituting the holdings (90) into $\widehat{\Lambda}$ gives $\widehat{\Lambda}(1+\sigma_r^2 S_J) = K + (\xi_I - \xi_H)\bar{r} P_J$, with $K$ the bracketed term below. Since $b = \omega_H(\xi_I - \xi_H)\sigma_r^2 P_J/(1+\sigma_r^2 S_J)$ by (91), the $\bar{r}$ terms cancel and

$$\omega_H \sigma_r^2 \widehat{\Lambda} - b\bar{r} = \frac{\omega_H \sigma_r^2}{1+\sigma_r^2 S_J}\left[\gamma_H\left(\sum_{T \notin J} \frac{A^M_{T-1} Q^M_T}{m_H} - \ell_H\right) - \gamma_I\left(\frac{\xi_I}{m_I}\sum_T A^C_{T-1} Q^C_T - \ell_I\right)\right].$$

Every term, and hence its tax derivative, is independent of $\bar{r}$ on the fixed branch. Common rate-risk compensation per unit of loading cancels from $y^M_\infty - R_\infty y^C_\infty$.

The joint holders' risk-premium coefficient is also independent of $\bar{r}$. With $\Gamma := m_H/\gamma_H + m_I/\gamma_I$, the value of $\Gamma^M_T$ on the jointly held block, clearing gives

$$\pi^M_\infty = \frac{\sigma_r^2}{\Gamma}\left(\sum_T A^M_{T-1} Q^M_T + \xi_I \sum_T A^C_{T-1} Q^C_T - m_H \ell_H - m_I \ell_I\right).$$

By (96) and the two previous displays, the mean-rate term in $\pi^C_\infty$ is $-b\bar{r}$. Since $1/A^C_\infty = 1 - \rho + b$, the mean-rate terms of $y^C_\infty$ sum to $\bar{r}(1 - bA^C_\infty) = A^C_\infty(1-\rho)\bar{r}$, so $y^C_\infty = A^C_\infty(1-\rho)\bar{r} + O(1)$. The bond-specific terms are $\iota^M_\infty = v^M_\infty(Q^M_\infty/\Gamma - \frac{1}{2})$ and $\iota^C_\infty = v^C_\infty(\gamma_I \xi_I Q^C_\infty/m_I - \frac{1}{2})$, where $Q^a_\infty$ and $v^a_\infty$ are the limiting supply and bond-specific variance in market $a$.

**Convexity in the long-end correction.** Dividing the relative yield compensation by $y^C_\infty$ isolates relative convexity, the only correction on the illustrated branch (Proposition A.6(iii)):

$$\Delta\tau_\infty := \frac{y^M_\infty}{y^C_\infty} - R_\infty = R_\infty(1-R_\infty)\chi_\infty + \psi_\infty,$$
$$\chi_\infty = \frac{\frac{1}{2}\sigma_r^2 (A^C_\infty)^2}{y^C_\infty}, \qquad \psi_\infty = \frac{A^M_\infty(\omega_H \sigma_r^2 \widehat{\Lambda} - b\bar{r}) + \iota^M_\infty - R_\infty \iota^C_\infty}{y^C_\infty}.$$

Since each implied tax is one minus its ratio, the long-end yield-implied tax equals the long-end loading-implied tax less $\Delta\tau_\infty$. The signed convexity contribution is at most $\chi_\infty/4$, since $R_\infty(1-R_\infty) \le 1/4$. The derivatives of the two terms sum to $\mathcal{P}^Y_i(\infty) - \mathcal{P}^A_i(\infty)$ in (98).

**What bond-specific risk changes.** The short end, the signs of the exposure responses, the positive loading gap and the rising loading ratio are the same on both branches. What changes is who holds the longest bonds. On the illustrated branch, the lower-tax investor alone holds them, so the long-end loading ratio is $\xi_I$ and long-end loading pass-throughs are $(0, 1)$. On a jointly held block that reaches the longest maturity, both investors hold them, the long-end response is the tax mix $\xi^*$, and the household's higher unit cost of rate risk adds the term-premium excess in (97). The two branches give close long-end responses when the threshold maturity's loading is near its long-end value. On the illustrated branch, $bA^M_{k-1} = \xi_I - \xi^*$, so the long-end response $\xi_I - bA^M_\infty$ is near $\xi^*$ when $A^M_\infty$ is near $A^M_{k-1}$. In the illustrations, the long-end municipal response is 0.663 on the illustrated branch against $\xi^* = 0.665$ with bond-specific risk, and the long-end loading pass-throughs are $(0, 1)$ against $(-0.012, 1.013)$.

### A.2.9 Illustration with bond-specific risk

Adding bond-specific risk to the main-text illustration broadens joint participation without materially changing the return gap. I report the holdings, return-gap channels and loading decomposition plotted in Figure 14, the tax pass-throughs, and checks of the conditions used in Appendix A.2.8.

**Setup and intuition.** I keep the other primitives of Appendix A.2.7 and add a municipal log-return standard deviation of $0.004[1 + 7\sqrt{(T-1)/29}]$ for $2 \le T \le 30$. Taxable bonds have no bond-specific shock, and the one-period claims remain riskless. Both investors now hold municipal maturities 17 through 30, while without the shock they share only maturity 23. With rate risk alone, joint holdings require both investors to demand the same return. Bond-specific risk adds a cost that rises with each investor's holding of that bond and can offset a small difference in the returns the investors require for taxes and rate risk. In this illustration, those requirements are close across long maturities, where rate loadings flatten, so a small bond-specific cost supports joint holdings over a broad maturity range while adding little to required returns.

**Return-gap channels.** Bond-specific compensation is at most 2.7% of the return gap. At thirty periods, the gap is about 205 basis points, comprising 92 of exemption benefit, 109 of rate loading and 4 of bond-specific compensation. At the last household-only maturity, $T = 16$, the 182-basis-point gap comprises 88 of duration cost, 91 of rate loading and 3 of bond-specific compensation. Removing the shock changes the return gap by at most 3.91 basis points over all thirty maturities, municipal yields by at most 0.33 basis points and taxable yields by at most 0.07 basis points.

**A unique loading solution.** For $a_{\min} = \xi_H \sum_{q=0}^{15} \rho^q$, next-period loadings on the jointly held block satisfy $A^M_{T-1} \ge a_{\min}$ and $P_J/S_J \le 1/a_{\min}$. The exposure map $\mathcal{F}_J$ of Appendix A.2.8 therefore maps $[0, \infty)$ into $[0, d_{\max}]$, where

$$d_{\max} = \frac{\xi_I - \xi_H}{(\gamma_H + \gamma_I)\sigma_r^2 a_{\min}} < 13.77.$$

Interval arithmetic over $[0, d_{\max}]$, which also bounds rounding error, gives $\sup |\mathcal{F}_J'| < 0.066 < 1$, so by the contraction mapping theorem the map has exactly one fixed point, $d^* \approx 11.20$. The resulting $A^M_{16} \approx 3.147$ is below $\xi^*/(1-\rho) = 3.325$, consistent with the bound (M) in (94). The left side of condition (S) in (72) is about 0.209, below $\rho = 0.8$.

**Levels and the portfolio conditions.** The liability hedges above give $\widehat{\Lambda} \approx 4.53$, which satisfies the participation inequalities selecting $T^* = 17$ under Theorem A.2. At the mean rate state, the holders' portfolio conditions, market clearing and the return-gap identity hold to numerical precision, and every non-holder's portfolio condition is strictly slack.

**Pass-through across maturities.** Re-solving the exposure fixed point, loadings and holdings for tax changes gives household yield pass-through declining from 1 to 0.230 and lower-tax pass-through rising from 0 to 0.739. Lower-tax pass-through is 0.06 at $T = 2$, a maturity the lower-tax investor does not hold, because its tax changes both investors' net rate exposures, which affect their required returns at every risky maturity. Interval arithmetic confirms that household pass-through falls and lower-tax pass-through rises at every successive maturity. Their sum ranges from about 0.969 to 1, and the threshold maturity remains $T^* = 17$.

**Long-end limit.** I extend the price recursions to longer maturities with the solved economy's exposure responses and risk-premium coefficients, holding municipal supply and bond-specific variance at their thirty-period values, $Q^M_\infty = 0.1$ and $v^M_\infty = 0.001024$. This gives $A^M_\infty = 3.325$ and $s^C_\infty = 0.851$. Loading-implied pass-throughs are $(-0.012, 1.013)$ and yield-implied pass-throughs $(-0.006, 0.959)$. The tax-difference condition of Proposition A.12 has left side $0.506 < 1$. Portfolio responses in both markets move both sets of pass-throughs away from the risk-tolerance shares, in the directions that part (i) of that proposition predicts.

**Equilibrium selection.** To select each illustration's participation pattern, I enumerate every affine configuration in which the household alone holds the municipal maturities below a cutoff, both investors hold those from the cutoff through a last joint maturity, and the lower-tax investor alone holds the longer ones and all risky taxable bonds. There are 29 such configurations without bond-specific risk, one per cutoff from 2 to 30, which is then the only joint maturity, and 435 with bond-specific risk, one per cutoff and last joint maturity. I keep a configuration only when its portfolio conditions at the mean rate state and clearing, a linear system, have a unique feasible solution. In each illustration exactly one configuration passes, so the check selects a unique participation pattern within this family at the mean rate state (Appendix A.2.8). Full 45-market nonlinear solutions pass the independent grid checks in Appendix B.7.2, and their maturity patterns are numerical results.

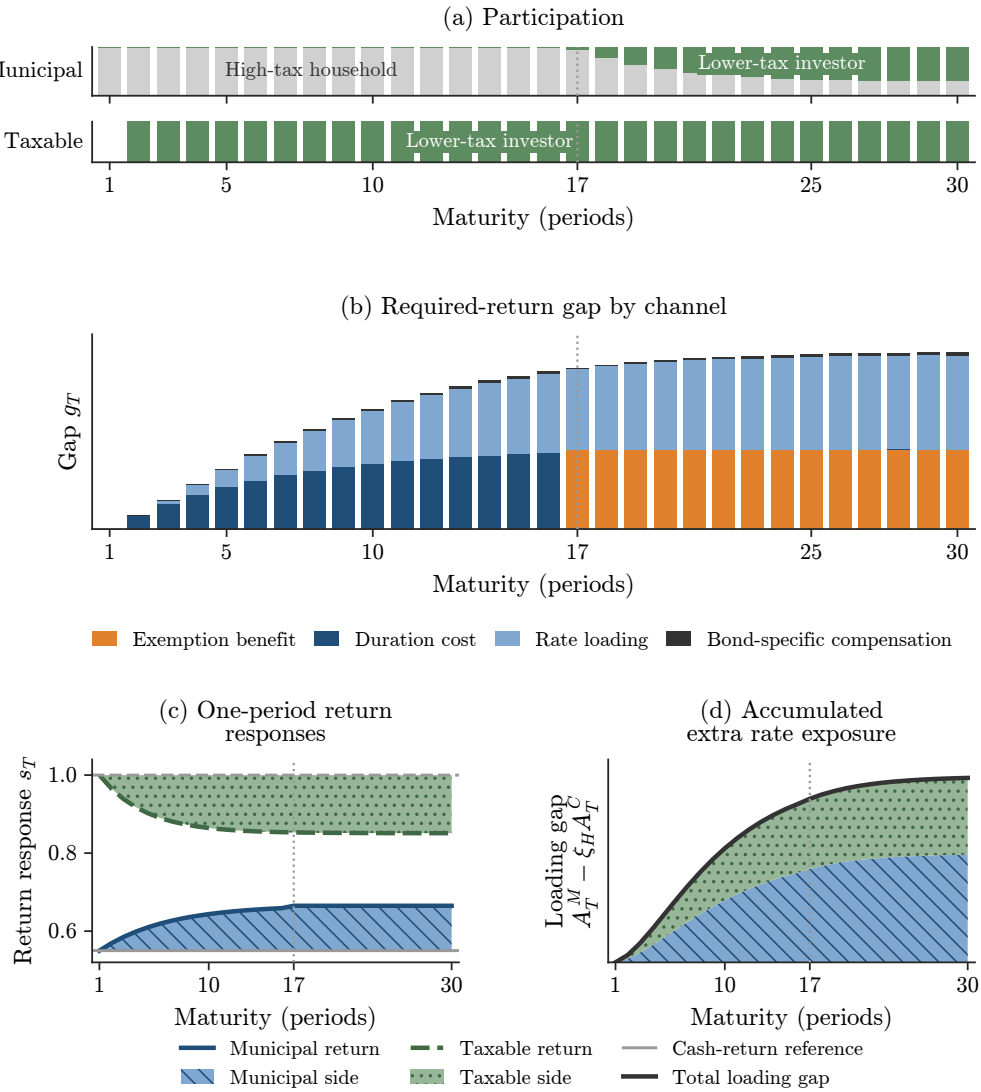



Figure 14: The analytical illustration with bond-specific risk. All other primitives match Appendix A.2.7. Panel (a) shows ownership. Panel (b) gives the three return-gap channels of Figure 5, with small bond-specific compensation in gray. Panels (c) and (d) use the grouping of Figure 6. The municipal side combines the tax mix and municipal portfolio response, while the taxable side reflects the taxable portfolio response. Panel (c) shades $s_T^M - \xi_H$ and $1 - s_T^C$, measured from the cash-return references $\xi_H$ and 1 (gray lines). Panel (d) accumulates $s_T^M - \xi_H$ and $\xi_H(1 - s_T^C)$ into the total loading gap. The dotted line marks joint municipal holdings from $T^* = 17$.

## A.3 Other motives for lower-tax participation

Lower trading costs or better default diversification can induce long-maturity participation without a liability-hedging advantage (Section 4.7). The pricing mechanism extends to any such motive that is neutral, meaning that by itself it creates no gap relative to the household benchmark when the household holds both bonds. Participation then brings the lower-tax investor's smaller exemption benefit into the return comparison. Default diversification is neutral only when municipal and taxable default risks load on one shock. With separate shocks, the gap net of holder-specific default compensation equals $(\xi_I - \xi_H)r$ wherever the lower-tax investor holds the municipal bond (Proposition A.15(iii)). A municipal-specific benefit has the opposite pricing implication. Throughout, $g_T = \mu_T^M - \xi_H \mu_T^C$ is the one-period return gap and $\xi_H < \xi_I$.

**Proposition A.13** (Participation creates the gap)**.** *In the two-investor affine model without bond-specific risk, suppose every non-tax characteristic other than rate risk is neutral (Definition A.1) and zero at the one-period*

*claim. Each is a risk common to both markets, independent of rate risk, with municipal exposure $\xi_H$ times the taxable exposure, or a cost that reduces after-tax wealth by the same amount in both markets.*

*(i) If the household alone holds every municipal and taxable bond and its shorting constraint on cash is slack, $g_T = 0$ at every maturity.*

*(ii) Wherever the lower-tax investor holds the maturity-$T$ municipal bond,*

$$g_T \;\geq\; \underbrace{(\xi_I - \xi_H)r}_{\text{exemption benefit}} \;+\; \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I \big(A^M_{T-1} - \xi_H A^C_{T-1}\big)}_{\text{rate loading}}.$$

Part (i) is proved in Appendix A.3.2, and part (ii) follows from Proposition A.14. Without additional common return characteristics, on the illustrated branch of Proposition 2 under condition (S) in (72), the rate-loading term is zero at $T \leq 2$ and positive at every $T \geq 3$ when the lower-tax investor's net rate exposure is positive.

### A.3.1 Non-tax required returns and neutral characteristics

Add a differentiable concave term $\Phi_i(x_i)$, in the same after-tax units, to investor $i$'s objective in (36). Dividing each portfolio condition (37) by its after-tax share gives, under the normalized municipal–taxable economy,

$$\mu^M_T \leq \xi_i r + K^M_{i,T}, \qquad \mu^C_T \leq r + K^C_{i,T}, \qquad \text{with equality for a positive holding,} \tag{99}$$

where

$$K^a_{i,T} := \gamma_i \operatorname{Cov}_z\big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\big) + \frac{\kappa_i}{\xi^a_i} - \frac{1}{\xi^a_i}\frac{\partial \Phi_i}{\partial x^a_{i,T}}.$$

Here $\widetilde{\Pi}_i(t+1)$ is the after-tax portfolio innovation in (35). With additive shocks, the non-tax required return $K^a_{i,T}$ splits into components $k^a_{ij,T}$ for rate risk, bond-specific risk, each additional shock or wealth characteristic, and the shorting constraint on cash. Additional shocks are independent of the rate and bond-specific shocks.

A common return characteristic is a unit-variance shock $q$ that adds $e^a_T q_{t+1}$ to the innovation $u^a_{T,t+1}$ in both markets, with nonnegative maturity-$T$ exposures $e^M_T$, $e^C_T$, and is taxed with returns. Its component is $k^a_{iq,T} = c_{iq} e^a_T$, where the cost per unit of exposure, $c_{iq} := \gamma_i \operatorname{Cov}_z(q_{t+1}, \widetilde{\Pi}_i(t+1))$, is the same in both markets. Affine rate risk is such a characteristic, with $q = -\varepsilon$, $e^a_T = \sigma_r A^a_{T-1}$, $c_{iq} = \gamma_i \sigma_r \widehat{D}_i$ and component $\gamma_i \sigma_r^2 A^a_{T-1} \widehat{D}_i$.

A wealth characteristic, such as a trading cost, reduces after-tax wealth by $w^a_T$ per dollar, scaled by investor $i$'s intensity $\varsigma_i \geq 0$. It is the state-independent holding benefit of Appendix A.2.4 with $\vartheta^a_{i,T} = -\varsigma_i w^a_T$. For $\Phi_i = -\varsigma_i \sum_{a,T} w^a_T x^a_{i,T}$, the component is $k^a_{iw,T} = \varsigma_i w^a_T / \xi^a_i$, and the shorting constraint on cash similarly gives $\kappa_i / \xi^a_i$.

**Definition A.1** (Neutral characteristic)**.** *A common return characteristic is neutral if $e^M_T = \xi_H e^C_T$, which equalizes the household's after-tax exposure per dollar across markets, and a wealth characteristic is neutral if $w^M_T = w^C_T$, at every maturity. For a household that bears it, a neutral characteristic satisfies $k^M_{Hj,T} = \xi_H k^C_{Hj,T}$ for its component $j$, so it creates no gap when the household holds both bonds.*

**Neutrality for each investor.** Neutrality is stated for the household because the puzzle is defined against the household benchmark. A neutral common return characteristic gives $k^M_{iq,T} = c_{iq} e^M_T = c_{iq} \xi_H e^C_T = \xi_H k^C_{iq,T}$ for every investor $i$. A neutral wealth characteristic, with common cost $w_T$, gives $k^M_{iw,T} = \varsigma_i w_T = \xi_i (\varsigma_i w_T / \xi_i) = \xi_i k^C_{iw,T}$, so

$$k^M_{Iw,T} - \xi_H k^C_{Iw,T} = (\xi_I - \xi_H) k^C_{Iw,T}. \tag{100}$$

Rate risk is neutral in the affine model where the household alone holds both bonds. Clearing fixes the household's exposure, so differentiation gives the return responses $s^M_T = \xi_H$ and $s^C_T = 1$. Starting at $A^M_1 = \xi_H$, $A^C_1 = 1$, recursion (53) then gives $A^M_{T-1} = \xi_H A^C_{T-1}$ at every maturity by induction, as in Proposition A.5.

### A.3.2 The gap without and with participation

**No one-period return gap when the household holds both bonds.** Suppose the household alone holds both bonds at every maturity, its shorting constraint on cash is slack, there is no bond-specific risk and every additional characteristic is neutral and zero at the one-period claim. Both household conditions in (99)

bind, so $g_T = \sum_j (k^M_{Hj,T} - \xi_H k^C_{Hj,T})$. Definition A.1 removes the additional components. In the affine model, the neutrality of rate risk makes the rate contribution $\gamma_H \sigma_r^2 \widehat{D}_H (A^M_{T-1} - \xi_H A^C_{T-1})$ zero. In the constant-rate case (c) of Proposition A.15, there is no rate component. At the riskless maturity $T = 1$, the municipal condition gives $\mu^M_1 = \xi_H r$ and cash pays $\mu^C_1 = r$, so $g_1 = 0$. Hence

$$g_T = 0 \qquad \text{at every maturity.} \tag{101}$$

A binding shorting constraint on cash would add $\kappa_H$ to the municipal condition and $\kappa_H/\xi_H$ to the taxable condition, which cancel in $g_T$ at risky maturities and leave $g_1 = \kappa_H$. The yield gap $y^M_T - \xi_H y^C_T$ also includes the convexity difference in (75), so it need not be zero.

**Participation comparison for any motive.** Evaluate the components at an equilibrium allocation. Both investors' municipal conditions in (99) share the expected return $\mu^M_T$, so the lower-tax investor's required return is no higher than the household's exactly when

$$\Delta_T := K^M_{H,T} - K^M_{I,T} \;\geq\; (\xi_I - \xi_H)\, r. \tag{102}$$

The lower-tax investor can hold the maturity-$T$ municipal bond only if (102) holds, and the household only if the reverse weak inequality holds. Rate risk gives $\Delta_T = \sigma_r^2 A^M_{T-1} \widehat{\Lambda}$, the duration benefit in (59), and Theorem A.1 gives the resulting sorting. Advantages from several motives add. Suppose $r > 0$, $\Delta_1 = 0$ and $\Delta_T$ increases strictly in $T$. Then the lower-tax investor holds either no municipal maturity or all maturities at or above a cutoff longer than one period, and the household holds every shorter maturity. A strictly increasing $\Delta_T$ crosses the maturity-invariant difference in exemption benefits at most once, and $\Delta_1 = 0 < (\xi_I - \xi_H) r$. Proposition A.15(i) applies this to the three motives below. With bond-specific risk, both investors' municipal equalities give $\Delta_T = (\xi_I - \xi_H) r$ at every jointly held maturity, so $\Delta_T$ is not strictly increasing there. For that case, Theorem A.2 uses the index $\Psi_T$, which adds to the duration benefit the household's bond-specific compensation if it held the whole supply, $\gamma_H v^M_T Q^M_T / m_H$.

**Proposition A.14** (The gap when the lower-tax investor participates)**.** *In the normalized municipal–taxable economy of Appendix A.1.1, with one municipal market, suppose every additional characteristic is a neutral common return or wealth characteristic. If the lower-tax investor holds the maturity-$T$ municipal bond at state $z$, then*

$$g_T \;\geq\; (\xi_I - \xi_H)\, r + g^{\text{risk}}_T + (\xi_I - \xi_H)\Big(\sum_w k^C_{Iw,T} + \frac{\kappa_I}{\xi_I}\Big),$$

$$g^{\text{risk}}_T := \gamma_I \Big[\operatorname{Cov}_z\big(u^M_{T,t+1}, \widetilde{\Pi}_I(t+1)\big) - \xi_H \operatorname{Cov}_z\big(u^C_{T,t+1}, \widetilde{\Pi}_I(t+1)\big)\Big],$$

*where the covariances run over the rate and bond-specific components of the innovations and $w$ runs over the wealth characteristics, with equality if this investor also holds the maturity-$T$ taxable bond. In the affine model without bond-specific risk, $g^{\text{risk}}_T = \gamma_I \sigma_r^2 \widehat{D}_I (A^M_{T-1} - \xi_H A^C_{T-1})$ and the shorting constraint on cash is slack. With nonnegative costs, this gives Proposition A.13(ii).*

**Proof.** The lower-tax investor's municipal condition binds, $\mu^M_T = \xi_I r + K^M_{I,T}$, and its taxable portfolio condition gives $\mu^C_T \leq r + K^C_{I,T}$, with equality when taxable holdings are positive. Subtracting $\xi_H$ times the inequality from the equality gives

$$g_T \;\geq\; (\xi_I - \xi_H) r + \sum_j \big(k^M_{Ij,T} - \xi_H k^C_{Ij,T}\big).$$

Rate and bond-specific components give $g^{\text{risk}}_T$. Neutral common-return components cancel, since $k^M_{iq,T} = \xi_H k^C_{iq,T}$. Neutral wealth costs give $(\xi_I - \xi_H) k^C_{Iw,T}$ by (100). The terms $\kappa_I$ and $\kappa_I/\xi_I$ from the shorting constraint on cash give $(\xi_I - \xi_H)\kappa_I/\xi_I$. In the affine model without bond-specific risk, the rate components $\gamma_I \sigma_r^2 A^a_{T-1} \widehat{D}_I$ give the stated $g^{\text{risk}}_T$. □

The third term arises because wealth costs and a binding shorting constraint on cash raise the lower-tax investor's required municipal return by $\xi_I$ times their effect on its required taxable return, while the household benchmark multiplies taxable returns by $\xi_H$. Nonnegative $g^{\text{risk}}_T$ and nonnegative costs suffice for $g_T \geq (\xi_I - \xi_H) r$.

### A.3.3 Three motives that produce the long-end gap

**Proposition A.15** (Three motives that produce the long-end gap)**.** *Let $r > 0$ and consider one of the three cases below. Cases (b) and (c) are set in the two-investor economy with a constant cash rate, no bond-specific*

*risk and a slack shorting constraint on cash, so the stated characteristic is the only non-tax component. In each case the characteristic is zero at the one-period claim. In (b), investors have no outside exposure to the default shocks.*

*(a) Liability hedging and risk-bearing capacity: the two-investor affine model at an equilibrium with strictly increasing municipal next-period loadings $A^M_{T-1}$. Bond-specific risk is allowed only in parts (ii) and (iii).*

*(b) Default diversification: municipal default risk loads on one unit-variance shock and taxable default risk on another, with correlation $\varrho \in [0,1]$ and exposures $\xi_H e_T$ and $e_T$, respectively, where $e_T$ is strictly increasing for $T \geq 1$. Write $X^M_i := \xi_H \sum_T e_T x^M_{i,T}$ and $X^C_i := \xi_i \sum_T e_T x^C_{i,T}$ for investor $i$'s municipal and taxable default exposures, and $k^M_{id,T} := \gamma_i \xi_H e_T (X^M_i + \varrho X^C_i)$ and $k^C_{id,T} := \gamma_i e_T (X^C_i + \varrho X^M_i)$ for its default components.*

*(c) Trading costs: the same round-trip cost $w_T \geq 0$ in both markets, strictly increasing for $T \geq 1$, and trading intensities $\varsigma_I < \varsigma_H$.*

*Then:*

*(i) at every equilibrium allocation, the participation advantage $\Delta_T$ in (102) is zero at $T = 1$. If the lower-tax investor holds a municipal bond, $\Delta_T$ increases strictly with maturity, and this investor holds exactly the municipal maturities at or above some cutoff longer than one period;*

*(ii) $g_1 = 0$ because the household anchors the one-period claim;*

*(iii) at every maturity at which the lower-tax investor holds the municipal bond, $g_T \geq (\xi_I - \xi_H) r + g^{\text{risk}}_T$ in (a), with $g^{\text{risk}}_T$ of Proposition A.14, $g_T \geq (\xi_I - \xi_H) r$ in (c), and in (b)*

$$g_T = (\xi_I - \xi_H) r + k^M_{Id,T} - \xi_H k^C_{d,T},$$

*where $k^C_{d,T}$ is the default component of any investor who holds the taxable bond, so the gap net of these default components is $(\xi_I - \xi_H) r$;*

*(iv) with the household alone holding every municipal and taxable bond, $g_T = 0$ in (a) and (c), and $g_T = k^M_{Hd,T} - \xi_H k^C_{Hd,T}$ in (b).*

**Proof.** (i) In each case, $\Delta_T$ is a loading that is zero at $T = 1$, where the characteristic is zero, and strictly increasing in $T$, times a unit-cost difference common to all maturities. At an equilibrium allocation, (a) gives $\Delta_T = \sigma_r^2 A^M_{T-1} \widehat{\Lambda}$. In (b), define investor $i$'s unit municipal-default cost $c_{id} := \gamma_i (X^M_i + \varrho X^C_i)$, constant across maturities, so that $\Delta_T = \xi_H e_T (c_{Hd} - c_{Id})$. Case (c) gives $\Delta_T = (\varsigma_H - \varsigma_I) w_T$. If the lower-tax investor holds some maturity $T$, its equality and the household's inequality give $\Delta_T \geq (\xi_I - \xi_H) r > 0$, so $T \geq 2$, where the loading, $A^M_{T-1}$, $e_T$ or $w_T$, is positive, and the unit-cost difference, $\widehat{\Lambda}$, $c_{Hd} - c_{Id}$ or $\varsigma_H - \varsigma_I$, must then be positive. The advantage $\Delta_T$ therefore increases strictly with maturity and exceeds $(\xi_I - \xi_H) r$ at every longer maturity. There the household's condition cannot bind, so the lower-tax investor holds every longer maturity, and $\Delta_1 = 0$ excludes the one-period claim.

(ii) Every non-tax component vanishes at the one-period claim, so the municipal conditions reduce to $\mu^M_1 \leq \xi_i r$. Positive supply gives $\mu^M_1 = \xi_H r$, while $\mu^C_1 = r$, as in Proposition A.2.

(iii) Apply Proposition A.14 to (a), with no additional characteristic, and (c), with $g^{\text{risk}}_T = 0$ and third term $(\xi_I - \xi_H) \varsigma_I w_T / \xi_I \geq 0$. In (b), the holders' binding conditions give $\mu^M_T = \xi_I r + k^M_{Id,T}$ and $\mu^C_T = r + k^C_{d,T}$. Every taxable holder's equality has the cash term $r$ and no other non-tax component, so every taxable holder has the same default component. Subtraction proves the expression.

(iv) In (a) and (c) this is (101). In (b), the household's conditions give $\mu^M_T = \xi_H r + k^M_{Hd,T}$ and $\mu^C_T = r + k^C_{Hd,T}$. □

**The three cases.** In (a), $\widehat{\Lambda}$ is positive when liability hedges or lower risk aversion reduce the lower-tax investor's unit cost of rate risk. In (b), the lower-tax investor participates at maturities where its saving on default compensation, $\xi_H e_T (c_{Hd} - c_{Id})$, covers the difference in exemption benefits. Participation requires $c_{Id} < c_{Hd}$, that is, less exposure to the municipal default shock per unit of risk tolerance than the household. The lower-tax investor's taxable holdings add to this exposure only through $\varrho$. Raw gaps also contain holder-specific default-premium differences driven by relative risk supplies, which Appendix C.1.6 adjusts with credit default swap (CDS) spreads. In (c), trading costs reduce after-tax wealth (Appendix C.1.5). For matched long bonds at 26 to 30 years, institutional round-trip costs are 0.53% of price in both markets (Table 18), supporting equal costs at the long end. With combined motives, the decomposition of Proposition A.14 still applies, and

its bound $g_T \geq (\xi_I - \xi_H)r$ holds whenever $g_T^{\text{risk}}$ is nonnegative at the resulting equilibrium. Separate default shocks are treated as in (b).

### A.3.4 Two solved examples without rate risk

Consider two investors and three maturities with no bond-specific risk and a constant $r > 0$, which eliminates rate-risk advantages. Both investors have risk aversion $\gamma$, masses $m_H$, $m_I$, after-tax shares $\xi_H < \xi_I$ and slack shorting constraints on cash. Supplies are $Q_T^M$ for $T = 1, 2, 3$ and $Q_T^C$ for $T = 2, 3$. Write $M1$, $M2$, $M3$ for the municipal bonds and $C2$, $C3$ for the longer taxable bonds. Each example adds only its stated non-tax characteristic.

**Proposition A.16** (Participation for diversification or lower trading costs)**.** *Consider the economy just described.*

*(a) Default diversification. Municipal and taxable default risk load on independent unit-variance shocks, with exposures $\xi_H e_T$ and $e_T$ and $0 = e_1 < e_2 < e_3$. Let $X_H^M := \xi_H e_2 Q_2^M / m_H$ and $X_I^M := \xi_H e_3 Q_3^M / m_I$ be the investors' municipal default exposures when the household holds $M1$ and $M2$ and the lower-tax investor holds $M3$, and let*

$$X^C := \frac{e_2 Q_2^C + e_3 Q_3^C}{m_H/\xi_H + m_I/\xi_I}.$$

*There is an equilibrium with this municipal allocation if and only if*

$$\gamma\,\xi_H e_2 (X_H^M - X_I^M) \;\leq\; (\xi_I - \xi_H)r \;\leq\; \gamma\,\xi_H e_3 (X_H^M - X_I^M).$$

*Both investors then hold taxable bonds, each with taxable default exposure $X^C$, and*

$$g_1 = 0, \qquad g_2 = \gamma\,\xi_H e_2 (X_H^M - X^C), \qquad g_3 = (\xi_I - \xi_H)r + \gamma\,\xi_H e_3 (X_I^M - X^C).$$

*Net of the default components of the investors who hold each bond, the gaps are 0, 0 and $(\xi_I - \xi_H)r$.*

*(b) Trading costs. Each investor pays the after-tax cost $\varsigma_i w_T$ per period on either bond of maturity $T$, with $0 = w_1 < w_2 < w_3$ and $0 \leq \varsigma_I < \varsigma_H$. If $(\varsigma_H - \varsigma_I)w_2 < (\xi_I - \xi_H)r \leq (\varsigma_H - \varsigma_I)w_3$, there is an equilibrium in which the household holds $M1$ and $M2$, the lower-tax investor holds $M3$, $C2$ and $C3$, and*

$$g_1 = 0, \qquad g_2 = \Big(\varsigma_H - \frac{\xi_H}{\xi_I}\varsigma_I\Big)w_2, \qquad g_3 = (\xi_I - \xi_H)\Big(r + \frac{\varsigma_I w_3}{\xi_I}\Big),$$

*with $0 < g_2 < g_3$. The ratio $\mu_T^M/\mu_T^C$ equals $\xi_H$ at $T = 1$, lies strictly between $\xi_H$ and $\xi_I$ at $T = 2$ and equals $\xi_I$ at $T = 3$.*

**Proof.** (a) Independent default shocks separate market risks. Investor $i$'s portfolio conditions are $\mu_T^M \leq \xi_i r + \gamma\xi_H e_T X_i^M$ and $\mu_T^C \leq r + \gamma e_T X_i^C$, for municipal and taxable exposures $X_i^M$, $X_i^C$. An investor with lower taxable exposure would require less on every risky taxable bond, so the other could hold none and its exposure could not be the higher one. Clearing $\sum_i m_i X_i^C/\xi_i = e_2 Q_2^C + e_3 Q_3^C$ then gives $X_H^C = X_I^C = X^C$, which is positive, so both investors hold taxable bonds and $\mu_T^C = r + \gamma e_T X^C$. Supply-proportional holdings, one of possibly several splits across the two maturities, implement these exposures and clear both maturities.

The holders' binding municipal conditions give $\mu_1^M = \xi_H r$, $\mu_2^M = \xi_H r + \gamma\xi_H e_2 X_H^M$ and $\mu_3^M = \xi_I r + \gamma\xi_H e_3 X_I^M$. The nonholders' portfolio conditions require $\xi_H r < \xi_I r$ at $M1$, the left threshold inequality at $M2$ and the right at $M3$.

Subtracting returns gives the gaps. Remove the holder-specific municipal component $\gamma\xi_H e_T X_i^M$ and $\xi_H$ times the taxable component $\gamma e_T X^C$ to leave $\xi_H r - \xi_H r$ at $T = 2$ and $\xi_I r - \xi_H r$ at $T = 3$.

(b) Investor $i$ requires $\xi_i r + \varsigma_i w_T$ on municipal bonds and $r + \varsigma_i w_T/\xi_i$ on taxable bonds. With slack shorting constraints on cash, each bond's return equals the lowest required return. Since $\varsigma_I/\xi_I < \varsigma_H/\xi_H$, the lower-tax investor holds all taxable bonds. It holds the maturity-$T$ municipal bond when $\xi_I r + \varsigma_I w_T \leq \xi_H r + \varsigma_H w_T$, here only at $T = 3$. Subtracting returns gives the gaps, and $(\varsigma_H - \varsigma_I)w_2 < (\xi_I - \xi_H)r$ and $w_2 < w_3$ give

$$g_2 = (\varsigma_H - \varsigma_I)w_2 + (\xi_I - \xi_H)\frac{\varsigma_I w_2}{\xi_I} < (\xi_I - \xi_H)\Big(r + \frac{\varsigma_I w_2}{\xi_I}\Big) \leq g_3.$$

The first term of $g_2$ is positive, so $g_2 > 0$ and the $T = 2$ ratio exceeds $\xi_H$. The ratio is below $\xi_I$ because the household's required municipal return is below the lower-tax investor's, which equals $\xi_I \mu_2^C$. It is $\xi_H r/r = \xi_H$ at $T = 1$, and at $T = 3$ it is $(\xi_I r + \varsigma_I w_3)/(r + \varsigma_I w_3/\xi_I) = \xi_I$. □

Case (a) of Proposition A.16 is a solved example of case (b) of Proposition A.15 with $\varrho = 0$ and requires $X_H^M > X_I^M$. The intermediate raw gap $\gamma \xi_H e_2 (X_H^M - X^C)$ is a default-premium difference. Case (b)'s intermediate gap reflects higher household trading costs, independent of other holdings.

### A.3.5 A municipal-specific benefit

A municipal-specific benefit $\zeta_T > 0$, from a mandate, issuer relationship or favorable regulation at matched maturity and credit, lowers the required municipal return without changing the taxable alternative and so violates neutrality. For the lower-tax investor, it gives the participation advantage $\Delta_T = \zeta_T$ but reduces the return gap by $\zeta_T$. Unlike default compensation, it has no taxable counterpart to subtract. A preference for municipal bonds as such is a benefit of this kind, so by itself it does not explain yields above the household benchmark.

**Proposition A.17** (Same participation, opposite pricing)**.** *Consider the two-investor economy without rate risk or bond-specific risk, with a constant cash rate $r > 0$, a slack shorting constraint on cash and no additional household characteristic. A fixed per-dollar benefit lowers the lower-tax investor's required municipal return by $\zeta_T \geq 0$, with $\zeta_1 = 0$, $\zeta_T$ nondecreasing and $\zeta_T < \xi_H r$. Compare two benefits. A neutral benefit is paid and taxed with returns, as $\zeta_T$ on the municipal bond and $\zeta_T/\xi_H$ before tax on the taxable bond. It therefore also lowers the investor's required taxable return by $\zeta_T/\xi_H$. A municipal-specific benefit leaves the required taxable return unchanged.*

*(i) In both cases the lower-tax investor holds the maturity-$T$ municipal bond only if $\zeta_T \geq (\xi_I - \xi_H) r$, so participation begins at the same maturity.*

*(ii) With the neutral benefit, $\mu_T^C = r - \zeta_T/\xi_H$ and*

$$g_T = \min\{\zeta_T,\ (\xi_I - \xi_H) r\} \geq 0,$$

*so the gap equals the full difference in exemption benefits wherever the lower-tax investor participates. The ratio $\mu_T^M/\mu_T^C$ rises with $\zeta_T$.*

*(iii) With the municipal-specific benefit, $\mu_T^C = r$ and*

$$g_T = \min\{0,\ (\xi_I - \xi_H) r - \zeta_T\} \leq 0,$$

*and the ratio $\mu_T^M/\mu_T^C = \min\{\xi_H,\ \xi_I - \zeta_T/r\}$ is weakly below $\xi_H$ where the lower-tax investor participates and strictly below it wherever $\zeta_T > (\xi_I - \xi_H) r$.*

*Because the rate is constant, each yield averages the one-period returns $\mu_1^a, \ldots, \mu_T^a$. The yield gap $y_T^M - \xi_H y_T^C$ therefore has the sign of the $g_j$, and the yield-implied tax is nonincreasing in maturity in part (ii) and nondecreasing in part (iii).*

**Proof.** As in Proposition A.16(b), minimum required returns determine holdings. The household requires $\xi_H r$ and $r$. The lower-tax investor requires $\xi_I r - \zeta_T$ and either $r - \zeta_T/\xi_H$ or $r$. This proves (i).

With neutrality, $\mu_T^C = r - \zeta_T/\xi_H$, so $g_T = \mu_T^M - \xi_H r + \zeta_T$ equals $\zeta_T$ when $\xi_H r$ is minimal, otherwise $(\xi_I - \xi_H) r$. The ratios $\xi_H r/(r - \zeta_T/\xi_H)$ below the threshold and $(\xi_I r - \zeta_T)/(r - \zeta_T/\xi_H)$ above agree at the threshold and increase in $\zeta_T$, the second because its derivative has the sign of $(\xi_I r - \zeta_T)/\xi_H - (r - \zeta_T/\xi_H) = (\xi_I - \xi_H) r/\xi_H$, positive since $\xi_I > \xi_H$. A municipal-specific benefit gives $g_T = \mu_T^M - \xi_H r$, with $\mu_T^M = \min\{\xi_H r,\ \xi_I r - \zeta_T\}$.

Since $\zeta_T$ is nondecreasing, $\mu_j^M/\mu_j^C$ is nondecreasing in $j$ in (ii) and nonincreasing in (iii). With a constant rate, $y_T^M/y_T^C = \sum_{j \leq T} \mu_j^M / \sum_{j \leq T} \mu_j^C$ averages these ratios with weights proportional to $\mu_j^C > 0$, so adding a maturity whose ratio is at least (in (iii), at most) every earlier ratio cannot lower (raise) the yield ratio, which proves the yield statements. □

## A.4 State-specific exemptions across municipal markets

The quantitative model of Section 5 has separate state municipal markets. A household resident in state $s'$ pays its home state's income tax on interest from a bond issued by another state $s \neq s'$, while home-state municipal interest is exempt. Appendix D.2 distinguishes this common model convention from states' statutory exceptions. Non-household investor types have no modeled state-income-tax wedge on either bond, so state taxation is a household-specific channel. Statutory top state rates vary across states, with a supply-weighted mean of 5.8% over 2000–2017. This cross-state variation is both a calibration input and P1's source of identification.

**What changes.** The multi-state model replaces $\xi_i$ with $\xi_{i,s} = \xi_i^C/\xi_{i,s}^M$, which depends on the issuing state $s$ and on the investor, whose type includes a household's state of residence. Here $\xi_{i,s}^M$ is investor $i$'s after-tax share of state $s$'s municipal interest. Each investor uses one payoff normalization across all markets, with after-tax shares $\xi_i^a$, risk aversion $\gamma_i$ and liability hedge $\ell_i$ as in (33). Its net rate exposure in these units is $\widehat{D}_i = \sum_{a,T} \xi_i^a x_{i,T}^a A_{T-1}^a - \ell_i$, which reduces to (50) when the municipal after-tax shares all equal one.

**What carries over.** For state $s$'s market $M_s$, put $\Gamma_T^{M_s} := \sum_{i \in \mathcal{U}_T^{M_s}} m_i/\gamma_i$ and the risk-tolerance shares $\omega_{i,T}^s := (m_i/\gamma_i)/\Gamma_T^{M_s}$ (Proposition 2), where $\mathcal{U}_T^{M_s}$ is the set of holders of the maturity-$T$ bond. With slack shorting constraints on cash, the exact aggregation of the holders' portfolio conditions (65) applies market by market, with $\xi_{ij} = \xi_{i,s}$. Applying (67) coordinate by coordinate gives, in the affine model, the maturity-$T$ return response in market $M_s$,

$$\begin{aligned} s_T^{M_s} = &\sum_{i \in \mathcal{U}_T^{M_s}} \omega_{i,T}^s \xi_{i,s} + \frac{\sigma_r^2 A_{T-1}^{M_s}}{\Gamma_T^{M_s}} \sum_{i \in \mathcal{U}_T^{M_s}} m_i \partial_z \widehat{D}_i \\ &+ \frac{v_T^{M_s}}{\Gamma_T^{M_s}} \sum_{i \in \mathcal{U}_T^{M_s}} m_i \xi_{i,s}^M \partial_z x_{i,T}^{M_s}. \end{aligned} \tag{103}$$

The first two terms contain the holders' tax mix and exposure responses, the changes in net rate exposure as rates rise (Section 4.5). The final term vanishes under a common municipal after-tax share among holders, a single holder or zero bond-specific variance, as for the taxable counterpart before (68). Market clearing alone does not remove it, because clearing does not weight holdings by $\xi_{i,s}^M$.

**The state-tax anchor and pass-through.** Proposition A.18 is the state-tax counterpart of Proposition A.9. Its implied state tax is the model counterpart of (25), and the threshold maturity $T_s^*$ is the first at which investors other than the resident household hold state $s$'s bonds.

**Proposition A.18** (State-tax anchor and tax-mix pass-through)**.** *If the household $H_s$ resident in state $s$ uniquely minimizes $\xi_{i,s}$, its shorting constraint on cash is slack and $r > 0$, then the riskless one-period yields satisfy $y_1^{M_s}/y_1^C = \xi_{H_s,s}$. Under the multiplicative mapping $\xi_{H_s,s} = \xi^{\text{fed}}(1-\tau_s^{\text{st}})$, with federal after-tax share $\xi^{\text{fed}}$ and state income-tax rate $\tau_s^{\text{st}}$, define*

$$\tau_{s,T}^{\text{st,impl}} = 1 - \frac{y_T^{M_s}/y_T^C}{\xi^{\text{fed}}}, \qquad \tau_{s,T}^{A,\text{st}} = 1 - \frac{A_T^{M_s}/A_T^C}{\xi^{\text{fed}}}. \tag{104}$$

*Then $\tau_{s,1}^{\text{st,impl}} = \tau_s^{\text{st}}$, with unit pass-through if these conditions hold at nearby state tax rates. Switch off the exposure-response and bond-specific terms in the affine recursions, and suppose only the resident household's effective after-tax share changes, with its risk-tolerance share one below $T_s^*$ and constant at $\omega_{H_s}^s$ thereafter. Then the tax-mix derivative of the loading-implied state tax $\tau_{s,T}^{A,\text{st}}$ with respect to $\tau_s^{\text{st}}$ is $\Theta_S^s(T) + \omega_{H_s}^s \Theta_L^s(T)$, where $\Theta_S^s, \Theta_L^s$ are the weights in (85) with $T_s^*$ in place of $T^*$.*

**Proof.** With this household's constraint slack, (39) gives the riskless-claim return $\min_i \xi_{i,s} r$, which is $\xi_{H_s,s} r$, as in Proposition A.2. With $y_1^C = r$, equation (104) gives the anchor $\tau_{s,1}^{\text{st,impl}} = 1 - \xi_{H_s,s}/\xi^{\text{fed}} = \tau_s^{\text{st}}$. Because $\partial \xi_{H_s,s}/\partial \tau_s^{\text{st}} = -\xi^{\text{fed}}$ cancels the factor $1/\xi^{\text{fed}}$ in (104), $\partial \tau_{s,T}^{A,\text{st}}/\partial \tau_s^{\text{st}} = \partial (A_T^{M_s}/A_T^C)/\partial \xi_{H_s,s}$, the household's tax-mix pass-through of Proposition A.9. Applying (85) with $T_s^*$ in place of $T^*$ gives the tax-mix derivative. □

Changes in out-of-state households' or other clienteles' after-tax shares add their own derivatives. Other statutory mappings require a corresponding implied-tax transformation, and the one-period yield ratio remains $y_1^{M_s}/y_1^C = \xi_{H_s,s}$. Reported state-tax responses use the full multi-state model, which supplies P1's full-yield pass-through profile.

## A.5 Participation and tax pass-through with multiple investors

To allow for more clienteles, as in the six investor groups of the quantitative model, extend the two-investor affine model of Appendix A.2 to $N$ investor types with slack shorting constraints on cash. Order after-tax shares $\xi_1 < \cdots < \xi_N$, so type 1 is the household with the largest exemption benefit. Without bond-specific

risk, compare type $j > 1$'s municipal condition with type 1's at an equilibrium allocation, as in Appendix A.2.3. Type $j$'s required return on the maturity-$T$ municipal bond is no higher than type 1's exactly when

$$\underbrace{\sigma_r^2 A_{T-1}^M \widehat{\Lambda}_j}_{\text{duration benefit}} \geq \underbrace{(\xi_j - \xi_1) r}_{\text{difference in exemption benefits}}, \qquad \widehat{\Lambda}_j := \gamma_1 \widehat{D}_1 - \gamma_j \widehat{D}_j. \tag{105}$$

As in (59), net rate exposures are common to all maturities, so the duration benefit grows with the bond's rate loading, while the difference in exemption benefits is flat in maturity. Without bond-specific risk, type $j$ requires $\xi_j r + \sigma_r^2 A_{T-1}^M \gamma_j \widehat{D}_j$, which is affine in the next-period loading, and only the types with the lowest requirement hold each bond. When next-period loadings increase strictly with maturity, each participating type holds an interval of maturities, types with lower $\gamma_j \widehat{D}_j$ hold longer maturities, and two types share at most one maturity, as in Theorem A.1. Positive bond-specific risk adds the existing holder's bond-specific compensation to the comparison, as in the portfolio conditions (47) and the participation index (89), and lets several clienteles hold the same maturities.

**Assumption A.1** (Long-end participation, clientele by clientele)**.** *Bond-specific risk is positive, and type 1 holds every maturity, so its first held maturity is $T_1^* = 1$. For each type $j > 1$, $\widehat{\Lambda}_j > 0$, and the municipal maturities it holds are exactly $[T_j^*, T_{\max}]$ with $T_j^* \geq 2$. The thresholds need not be ordered by tax advantage.*

**Many-clientele tax-mix pass-through.** To isolate the tax mix under the nested holding intervals of Assumption A.1, switch off exposure-response and bond-specific terms and fix each maturity's holders and risk-tolerance shares, as in Proposition A.9. Let $\omega_{n,j}$ be type $n$'s share of the holders' risk tolerance at municipal maturity $j$, zero when it does not hold that bond. Then $A_T^C = G_T$ and $A_T^M = \sum_{j \leq T} \rho^{T-j} \sum_n \omega_{n,j} \xi_n$, where $G_T = \sum_{j=1}^T \rho^{T-j}$. Differentiating the loading-implied tax $\tau_T^A = 1 - A_T^M / G_T$ in $\xi_i$ gives its tax-mix pass-through $-\partial \tau_T^A / \partial \xi_i$,

$$\mathcal{D}_i(T) = \frac{\sum_{j=T_i^*}^T \rho^{T-j} \omega_{i,j}}{\sum_{j=1}^T \rho^{T-j}}, \tag{106}$$

with an empty numerator equal to zero. It is positive after participation begins and, with the recursions extended to infinite maturity, converges to $\omega_i$ if $\omega_{i,j}$ is eventually constant at $\omega_i$, because the finitely many maturities before then receive vanishing weight at the long end.

Participation in order of increasing after-tax shares makes the tax mix nondecreasing in maturity, because adding an after-tax share no smaller than the existing holders' does not lower their risk-tolerance-weighted mean. Such ordered participation does not by itself order total return responses, loading ratios $A_T^M / A_T^C$ or yields, since other terms in (67) and the taxable bond's response remain unrestricted. Clientele-specific pass-through and long-end-limit comparisons need no tax ordering.

Under Assumption A.1, the household anchors the riskless short end, and (106) gives each clientele's tax-mix derivative. Full yield responses also include relative compensation and convexity (Appendix A.2.5). The portfolio response can transmit a later clientele's tax to shorter risky bonds it never holds. P1 concerns the household, which holds every maturity, and P2 banks, a later-participating clientele whose tax-mix pass-through is zero below its threshold. Assumption A.1 restricts only this affine extension. The calibration does not impose it (Section 5).

## A.6 Applications to other bond markets

The mechanism operates beyond municipal bonds. It requires heterogeneous relative taxes, a shorting friction and a neutral participation motive that grows with maturity. Common rate risk supplies such a motive, and participation then also changes rate loadings. For a tax-favored bond and a comparable fully taxed bond, $\xi_i = (1 - \tau_i^{\text{tax}})/(1 - \tau_i^{\text{fav}})$, with investor $i$'s tax rates on the taxed and favored bonds. Symmetric taxation gives $\xi_i = 1$ and removes the relative tax incentive. When after-tax shares differ, the exact aggregation (65) applies. Positive taxable forwards and an increasing relative forward ratio imply a declining yield-implied tax (Proposition A.7). Condition (S) in (72) orders the affine loading ratio (Proposition A.6).

Applications include:

- state-exempt treasury versus corporate interest
- state-exempt Federal Home Loan Bank and Farm Credit issues versus Fannie Mae and Freddie Mac issues
- treaty-dependent withholding on cross-border sovereign debt

- the inflation accrual on treasury inflation-protected securities (TIPS), taxed before payment in taxable accounts
- taxable accounts, individual retirement accounts (IRAs) and pensions, which assign different after-tax payoffs to the same security at one market price

In each application, the maturity profile of the yield-implied tax depends on which clienteles hold the bonds. Municipal bonds, the central application, combine a large federal wedge with tight shorting restrictions.

# B Quantitative model, solution and calibration

This appendix explains the quantitative model, how I solve it and how I estimate the coefficients that are not directly measured. Measured taxes, portfolio assets, funding, bond supplies, household balance sheets and rate dynamics are fixed before estimation. Seventeen common coefficients are fitted to twenty-two ownership, insurer-portfolio and term-spread targets, with municipal yields left untargeted. One model period is two years, and calendar maturities are stated in years.

## B.1 Markets, investors and measured inputs

**Markets and maturities.** The model has 45 municipal markets and one taxable market. Each of the 43 states with a personal income tax and the District of Columbia has its own municipal market. The seven states without one (Alaska, Florida, Nevada, South Dakota, Texas, Washington and Wyoming) share a market because investors treat their bonds alike in the model. Each municipal market has fifteen maturities in two-year steps from two to thirty years. The taxable market has fourteen risky maturities, four through thirty years, and cash is its one-period claim. Each rate state therefore has $45 \times 15 + 14 = 689$ bond positions to clear. Cash and two-year municipal bonds are riskless. The taxable market pools treasury and corporate supply, is taxed like corporate interest and has its yields compared with the corporate curve.

**Time periods and the one-market estimation model.** Measured inputs vary across six three-year periods, 2000–2002 through 2015–2017, centered on the 2001, 2004, 2007, 2010, 2013 and 2016 Survey of Consumer Finances (SCF) waves. For tractability, estimation uses a smaller one-market model with one national municipal market, in which household tax treatment is averaged across states with municipal supply weights. This model has twenty-seven investor types and twenty-nine bond positions. After estimation, I restore all 45 municipal markets and solve again with the coefficients fixed.

**Investor groups, portfolio assets and funding.** The six investor groups are high-tax households, lower-tax households, property-and-casualty (P&C) insurers, life insurers, banks and a tax-neutral residual group. High-tax households face a federal marginal rate of at least 35% under the law of the SCF year. Table 3 reports the six groups' portfolio assets and gross exemption benefits in 2015–2017. For type $i$, baseline portfolio assets $m_i$ are its debt securities and cash in the Financial Accounts. A measured fraction $W_i$ is available for the modeled investments, giving funding $m_i W_i$. These investments are municipal bonds, corporate bonds excluding asset-backed securities, treasury notes and bonds, and cash excluding bank cash. Positions $x_i$ are measured per dollar of portfolio assets, so aggregate bond holdings are $m_i x_i$ and cash is $m_i x_i^0$, where $x_i^0 = W_i - \mathbf{1}' x_i$. The amount $m_i$ scales the investor's risk-bearing capacity. Neither $m_i$ nor $W_i$ is chosen to fit yields.

**Outside rate exposure.** Each type has an outside rate exposure $h_i = -\ell_i$, the negative of the liability hedge of Section 4, measured per dollar of portfolio assets on the same scale as a bond's rate loading. It summarizes how the rest of the balance sheet changes in value when rates move. Positive $h_i$ adds to the losses on the bond portfolio when rates rise. Negative $h_i$ offsets those losses, as when rising rates reduce the value of liabilities.

**Investor types.** Types within each group differ in outside rate exposure, so investors with the same tax treatment can hold different portfolios. Their asset shares divide each group's measured portfolio assets, so the types add no assets, and all types in a sector share the measured funding fraction $W_i$. The definitions below give $45 \times 2 \times 7 + 2 + 3 + 1 + 7 = 643$ types in the full 45-market model.

**Households.** High- and lower-tax households each have seven balance-sheet types in every municipal market. Each SCF wave supplies tax-unit income, filing status, outside financial assets and debt. Within each household group, I sort households by outside assets less debt relative to their bond and deposit holdings, which are

deposits, direct bonds and bond funds. Seven bins, one per type, each hold about one seventh of the group's portfolio assets. Let $Out_k$ and $Debt_k$ be bin $k$'s outside-asset and debt ratios. Its outside rate exposure is

$$h_k = a_H + \frac{b_A Out_k - b_D Debt_k}{2}.$$

The intercept $a_H$ gives common exposure, $b_A$ measures the risk added by outside assets and $b_D$ the hedge from debt. Halving converts $b_A$ and $b_D$ to the model's two-year units, and the intercept $a_H$ is stated in those units directly. These three coefficients are estimated jointly and apply to both household groups. The balance-sheet ratios and bin weights are measured, while each period's federal and state tax rules determine each group's after-tax shares. SCF direct and fund holdings divide household municipal ownership between the two household groups. The national household composition is repeated across states, with portfolio assets allocated by municipal supply shares.

**Property-and-casualty insurers.** Two types share the sector's tax treatment but differ in outside rate exposure. Their exposures are $a_P$ and $a_P + d_P$, and their shares of the sector's portfolio assets are $w_P$ and $1 - w_P$. Thus $a_P$ sets the lower exposure, $d_P$ the difference between types and $w_P$ their relative size. These three coefficients are estimated. Schedule D filings with the National Association of Insurance Commissioners (NAIC) supply the municipal and taxable portfolio composition and maturity moments used in estimation. I retain positive-value positions, assign each position its tax status and remaining maturity, and construct cross-sectional moments separately each year before averaging equally within each period.

**Life insurers.** Three types have outside exposures $a_L$, $a_L + t_L d_L$ and $a_L + d_L$. Their portfolio-asset shares are $w_L$, $(1-w_L)u_L$ and $(1-w_L)(1-u_L)$. The coefficients $a_L$ and $d_L$ set the exposure range, $t_L$ locates the middle type within it, and $w_L$ and $u_L$ divide the sector's assets across types. All five are estimated. NAIC Schedule D supplies portfolio moments constructed in the same way as for P&C insurers, and statutory life-insurer proration fixes tax treatment.

**Banks.** One type represents the sector, with a single estimated outside exposure $h_B$. In addition to statutory corporate taxes, banks incur the tax cost of disallowed interest deductions on municipal holdings. Call Reports supply interest expense divided by assets, which I average across banks using municipal-holdings weights. Multiplying this ratio by the statutory corporate tax rate gives the financing-tax cost when banks' municipal holdings are treated as non-bank-qualified.

**Tax-neutral residual investors.** The residual group consists mainly of foreign investors, pensions, governments, money market funds and nonprofits. Seven types approximate a normal distribution of outside exposures, $h_k = \mu_R + s_R \zeta_k$, where $\zeta_k$ are fixed standardized normal quadrature nodes. The mean $\mu_R$ and dispersion $s_R$ are estimated. The fixed quadrature weights divide the sector's portfolio assets across types, and all types have the same tax-neutral treatment.

**Household balance sheets.** In 2015–2017, high-tax households are about 5% of families and hold 46% of household portfolio assets. Table 7 reports the outside-asset and debt ratios used in the exposure formula, each formed at the tax-unit level before bin averaging.

**Household tax treatment.** Statutory rules determine after-tax shares. Household calculations use each SCF tax unit's income and filing status and include the net investment income tax and the additional tax on Social Security benefits induced by interest income.[24] Federal and state after-tax products are formed at the tax-unit level before aggregation. High-tax households face the home state's top rate, and lower-tax households face tax-unit bracket rates. The model applies a common home-state exemption in all income-tax jurisdictions and taxes out-of-state municipal interest at the resident state's rate. The baseline multiplies federal and state shares for taxable interest and includes no federal deduction of state tax on out-of-state municipal interest. Appendix D.2.1 varies this convention. The Social Security component, the extra tax on benefits caused by exempt interest, is approximated by the tax response to a small ordinary-income increment less the tax on that increment. It can make the after-tax share on home-state municipal interest slightly below one.

**Institutional tax treatment.** Institutions pay no state income tax in the model and choose across all markets. Independent bond-specific shocks, with the same variance in every state market at a given maturity, let them diversify geographically. Before 2018, institutions face a 35% corporate rate. P&C proration taxes

[24] For 209 of 792 state-years, missing lower-tax filing-status cutoffs or related bracket rules are imputed from nearest-year thresholds or simple proxies. Published rates and top rates are unchanged.

Table 7: Measured household outside assets and debt by balance-sheet type

| Bin | 2000–02 | 2003–05 | 2006–08 | 2009–11 | 2012–14 | 2015–17 |
|---|---|---|---|---|---|---|
| *High-tax households* | | | | | | |
| 1 | 0.93, 3.17 | 1.17, 4.21 | 0.83, 3.00 | 0.77, 2.85 | 0.63, 3.23 | 0.63, 2.93 |
| 2 | 0.39, 0.23 | 0.10, 0.07 | 0.19, 0.10 | 0.22, 0.18 | 0.26, 0.15 | 0.12, 0.08 |
| 3 | 0.62, 0.12 | 0.43, 0.12 | 0.45, 0.07 | 0.25, 0.06 | 0.57, 0.11 | 0.35, 0.12 |
| 4 | 1.18, 0.24 | 0.84, 0.12 | 0.91, 0.14 | 0.60, 0.11 | 0.87, 0.15 | 0.70, 0.05 |
| 5 | 1.98, 0.26 | 1.53, 0.26 | 1.35, 0.11 | 1.15, 0.19 | 1.46, 0.25 | 1.32, 0.16 |
| 6 | 3.77, 0.29 | 2.55, 0.38 | 2.19, 0.28 | 1.80, 0.16 | 2.50, 0.30 | 2.77, 0.23 |
| 7 | 18.49, 1.55 | 12.06, 1.28 | 10.56, 0.90 | 7.76, 0.81 | 11.08, 0.73 | 12.12, 0.94 |
| *Lower-tax households* | | | | | | |
| 1 | 1.42, 11.55 | 1.03, 12.34 | 1.32, 16.95 | 0.90, 12.88 | 0.75, 14.31 | 0.72, 13.74 |
| 2 | 0.30, 0.65 | 0.35, 0.69 | 0.39, 1.20 | 0.16, 0.26 | 0.31, 0.87 | 0.23, 0.88 |
| 3 | 0.06, 0.04 | 0.04, 0.02 | 0.08, 0.08 | 0.05, 0.02 | 0.01, 0.01 | 0.01, 0.02 |
| 4 | 0.30, 0.10 | 0.28, 0.12 | 0.19, 0.07 | 0.32, 0.12 | 0.17, 0.07 | 0.38, 0.18 |
| 5 | 0.78, 0.15 | 0.71, 0.16 | 0.63, 0.14 | 0.58, 0.08 | 0.71, 0.13 | 0.71, 0.10 |
| 6 | 1.73, 0.29 | 1.40, 0.19 | 1.50, 0.29 | 1.25, 0.19 | 1.57, 0.17 | 1.53, 0.15 |
| 7 | 12.82, 1.26 | 8.93, 0.91 | 9.85, 0.96 | 6.59, 0.71 | 8.70, 0.76 | 10.19, 0.80 |

*Notes.* Each cell gives two ratios, outside financial assets and then debt, each divided by bond and deposit holdings. These are $Out_k$ and $Debt_k$ in the household exposure formula of Appendix B.1. Outside financial assets exclude retirement accounts and the bond and deposit holdings, which are deposits, direct bonds and bond funds. Within each household group, bins are ordered by outside assets less debt relative to bond and deposit holdings, and each contains about one seventh of the group's portfolio assets. SCF waves 2001, 2004, 2007, 2010, 2013 and 2016 supply the six periods.

15% of municipal interest at that rate, giving relative after-tax share $\xi_P = 0.65/0.9475 = 0.686$. Life-insurer proration taxes one half of municipal interest, giving $\xi_L = 0.65/0.825 = 0.788$. Banks have $\xi_B = 0.65$. With their municipal holdings treated as non-bank-qualified (Appendix D.1), their annual financing-tax cost falls from 1.02% in 2000–2002 to 0.11% in 2015–2017. The residual group has $\xi = 1$.

**Data sources.** Portfolio assets, funding and aggregate ownership are from the Financial Accounts. I assign fund holdings to their owners, separate nonprofits and adjust sector stocks for tax-exempt holdings. Ownership is measured from stocks of holdings without using tax-exempt interest flows. Mergent's dated outstanding amounts of tax-exempt bonds give municipal supply by maturity and issuer state. Financial Accounts treasury and corporate totals give taxable supply, with corporate maturity shares from the Mergent Fixed Income Securities Database (FISD) and treasury maturities from the Monthly Statement of the Public Debt. Table 8 reports the resulting quantities.

**Rate process.** The annual three-month treasury bill rate for 1952–2024 gives persistence 0.854 and an innovation standard deviation of 1.63 percentage points, which become $\rho = 0.729151$ and $\sigma_r = 0.042778$ over two-year periods.[25] The A-or-above corporate zero curve sets the mean rate and the observed monthly states. The mean $\bar{r} = 0.060412$ is twice the 2000–2017 mean annualized two-year corporate zero yield, and the short rate is $r(z) = \max\{\bar{r} + z, 0\}$. Each month's observed state is twice that month's continuously compounded two-year corporate zero yield, less $\bar{r}$. Long yields therefore do not determine the rate states. The same curve supplies the ten-, twenty- and thirty-year yield targets in annual log units.

**The 2018–2020 extension.** To compare model and observed yields after 2017, I keep all estimated coefficients fixed and update measured inputs using the 2019 SCF composition, annually inflation-adjusted incomes and mean 2018–2020 end-year stocks. Institutions face the 21% corporate rate, P&C proration of 25% and a life-insurer policyholders' share of 30%. The annual bank financing-tax cost is 0.117%, and insurer portfolio fractions use available 2018–2019 holdings. This seventh period adds no calibration targets.

[25] Annual persistence is 0.853903 and the innovation standard deviation is 0.016266. The two-year state is twice the annualized rate deviation, so $\rho$ is the square of annual persistence and $\sigma_r$ is $2\sqrt{1+\rho}$ times the annual innovation standard deviation, both evaluated at the unrounded estimates.

Table 8: Inputs fixed from data in each calibration period

| | 2000–02 | 2003–05 | 2006–08 | 2009–11 | 2012–14 | 2015–17 |
|---|---|---|---|---|---|---|
| *Panel A. Group portfolio assets $m_g$ (trillions of current dollars)* | | | | | | |
| High-tax households | 2.02 | 1.97 | 3.47 | 3.82 | 5.72 | 6.97 |
| Lower-tax households | 3.99 | 5.94 | 6.50 | 8.44 | 7.97 | 8.09 |
| P&C insurers | 0.58 | 0.80 | 0.91 | 1.00 | 1.06 | 1.11 |
| Life insurers | 1.56 | 2.11 | 2.10 | 2.52 | 2.84 | 3.05 |
| Banks | 1.50 | 1.89 | 2.38 | 3.28 | 4.21 | 4.83 |
| Tax-neutral residual | 11.54 | 14.94 | 20.04 | 21.42 | 23.29 | 26.47 |
| *Panel B. Funding as a share of portfolio assets, $100W_g$* | | | | | | |
| Households (both tax groups) | 86.9 | 88.0 | 86.2 | 90.6 | 91.1 | 91.4 |
| P&C insurers | 73.9 | 71.3 | 71.2 | 73.5 | 73.3 | 72.6 |
| Life insurers | 73.2 | 68.9 | 66.4 | 69.7 | 69.8 | 71.5 |
| Banks | 30.1 | 25.6 | 24.4 | 24.4 | 25.2 | 28.0 |
| Tax-neutral residual | 50.9 | 52.0 | 50.1 | 57.9 | 68.8 | 72.5 |
| *Panel C. Tax inputs and national bond supplies* | | | | | | |
| High-tax household $\xi_i$ | 0.577 | 0.609 | 0.610 | 0.606 | 0.565 | 0.538 |
| Lower-tax household $\xi_i$ | 0.766 | 0.771 | 0.760 | 0.764 | 0.786 | 0.761 |
| Bank financing-tax cost (%/year) | 1.020 | 0.556 | 0.858 | 0.323 | 0.134 | 0.108 |
| Municipal supply ($ trillion) | 1.08 | 1.92 | 2.36 | 2.78 | 2.99 | 3.14 |
| Taxable supply ($ trillion) | 6.18 | 7.98 | 9.74 | 13.72 | 17.39 | 20.02 |

*Notes.* In Panel A, $m_g$ is group $g$'s total of type-level portfolio assets $m_i$, its measured debt securities and cash. Panel B is the measured funding for modeled bonds and cash divided by those portfolio assets. Dollar funding is $m_g W_g$. For example, banks' 2015–2017 funding is \$4.83 trillion times 0.280, or \$1.35 trillion. The same $W_g$ applies to every type within a sector and to both household groups. Panel C's household shares are the issuer-supply-weighted averages used in the one-market estimation. The full 45-market model retains each state's tax shares. Institutional relative after-tax shares are constant before 2018: 0.686 for P&C, 0.788 for life insurers, 0.650 for banks and 1 for the residual group. Banks' financing-tax cost is additional to this relative share. Stocks are mean end-year values in each period. Taxable supply includes treasury and corporate debt.

## B.2 Solving equilibrium at given coefficients

An equilibrium is a set of municipal and taxable log price functions of the rate state, together with the portfolios they support at every state. Today's portfolios depend on the risk of reselling bonds next period, and that risk depends on tomorrow's prices, so equilibrium is a fixed point in the price functions. I compute it with two nested steps, summarized in Algorithm 1. The outer loop iterates on the price functions until the prices investors use to form expectations are the prices that clear markets. Within each outer iteration, the candidate prices fix the distribution of every bond's next-period price, so each rate state becomes a static problem. At each state, one Newton solve finds every investor's portfolio, including which bonds it holds, and the returns that clear all markets.

**Price functions and expectations.** Log prices $p_j(z) = \log P_j(z)$ are stored at eighteen rate states $\{z_g\}$, which are fifteen evenly spaced states and the period's three December states (Appendix B.7). Position $j$ identifies a market and a maturity, and $p_{j,-}(Z')$ is the same bond's next-period log price, with one period less to maturity. Between grid states, log prices are linear in $z$. The next state is $Z' = \rho z + \sigma_r U$ with $U \sim N(0,1)$, so expectations over $Z'$ are exact integrals of the normal density over linear segments, with no quadrature. From the candidate price functions, I compute at each grid state

$$\mathcal{M}_j(z) = \log \mathbb{E}_z \exp\{p_{j,-}(Z')\} + \tfrac{1}{2} v_j, \tag{107}$$

$$\Sigma_{jk}(z) = \mathrm{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j. \tag{108}$$

Here $v_j$ is the bond-specific return variance. The first expression is the log expected resale price plus the bond-specific payoff term $\frac{1}{2} v_j$, so the mean of the approximated return of Appendix A.1.1 is $\mu_j = \mathcal{M}_j - p_j = \log \mathbb{E}_z[1 + R_j]$. The covariance matrix $\Sigma$ measures risk in unexpected log returns, and the liability credit from the liability hedge $\ell_i = -h_i$ is $c^L_{ij} = -\mathrm{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r U)$, as in (46).

**Inner step: portfolios and clearing at one state.** At grid state $z_g$, the moments $\mathcal{M}$, $\Sigma$ and $c^L$ are held fixed and the cash rate is $r(z_g) = \max\{\bar{r} + z_g, 0\}$. The unknowns are the return vector $\mu$, every investor's

**Algorithm 1** Equilibrium prices and portfolios at given coefficients, in the notation of equations (107)–(112)

**Require:** Measured taxes, portfolio assets, funding, supplies, rate dynamics, state grid and candidate coefficients

Construct investor exposures and bond-specific variances from the coefficients
Initialize portfolios and the log price functions on the grid
**repeat** ▷ outer loop over the price functions
  From the candidate prices, compute $\mathcal{M}$, $\Sigma$ and $c^L$ at every grid state
  **for all** grid states $z_g$, independently **do** ▷ inner step
    Solve (110) and (111) jointly by Newton's method for $\mu(z_g)$, holdings and cash
  **end for**
  Form $\widehat{p} = \mathcal{M} - \mu$ and record $\max_{j,g} |\widehat{p}_j(z_g) - p_j(z_g)|$
  If not converged, update all candidate price functions from $\widehat{p}$ by a damped, accelerated step
**until** the recorded maximum difference is below $10^{-10}$
Recompute the moments and check portfolio conditions, clearing and the shorting constraint on cash
Evaluate yields at monthly states and holdings at December states

bond holdings $x_i$ and its cash $x_i^0$. Portfolio equations use the normalized payoff units of Section 4.2, which divide a household's payoffs by its home-state municipal after-tax share and an institution's payoffs by its municipal after-tax share. These shares are then one, and risk aversion and outside exposures are stated in these units. Let $K_i = \mathrm{diag}(\xi_i^{a(j)})$ collect the after-tax shares for the market–maturity positions $j$, where $a(j)$ is the market of position $j$. The taxable share $\xi_i^C$, which equals the relative after-tax share $\xi_i$ in these units, also applies to cash. For a household, the normalized share on out-of-state municipal interest is one minus its state rate. Fiscal calculations use absolute federal and state tax rates (Appendix D.4). Given returns $\mu$, investor $i$'s portfolio solves

$$\max_{x_i \geq 0,\, \mathbf{1}'x_i \leq W_i} x_i'\big(K_i\mu - \xi_i^C r\mathbf{1} - \tau_i^{\mathrm{fin}}\big) + \gamma_i x_i' K_i c_i^L - \frac{\gamma_i}{2} x_i' K_i \Sigma K_i x_i. \tag{109}$$

The three terms are the return above cash, the liability credit and the cost of portfolio risk. The financing-tax vector $\tau_i^{\mathrm{fin}}$ is zero except on banks' municipal positions, where it equals two years of their measured annual cost. Cash is the uninvested funding $x_i^0 = W_i - \mathbf{1}'x_i$, and $\kappa_i$ is the multiplier on the shorting constraint on cash, $x_i^0 \geq 0$. The portfolio conditions are

$$\begin{aligned} \widetilde{\nu}_i &= \gamma_i K_i[\Sigma K_i x_i - c_i^L] - K_i\mu + \xi_i^C r\mathbf{1} + \tau_i^{\mathrm{fin}} + \kappa_i\mathbf{1}, \\ \widetilde{\nu}_{ij} &= \frac{\epsilon w_j}{x_{ij}}, \qquad \kappa_i = \frac{\epsilon}{x_i^0}, \qquad x_{ij} > 0, \quad x_i^0 > 0, \end{aligned} \tag{110}$$

and returns must clear every market,

$$\sum_i m_i x_i(z_g) = Q, \tag{111}$$

where $Q$ is the fixed vector of bond supplies. The first line of (110) gives $\widetilde{\nu}_{ij}$, the required minus the expected after-tax return on each position.

**How participation is determined.** The second line of (110) replaces exact complementarity with a small log barrier, $\epsilon = 10^{-9}$. Municipal barrier weights $w_j$ equal issuer-state supply shares, and taxable weights equal one, so splitting the national municipal market leaves its total barrier weight unchanged. A position whose required return is well above its expected return, so that $\widetilde{\nu}_{ij}$ is well above zero, is held only in the negligible amount $x_{ij} = \epsilon w_j / \widetilde{\nu}_{ij}$. A position the investor values has $\widetilde{\nu}_{ij}$ close to zero and a holding set by its portfolio condition. Which investors hold which bonds is therefore an output of each state's solve, and it can differ across rate states and change from one outer iteration to the next.

**Outer step: updating the price functions.** Through the resale relation (40), the returns that clear markets imply updated current log prices

$$\widehat{p}_j(z_g) = \mathcal{M}_j(z_g) - \mu_j(z_g). \tag{112}$$

If the candidate prices were an equilibrium, $\widehat{p}$ would equal them. Otherwise I update the candidate price functions from $\widehat{p}$ by a damped, Anderson-accelerated step, recompute the moments and solve every grid state again. All grid states in an iteration use the same candidate prices, and all markets and maturities are updated together because covariances and funding link them. Two boundary conditions close the recursion. The one-period municipal claim pays one, so its continuation log price and return variance are zero. Cash has log price $-r$, which supplies the continuation price of the four-year taxable bond. The iteration stops when the maximum difference between the candidate prices and the prices $\widehat{p}$ implied by clearing is below $10^{-10}$. Because $\widehat{p}$ is computed before damping, a small damped step alone cannot trigger convergence.

## B.3 Coefficients, targets and criterion

The data do not directly measure risk aversion or how much rate risk an investor's outside assets and liabilities create. Seventeen coefficients, collected in $\theta$ and common across periods and issuer states, govern these objects and the small bond-specific risk. I estimate them jointly from the twenty-two targets below.

**Coefficients.** Three coefficients govern portfolio risk: common risk aversion $\gamma_i = \gamma$ in the normalized payoff units, the thirty-year bond-specific variance $v_{30}$ and its maturity exponent $p_v$ (Table 9). For $n = 4, 6, \ldots, 30$ calendar years, the two-year return variance is

$$v_n = v_{30}(n/30)^{p_v}, \qquad v_2 = 0.$$

The same variance applies to municipal and taxable bonds at a given maturity, in both the portfolio problem and the price recursion. Appendix B.5 discusses the bond-specific shock and its mean. The remaining fourteen coefficients set outside exposures and their distribution. Three map measured household balance sheets into exposure, eight describe the insurer types, one sets bank exposure and two set residual-group exposure (Table 9).

**Targets.** Twenty-two targets compare model outcomes with data. Nine concern ownership: municipal stocks held by the two household groups, P&C insurers, life insurers and banks, and taxable stocks held by households and those three institutional sectors. Four are the mean maturities of P&C and life-insurer municipal and taxable portfolios. Six describe insurer portfolio composition. Two, one per insurer sector, measure how unevenly municipal bonds are spread across insurers, and four are the fractions of municipal and taxable holdings that mature beyond twenty years. Three are taxable term spreads over the two-year yield at ten, twenty and thirty years. Municipal yields, the municipal puzzle, the empirical tax-pass-through coefficients and the predictions P1–P3 are not targeted. Section 5.2 explains which targets inform which estimated coefficients, and the panel headings of Table 9 name the most informative targets for each group of coefficients.

**Model-minus-data deviations.** Let $\mathcal{C}_{kj}(\theta)$ be the signed model-minus-data deviation for target $j$ in period $k$, after solving that period's equilibrium, and let $o_{kj} > 0$ be the observed normalizer that scales it in the criterion. For ownership, $\mathcal{C}$ is the model stock less the observed stock, and $o$ is the observed stock. For insurer mean maturity, $\mathcal{C}$ is $B(\overline{T} - \overline{T}^{\text{obs}})$ and $o = B^{\text{obs}}\overline{T}^{\text{obs}}$, where $B$ is the value of the model portfolio, $\overline{T}$ its holdings-weighted maturity and the superscript obs marks observed values. Weighting by the model portfolio prevents a nearly empty portfolio from counting as much as a large one. Ownership and maturity targets include the riskless two-year municipal stock.

The six composition targets use the risky maturities of four through thirty years. For insurer $i$, let $M_i$ and $C_i$ be municipal and taxable holdings at these maturities and $B_i = M_i + C_i$. The municipal-share concentration $\sum_i (M_i^2/B_i)/\sum_i B_i$ averages each insurer's squared municipal portfolio share with portfolio-size weights. At a given aggregate municipal share, a larger value means greater concentration in portfolios with high municipal shares. Its deviation is the model value less the observed value. For the fraction of a sector's municipal or taxable holdings that matures beyond twenty years, let $B$ be the sector's model holdings of that asset at four to thirty years, $L$ its holdings beyond twenty years and $f^{\text{obs}}$ the observed fraction. The deviation is

$$\mathcal{C}^{\text{long}} = \frac{L - f^{\text{obs}}B}{B^{\text{obs}}_{\text{window}}}.$$

The observed holdings $B^{\text{obs}}_{\text{window}}$ equal the national sector stock times the measured share of its holdings at four to thirty years. The normalizers are the observed concentrations and fractions.

Each holdings deviation, including the products and ratios above, is computed at each of the period's three observed December states, and the three values are averaged. Yield deviations use annual log yields at all 36 monthly states. Because observed two-year yields set those states, the ten-, twenty- and thirty-year yield

deviations are also deviations in spreads over the two-year yield. Their normalizers are the observed period-mean term spreads.

**The criterion.** Each signed deviation is averaged across the six periods and scaled by one tenth of its mean normalizer,

$$f_j(\theta) = \frac{6^{-1}\sum_{k=1}^{6} \mathcal{C}_{kj}(\theta)}{0.1\,(6^{-1}\sum_{k=1}^{6} o_{kj})}, \qquad \mathcal{L}(\theta) = \frac{1}{2}\sum_{j=1}^{22} f_j(\theta)^2. \tag{113}$$

A deviation equal to 10% of the mean observed normalizer gives $|f_j| = 1$. Each target's weight in $\mathcal{L}$, the inverse square of one tenth of its mean normalizer, depends on the data alone. Estimation searches over the common coefficient vector $\theta$ to reduce $\mathcal{L}$, solving the model in all six periods with Algorithm 1 at each trial $\theta$ and holding measured inputs fixed.

Table 9: Measured rate parameters and estimated coefficients, with search settings

| Symbol | Economic role | Value | Search interval | Start | Scale |
|---|---|---|---|---|---|
| *Measured rate process, fixed before calibration* | | | | | |
| $\bar{r}$ | Central short-rate level, twice the mean annualized two-year corporate zero yield | 0.06041 | – | – | – |
| $\rho$ | Rate-state persistence, estimated from historical short rates | 0.7292 | – | – | – |
| $\sigma_r$ | Rate-state innovation standard deviation, estimated from historical short rates | 0.04278 | – | – | – |
| *Risk. Main information: taxable term spreads and the allocation of long bonds* | | | | | |
| $\gamma$ | Common risk-aversion curvature, per unit of portfolio assets | 9.6969 | $[1, 10]$ | 9 | 1 |
| $v_{30}$ | Bond-specific variance of a thirty-year bond's two-year return | 0.000722 | $[0.000020, 0.006000]$ | 0.0015 | 0.0004 |
| $p_v$ | Maturity exponent of the bond-specific variance | 1.2597 | $[0, 3]$ | 1.15 | 0.3 |
| *Households. Main information: household municipal and taxable stocks* | | | | | |
| $a_H$ | Outside rate-exposure intercept | 1.3045 | $[-3, 3]$ | 1 | 0.3 |
| $b_A$ | Loading on the outside-asset ratio | 2.704 | $[0, 12]$ | 2 | 0.4 |
| $b_D$ | Loading on the debt ratio | 0.8379 | $[0, 20]$ | 0.5 | 0.5 |
| *P&C insurers. Main information: P&C stocks, maturities, composition and long fractions* | | | | | |
| $a_P$ | Low type's outside rate exposure | −1.961 | $[-10, 2]$ | −1.5 | 0.5 |
| $d_P$ | High-minus-low outside rate exposure | 3.0311 | $[0.02, 12]$ | 2.4 | 0.6 |
| $w_P$ | Low type's share of portfolio assets | 0.0802 | $[0.01, 0.99]$ | 0.22 | 0.08 |
| *Life insurers. Main information: life-insurer stocks, maturities, composition and long fractions* | | | | | |
| $a_L$ | Low type's outside rate exposure | −5.3927 | $[-12, 2]$ | −4 | 0.6 |
| $d_L$ | High-minus-low outside rate exposure | 6.3007 | $[0.02, 15]$ | 3.2 | 0.8 |
| $w_L$ | Low type's share of portfolio assets | 0.2616 | $[0.01, 0.99]$ | 0.1 | 0.08 |
| $t_L$ | Middle type's location within the exposure span | 0.8625 | $[0.01, 0.99]$ | 0.75 | 0.08 |
| $u_L$ | Middle type's share of portfolio assets excluding the low type | 0.0162 | $[0.01, 0.99]$ | 0.2 | 0.08 |
| *Banks and residual group. Main information: bank stocks and taxable term spreads* | | | | | |
| $h_B$ | Bank outside rate exposure | 1.2045 | $[-3, 3]$ | 0.6 | 0.3 |
| $\mu_R$ | Residual-group mean outside rate exposure | 1.0683 | $[-4, 4]$ | 1 | 0.4 |
| $s_R$ | Residual-group outside rate-exposure dispersion | 1.7168 | $[0.01, 5]$ | 2.2 | 0.4 |

*Notes.* Values use the model's two-year period. The seventeen estimated coefficients are fitted jointly to the twenty-two averaged targets of Table 10, then held fixed in the full 45-market model. Risk aversion and outside exposures $h_i = -\ell_i$ are in the normalized payoff units of Appendix B.2, and exposures are in units of a bond's rate loading. Positive values add rate risk and negative values hedge it, as explained in Appendix B.1. Portfolio-asset shares divide each insurer sector's measured debt securities and cash across its types. The last three columns give the search interval, the starting value and the step scale, which converts a unit move of the search into a change in the coefficient.

Table 10: Calibration targets and fitted quantities, 2000–2017

| Quantity | Data | Model |
|---|---|---|
| *Ownership stocks (billions of dollars)* | | |
| High-tax households, municipal | 906.8 | 1,135.1 |
| Lower-tax households, municipal | 903.6 | 662.2 |
| P&C insurers, municipal | 295.0 | 289.1 |
| Life insurers, municipal | 21.7 | 22.0 |
| Banks, municipal | 251.6 | 266.5 |
| Households, taxable | 1,635.7 | 1,544.0 |
| P&C insurers, taxable | 335.8 | 244.7 |
| Life insurers, taxable | 1,589.7 | 1,624.4 |
| Banks, taxable | 537.3 | 522.4 |
| *Insurer mean maturities (years)* | | |
| P&C insurers, municipal | 12.1 | 15.1 |
| P&C insurers, taxable | 7.6 | 6.3 |
| Life insurers, municipal | 14.6 | 17.9 |
| Life insurers, taxable | 12.3 | 11.7 |
| *Taxable term spreads (basis points)* | | |
| 10-year taxable spread | 209.0 | 214.7 |
| 20-year taxable spread | 279.9 | 284.6 |
| 30-year taxable spread | 299.3 | 303.6 |
| *Insurer portfolio composition* | | |
| P&C municipal-share concentration | 0.4374 | 0.5613 |
| Life municipal-share concentration | 0.0065 | 0.0063 |
| P&C municipal holdings beyond twenty years | 11.2% | 19.8% |
| P&C taxable holdings beyond twenty years | 7.1% | 4.1% |
| Life municipal holdings beyond twenty years | 24.9% | 28.4% |
| Life taxable holdings beyond twenty years | 25.6% | 26.4% |

*Notes.* Model values are from the one-market model used to estimate the common coefficients. Entries average the six periods equally. Stocks and mean maturities include two-year municipal bonds. Concentration and holdings beyond twenty years use the risky maturities of four through thirty years. Municipal-share concentration, the portfolio-size-weighted average of squared municipal shares, measures how unevenly municipal bonds are spread across insurers (Appendix B.3). Term spreads are in annual log-yield units over the two-year yield. The criterion first computes portfolio-weighted deviations in maturity and in holdings beyond twenty years at each date and then averages them, as in (113), so its terms are not differences of these displayed averages.

## B.4 Estimated coefficients and fit

**Estimates.** Estimation starts from the initial values in Table 9 and reaches $\mathcal{L} = 36.3575$. A restart with smaller finite-difference steps lowers the criterion to 36.2870, and the coefficients at this point are the estimates reported in Table 9.[26]

**Fit to the targets.** Table 10 reports the observed and fitted quantities behind all twenty-two targets in the one-market model used for estimation. Fourteen targets are matched to within 10% of their observed normalizers.

**Sensitivity of the estimates.** The best small move from the estimates, in one coefficient or several together, lowers the criterion by less than 0.001% and changes the average thirty-year gap by 0.004 basis points. This gap is measured on the model's own taxable curve, in annual continuously compounded yields averaged over each period's 36 monthly rate states and then equally across the six periods. Larger changes in any one group of coefficients, holding the others and measured inputs fixed, all worsen the fit (Table 11). Lowering risk aversion by 20% has the largest effect on yields (Section 5.2). The taxable and municipal curves fall nearly in proportion, so the gap per unit of taxable yield falls only from 0.170 to 0.166 and the model's implied tax barely changes.

[26] The coefficient search is a bounded trust-region method. Central differences measure how the averaged deviations respond to coefficients, with feasible one-sided differences at bounds. The search compares damped Gauss–Newton, backtracked and projected-gradient steps, using a secant estimate of the curvature that Gauss–Newton omits. The initial search and the restart stop when the trust-region radius falls below $10^{-5}$, with final improvements below 0.001.

Table 11: Sensitivity of the one-market calibration to estimated coefficients

| Coefficient change | 30-year gap (bp) | Change (bp) | Criterion $\mathcal{L}$ |
|---|---|---|---|
| Estimated coefficients | 102.6 | | 36.29 |
| Risk aversion $\gamma \times 0.8$ | 89.0 | −13.6 | 55.02 |
| Risk aversion $\gamma = 10$ (upper bound) | 104.7 | +2.1 | 36.74 |
| Bond-specific variance $v_{30} \times 0.8$ | 102.7 | 0.0 | 36.61 |
| Bond-specific variance $v_{30} \times 1.2$ | 102.7 | 0.0 | 36.59 |
| Maturity exponent $p_v - 0.2$ | 102.6 | 0.0 | 36.55 |
| Maturity exponent $p_v + 0.2$ | 102.7 | 0.0 | 36.47 |
| Household exposure intercept $a_H - 0.25$ | 99.0 | −3.7 | 78.82 |
| Household exposure intercept $a_H + 0.25$ | 106.7 | +4.1 | 160.84 |
| Household asset loading $b_A \times 0.8$ | 100.9 | −1.7 | 62.45 |
| Household asset loading $b_A \times 1.2$ | 105.2 | +2.6 | 400.24 |
| Household debt loading $b_D \times 0.8$ | 106.2 | +3.5 | 6,720 |
| Household debt loading $b_D \times 1.2$ | 103.9 | +1.3 | 82.17 |
| P&C exposures ×0.8 | 102.2 | −0.4 | 56.35 |
| P&C exposures ×1.2 | 103.3 | +0.6 | 62.51 |
| Life-insurer exposures ×0.8 | 101.9 | −0.7 | 59.68 |
| Life-insurer exposures ×1.2 | 103.4 | +0.8 | 46.24 |
| Bank exposure ×0.8 | 100.5 | −2.1 | 119.11 |
| Bank exposure ×1.2 | 104.5 | +1.8 | 241.78 |
| Residual exposure mean ×0.8 | 102.6 | −0.1 | 538.79 |
| Residual exposure mean ×1.2 | 101.7 | −0.9 | 65.73 |
| Residual exposure s.d. ×0.8 | 99.3 | −3.3 | 92.74 |
| Residual exposure s.d. ×1.2 | 105.8 | +3.1 | 1,687 |

*Notes.* Results average the six periods of 2000–2017. The $\gamma = 10$ row uses the upper bound of the search interval in Table 9. The P&C, life-insurer and bank rows scale every type's exposure, and the residual rows scale the mean or the dispersion of exposures. The 30-year gap is the thirty-year municipal yield less the household benchmark on the model's own taxable curve, before the corporate normalization. The criterion is (113).

**From the one-market model to the full model.** In the six periods, the thirty-year gap of the full 45-market model less that of the one-market model is $(-0.85, -0.72, 2.48, -0.22, -1.15, -0.93)$ basis points, and thirty-year taxable yields differ by at most 0.52 basis points. The full 45-market equilibria supply the municipal yield curves, cross-state pass-through and policy calculations. With the corporate normalization and the empirical sample weights, the model explains 83% of the 117.5-basis-point thirty-year puzzle (Figure 8).

## B.5 The bond-specific return shock

The calibrated model adds the independent shock in (32) to the main-text return. The gap barely responds to its variance or to scaling the shock to mean one.

**Role.** The shock can represent an issue-specific change in trading demand that affects sale proceeds without moving aggregate rates. Its variance cost grows with position size, which makes bonds imperfect substitutes and supports joint holdings by several investors, as Figure 14 illustrates. In the one-market calibration, 20% changes in its variance move the thirty-year gap (Table 11) and the thirty-year taxable yield by less than 0.1 basis points.

**Specification.** Writing the shock into realized resale proceeds, $P^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta}$, is algebraically identical to multiplying the holding return by it. Every return, conditional moment and portfolio condition is therefore the same under either form, and the shock adds no persistent price state.

**A mean-one shock.** The lognormal shock factor can also be scaled to have mean one without changing holding returns. Replace the shock factor by $e^{\sigma^a_{\eta,T}\eta-(\sigma^a_{\eta,T})^2/2}$ and shift log prices to

$$\check{p}^a_T(z) = p^a_T(z) - \frac{1}{2}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2, \qquad \check{P}^a_T(z) = e^{\check{p}^a_T(z)}.$$

The shifts at adjacent maturities exactly offset this scaling,

$$\frac{\check{P}^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta-(\sigma^a_{\eta,T})^2/2}}{\check{P}^a_T(z)} = \frac{P^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta}}{P^a_T(z)}.$$

With fixed dollar supplies, portfolios, optimality and clearing are unchanged. For $T$ periods of $\Delta = 2$ years, the annualized yield shift is

$$\check{y}^a_T - y^a_T = \frac{1}{2T\Delta}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2.$$

The fitted variance profile is common to both markets, so both thirty-year continuously compounded yields rise by 0.855 basis points and their difference is unchanged. The corporate normalization uses their ratio, so the normalized thirty-year gap rises by 0.20 basis points, from 97.39 to 97.59. The yields on cash and the two-year municipal claim are unchanged.

## B.6 Bond-supply sensitivity

The calibration and the puzzle accounting of Section 5.3 hold bond quantities fixed. Here supply responds to yields in both markets, with the estimated coefficients fixed. This exercise uses the one-market model and reports gaps on the model's own taxable curve, before the corporate normalization used for the full 45-market comparison. Elastic supply leaves the thirty-year gap close to its fixed-supply value.

For market $a \in \{M, C\}$ and maturity $T$, supply follows

$$Q^a_T(z) = \overline{Q}^a_T\left(\frac{y^a_T(z)}{\overline{y}^a_T}\right)^{-\chi_{a,T}},$$

where $y^a_T$ is the annual continuously compounded yield and $\chi_{a,T}$ is the yield elasticity of supply in absolute value. Each schedule passes through the fixed-supply quantity $\overline{Q}^a_T$ and yield $\overline{y}^a_T$ at the mean rate state. This reference point stays fixed when both yield curves are re-solved. Risky maturities from four to thirty years have elastic supply. The two-year riskless claims retain their baseline treatment.

St. Clair (2024) estimates that a one-percent increase in borrowing costs reduces municipal issuance by 0.28 to 0.59 percent. I use 0.45, the midpoint of the rounded range 0.3–0.6, and impose the same elasticity in both markets. The estimate concerns annual borrowing near the bank-qualified threshold. With other financing terms fixed, a proportional change in borrowing cost is locally a proportional change in yield. The comparative-static exercise applies that response to the model's outstanding supply.

At equal elasticities of 0.3, 0.45 and 0.6, the average thirty-year gap over the six calibration periods lies between 101.0 and 102.0 basis points, compared with 102.6 under fixed supply (Table 12). At 0.45, municipal yields fall by 14.1 basis points and the household benchmark by 13.4, so the gap changes by less than one basis point.

I also consider a larger taxable response. The November 2022 estimates of Siani (2022) imply about 1.5% less corporate issuance per ten-basis-point increase in credit spreads, holding the benchmark rate fixed. I match that local slope by setting $\chi_{C,T} = 15\overline{y}^C_T$, while keeping $\chi_{M,T} = 0.45$. The thirty-year taxable elasticity is 0.86–0.97 across periods. Applying this corporate issuance response to the pooled taxable supply is an additional sensitivity assumption. The average thirty-year gap is 99.2 basis points, retaining 96.7% of its fixed-supply value. Municipal yields fall by 19.5 basis points and the household benchmark by 16.1. A larger taxable supply elasticity therefore need not widen the gap.

## B.7 Numerical implementation and accuracy

The baseline equilibria satisfy their conditions to tight tolerances, and the 2015–2017 thirty-year gap barely moves when the grid or the barrier changes.

### B.7.1 Grid, integration and convergence

The fifteen equally spaced states of each period's grid span $[-6\sigma_z, 6\sigma_z]$ and include zero, where $\sigma_z = \sigma_r/\sqrt{1-\rho^2}$. Beyond the end nodes, log prices are extrapolated along the end segments, and all moments integrate the resulting piecewise-linear function analytically. For a standard normal $U$ and a log-price segment $a + bU$ on $l < U < u$,

$$\mathbb{E}[e^{a+bU}\mathbf{1}_{\{l<U<u\}}] = e^{a+b^2/2}\{\Phi(u-b) - \Phi(l-b)\}.$$

Table 12: Yield gaps with elastic supply in both markets

| Municipal elasticity | Taxable elasticity | 10 years | 20 years | 30 years |
|---|---|---|---|---|
| 0 | 0 | 35.7 | 84.9 | 102.6 |
| 0.30 | 0.30 | 36.4 | 84.1 | 101.9 |
| 0.45 | 0.45 | 36.7 | 84.0 | 102.0 |
| 0.60 | 0.60 | 36.8 | 83.4 | 101.0 |
| 0.45 | $15\overline{y}_T^C$ | 37.2 | 83.2 | 99.2 |

*Notes.* Gaps are on the model's own taxable curve, in basis points, averaged over the observed monthly rate states within each period and equally across the six periods of 2000–2017. All other coefficients and inputs remain fixed. Elasticities are reported as positive magnitudes, with higher borrowing costs reducing supply. The last row converts a corporate spread semi-elasticity to a yield elasticity at each fixed-supply yield, with yields in annual decimal units.

The local rate loadings $A_T^a = -\partial p_T^a/\partial z$ satisfy equation (45) under its differentiation conditions. The baseline grid has no node exactly at the rate floor $z = -\bar{r}$, and adding one is checked below. The comparisons in Figures 7 and 8 read each bond's issuer-state yield at its issue-quarter-end state and use empirical bond, rating and quarter weights.

During calibration, the solution at a nearby coefficient vector supplies starting prices and portfolios, and the full 45-market solve starts from the one-market equilibrium at the same coefficients. At each node, Newton's method solves portfolios and clearing jointly until the residuals of (110) and (111) are below $10^{-11}$. In the first outer iteration, each node approaches the barrier $\epsilon = 10^{-9}$ gradually from $10^{-5}$. Later iterations start each node from its previous solution and solve at $\epsilon$ directly. The price update is damped and uses Anderson acceleration over the last seven iterates, with each step capped at 0.025 in log price. In the six baseline periods, the maximum difference between candidate prices and $\widehat{p}$ falls from about $10^{-2}$ to $10^{-6}$ within forty iterations and below $10^{-10}$ within seventy.

### B.7.2 Equilibrium accuracy

Four diagnostics check prices against resale values ($R_p$), market clearing ($R_c$), portfolio optimality ($R_s$) and complementarity ($R_{\text{comp}}$):

$$\begin{aligned} R_p &= \|p - \mathcal{M} + \mu\|_\infty, \qquad & R_c &= \|Q - \textstyle\sum_i m_i x_i\|_\infty, \\ R_s &= \|\widetilde{\nu} - \epsilon w/x\|_\infty, \qquad & R_{\text{comp}} &= \max\{\|x \odot \widetilde{\nu}\|_\infty, \|\kappa \odot x^0\|_\infty\}. \end{aligned} \tag{114}$$

Residuals recomputed separately for the full 45-market equilibria at the eighteen nodes of each of the six periods give maxima of $R_p < 9.3 \times 10^{-11}$, $R_s < 9.9 \times 10^{-12}$, $R_c < 2.3 \times 10^{-16}$ and $R_{\text{comp}} < 1.01 \times 10^{-9}$. No solution violates the shorting constraint on cash or nonnegativity, and riskless claims have zero variance. Acceptance requires $R_p < 3 \times 10^{-7}$, $R_s < 10^{-7}$, $R_c < 10^{-8}$ and a violation of the shorting constraint on cash below $10^{-10}$. Complementarity is reported separately because at an exact barrier solution each of its products equals $\epsilon w_j$, or $\epsilon$ for cash. The 2018–2020 period, solved at the same coefficients, gives $R_p = 4.52 \times 10^{-11}$, $R_s = 7.78 \times 10^{-12}$, $R_c = 2.22 \times 10^{-16}$ and complementarity $1.00 \times 10^{-9}$. The grid comparison in Appendix B.7.3 assesses accuracy between nodes.

**Riskless municipal returns.** Each issuer's riskless one-period municipal return is the lowest return required by any eligible investor, including that investor's multiplier on its shorting constraint on cash, as in (39). The lowest return that households require is

$$\min_{i \in \mathcal{H}} \frac{\xi_i^C r(z) + \kappa_i(z)}{\xi_i^{M_s}},$$

where $\mathcal{H}$ contains all 630 household types and $s$ identifies the issuer. Every issuer's one-period return is within 0.01 basis points of this lowest household required return at every grid node. The minimum can differ from the household benchmark of the issuing state's high-tax residents when the shorting constraint on cash binds for all their types or another household type requires less. In the six calibration periods, the national supply-weighted one-period municipal yield is also within 0.01 basis points of the household benchmark at every observed month. Household taxes therefore anchor the riskless short end in the solved model, as in (12).

### B.7.3 Grid and barrier sensitivity

Three 2015–2017 re-solves, each with separately recomputed residuals, test whether the baseline grid and barrier are adequate. Each keeps coefficients and measured inputs fixed and makes one change. One refines the equally spaced grid from fifteen to twenty-nine nodes, one lowers the barrier from $10^{-9}$ to $10^{-10}$, and one adds a node exactly at the rate floor (Table 13).

Table 13: Numerical sensitivity at fixed coefficients, 2015–2017

| Comparison | Nodes | $\epsilon$ | Δ 30-year gap (bp) | Δ lower-tax household municipal holdings (%) | Δ bank taxable holdings (%) |
|---|---|---|---|---|---|
| Finer grid | 32 | $10^{-9}$ | +0.199 | −4.02 | −2.59 |
| Smaller barrier | 18 | $10^{-10}$ | +0.0001 | −0.004 | +0.001 |
| Node at the rate floor | 19 | $10^{-9}$ | −0.228 | −1.69 | −2.80 |

*Notes.* Each row compares one change with the baseline eighteen-node, $10^{-9}$-barrier solution. Nodes include the three December states, and $\epsilon$ is the barrier of (110). The thirty-year gap is the 45-market model's national supply-weighted municipal yield minus the household benchmark on its own taxable curve, in continuously compounded yields over the period's 36 monthly states. Its baseline is 104.27 bp. Holdings are measured at the three December nodes, and changes are relative to each group's baseline holdings. The rows are numerical comparisons, distinct from the empirically weighted gap of Table 4. The rate floor affects prices in every row.

The thirty-year gap moves by less than 0.23 basis points in every case. Holdings move more. Grid refinement reduces lower-tax household municipal holdings by 4.02%, and adding the floor node lowers those holdings by 1.69% and banks' taxable holdings by 2.80%. Lowering the barrier changes risky positions at December nodes by at most 0.071% of investor funding. With fixed prices and portfolio optimality met to tolerance, concavity bounds the barrier portfolio's utility loss by $\epsilon(\sum_j w_j + 1)$. Cash and a riskless bond can offer the same after-tax return, so prices leave the split between them undetermined. At December nodes, lowering the barrier moves this split by up to 24.84% of an investor's funding, while complementarity falls from about $10^{-9}$ to $10^{-10}$. The smaller barrier changes the two group holdings in Table 13, both ownership targets, by at most 0.004% of their baseline values.

### B.7.4 Policy calculations

The counterfactuals of Section 7 use the same price-and-portfolio algorithm. The home-state repeals, seven states in each of six periods, use the baseline barrier and tolerances. Separately recomputed residuals give maximum $R_p = 9.79 \times 10^{-11}$, $R_s = 9.20 \times 10^{-11}$ and $R_c = 3.33 \times 10^{-16}$, with zero variance for riskless claims and no violations of the shorting constraint on cash or of nonnegativity. Riskless municipal returns stay within $1.83 \times 10^{-5}$ basis points of the lowest household required return at every observed month. Separate checks verify the changed tax rules and fiscal identities.

The equal-treatment reform, which removes 99.9% of tax distinctions in every period, is solved first for 2015–2017, starting from the solution with 99% removal. The 2015–2017 solution initializes the other five periods, each checked separately. The equal-treatment solves use tolerances of $2 \times 10^{-7}$ on the price difference, $2 \times 10^{-8}$ on the inner solve, $5 \times 10^{-9}$ on each holding times its portfolio-optimality residual and $2 \times 10^{-9}$ on linear solves. In a smaller version of the model that combines the states into three municipal markets, moving from 99% to 99.9% removal changes thirty-year municipal and taxable yields by 1.9 and 0.2 basis points and each group's maturity shares by at most 0.2 percentage points. Near full removal, municipal and taxable bonds become close substitutes, and within-maturity allocations between them can change by 9.5 percentage points of group assets. Table 37, Panel B, therefore reports riskless shares and the maturity composition of risky holdings, which combine municipal and taxable bonds of each maturity.

# C Adjustments and empirical tests

This appendix first constructs the five adjustments used to measure and account for the municipal puzzle. Comparisons with issuer-matched Build America Bonds and with high-grade general-obligation bonds then check whether the puzzle depends on the taxable comparison or the security sample. Appendix C.4 relates the puzzle to the single-tax-rate benchmark of Green (1993). The remaining subsections report the tax and interest-rate tests P1 to P3, their robustness checks and ownership evidence by tax status.

## C.1 Comparability and bond-side adjustments

Before the municipal spread over the household benchmark can be compared with the model, five effects outside the model must be removed. Two concern comparability. Municipal yields are measured at issuance, and almost all long-maturity municipal bonds are callable, while the corporate curve describes seasoned, option-free bonds. Removing the value of the issuer's call right and the issuance premium puts the two curves on the same basis and defines the municipal puzzle. The other three are bond-side differences in capital-gains taxation, liquidity and default risk. Each could explain part of the puzzle independently of the model's mechanism, so I subtract them before comparing the puzzle with the model.

Table 14 reports each adjustment's contribution by maturity, along with the puzzle and the residual before the model. At thirty years, the call and issuance-premium adjustments reduce the 159.8-basis-point spread to a puzzle of 117.5 basis points. The bond-side adjustments explain 14.5 basis points, or 12% of the puzzle, and leave a residual before the model of 103.0 basis points. Taking the upper end of every adjustment's range still leaves 85.1 basis points. As a share of the puzzle, the bond-side adjustments matter more at shorter maturities and explain 40% of the ten-year puzzle. Their sum stays near 12–16 basis points from twelve years onward, while the puzzle widens.

Table 14: What explains the municipal spread over the household benchmark: contributions in basis points to Spread($T$) of equation (115). The two comparability columns remove the call option and the issuance premium from the municipal yield itself, giving the puzzle of Figure 2. The three bond-side differentials are municipal-minus-corporate.

| | | Comparability | | | Bond-side differentials | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | Spread | Call | Issuance premium | Puzzle | Capital gains | Liquidity | Default | Bond-side adjustments | Share of puzzle | Residual before the model |
| 2 | 9.9 | 0.1 | 2.9 | 6.9 | 0.0 | −0.4 | 14.7 | 14.3 | | −7.3 |
| 6 | 27.9 | 0.7 | 3.3 | 23.9 | 10.1 | −0.4 | 8.5 | 18.2 | 76% | 5.8 |
| 10 | 52.2 | 4.6 | 4.0 | 43.6 | 9.9 | −0.3 | 8.1 | 17.6 | 40% | 26.0 |
| 14 | 87.0 | 17.9 | 7.2 | 61.9 | 7.1 | −0.1 | 7.6 | 14.6 | 24% | 47.3 |
| 18 | 110.3 | 23.9 | 9.6 | 76.9 | 4.7 | −0.2 | 7.9 | 12.4 | 16% | 64.5 |
| 22 | 128.8 | 27.0 | 11.4 | 90.5 | 3.6 | 2.8 | 9.0 | 15.4 | 17% | 75.1 |
| 26 | 145.0 | 28.5 | 12.5 | 104.0 | 2.9 | 1.5 | 11.3 | 15.7 | 15% | 88.3 |
| 30 | 159.8 | 29.5 | 12.8 | 117.5 | 2.3 | 0.1 | 12.2 | 14.5 | 12% | 103.0 |

*Notes.* The bond-side column sums the three adjustments, and the share divides that sum by the puzzle (omitted where the puzzle is below 20 basis points). The residual before the model is the puzzle less this sum. Thirty-year ranges, in basis points: call $[22.8, 36.3]$, issuance premium $[8.1, 17.3]$, capital gains $[1.2, 2.5]$, liquidity $[-2.2, 2.3]$, default $[6.1, 16.4]$. Combining the range endpoints gives a thirty-year residual before the model between 85.1 and 123.8 basis points.

### C.1.1 The common comparison

Every adjustment is a basis-point contribution to the spread

$$\text{Spread}(T) = y_T^{M,\text{raw}} - \bar{\xi}_H(T)\, y_T^C, \tag{115}$$

where $y_T^{M,\text{raw}}$ is the observed municipal zero-coupon curve before either comparability adjustment and $y_T^C$ is the A-or-above High Quality Market (HQM) corporate zero-coupon curve of Section 3.6. I construct $y_T^{M,\text{raw}}$ from each bond's offering price and contractual cash flows, treating the bond as a noncallable bullet. A flat spread over a smooth municipal reference curve prices these cash flows at the offering price, and the bond's zero-coupon yield is the reference yield at its maturity plus this spread. A parallel shift of the reference curve moves the spread by the opposite amount, so the zero-coupon yield depends on the reference curve only through its shape. The household benchmark $\bar{\xi}_H(T)y_T^C$ averages the household's after-tax corporate yield at the federal and state tax rates of each municipal bond's state and date. Dividing this average by the averaged corporate yield gives $\bar{\xi}_H(T)$. The tax rates include the net investment income tax from 2013 and, from 2018, the cap on state and local tax deductions. The household wedge $1 - \bar{\xi}_H(T)$ averages 41.3%.

All curves use the same three-stage averaging: equal weights within maturity × issue-quarter × half-notch rating cells, then across ratings within quarters, then across quarters. Each adjustment is estimated on the observations that identify it and is then carried to the same maturity grid. The two comparability adjustments change the municipal yield itself. The three bond-side adjustments are municipal-minus-corporate differences in compensation, measured in the household's after-tax yield units, so compensation common to both markets

cancels. The contributions are estimated separately and add up as in (24). The combined range adds the endpoints of the separate ranges and is not a joint confidence interval.

### C.1.2 Adjustment I: embedded call options

**What it measures.** An issuer's call right lets it repay a bond before maturity at a stated call price. When rates fall, the issuer can refinance, so the holder loses part of the price gain that a noncallable bond would deliver. Investors therefore pay less for callable bonds, and callable yields exceed the yields on otherwise identical option-free bonds. More than 95% of municipal bonds beyond twenty years are callable, so dropping them would leave almost no long-maturity sample. I therefore value each bond's call right and remove it.

**Calculation.** I value each bond's contractual coupons, principal and call schedule on a grid of future rate states. Working backward from the final coupon and principal at maturity, I set the value at each earlier coupon date to the coupon plus the discounted continuation value, with the continuation value capped at the call price on call dates:

$$C_k(z) = P_\Delta(z)e^{-s^{\text{fit}}\Delta}\sum_j \Pi_{zj}V_{k+1}(z_j), \qquad V_k(z) = 100ca_k + \min\{C_k(z), K_k\}. \tag{116}$$

Here $V_k(z)$ and $C_k(z)$ are the value and the discounted continuation value at coupon date $k$ in rate state $z$, $c$ is the annual coupon rate, $a_k$ the accrual fraction, $P_\Delta(z)$ the reference discount price over one coupon interval $\Delta$, $\Pi_{zj}$ the transition probabilities between rate states and $K_k$ the call price, with $K_k = \infty$ at dates without a call right. The constant spread $s^{\text{fit}}$ is set so that the callable value at the bond's starting rate state, defined below, equals its observed price. Holding this fitted spread and the cash flows fixed, I remove the cap. The difference between the two values, the uncapped (bullet) value $V_b^{\text{bul}}$ and the callable value $V_b^{\text{call}}$ of bond $b$, is the call value,

$$CV_b \;=\; V_b^{\text{bul}} - V_b^{\text{call}} \;\geq\; 0. \tag{117}$$

Adding $CV_b$ to the observed price gives the option-free price. I convert the observed price, treated as a bullet, and the option-free price into zero-coupon yields with the same bullet cash flows and reference curve. For each price, a flat continuously compounded spread $\hat{s}$ over the reference discount curve $D_0(t)$ at rate state $z = 0$ prices the bond's coupons and principal at that price. The zero-coupon yield is $200[\exp\{(y_0(T_b) + \hat{s})/2\} - 1]$ percent, a nominal semiannual yield, with $y_0(T_b) = -\log D_0(T_b)/T_b$ and $T_b$ the bond's maturity. This conversion spread only changes units and is separate from the fitted spread $s^{\text{fit}}$. The difference between the two resulting zero-coupon yields is the bond's call spread, which is nonnegative because the option-free price is no lower. Because the fitted spread is held fixed, the adjusted yield keeps its compensation for credit, liquidity and other features, and only the call right is removed. Noncallable bonds have a zero call spread. The call adjustment Call($T$) is the three-stage average of the call spreads in the pricing sample described below, and I subtract it from the full observed curve in forming the puzzle.

The reference prices come from the calibrated model of the bond's issue period, one of the six calibration periods or the 2018–2020 period. At each maturity and rate state, I average the model's municipal log prices across state markets with supply weights. The rate state follows the calibrated AR(1) process, converted from two-year periods to half-year steps for the coupon dates. At each maturity, I scale the deviation of the log price from its value at $z = 0$ so that the volatility of twelve-month yield changes equals measured municipal yield volatility times a scale $m$, leaving the curve at $z = 0$ unchanged. Measured volatility at maturity $T$ is the standard deviation of twelve-month changes in the $T$-year AA-and-above municipal zero-coupon yield, so $m = 1$ prices the call at measured volatility. A pricing drift is then fitted at the model's rate-state nodes so that discounting along the grid reproduces the scaled prices. Each bond starts at the observed rate state in the final month of its issue quarter, $z_t = r_t - \bar{r}$. Here $\bar{r}$ is the calibrated mean rate before the rate floor in Section 4.1. The observed one-period cash rate $r_t$ is twice the continuously compounded two-year corporate zero yield, because a model period lasts two years. The pricing sample contains 1,828,379 A-or-above fixed-rate and original-issue-discount bonds. Calendar and price screens exclude 16,295 bonds with irregular first coupons, unsupported conventions or unusable call schedules, and the eligible bonds carry 97.6%–99.8% of each maturity's weight. These screens change three-stage municipal yields by at most 1.1 basis points and the household benchmark by less than 0.1 basis points.

**Why the estimate is reliable.** Three features discipline the option value. First, the contract fixes the payoff. The coupon, call dates and call prices determine what the issuer gains by calling. Second, market prices of callability discipline the volatility scale $m$, which determines how often calling pays. I fit $m$ so that model call

spreads match observed callable-minus-noncallable yield differences among AA-and-above bonds matched by maturity, issue quarter, rating and half-point coupon at six to twenty years. I then apply it to A-or-above bonds without re-estimation. If callability is the only difference within these cells, the correct $m$ equates the model and market differences. The estimate is $\hat{m} = 1.077$. Its conditional bootstrap 95% band, $[0.964, 1.189]$, comes from 2,000 draws that resample issue quarters independently, preserving within-quarter dependence across maturities. Each draw holds fixed the model's matched-cell differences at $m = 1$, their elasticities with respect to $m$ and the fitting weights, which makes the band conditional. The band contains measured volatility ($m = 1$). The municipal spread over the household benchmark is not a fitting target.

Third, the adjusted curve matches bonds that never needed an adjustment. Figure 15 compares the call-adjusted curve with yields on bonds that were noncallable from issuance, in shared maturity, quarter and rating cells. The two curves differ by less than three basis points through ten years. The largest difference on the two-year grid is eight basis points, at eighteen years, and at twenty years the adjusted curve lies 3.5 basis points below the noncallable curve. Relative to the sample used to fit $m$, this comparison adds A-rated bonds, covers two to twenty years and does not match coupons. Beyond twenty years, noncallable bonds become scarce, fewer than 200 at each maturity, so the comparison stops at twenty years.

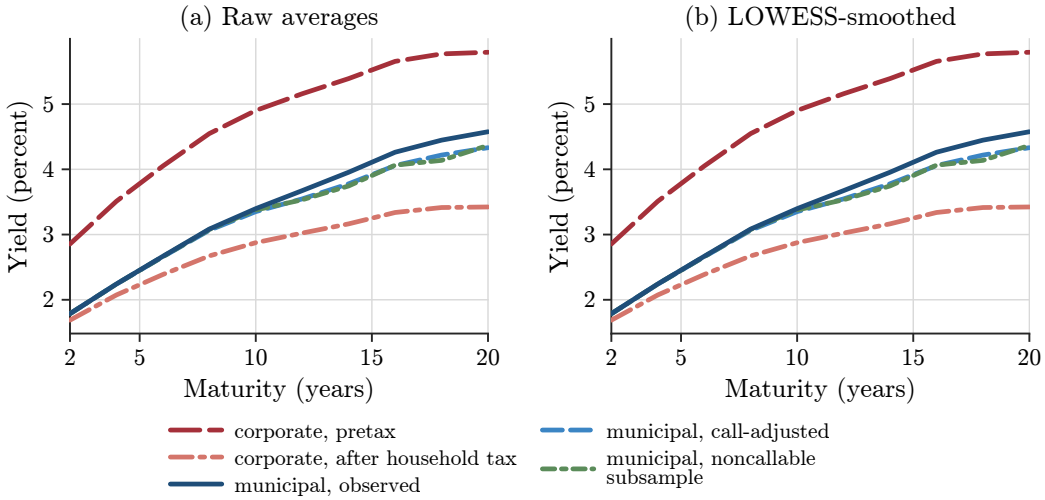



Figure 15: Noncallable validation of the call adjustment: raw means (a) and locally weighted scatterplot smoothing (LOWESS) (b). Lines show pretax and household after-tax corporate yields alongside observed, call-adjusted and noncallable municipal yields. The call-adjusted and noncallable curves share A-or-above maturity–quarter–rating cells containing noncallables, with fifty or more quarters per maturity and no extrapolation past twenty years.

**Result and sensitivity.** The call adjustment is 4.6 basis points at ten years and 29.5 at thirty (Figure 16(a)). Repricing at the endpoints of the bootstrap band for $\hat{m}$ gives a thirty-year range of $[22.8, 36.3]$ basis points, and pricing at measured volatility gives 24.9. Panel (b) compares the market callable-minus-noncallable differences at the fitting maturities with model differences at measured and fitted volatility. Panel (c) shows the elasticity of the adjustment with respect to $m$, 2.2 at thirty years, which is why the range is wider in percentage terms than the band for $\hat{m}$. A different option model gives nearly the same adjustment. In a one-factor Hull–White model (Hull and White, 1994), the drift fits each dated discount curve exactly, annual mean reversion is fixed at 0.10, and volatility is fitted to the same callable-minus-noncallable differences. This model gives 4.7, 23.4 and 28.5 basis points at ten, twenty and thirty years, against 4.6, 25.3 and 29.5 in the baseline. It fits the market differences with a root-mean-square error of 3.7 basis points, against 4.2 for the baseline model.

### C.1.3 Adjustment II: the issuance premium

**What it measures.** Buyers at the offering receive a higher yield than buyers of the same bond shortly afterward, whether because of placement costs, distribution, liquidity seasoning or price discovery. The issuance premium is this difference, and removing it puts the municipal curve on the seasoned basis of the corporate curve.

**Calculation.** For bond $b$, let $y_b^{\text{off}}$ be its offering yield and $y_{bj}^{\text{tr}}$ the yield on customer purchase $j$ within thirty days of the offering. Let $\bar{y}_k^{\text{seas}}(w)$ be the yield on seasoned bonds in the same maturity bucket $k$ and week $w$,

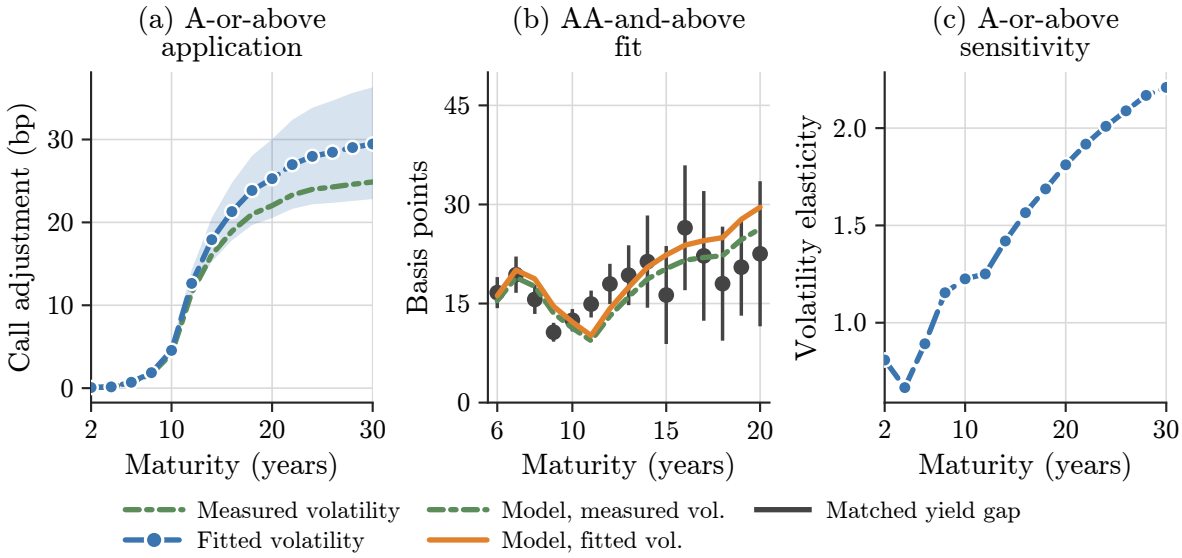



Figure 16: Call adjustment and volatility calibration. (a) A-or-above call adjustments at measured and fitted volatility, with direct repricing at the endpoints of the bootstrap band for $\hat{m}$. (b) AA-and-above callable-minus-noncallable market differences in matched cells (matched yield gap, ±1.96 standard errors) and model differences at the fitting maturities, 6–20 years. (c) Elasticity of the A-or-above call adjustment with respect to the volatility scale $m$.

with $w_b^{\text{off}}$ and $w_{bj}$ the weeks of the offering and of purchase $j$. The premium is the offering-to-trade yield decline less the contemporaneous decline in seasoned yields:

$$\text{Premium}_{bj} = (y_b^{\text{off}} - y_{bj}^{\text{tr}}) - [\bar{y}_k^{\text{seas}}(w_b^{\text{off}}) - \bar{y}_k^{\text{seas}}(w_{bj})]. \tag{118}$$

Comparing the bond with itself removes differences in issuer and contract terms, and subtracting the seasoned-yield change removes market-wide rate movements. The estimate uses neither the corporate yield nor the household benchmark. The early trade is a customer purchase, like the offering itself, so the comparison avoids a purchase-to-sale bid–ask difference that would overlap with the liquidity adjustment. Early and seasoned trades must have at least $250,000 par, and when-issued trades are excluded. The seasoned benchmark is the par-weighted purchase yield of bonds older than three years, with at least 25 prints in the bucket and week.

Such an early institutional-size purchase exists for 76,763 of the 1,823,080 curve bonds with an offering yield and date, or 4%, and an early purchase of any size exists for 16%. For each of the 76,763 bonds, I average the premia over its early purchases. I then average the bonds within each maturity bucket with the three-stage weights and apply the bucket estimates to the whole curve. Issues from 2000–2004, before the trade tape begins, receive the estimate from later issues. Bucket estimates are interpolated from bucket midpoints to the two-year grid, with end values held flat, giving Prem($T$). The premium is a yield change between the offering and early trades, so I subtract it from the zero-coupon curve basis point for basis point. Offering and zero-coupon yields move closely together, with correlations of 0.998, 0.993 and 0.937 at five, ten and thirty years, and their level differences come from the zero-coupon conversion.

**Result and robustness.** The issuance premium rises from 2.9 basis points at two years to 12.8 at thirty (Table 15 and Figure 17(b)). A block bootstrap over issue quarters gives a thirty-year interval of $[8.1, 17.3]$ basis points, and the intervals exclude zero at every maturity.

The rising maturity profile holds across trade windows, trade sizes and coupon types. In every trade-age window of Table 15 and Figure 17(a), premia beyond twelve years exceed those below twelve years, and from fourteen days on the two longest buckets give 9.3–12.8 basis points. In those buckets, par and discount bonds give 10.9 and 13.2 basis points and premium bonds 9.3 and 10.9, so yield-to-worst conventions for premium bonds do not drive the gradient. Including trades of all sizes raises the thirty-year estimate to 21.1 basis points. Long-end premia remain positive in every trade-size decile and are smallest for the largest trades (Figure 17(c)), consistent with small-trade markups raising the all-size estimates.

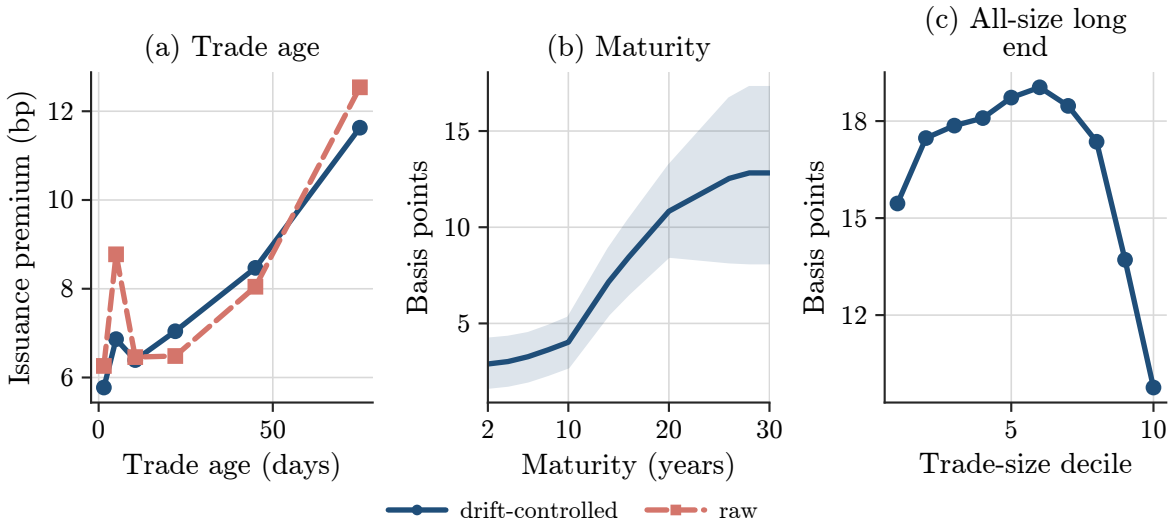



Figure 17: Issuance-premium diagnostics. (a) Institutional-size estimates by trade age, with (drift-controlled) and without (raw) the market-movement control. (b) Institutional-size maturity profile with block-bootstrap band. (c) All-size long-end estimates by trade-size decile.

Table 15: Issuance premium by maturity bucket (bp) under alternative trade-age windows, in days. The baseline window is thirty days. Panel A, the baseline, requires prints of at least $250,000 par in both the early print and the seasoned benchmark. Panel B uses prints of all sizes.

| maturity bucket (yr) | 3 | 7 | 14 | 30 | 60 |
|---|---|---|---|---|---|
| *Panel A: institutional-size prints (baseline)* | | | | | |
| 1–5 | 3.7 | 3.6 | 1.4 | 2.9 | 3.3 |
| 5–8 | 2.8 | 5.2 | 3.0 | 3.3 | 4.0 |
| 8–12 | 3.4 | 5.2 | 3.9 | 4.0 | 4.6 |
| 12–17 | 6.2 | 9.0 | 7.7 | 7.5 | 7.8 |
| 17–23 | 9.1 | 7.3 | 9.3 | 10.8 | 12.2 |
| 23–31 | 9.5 | 7.2 | 11.9 | 12.8 | 12.0 |
| *Panel B: prints of all sizes* | | | | | |
| 1–5 | 4.5 | 6.7 | 5.6 | 3.6 | 2.2 |
| 5–8 | 5.9 | 10.1 | 7.6 | 6.1 | 5.0 |
| 8–12 | 7.2 | 10.5 | 9.0 | 7.9 | 7.4 |
| 12–17 | 15.5 | 16.2 | 13.8 | 12.8 | 12.4 |
| 17–23 | 15.0 | 17.3 | 16.7 | 16.6 | 16.7 |
| 23–31 | 18.1 | 18.7 | 20.0 | 21.1 | 21.0 |

*Notes.* Each cell is the premium for bonds with at least one print inside the window, averaged first over each bond's prints inside the window and then over bonds with the curve's three-stage weights (Appendix C.1.1).

### C.1.4 Adjustment III: capital-gains taxation

**What it measures.** A bondholder earns interest and, if the bond is sold before maturity, a price gain or loss. The model applies the interest-income tax treatment to the whole holding return. Under federal tax, it therefore treats municipal price gains as exempt and corporate price gains as ordinary income. Actual rules tax the two parts differently. Municipal interest is exempt, but a resale gain can be taxable, and a corporate resale gain can face the lower capital-gains rate. Consider a bond bought at par and sold more than a year later for a $1 gain. At the federal rates used here, the model assigns no tax to the municipal gain and 37 cents to the corporate gain, while capital-gains treatment assigns 20 cents to each. Capital-gains treatment therefore reduces the municipal bond's after-tax advantage when prices rise. Under the full loss offsets assumed below, losses have the opposite effect. The sale price also reflects the next buyer's taxes. At a given required after-tax return, a buyer who owes market-discount tax at redemption pays less for the bond.

**Tax rules.** Table 16 compares the model treatment $B$ with the detailed treatment $A$. Coupons and original-issue discount (OID) are treated the same way in both, exempt for municipal bonds and ordinary income for corporate bonds. OID raises tax basis as it accrues, so accrued interest is not taxed again as a gain. The original buyer's basis at a sale date $h$ is the adjusted issue price $\mathrm{AIP}_h$, which moves from the issue price to face at constant yield. A secondary buyer's basis starts at the purchase price, with premium adjustments following Appendix D.3. Market discount is the shortfall of a secondary purchase price below the adjusted issue price for OID bonds or below face otherwise. It is ordinary income at disposition, or a capital gain if it is below the de minimis threshold. With $n$ remaining years and face 100, the threshold is

$$\bar{D}_n = 0.0025 \times 100 \times n. \tag{119}$$

Both treatments use the same federal rates in all seven periods, $\tau_{oi} = 0.37$ on ordinary income and $\tau_{cg} = 0.20$ on capital gains, excluding the net investment income tax and state tax. Under treatment $A$, residual gains and losses face the capital-gains rate with full offsets for losses, so the tax on a residual gain $G$, negative for a loss, is

$$\mathcal{T}(G) = \tau_{cg} G. \tag{120}$$

Table 16: Tax treatments compared in the capital-gains adjustment

| Return component | Model-based treatment $B$ | Detailed treatment $A$ |
|---|---|---|
| Municipal coupons and OID | Exempt | Exempt |
| Corporate coupons and OID | Ordinary income | Ordinary income |
| Original municipal buyer's residual resale gain | Exempt | Capital gain |
| Original corporate buyer's residual resale gain | Ordinary income | Capital gain |
| Municipal secondary buyer's market discount | Exempt | Capital gain if de minimis, otherwise ordinary income |
| Corporate secondary buyer's market discount | Ordinary income | Capital gain if de minimis, otherwise ordinary income |

*Notes.* Residual gains are measured relative to adjusted basis. Ordinary-income and capital-gains rates are 37% and 20%. A secondary buyer holds to maturity, with market-discount tax due at redemption.

**Calculation.** I follow an investor who buys at issuance and first sells after $h$ years. For each sale horizon, I find the municipal issuance yield that gives this investor the same expected after-tax holding return as the corporate bond, averaging over sale-date rate states. I do this first under the model treatment $B$ and then under the detailed treatment $A$, holding corporate prices, the rate process and the reinvestment curve fixed. Terminal wealth at the sale date is the after-tax sale proceeds plus interim coupons, net of tax, reinvested to that date on the unshifted municipal curve. The holding return is the annualized log of expected terminal wealth over the offering price. Resale prices are those that give the next buyer, who holds to maturity, the corporate bond's after-tax return under the same treatment, including the market-discount tax due at redemption. With $\delta_B$ and $\delta_A$ the two required changes in the municipal issuance yield, the correction for a maturity-$T$ bond sold after $h$ years is

$$\Delta_{\mathrm{cg}}(h, T) = 10^4 \big(\delta_A - \delta_B\big) \text{ basis points.} \tag{121}$$

Both yield changes are parallel shifts of the municipal log-discount curve in annual continuously compounded units. For a par bond, a higher issuance yield raises the coupon at an issue price of 100, and the municipal resale price is recalculated for the changed payments. Subtracting $\delta_B$ removes any return difference already present on the model curves, so $\Delta_{\mathrm{cg}}$ isolates the difference in tax treatment. A positive correction means that the detailed treatment $A$ requires a higher municipal issuance yield than the model treatment $B$, so it explains part of the puzzle. The additive accounting uses it as a basis-point approximation to the corresponding nominal semiannual shift.

**Averaging across bonds and sale dates.** Representative par and zero-coupon bonds are built from each period's supply-weighted municipal and corporate model curves at the rate state $z = 0$. Sale-date rate states follow the calibrated AR(1) distribution $h$ years ahead, truncated to the model grid. Each corporate–municipal pair of instruments is weighted by the product of each instrument's share of issuance in its market, and at thirty years the par/par pair receives 98.7% of the weight.

The correction applies only to bonds that are sold before maturity, so I weight each horizon by the probability $\pi_h(T)$ that the original buyer first sells then:

$$\mathrm{CG}(T) = \sum_h \pi_h(T)\,\Delta_{\mathrm{cg}}(h,T), \tag{122}$$

with a zero correction for bonds held to maturity. Total turnover overstates these probabilities. The same dollar of bonds can trade repeatedly, and the next buyer's tax is already in the resale price, so weighting every later trade would count that tax again. I therefore let a trading segment, a share $\phi(T)$ of par, carry all resales, while the remaining par is held to maturity.

Turnover $t_a$ is seasoned traded par over outstanding par at bond age $a$ in the MSRB customer tape. Holding aggregate turnover fixed, the first-sale hazard within the trading segment at age $a$ is $\lambda_a = \min(t_a/\phi, 0.99)$, and the probability of a first sale at age $a$ is $f_\phi(a) = \phi\lambda_a \prod_{j<a}(1-\lambda_j)$. Sale horizons lie on the two-year grid, so the weight $\pi_h(T)$ sums $f_\phi(a)$ over ages $h-1$ and $h$. The first horizon also receives age 0, and the last also receives age $T-1$. I set $\phi$ so that the probability of a first sale within the observation window equals the measured ceiling on the share of par ever sold, which is cumulative seasoned volume per CUSIP capped at outstanding par. The ceiling uses issues offered from 2005 with at least three observable years, and each bond's observation window is the shorter of its original maturity and its years from offering to 2021. If a homogeneous hazard ($\phi = 1$) stays below the ceiling, the segment share is one, as at four years (Table 17). At thirty years, almost all of the trading segment sells within the window, so $\phi$ equals the ceiling, 0.602, and in this first-sale model the original buyer holds 40% of thirty-year par to maturity. Results average the seven periods equally.

**Why the adjustment is small.** The adjustment is 10.7 basis points at eight years and 2.3 at thirty (Table 17 and Figure 18). Four features keep it small at long maturities. Coupons and OID are taxed identically in both treatments, so only price gains and losses matter. Gains and losses partly offset across rate states. Bonds held to maturity, 40% of thirty-year par, have no residual gain. Finally, the resale-tax wedge is spread over the bond's remaining life, and the de minimis threshold, below which market discount is taxed at the lower capital-gains rate, rises with that life.

**Sensitivity.** The reported adjustment uses the full tape window, the measured resale ceiling and issuance weights matched to each maturity. Its thirty-year range is $[1.2, 2.5]$ basis points, the envelope of eighteen variants. They cross three tape windows (the full window and its two halves) with three scalings of the measured resale ceiling (0.5, 0.75 and 1.0), and the segment share is refitted for each scaling. Each of these nine combinations is computed under two issuance weightings of the par and zero-coupon instrument pairs, matched to each maturity or pooled across maturities. Two diagnostics that count more resales also leave the adjustment small. Weighting horizons by the ages of observed trades (conditional on observed sale), which include repeat sales and exclude bonds held to maturity, gives 4.8 basis points at thirty years. A homogeneous hazard in which every bond can trade ($\phi = 1$) gives 3.9. It lies above the range because it sells more long-maturity par than the measured resale ceiling allows.

Table 17: Capital-gains adjustment at selected maturities (bp, except $\phi$ and $P$(hold)). The reported column weights sale horizons by first-sale probabilities, with the eighteen-variant envelope as its range. The cond. on sale and homogeneous columns are the diagnostics. The max column is the largest correction $\Delta_{\mathrm{cg}}(h,T)$ across sale horizons. Columns $\phi$ and $P$(hold) give the trading-segment share and the held-to-maturity probability. At two years no sale horizon precedes maturity, so the adjustment is zero.

| $T$ | cond. on sale | max over $h$ | $\phi$ | $P$(hold) | homogeneous | reported | range |
|---|---|---|---|---|---|---|---|
| 2 | 0.0 | 0.0 | – | 1.00 | 0.0 | 0.0 | [0.0, 0.0] |
| 4 | 17.2 | 17.2 | 1.00 | 0.60 | 6.9 | 6.9 | [4.4, 7.0] |
| 8 | 17.7 | 19.9 | 0.88 | 0.40 | 11.3 | 10.7 | [5.2, 11.3] |
| 10 | 15.7 | 19.0 | 0.82 | 0.37 | 10.8 | 9.9 | [4.5, 10.5] |
| 20 | 8.8 | 11.6 | 0.57 | 0.43 | 7.2 | 4.1 | [2.0, 4.4] |
| 30 | 4.8 | 6.6 | 0.60 | 0.40 | 3.9 | 2.3 | [1.2, 2.5] |

### C.1.5 Adjustment IV: liquidity

**What it measures.** Investors who expect to trade require compensation for trading costs. If municipal bonds cost more to trade than comparable corporate bonds, part of the municipal spread compensates for liquidity. The adjustment is small because comparable municipal and corporate bonds cost about the same to

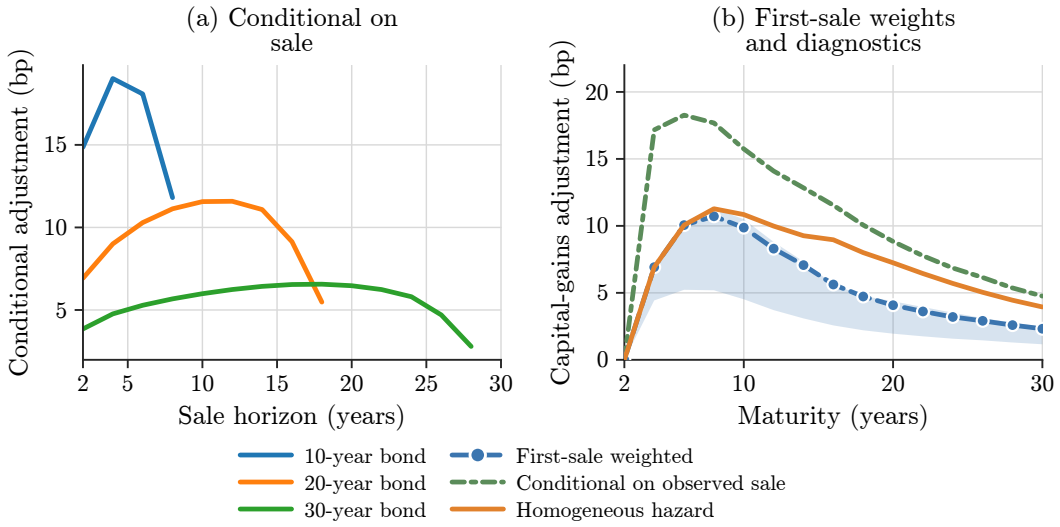



Figure 18: Capital-gains adjustment, averaged over the seven periods. (a) Issuance-weighted correction $\Delta_{\text{cg}}(h, T)$ conditional on a sale at horizon $h$, for ten-, twenty- and thirty-year bonds. (b) Adjustment weighted by first-sale probabilities, with the range of its eighteen variants, and the conditional-on-observed-sale and homogeneous-hazard diagnostics.

trade. Municipal costs are slightly lower through twenty years, higher at twenty-one to twenty-five years and almost identical at twenty-six to thirty years.

**Calculation.** I compare institutional-size trades of at least $250,000 in MSRB municipal and TRACE corporate data for 2005–2020, made more than three months after issue and more than one year before maturity. The imputed round-trip cost is (max − min)/ max price within CUSIP × day × size cells that contain both customer buys and customer sells. Costs are matched by month × rating family × maturity bin, with at least ten bonds per market in each cell. A round-trip cost is paid once per holding period, while a yield is earned every year. Hence, with matched municipal and corporate round-trip costs $\text{RTC}^M$ and $\text{RTC}^C$ in percent of price and a holding period of $h$ years, the yield differential in basis points is

$$\Delta y^{\text{liq}} = \frac{100(\text{RTC}^M - \text{RTC}^C)}{h}.$$

The baseline uses $h = 5$ in both markets, so each percentage point of round-trip cost is worth 20 basis points of annual yield. Trading costs reduce after-tax wealth in both markets, so the differential is not multiplied by an interest-income after-tax share.

**Why the estimate is reliable.** Two regression designs support the conversion. A regression of A-or-above municipal offering yields on bond-average round-trip costs, within maturity-bin × rating × offer-year cells, gives 15.4 basis points per percentage point of cost, close to the 20 of the baseline conversion. Issuer-level regressions (Table 19) show that liquidity is priced within the municipal market. At twenty-six to thirty years, more liquid issuers have offering yields about 8–15 basis points lower, and the cost coefficient implies 27.3 basis points per percentage point. Applied to the 26–30 year cost difference of 0.0025 percentage points, even this larger slope gives about 0.07 basis points.

**Result and sensitivity.** At twenty-six to thirty years, municipal and corporate costs are 0.5335% and 0.5310% of price (Table 18). The difference gives 0.050 basis points annually under the baseline, reported as 0.1, and 0.04 under the regression map. The thirty-year band, $[-2.2, 2.3]$ basis points, spans holding periods of two and ten years, the regression map and ±1.96 standard errors. The largest municipal cost excess is in the twenty-one-to-twenty-five-year bin, where the adjustment is 3.57 basis points, about 1.5 standard errors from zero. To obtain $\text{Liq}(T)$, I place the six bin estimates at 3, 8, 13, 18, 23 and 28 years and interpolate linearly to the two-year grid, holding end values flat. The thirty-year adjustment is therefore the 26–30 year estimate. The interpolated adjustment peaks at 2.9 basis points at twenty-four years (Figure 19). The ten-bond minimum per cell avoids thin cells. With a one-bond minimum, the 26–30 year estimate becomes −6.53 basis points, which would raise the residual before the model, while a twenty-bond minimum gives 0.30.

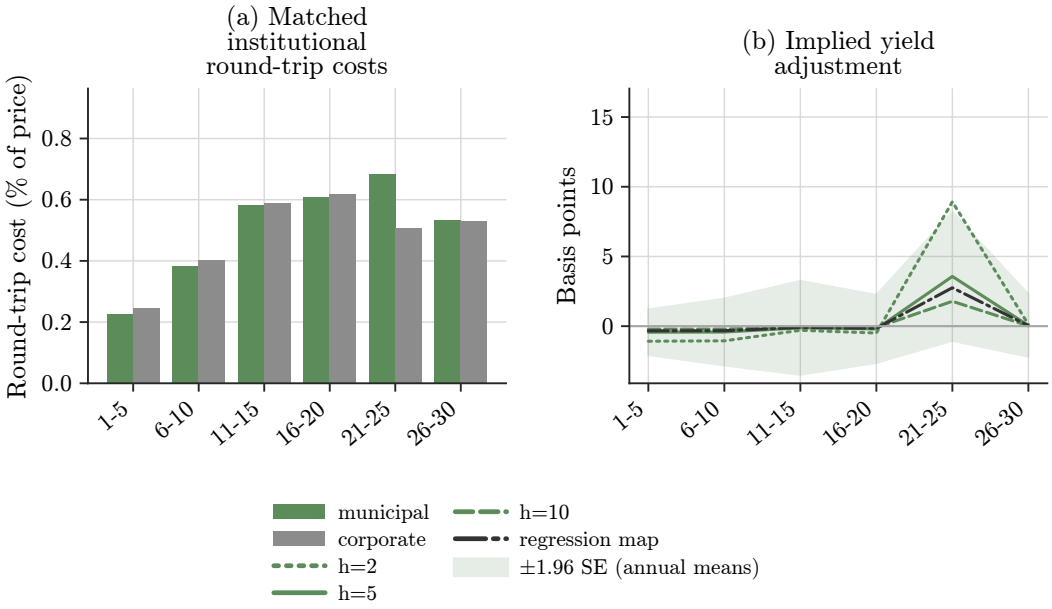



Figure 19: Liquidity adjustment from matched institutional-size customer trades of at least $250,000, 2005–2020. Panel (a): imputed round-trip costs in common month, rating and maturity cells, percent of price. Panel (b): the municipal-minus-corporate cost difference annualized under three holding periods and under the regression map (15.4 basis points per percentage point of cost). Shading is ±1.96 standard errors calculated across annual means at $h = 5$.

Table 18: Liquidity by remaining-maturity bin (years), matched trades of 2005–2020. Round-trip costs (RTC) and their municipal-minus-corporate difference in percent of price, and adjustments in bp under three holding periods $h$ and the regression map (regr., 15.4 bp per percentage point of cost).

| Bin | Municipal RTC | Corporate RTC | Difference | $h = 2$ | $h = 5$ | $h = 10$ | regr. |
|---|---|---|---|---|---|---|---|
| 1–5 | 0.225 | 0.246 | -0.022 | -1.08 | -0.43 | -0.22 | -0.33 |
| 6–10 | 0.382 | 0.403 | -0.021 | -1.05 | -0.42 | -0.21 | -0.33 |
| 11–15 | 0.582 | 0.588 | -0.006 | -0.30 | -0.12 | -0.06 | -0.09 |
| 16–20 | 0.609 | 0.619 | -0.010 | -0.49 | -0.20 | -0.10 | -0.15 |
| 21–25 | 0.684 | 0.506 | 0.178 | 8.92 | 3.57 | 1.78 | 2.75 |
| 26–30 | 0.533 | 0.531 | 0.003 | 0.13 | 0.05 | 0.03 | 0.04 |

**Issuer-level evidence.** Table 19 asks whether more liquid municipal issuers have lower offering yields, controlling for maturity, rating, state, issue month and call status. Trading frequency averages, across an issuer's bonds, each bond's mean daily MSRB trades from its dated date to maturity. The spread measure averages each bond's mean imputed round-trip cost in the same way, in price units, so 0.01 is 1% of price. The coefficients are not added as a separate adjustment.

### C.1.6 Adjustment V: default

**What it measures.** Both curves contain compensation for default risk. If municipal credit compensation exceeds corporate credit compensation, part of the municipal spread is a default premium. I measure each market's credit compensation with credit default swap (CDS) spreads, the annual premium for protection against default.

**Calculation.** A CDS premium is taxable income to the protection seller, so CDS spreads measure credit compensation in pretax units. Following the taxable-equivalent convention of Schwert (2017), I therefore treat them as credit compensation in the comparison $y^M/\xi_H - y^C$. Multiplying this comparison by the household's after-tax share gives the puzzle, $y^M - \xi_H y^C = \xi_H(y^M/\xi_H - y^C)$, so the CDS difference receives the same multiplier. With matched municipal and corporate spreads $s_b^M$ and $s_b^C$ for curve observation $b$, and its dated household after-tax share $\xi_{H,b}$, the adjustment is

$$\mathrm{Def}(T) = \mathrm{Avg}_T[\xi_{H,b} s_b^M] - \mathrm{Avg}_T[\xi_{H,b} s_b^C], \tag{123}$$

Table 19: Issuer liquidity and offering yield, by maturity bin.

| | Trading frequency | | Transaction-cost spread | |
|---|---|---|---|---|
| | (1) Above corp median | (2) Top 25% trade | (3) Issuer-avg spread (cont.) | (4) Lowest-spread quartile |
| Maturity 1–5y | $-0.046^{***}$ | $-0.028^{***}$ | $13.446^{***}$ | $-0.057^{***}$ |
| | (0.009) | (0.004) | (1.200) | (0.003) |
| Maturity 6–10y | −0.007 | $0.021^{***}$ | $11.624^{***}$ | $-0.048^{***}$ |
| | (0.009) | (0.005) | (1.415) | (0.005) |
| Maturity 11–15y | −0.011 | $0.041^{***}$ | $11.860^{***}$ | $-0.040^{***}$ |
| | (0.009) | (0.005) | (0.830) | (0.005) |
| Maturity 16–20y | $-0.035^{***}$ | $0.024^{***}$ | $13.447^{***}$ | $-0.043^{***}$ |
| | (0.010) | (0.006) | (1.084) | (0.006) |
| Maturity 21–25y | $-0.091^{***}$ | $-0.041^{***}$ | $20.427^{***}$ | $-0.104^{***}$ |
| | (0.019) | (0.015) | (2.588) | (0.016) |
| Maturity 26–30y | $-0.093^{***}$ | $-0.082^{***}$ | $27.262^{***}$ | $-0.146^{***}$ |
| | (0.023) | (0.027) | (4.383) | (0.026) |
| Fixed effects | Yes | Yes | Yes | Yes |
| Issuer cluster | Yes | Yes | Yes | Yes |
| Observations | 785,256 | 785,256 | 779,988 | 785,256 |

*Notes.* AA-and-above federally exempt issues, 2000–2021. The outcome is the offering yield in percent, so −0.046 is −4.6 bp. Each row reports, for that maturity bin, the coefficient on the column's issuer-average liquidity measure: (1) an indicator that daily MSRB trades per bond exceed the TRACE median for plain-vanilla corporate debentures and medium-term notes; (2) an indicator for the top quartile of municipal trade frequency; (3) the continuous (cont.) spread, the imputed round-trip cost in price units, so 13.4 is 13.4 bp per 0.01 of spread; (4) an indicator for the lowest-spread quartile. Fixed effects interact rounded maturity, rating rounded down, state, issue year–month and call status. Standard errors are clustered by issuer (20,346 issuers; 19,806 in column (3)). $^{*}\,p<0.10$, $^{**}\,p<0.05$, $^{***}\,p<0.01$.

where $\mathrm{Avg}_T$ is the three-stage averaging of Appendix C.1.1 with whole-notch in place of half-notch rating cells. A positive adjustment means that municipal credit compensation explains part of the puzzle.

State general-obligation (GO) Markit CDS supply the municipal spread. Each state GO issue is matched to its own state's CDS spread at the issue's maturity in its offering month, and the matched spreads are averaged within quarter × rating × maturity cells. A-or-above curve observations then receive the spread of their cell, whatever their state, when their quarter and rating contain a matched issue. Beyond a state's longest quoted CDS tenor, the cumulative spread $T\,\mathrm{CDS}(T)$ is extrapolated linearly. Within a quarter and rating, maturities without a matched issue are filled by linear interpolation of $T\,\mathrm{CDS}(T)$ between matched maturities and by its linear extrapolation beyond the longest. Below the shortest matched maturity, and at every maturity when only one is matched, the spread is held flat. Exactly observed cells, those with a matched issue, cover 49% of the panel at 1–10 years, 47% at 11–20 and 13% at 21–30 (Figure 20(c)). The corporate spread weights pooled AAA, AA and A Markit CDS by each year's par shares of corporate bonds with at least $250 million outstanding (on average 8.6%, 22.8% and 68.6% over 2000–2020), so the credit comparison has the same rating mix as the corporate yield curve. The CDS sample begins in 2004 and contains 1,559,971 curve observations. On this sample, the thirty-year spread over the household benchmark is 165.7 basis points, against 159.8 in the full sample. The full-sample accounting uses the adjustment estimated on this sample, maturity by maturity, without rescaling.

**Result and sensitivity.** The default adjustment is positive at every maturity and explains 12.2 basis points of the thirty-year puzzle (Figure 20(b)). It is largest at two years, 14.7 basis points, where state GO CDS spreads average 64 basis points against 38 for the corporate blend. Its range, $[6.1, 16.4]$ basis points, spans five variants: the baseline, cells with at least half of their observations quoted or interpolated, the sample from 2007 and CDS-to-cash basis factors of 0.75 and 0.50, which scale both CDS proxies (Table 20). The baseline lies between a pooled AA corporate leg, which gives 16.5 basis points, and a pooled A leg, which gives 6.9.

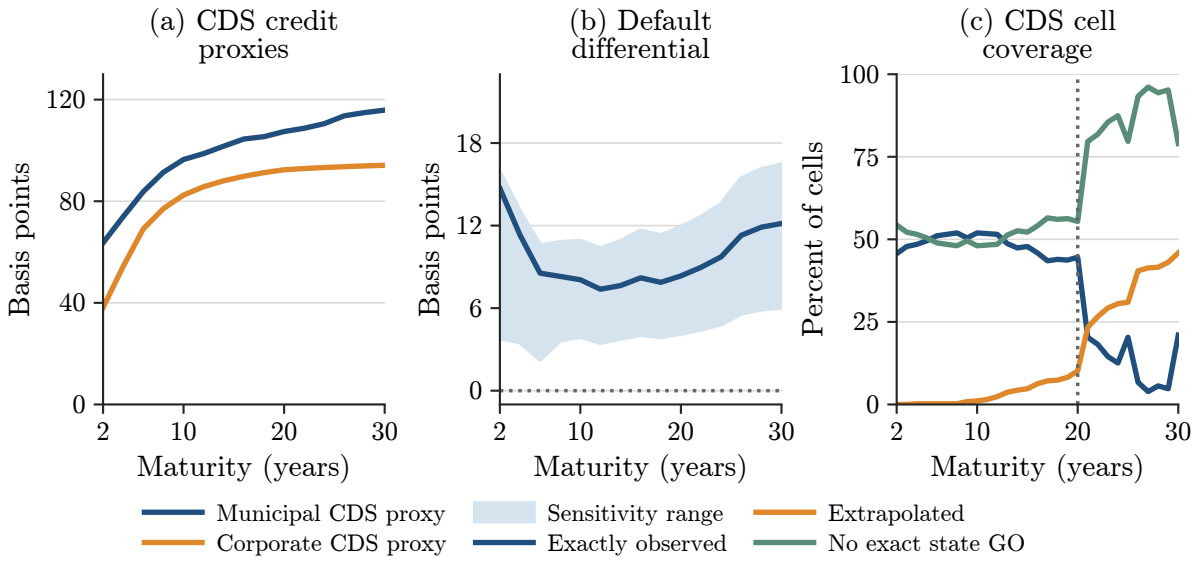



Figure 20: Default adjustment on the A-or-above CDS panel. (a) Municipal and corporate CDS proxies in taxable-equivalent units. (b) Their differential, with each leg multiplied by the dated household after-tax share as in (123), and its five-variant sensitivity range. (c) Shares of rating-matched CDS cells that are exactly observed, extrapolated or without a state GO observation (no exact state GO). The last two overlap, so the three shares are not a partition. The vertical rule marks twenty years.

Table 20: Default sensitivity in basis points. Every column multiplies both CDS legs by the dated household after-tax share, as in (123). The 2007+ column starts the sample in 2007, the basis columns scale both CDS proxies by the stated factor, and the quality-capped column keeps cells with at least half of their source observations quoted or interpolated.

| $T$ | 2007+ | basis 0.50 | basis 0.75 | baseline | quality-capped |
|---|---|---|---|---|---|
| 2 | 15.79 | 7.35 | 11.03 | 14.71 | 3.85 |
| 6 | 10.53 | 4.27 | 6.41 | 8.54 | 2.31 |
| 10 | 10.85 | 4.03 | 6.05 | 8.06 | 3.95 |
| 14 | 10.83 | 3.82 | 5.73 | 7.64 | 4.00 |
| 18 | 11.25 | 3.94 | 5.91 | 7.87 | 4.53 |
| 22 | 12.59 | 4.48 | 6.72 | 8.97 | 6.84 |
| 26 | 15.37 | 5.65 | 8.47 | 11.29 | 12.64 |
| 30 | 16.40 | 6.08 | 9.11 | 12.15 | 13.85 |

### C.1.7 Comparing the model with the adjusted data

The corporate normalization of Section 5.3 applies the model's municipal-to-taxable yield ratio to the observed corporate yield and subtracts the observed household benchmark. With bars denoting sample averages and superscripts $m$ for model and $d$ for data,

$$\widetilde{y}_T^M = \frac{\bar{y}_T^{M,m}}{\bar{y}_T^{C,m}} \bar{y}_T^{C,d}, \qquad \widetilde{\mathcal{G}}_T = \widetilde{y}_T^M - B_T^d.$$

Here $\bar{y}_T^{C,d}$ is the corporate curve $y_T^C$ of (115), and $B_T^d = \bar{\xi}_H(T)\bar{y}_T^{C,d}$ is the observed household benchmark. All yields use nominal semiannual units, and dated after-tax yields are formed before the averaging of Appendix C.1.1. Figure 8 and Table 4 report the normalized comparison. With the same averaging and issue-quarter rate states, the unnormalized gaps, on the model's own taxable curve, are −0.2, 36.2 and 107.4 basis points at two, ten and thirty years. The thirty-year value differs from the 102.4 basis points of Section 5.2, which averages the six calibration periods equally, because this averaging follows the 2000–2020 bond sample.

Across the fifteen maturities, the normalized root-mean-square unexplained residual is 7.53 basis points (12.94 unnormalized), and the maximum absolute residual is 10.72 at sixteen years. Fourteen of fifteen absolute residuals are below 10 basis points. The signed residuals are −7.83 at twelve years, −6.72 at twenty and +5.62 at thirty.

## C.2 Issuer-matched Build America Bonds and tax-exempt bonds

Build America Bonds (BABs) are taxable bonds sold by the same issuers as tax-exempt municipal bonds. Section 3.8 compares the two to check whether the widening gap over the household benchmark survives within issuers when the BAB yield replaces the corporate benchmark. The gap widens with maturity in every specification.

**Design and estimating equation.** The Mergent offering sample, with one observation per CUSIP, contains non-AMT bonds issued in 2009Q2–2010Q4. Table 21 uses AA-and-above (agency score $\leq 3$) and A-or-above ($\leq 7$) issues. Figure 3 uses the AA-and-above issues and averages yields by bond type and integer maturity $n \in \{1, \ldots, 30\}$. Multiplying BAB offering yields by 0.65, one minus the period's top federal household rate, puts them on the top-bracket household's federal after-tax scale. With $y_b$ the tax-exempt offering yield or the scaled BAB yield, $EXMP_b$ a tax-exempt indicator and $n_b$ maturity in years, I estimate

$$y_b \;=\; \alpha_1\, EXMP_b \;+\; \beta\,(EXMP_b \times n_b) \;+\; \varphi\,(0.65\, y^{C}_{n_b,t_b}) \;+\; \lambda_{j(b)\times n_b \times q_b} \;+\; \varepsilon_b. \tag{124}$$

The fixed effect compares bonds of issuer $j(b)$ with the same maturity and issue quarter $q_b$, so $\beta$ is the change per year of maturity in the tax-exempt yield minus the scaled BAB yield. The corporate yield $y^{C}_{n_b,t_b}$, measured at the bond's maturity and issue month $t_b$, varies across issue months within matched quarters, which identifies $\varphi$. Equation (4) of Section 3.8 uses the unscaled corporate yield as its control, so the estimates differ only in that control's coefficient, $\varphi_C = 0.65\varphi$, and its standard error.

**Identification.** Within issuers, the widening gap supports the participation mechanism if the 0.65 scaling is accurate and the two bond types have comparable security design and issuance premia. In the AA-and-above sample, optional calls appear on 49% of BABs and 39% of tax-exempt bonds, so the call-matched columns compare only bonds with the same call status.

**Results.** All maturity interactions in Table 21 are positive and significant at one percent, and the Panel B bin coefficients rise with maturity.

**Robustness.** Call matching lowers the longest-bin coefficient by about 15% in the AA-and-above sample and leaves the $[5, 10)$ coefficient unchanged. An exact after-tax internal rate of return would treat original-issue discount, acquisition premium, interest income and basis separately for BABs and exempt bonds. Near-par bonds have little discount or premium, so I also check the 0.65 scaling on them. Restricting the call-matched AA-and-above sample to offering prices of 99–101 leaves a maturity gradient of 2.1 basis points per year (s.e. 0.6), against 3.1 in the full call-matched sample. The 98–102 restriction gives 2.7 basis points (s.e. 0.4). Both remain significant at one percent. The scaling uses the 35% top federal rate, below the 41.1% average household wedge of the AA-and-above curve (Appendix Figure 27). Scaling BAB yields by the lower after-tax share implied by that wedge would widen the gap.

## C.3 The municipal puzzle on unlimited-tax general-obligation bonds

As Section 3.6 reports, the implied tax also declines with maturity among high-grade general-obligation (GO) bonds.

**Design.** Unlimited-tax GO bonds pledge the issuer's full faith and credit and unlimited taxing power, while revenue bonds pledge project revenues. Figure 21 restricts the comparison to AA-and-above unlimited-tax GO bonds[27] passing the call adjustment's calendar and price screens (Appendix C.1.2) and applies only the call and issuance-premium adjustments. Appendix C.1.6 separately assesses the aggregate default adjustment, which uses state-government CDS as a proxy.

**Result.** The adjusted GO implied tax falls from 43.1% at two years to 26.0% at thirty. The 17.1-point decline matches that of the A-or-above sample and exceeds the 14.6-point decline of the AA-and-above sample. The result agrees with earlier evidence that the maturity pattern persists after default and liquidity adjustments and in high-grade or GO samples (Longstaff, 2011; Fleckenstein and Longstaff, 2025; Chalmers, 1998).

**Robustness.** The screens exclude 4,029 GO bonds (0.86%). Holding the issuance premium fixed, they change the decline by 0.49 points relative to the unscreened call-adjusted GO series, almost entirely at the short end.

[27] Mergent security code "K".

Table 21: Issuer-matched tax-exempt and BAB yields: the gap widens with maturity.

| | AA and above (score $\leq 3$) | | A or above (score $\leq 7$) | |
|---|---|---|---|---|
| | (1) base | (2) call-matched | (3) base | (4) call-matched |
| *Panel A. Continuous maturity* | | | | |
| Tax-exempt | $-0.102^{***}$ | $-0.091^{**}$ | $-0.062^{**}$ | $-0.043$ |
| | (0.034) | (0.036) | (0.027) | (0.030) |
| Tax-exempt $\times\, n_b$ | $0.034^{***}$ | $0.031^{***}$ | $0.033^{***}$ | $0.030^{***}$ |
| | (0.003) | (0.003) | (0.002) | (0.003) |
| After-tax corporate yield | $0.379^{***}$ | $0.376^{***}$ | $0.264^{*}$ | $0.261^{*}$ |
| | (0.138) | (0.135) | (0.134) | (0.130) |
| *Panel B. Five-year maturity bins* | | | | |
| Tax-exempt | $-0.048$ | $-0.050$ | $-0.007$ | $-0.008$ |
| | (0.033) | (0.033) | (0.024) | (0.024) |
| Tax-exempt $\times\, [5, 10)$ y | $0.192^{***}$ | $0.191^{***}$ | $0.192^{***}$ | $0.190^{***}$ |
| | (0.022) | (0.022) | (0.017) | (0.017) |
| Tax-exempt $\times\, [10, 15)$ y | $0.396^{***}$ | $0.364^{***}$ | $0.368^{***}$ | $0.342^{***}$ |
| | (0.033) | (0.033) | (0.026) | (0.030) |
| Tax-exempt $\times\, [15, 20)$ y | $0.450^{***}$ | $0.390^{***}$ | $0.419^{***}$ | $0.403^{***}$ |
| | (0.041) | (0.039) | (0.029) | (0.039) |
| Tax-exempt $\times\, [20, 25)$ y | $0.576^{***}$ | $0.540^{***}$ | $0.593^{***}$ | $0.510^{***}$ |
| | (0.061) | (0.066) | (0.071) | (0.066) |
| Tax-exempt $\times\, [25, 30]$ y | $0.755^{***}$ | $0.639^{***}$ | $0.996^{***}$ | $0.804^{***}$ |
| | (0.079) | (0.039) | (0.074) | (0.060) |
| After-tax corporate yield | $0.380^{***}$ | $0.378^{***}$ | $0.264^{*}$ | $0.262^{**}$ |
| | (0.138) | (0.135) | (0.134) | (0.130) |
| Issuer $\times$ maturity $\times$ quarter FE | Yes | Yes | Yes | Yes |
| $\times$ call status | No | Yes | No | Yes |
| Jurisdiction cluster | Yes | Yes | Yes | Yes |
| Observations | 29,839 | 28,890 | 61,816 | 59,886 |

*Notes.* FE denotes fixed effects. Offering yields in percent, 2009Q2–2010Q4, excluding AMT bonds. BAB yields are multiplied by 0.65. The corporate-yield row reports $\varphi$, the coefficient on the after-tax control in equation (124). The omitted bin is $[1, 5)$ years. Standard errors are clustered by issuing jurisdiction (51 in columns (1)–(2), 52 in columns (3)–(4)). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## C.4 The single-tax-rate benchmark of Green (1993)

Sections 2 and 3.7 cite Green (1993), whose single-tax-rate explanation attributes rising municipal-to-taxable yield ratios to the timing of tax on taxable bonds. His pricing relation need not bind, so I keep the household benchmark. Participation in the model also answers his critique of clientele explanations.

**Green's mechanism.** Green's package, which defers tax, buys two five-percent market-discount bonds and shorts one ten-percent par bond of equal maturity. The coupons cancel, so after purchase the package's only net cash flow is one dollar of principal at maturity. At tax rate $\tau$, the package costs $P_T$ and pays $1 - \tau(1 - P_T)$ after tax at maturity. With $d_T$ the price of a dollar paid at $T$,

$$P_T = d_T\big[1 - \tau(1 - P_T)\big], \qquad \text{so} \qquad P_T = \frac{(1-\tau)\, d_T}{1 - \tau d_T}.$$

Asymmetric taxation of long and short positions limits the strategy, because borrowing and short-sale deductions cannot exceed the investment income they support. Dealers face no such limit and tie taxable par-bond prices to the package, which lowers long taxable yields below fully taxed returns. Green's high-tax investor prices both municipal bonds and the package, so one investor can price municipal bonds at every maturity while the yield ratio rises. In Fama (1977), different investors set prices at different maturities.

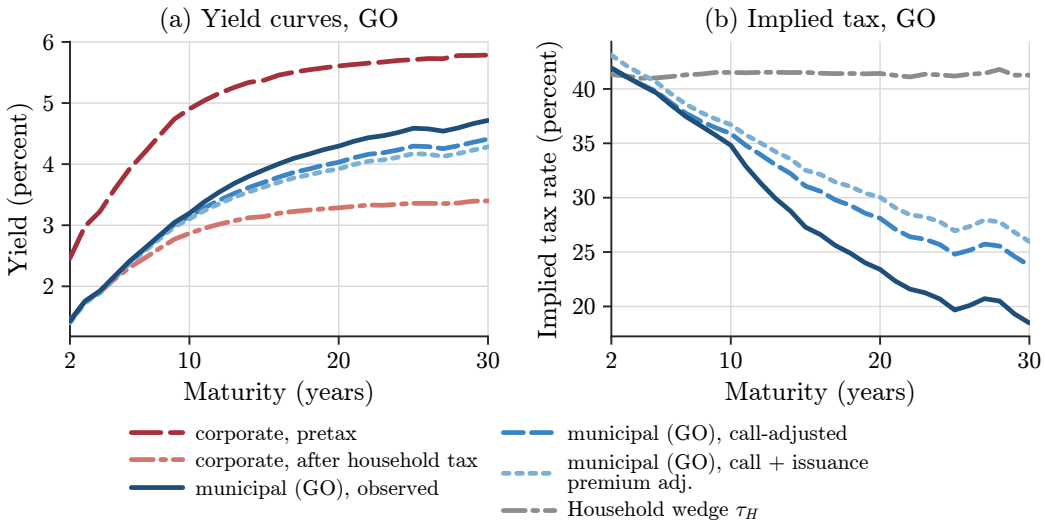



Figure 21: The municipal puzzle on AA-and-above unlimited-tax GO bonds: yields in panel (a) and the implied tax against the household wedge in panel (b). The 465,258 bonds pass the call adjustment's calendar and price screens, and curves use the sample's own weights and household benchmark. Call adjustments use the calibrated model. Few GO bonds have an early institutional-size purchase, so the issuance-premium adjustment applies the estimate from all bonds (Appendix C.1.3).

**Why the relation need not bind.** Deep market-discount bonds can be scarce, annual accrual of original-issue discount removes the deferral, and shorts consume deduction capacity. Households that forgo the package can accept the household benchmark $\xi_H y_T^C$, which lies below Green's relation. When they can absorb the supply, these households set municipal prices, and investors using the strategy do not. Raising municipal yields to Green's relation would require shorting municipal bonds, which faces the frictions in Appendix D.5. Green's own 1987–1990 estimates also leave thirty-year municipal yields about twenty taxable-equivalent basis points above his relation, with a gap that increases with maturity.

**Green's critique of clientele explanations.** Green rejects earlier clientele explanations because long municipal yields remained high relative to after-tax taxable yields after the Tax Reform Act of 1986 reduced banks' tax advantages and municipal holdings. In the model, participation depends on exemption benefits and the duration benefit (Section 4.3), so the investors holding each maturity can change. Households alone hold short maturities and set their yields. Institutions with smaller exemption benefits also hold long maturities, so long yields reflect these institutions' tax treatment too. Implied taxes therefore respond to supply and risk-bearing capacity. Section 3.4 documents ownership across exemption benefits. I keep the household benchmark $\xi_H y_T^C$ and measure tax timing separately in the capital-gains adjustment (Appendix C.1.4). To distinguish participation from tax timing alone, P1 tests household state-tax pass-through and P2 tests a bank-only benefit (Sections 6.1 and 6.2).

## C.5 State-tax pass-through: full regression tables

The P1 estimates of Section 6.1 use the sample, outcome and inference described below. Two tables then check whether the declining state-tax pass-through depends on the yield construction or on state credit risk. Both keep specification (26), and the decline survives both checks.

**Design.** The main sample is federally exempt AA-and-above issuance (agency score $\leq 3$) in 2000–2017 from the 47 states retained in Section 6.1, counting the District of Columbia as a state. It excludes zero-maturity bonds and groups maturities into $[0, 5)$, $[5, 10)$, $[10, 15)$, $[15, 20)$ and $[20, 30]$ years. Every regression keeps the controls of equation (26). The six lagged state controls and the rounded average rating form $Z_{b,t}$ and are interacted with all five bins, and log offering par and security-type indicators form $V_b$. Controls built from outstanding and newly issued bonds use all observed federally exempt bonds, regardless of rating. Outstanding par equals the offering amount until the date of the reported outstanding amount, and that amount afterward. Fixed-effect cells are defined by issue year, issue month, rounded maturity and callability. Observations with

missing controls are dropped without imputation, and singleton cells are then removed. The zero-coupon-yield columns have 681,838 observations, and six missing offering yields leave 681,832 in the offering-yield columns.

**Call-adjusted outcome.** The main outcome uses call-adjusted zero-coupon yields. As in the call adjustment (Appendix C.1.2), a constant spread over the calibrated model's supply-weighted municipal curve for the bond's issue period is fitted to each bond's observed callable price. Each bond starts at the observed rate state in the final month of its issue quarter. Unlike that adjustment, which follows the contractual call calendar, P1 places cash flows on a regular semiannual grid. Observed call dates map to the nearest grid date within 0.27 years, and the call is exercisable on every grid date from the grid date nearest the first call date. Callable bonds are excluded when the first call date lies more than 0.27 years before the first semiannual grid date or the call schedule lacks a usable date and price.

**Identification and inference.** Unweighted OLS is the main estimator. For the IV comparison, the 1990 rate interacted with each maturity bin instruments the corresponding current-rate interaction. The exclusion restriction is that, conditional on the controls and fixed effects, the historical rate affects current implied state taxes only through the current rate. The first stage gives a Kleibergen–Paap (KP) rk Wald $F$ of 10.4 and Sanderson–Windmeijer (SW) conditional statistics of 76.1, 115.9, 104.4, 155.7 and 230.4 for the current-rate interactions from the shortest to the longest bin. Standard errors are clustered by state, with $t(G-1)$ critical values for $G$ states ($t(46)$ in the main sample). In each table, the $p$-value of the short-to-long decline tests $\beta_1 = \beta_5$.

### C.5.1 Comparison across yield constructions

Table 22 compares Mergent offering yields, which for premium callable bonds can be yields to worst, with zero-coupon yields from offering prices, with and without call adjustment. For every construction, the short-to-long decline is significant at one percent under both OLS and IV, so neither the call adjustment nor the zero-coupon conversion drives it.

Table 22: P1 across yield constructions, 2000–2017.

| | Offering yield | | Call-adjusted zero | | Unadjusted zero | |
|---|---|---|---|---|---|---|
| | OLS | IV | OLS | IV | OLS | IV |
| $\beta_1$: $[0,5)$ years | $0.985^{***}$ | $1.084^{***}$ | $1.069^{***}$ | $1.088^{***}$ | $1.080^{***}$ | $1.102^{***}$ |
| | (0.238) | (0.249) | (0.195) | (0.225) | (0.196) | (0.227) |
| $\beta_2$: $[5,10)$ years | $0.635^{***}$ | $0.648^{***}$ | $0.677^{***}$ | $0.694^{***}$ | $0.683^{***}$ | $0.693^{***}$ |
| | (0.075) | (0.093) | (0.069) | (0.086) | (0.069) | (0.087) |
| $\beta_3$: $[10,15)$ years | $0.414^{***}$ | $0.410^{***}$ | $0.484^{***}$ | $0.447^{***}$ | $0.515^{***}$ | $0.479^{***}$ |
| | (0.053) | (0.075) | (0.096) | (0.118) | (0.100) | (0.117) |
| $\beta_4$: $[15,20)$ years | $0.268^{***}$ | $0.257^{***}$ | $0.390^{***}$ | $0.326^{***}$ | $0.428^{***}$ | $0.360^{***}$ |
| | (0.060) | (0.067) | (0.102) | (0.116) | (0.108) | (0.121) |
| $\beta_5$: $[20,30]$ years | $0.193^{***}$ | $0.152^{**}$ | $0.305^{***}$ | $0.229^{***}$ | $0.335^{***}$ | $0.255^{***}$ |
| | (0.052) | (0.057) | (0.075) | (0.081) | (0.078) | (0.085) |
| Decline $\beta_1 - \beta_5$ | $0.791^{***}$ | $0.933^{***}$ | $0.764^{***}$ | $0.859^{***}$ | $0.745^{***}$ | $0.847^{***}$ |
| $p$-value of decline | 0.004 | $< 0.001$ | 0.002 | $< 0.001$ | 0.003 | 0.001 |
| Issue month × maturity × call FE | Yes | Yes | Yes | Yes | Yes | Yes |
| State cluster | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 681,832 | 681,832 | 681,838 | 681,838 | 681,838 | 681,838 |
| First-stage $F$ (Kleibergen–Paap) | — | 10.4 | — | 10.4 | — | 10.4 |

*Notes.* FE denotes fixed effects. Outcomes are implied state taxes (25) from Mergent offering yields or from zero-coupon yields with and without the call adjustment. The call-adjusted zero is the main outcome. Controls, fixed effects and state-clustered inference (47 states) follow Appendix C.5, 2000–2017. IV uses bin-interacted 1990 rates. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### C.5.2 State credit-risk controls

Table 23 starts from the main sample, restricts it to pre-2018 observations that have a five-year state-government CDS quote in the preceding month and then adds that quote interacted with the maturity bins. Because the CDS quotes begin in 2004, this sample covers 2004–2017. On this common sample of 345,877 bonds from 25 states, the CDS controls raise the longest-bin OLS estimate from 0.335 to 0.416 and the IV estimate from 0.322 to 0.368. In every row, both estimators keep the longest-bin estimate far below the shortest-bin estimate, and both short-to-long declines are significant at five percent.

Table 23: P1 sample and credit controls, 2000–2017.

| Sample | Obs. | States | $\beta_1$, OLS | $\beta_5$, OLS | Decline $p$ | $\beta_1$, IV | $\beta_5$, IV | Decline $p$ |
|---|---|---|---|---|---|---|---|---|
| Main sample | 681,838 | 47 | $1.069^{***}$ | $0.305^{***}$ | 0.002 | $1.088^{***}$ | $0.229^{***}$ | $< 0.001$ |
| | | | | (0.075) | | | (0.081) | |
| Observed CDS sample | 345,877 | 25 | $1.210^{***}$ | $0.335^{***}$ | 0.009 | $1.349^{***}$ | $0.322^{***}$ | 0.009 |
| | | | | (0.081) | | | (0.098) | |
| CDS sample + CDS × bin | 345,877 | 25 | $1.400^{***}$ | $0.416^{***}$ | 0.011 | $1.391^{***}$ | $0.368^{***}$ | 0.025 |
| | | | | (0.068) | | | (0.087) | |

*Notes.* Controls, fixed effects and state-clustered inference follow Appendix C.5, 2000–2017. The observed-CDS sample keeps bonds with a prior-month five-year state-government CDS quote, without imputation, and covers 2004–2017. The last row adds that quote interacted with the maturity bins. IV uses bin-interacted 1990 rates. Parentheses give standard errors of $\beta_5$, and decline $p$-values test $\beta_1 = \beta_5$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## C.6 State-tax pass-through: instrument vintages, weighting and further checks

These checks ask whether P1 depends on the historical tax-rate instrument, the weight assigned to each state, fixed state tax levels or any single state, and they report the post-2018 tax regime separately. All retain the controls, fixed effects and state-clustered inference of Appendix C.5, and every check shows a declining profile.

**Instrument vintages.** The baseline IV instrument is the state's 1990 top income-tax rate, with the exclusion restriction of Appendix C.5. Tables 24 and 25 compare instruments from different years (vintages) in 1945–2000 on the same pre-2018 sample. Table 24 replaces the five current-rate bin interactions in (26) with the current rate and its interaction with maturity in years, $n_b$. The interaction's coefficient is the pass-through gradient per year of maturity. IV instruments the two current-rate terms with the historical rate and its maturity interaction. OLS and all instrument vintages give negative gradients, unweighted and with equal total weight per state. Unweighted, every vintage's gradient is significant at five percent, with KP $F$ statistics of 7.8 to 65.1. Table 25 compares the five-bin IV profiles with OLS, and Figure 22 plots them. Under every vintage, the short-to-long decline is significant at five percent.

Table 24: P1 maturity gradients by instrument vintage, 2000–2017.

| | | Unweighted | | | | Equal per state | |
|---|---|---|---|---|---|---|---|
| Vintage | Corr. | Gradient | s.e. | KP $F$ | SW $F$ | Gradient | KP $F$ |
| OLS | — | $-0.0392^{***}$ | (0.0131) | — | — | $-0.0326^{**}$ | — |
| 1945 | 0.40 | $-0.0535^{**}$ | (0.0251) | 8.0 | 19.2 | $-0.0440$ | 2.8 |
| 1950 | 0.31 | $-0.0607^{**}$ | (0.0273) | 7.8 | 18.4 | $-0.0878$ | 1.8 |
| 1960 | 0.38 | $-0.0620^{**}$ | (0.0254) | 9.0 | 28.8 | $-0.0997^{*}$ | 2.5 |
| 1970 | 0.51 | $-0.0574^{***}$ | (0.0201) | 11.5 | 45.2 | $-0.0795^{***}$ | 7.3 |
| 1980 | 0.76 | $-0.0424^{**}$ | (0.0164) | 10.6 | 103.4 | $-0.0385^{**}$ | 26.9 |
| 1990 | 0.82 | $-0.0440^{***}$ | (0.0127) | 30.1 | 404.1 | $-0.0307^{**}$ | 25.9 |
| 2000 | 0.88 | $-0.0399^{***}$ | (0.0124) | 65.1 | 998.5 | $-0.0258^{*}$ | 36.5 |

*Notes.* Controls, fixed effects and state-clustered inference follow Appendix C.5, 2000–2017. The gradient is the coefficient on the current rate times maturity $n_b$, the change in pass-through per year of maturity. IV instruments the current rate and its maturity interaction with the vintage's rate and its maturity interaction. KP $F$ is the Kleibergen–Paap statistic and SW $F$ the Sanderson–Windmeijer conditional statistic for the interaction. Corr. is the cross-state correlation between each state's mean current top rate over the estimation sample and its vintage-year top rate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 25: P1 five-bin profiles by instrument vintage, 2000–2017.

| Estimator/vintage | $\beta_1$ | $\beta_2$ | $\beta_3$ | $\beta_4$ | $\beta_5$ | Decline $p$ | KP $F$ |
|---|---|---|---|---|---|---|---|
| OLS (main) | $1.069^{***}$ | $0.677^{***}$ | $0.484^{***}$ | $0.390^{***}$ | $0.305^{***}$ | 0.002 | — |
| | (0.195) | (0.069) | (0.096) | (0.102) | (0.075) | | |
| 1945 | $1.257^{***}$ | $0.868^{***}$ | $0.551^{***}$ | $0.363^{**}$ | $0.245^{**}$ | 0.027 | 3.1 |
| | (0.421) | (0.158) | (0.158) | (0.143) | (0.103) | | |
| 1950 | $1.453^{***}$ | $0.942^{***}$ | $0.622^{***}$ | $0.428^{***}$ | $0.313^{***}$ | 0.024 | 3.1 |
| | (0.473) | (0.181) | (0.153) | (0.143) | (0.116) | | |
| 1960 | $1.483^{***}$ | $0.942^{***}$ | $0.627^{***}$ | $0.432^{***}$ | $0.332^{***}$ | 0.014 | 3.6 |
| | (0.439) | (0.167) | (0.147) | (0.140) | (0.113) | | |
| 1970 | $1.394^{***}$ | $0.858^{***}$ | $0.566^{***}$ | $0.400^{***}$ | $0.325^{***}$ | 0.006 | 4.4 |
| | (0.350) | (0.125) | (0.143) | (0.141) | (0.114) | | |
| 1980 | $1.084^{***}$ | $0.756^{***}$ | $0.528^{***}$ | $0.374^{***}$ | $0.280^{***}$ | 0.010 | 3.8 |
| | (0.273) | (0.083) | (0.121) | (0.120) | (0.084) | | |
| 1990 | $1.088^{***}$ | $0.694^{***}$ | $0.447^{***}$ | $0.326^{***}$ | $0.229^{***}$ | $< 0.001$ | 10.4 |
| | (0.225) | (0.086) | (0.118) | (0.116) | (0.081) | | |
| 2000 | $1.072^{***}$ | $0.678^{***}$ | $0.480^{***}$ | $0.378^{***}$ | $0.280^{***}$ | 0.001 | 23.5 |
| | (0.215) | (0.079) | (0.095) | (0.096) | (0.069) | | |

*Notes.* Controls, fixed effects and state-clustered inference follow Appendix C.5, 2000–2017. IV uses the vintage-year top rate interacted with each bin. KP $F$ is the Kleibergen–Paap statistic, and decline $p$-values test $\beta_1 = \beta_5$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

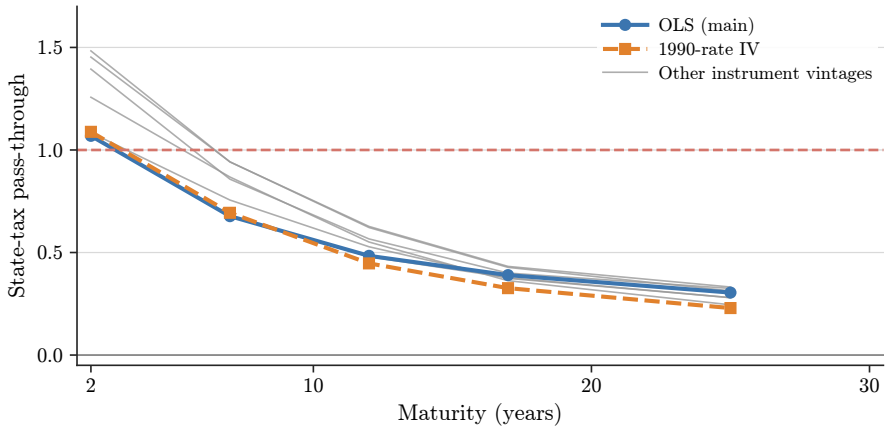



Figure 22: State-tax pass-through using historical tax-rate instruments, 2000–2017. Blue: main OLS profile. Orange: IV using the 1990 rate. Gray: IV using the 1945, 1950, 1960, 1970, 1980 and 2000 rates. Dashed red: representative-tax benchmark. Table 25 reports state-clustered standard errors.

**Weighting.** Unweighted OLS gives states with more bonds more influence. The alternatives in Table 26 give equal total weight to each state or to each state-year, or weight each bond by $1/\sqrt{n_{\text{state}}}$, with $n_{\text{state}}$ the number of observations in the bond's state. All four weightings retain a declining OLS profile, with the short-to-long decline significant at one percent.

**State fixed effects.** State fixed effects absorb fixed tax levels. Adding them to the pre-2018 OLS specification, on 681,838 bonds, gives a short-bin coefficient of 0.665 (s.e. 0.201) and a long-bin coefficient of −0.112 (0.212). The decline is 0.777 (0.220, $p = 0.001$). IV is not identified with state effects. The five excluded instruments, the 1990 rate interacted with each maturity bin, sum to the state's 1990 rate, which is collinear with the state effects.

**Single-state omissions.** All 47 single-state omissions retain a positive short-to-long decline under both OLS and IV (Figure 23). OLS long-bin estimates span 0.232–0.482, with declines 0.452–0.889. IV long-bin estimates span 0.116–0.315, with declines 0.669–1.004. The largest $p$-values of the decline are 0.062 for OLS and 0.036 for IV.

Table 26: P1 OLS estimates under four weighting schemes, 2000–2017.

| Weighting | $\beta_1$ | $\beta_2$ | $\beta_3$ | $\beta_4$ | $\beta_5$ | Decline $p$ |
|---|---|---|---|---|---|---|
| Unweighted (main) | $1.069^{***}$ | $0.677^{***}$ | $0.484^{***}$ | $0.390^{***}$ | $0.305^{***}$ | 0.002 |
| | (0.195) | (0.069) | (0.096) | (0.102) | (0.075) | |
| Equal per state | $0.863^{***}$ | $0.543^{***}$ | $0.374^{***}$ | $0.308^{***}$ | $0.234^{**}$ | 0.006 |
| | (0.191) | (0.082) | (0.101) | (0.111) | (0.088) | |
| Equal per state-year | $0.911^{***}$ | $0.509^{***}$ | $0.317^{***}$ | $0.261^{***}$ | $0.211^{***}$ | 0.002 |
| | (0.190) | (0.069) | (0.080) | (0.090) | (0.074) | |
| $1/\sqrt{n_{\text{state}}}$ | $0.990^{***}$ | $0.617^{***}$ | $0.420^{***}$ | $0.344^{***}$ | $0.282^{***}$ | 0.006 |
| | (0.202) | (0.080) | (0.109) | (0.121) | (0.091) | |

*Notes.* OLS with the controls, fixed effects and state-clustered inference of Appendix C.5, 2000–2017. The equal-weight rows give each state or state-year the same total weight, and $n_{\text{state}}$ counts the observations in the bond's state, all measured after dropping observations with missing controls and singleton cells. Decline $p$-values test $\beta_1 = \beta_5$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

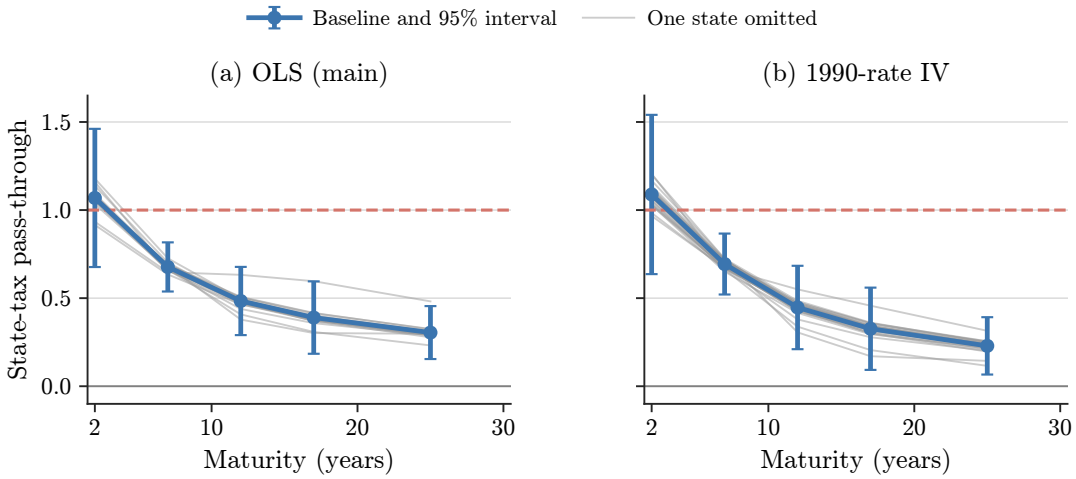



Figure 23: P1 profiles with single-state omissions, 2000–2017. Gray: each of the 47 state omissions. Blue: baseline estimates with state-clustered 95% intervals. Panel (a) uses OLS and panel (b) the 1990-rate instruments. Dashed: representative-tax benchmark.

**The two tax regimes.** The main P1 sample ends in 2017, so its after-tax share of taxable interest, $\xi_s^{\text{st}}\xi_t^{\text{fed}}$, follows the same deduction convention as the model's six pre-2018 periods in Figure 9. From 2018, the TCJA capped state and local tax (SALT) deductions at \$10,000 per return. When the cap binds, an additional dollar of state tax is no longer federally deductible. The combined after-tax share on taxable interest then becomes $\xi_t^{\text{fed}} - \tau_s^{\text{st}}$, with $\xi_t^{\text{fed}} = 0.63$ from the 37% top rate (notation of Section 6.1). For a household whose cap binds, representative-tax indifference gives $y_{s,t}^M(T) = (\xi_t^{\text{fed}} - \tau_s^{\text{st}})\, y_t^C(T)$. The implied state tax for this regime, which equals the statutory rate under the benchmark, is[28]

$$\mathit{ImpliedStateTax}_{s,t}^{\text{cap}}(T) \;\equiv\; \xi_t^{\text{fed}} - \frac{y_{s,t}^M(T)}{y_t^C(T)}. \tag{125}$$

Table 27 estimates each regime separately. In each period, the short-to-long decline is significant at one percent under both estimators.

## C.7 The bank-qualified discontinuity: construction and robustness

Bank qualification changes banks' tax benefit without changing households' exemption, so comparing bank-qualified with nonqualified bonds measures what share of this bank-only benefit is reflected in yields at each maturity. P2 predicts that this capitalization share rises with maturity. In Table 28, the 2SLS designation

[28] See the IRS, 2017 Schedule A instructions, and its summary of the 2018 changes.

Table 27: P1 before and after the 2018 SALT deduction cap.

| | 2000–2017 (main) | | 2018–2020 | |
|---|---|---|---|---|
| | OLS | IV | OLS | IV |
| $\beta_1$: $[0, 5)$ years | $1.069^{***}$ | $1.088^{***}$ | $0.948^{***}$ | $0.909^{***}$ |
| | (0.195) | (0.225) | (0.182) | (0.246) |
| $\beta_2$: $[5, 10)$ years | $0.677^{***}$ | $0.694^{***}$ | $0.834^{***}$ | $0.879^{***}$ |
| | (0.069) | (0.086) | (0.121) | (0.143) |
| $\beta_3$: $[10, 15)$ years | $0.484^{***}$ | $0.447^{***}$ | $0.446^{***}$ | $0.466^{***}$ |
| | (0.096) | (0.118) | (0.061) | (0.074) |
| $\beta_4$: $[15, 20)$ years | $0.390^{***}$ | $0.326^{***}$ | $0.312^{***}$ | $0.323^{***}$ |
| | (0.102) | (0.116) | (0.078) | (0.105) |
| $\beta_5$: $[20, 30]$ years | $0.305^{***}$ | $0.229^{***}$ | 0.088 | 0.082 |
| | (0.075) | (0.081) | (0.091) | (0.109) |
| Decline $\beta_1 - \beta_5$ | $0.764^{***}$ | $0.859^{***}$ | $0.860^{***}$ | $0.827^{***}$ |
| $p$-value of decline | 0.002 | $< 0.001$ | $< 0.001$ | 0.008 |
| Issue month × maturity × call FE | Yes | Yes | Yes | Yes |
| State cluster | Yes | Yes | Yes | Yes |
| Observations | 681,838 | 681,838 | 146,174 | 146,174 |
| First-stage $F$ (Kleibergen–Paap) | — | 10.4 | — | 7.9 |

*Notes.* FE denotes fixed effects. Controls, fixed effects and state-clustered inference (47 states) follow Appendix C.5. IV uses bin-interacted 1990 rates. The outcome is (25) before 2018 and, for a household whose SALT deduction cap binds, (125) from 2018. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

premium in offering yields is 1.8 basis points at 1–2 years, indistinguishable from zero, and 18.4 at 10–20 years. With each yield divided by its own expected annual benefit before estimation, the estimated capitalization shares are 0.17 and 0.52. The gradient survives alternative windows, the construction choices of Dagostino (2025) and splits by benefit regime. A placebo cutoff at \$5 million shows essentially no designation jump.

### C.7.1 Capitalization of the bank benefit

The designation premium $\varpi_k = y_k^N - y_k^Q$, the nonqualified minus the bank-qualified yield, measures how much of the bank benefit $X$ is capitalized into maturity-$k$ yields. It is zero in the representative-tax benchmark, $X$ in the bank-only benchmark and $\theta_k X$ in general, where the capitalization share $\theta_k$, derived below, is the yield reduction as a fraction of the benefit.

**The two benchmarks.** Both benchmarks compare banks' annual after-tax income per dollar of tax basis on otherwise identical bonds, in the notation of Section 6.2, with dates suppressed and taxable yield $y_k^C > 0$ at maturity $k$.

In the representative-tax benchmark, $y_k^N = (1 - \tau_H) y_k^C$, where $\tau_H = 1 - \xi_H$ is the household wedge. At this yield, a designated bond earns the bank $y_k^N - 0.2\,\tau_B c$ after the tax cost of its disallowed interest deduction. A taxable bond earns $(1 - \tau_B) y_k^C$, so the taxable bond has an advantage of

$$\underbrace{(1 - \tau_B) y_k^C}_{\text{taxable alternative}} - \underbrace{\left[(1 - \tau_H) y_k^C - 0.2\,\tau_B c\right]}_{\text{qualified bond at the household yield}} = (\tau_H - \tau_B) y_k^C + 0.2\,\tau_B c. \tag{126}$$

The advantage is positive whenever $\tau_H > \tau_B - 0.2\,\tau_B c / y_k^C$, in particular when household rates are no lower than banks' and financing costs are positive. Banks then hold neither municipal bond, and designation leaves household taxes unchanged, so $y_k^Q = y_k^N$ and $\varpi_k = 0$ at every maturity.

In the bank-only benchmark, each municipal bond gives banks the same after-tax income as taxable bonds:

$$\begin{aligned} y_k^N - \underbrace{\tau_B c}_{\text{full disallowance}} &= (1 - \tau_B) y_k^C, \\ y_k^Q - \underbrace{0.2\,\tau_B c}_{\text{qualified disallowance}} &= (1 - \tau_B) y_k^C. \end{aligned} \tag{127}$$

Table 28: Bank-qualified designation and yield premia by maturity.

| | (1) | (2) | (3) |
|---|---|---|---|
| Dependent variable | BQ designation | Offering yield (bp) | Yield / expected benefit |
| Estimator | OLS | 2SLS | 2SLS |
| Regressor, interacted with each maturity bin | Below-cap indicator | BQ designation | BQ designation |
| Reported effect | Designation increase | Yield reduction | Scaled yield reduction |
| 1–2 years | 0.42*** | 1.8 | 0.17 |
| | (0.015) | (2.2) | (0.11) |
| 2–5 years | 0.42*** | 8.7*** | 0.31*** |
| | (0.015) | (2.2) | (0.08) |
| 5–10 years | 0.44*** | 13.8*** | 0.41*** |
| | (0.015) | (2.2) | (0.06) |
| 10–20 years | 0.45*** | 18.4*** | 0.52*** |
| | (0.017) | (2.3) | (0.07) |
| Over 20 years | 0.43*** | 18.2*** | 0.47** |
| | (0.027) | (6.3) | (0.19) |
| 10–20 minus 1–2 years | | 16.5*** | 0.34*** |
| | | (2.7) | (0.11) |
| Fixed effects | Yes | Yes | Yes |
| Issuer cluster | Yes | Yes | Yes |
| Observations | 355,372 | 355,372 | 355,372 |
| First-stage $F$ (Kleibergen–Paap) | | 156.0 | 156.0 |

*Notes.* Each column is a separate pooled regression containing all five maturity interactions. Columns (2)–(3) instrument designation by below-cap status within each bin, as in (28), and report $-\beta_g$, so positive numbers mean lower yields on designated bonds. Column (3) divides each bond's yield by its expected annual benefit $\bar{X}_b = 0.8\,\tau_{B,t}\hat{c}_b$ before estimation (Appendix C.7.2), so its entry in each maturity bin estimates the capitalization share. Contrasts use unrounded estimates. Parentheses give standard errors clustered by issuer (15,889 issuers). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

*Fixed effects in every column.* Maturity bucket × year, state × year, state × offering year–month, rating category × year, security type × year, use of proceeds, coupon bin × maturity bucket and callability × maturity bucket × year. Other controls are log tranche par, maturity-bin indicators and separate linear issuance slopes on each side of the cap within each bin.

*Sample.* Tax-exempt new issues in 2000–2021 within \$0.5–4 million of the statutory cap (\$30 million in 2009–10, \$10 million otherwise). Maturity is measured at issuance. Appendix C.7.2 defines the fixed-effect buckets.

Subtracting the two conditions cancels the taxable yield and gives $\varpi_k = 0.8\,\tau_B c = X$. With constant financing costs, the premium in basis points is flat in maturity.

**Derivation of the designation premium.** The derivation compares otherwise identical bonds of maturity $k$ in supply $q_k > 0$, with outside opportunities fixed. The bank's designation benefit $u$ is zero for a nonqualified bond and $X$ for a bank-qualified one. At yield $y$, household demand is $d_k^H(y)$ and bank demand is $d_k^B(y+u)$, both reflecting baseline financing costs, taxes and risk compensation. For each $u \in [0, X]$, assume that a market-clearing yield exists, that demands are continuous and nondecreasing everywhere and continuously differentiable except at finitely many points, and that total demand has a strictly positive slope at that yield. Then the clearing yield $y_k(u)$ satisfies

$$d_k^H(y_k(u)) + d_k^B(y_k(u) + u) = q_k, \qquad y_k^N = y_k(0), \quad y_k^Q = y_k(X). \tag{128}$$

Differentiating (128) on a smooth segment gives

$$-\frac{dy_k(u)}{du} = \frac{\partial d_k^B/\partial y}{\partial d_k^H/\partial y + \partial d_k^B/\partial y} \equiv w_{Bk}(u) \in [0, 1], \tag{129}$$

with household and bank demand slopes evaluated at $y_k(u)$ and $y_k(u)+u$, so $w_{Bk}(u)$ is the banks' share of the total demand response. A larger benefit raises bank demand, and the yield falls to offset it. Integrating across participation boundaries, where $y_k(u)$ is continuous, gives

$$\varpi_k = y_k(0) - y_k(X) = \int_0^X w_{Bk}(u)\,du = \theta_k X, \qquad \theta_k = \frac{1}{X}\int_0^X w_{Bk}(u)\,du. \tag{130}$$

If banks never participate, $\partial d_k^B/\partial y = 0$ and $\varpi_k = 0$, the representative-tax benchmark. If only banks hold the bond, $\partial d_k^H/\partial y = 0$ and $\varpi_k = X$, the bank-only benchmark. When both respond, capitalization lies between them. The capitalization share $\theta_k$ averages banks' share of the total demand response, which can differ from their share of holdings.

The bank benefit $X$ is an annual tax saving per dollar of tax basis. At issue, a bank's tax basis equals the price it pays, so $X$ and offering yields are comparable.

### C.7.2 Design and the benefit measures

**Running variable, data and window.** The running variable $R_{jt}$ of Section 6.2 is issuance minus the year's cap. Issuance is all tax-exempt par issued under a six-digit CUSIP issuer prefix in an offering calendar year, measured before sample restrictions. Designation and other bond characteristics come from the Mergent bond master files. The baseline window keeps issuer-years with $0.5 \le |R_{jt}| \le 4$, and the below-cap indicator is $E_{jt} = \mathbf{1}\{R_{jt} \le 0\}$.

**Controls.** In (28), $Z_b$ contains the fixed effects and controls listed under Table 28. The fixed effects use seven maturity buckets with upper limits of 2, 5, 10, 15, 20 and 30 years and a seventh beyond 30 years. These buckets differ from the five maturity bins $g$ of (28).

**Financing costs and the current-year benefit.** Section 265(b) allocates a bank's interest expense across its assets in proportion to their average adjusted bases, without matching each bond to borrowing of the same maturity. A dollar of municipal basis held throughout the year is therefore allocated $c_{nt} = I_{nt}/\bar{A}_{nt}$ of interest expense. Here $I_{nt}$ is bank $n$'s annual interest expense in its December report and $\bar{A}_{nt}$ its average assets, both from Call Report data in the BankFind database of the Federal Deposit Insurance Corporation (FDIC). This expense-to-assets ratio is neither a deposit rate nor a maturity-matched financing cost. Designation reduces the nondeductible fraction from one to 0.2 and saves $X_{nt} = 0.8\,\tau_{B,t}c_{nt}$ at the federal bank tax rate $\tau_{B,t}$, 35% before 2018 and 21% thereafter. Weighting banks by their municipal holdings $M_{nt}$, held to maturity or available for sale and bank-qualified or not, gives the financing cost $c_t$ and the current-year benefit $X_t$

$$c_t = \frac{\sum_{n\in\mathcal{B}_t} M_{nt}(I_{nt}/\bar{A}_{nt})}{\sum_{n\in\mathcal{B}_t} M_{nt}}, \qquad X_t = 0.8\,\tau_{B,t}c_t. \tag{131}$$

Here $\mathcal{B}_t$ is the set of banks with positive holdings, positive average assets, nonmissing interest expense and an expense-to-assets ratio from zero to below 15%. Mean $c_t$ is 2.34% in 2000–04, 2.31% in 2005–08, 1.09% in 2009–10, 0.51% in 2011–13, 0.31% in 2014–17 and 0.66% in 2018–19, giving current-year benefits $X_t$ of about 66, 65, 30, 14, 9 and 11 bp.

The measure uses book assets and interest expense because the statute's adjusted tax bases and otherwise-deductible interest are not observed bank by bank, and the IRS permits specified book-asset averaging conventions. With these observable inputs, $X_t$ measures a C-corporation bank's benefit under the ordinary Section 265(b) disallowance.[29] In 2009–10, the Section 265(b)(7) exception for holdings up to two percent of assets can reduce the disallowance on nonqualified bonds and with it designation's incremental benefit, so I report those years separately.

**The expected annual benefit.** A bond's benefit accrues over its life. For a bond $b$ issued in month $t$, the expected annual benefit is $\bar{X}_b = 0.8\,\tau_{B,t}\,\hat{c}_b$, where $\hat{c}_b$ is the expected annual financing cost over the bond's life. Regressing $c_t$ on one-year expected rates over 2000–2021 ($R^2 = 0.75$) gives the fitted relation $\hat{c}_b = 0.38 + 0.55\,r_b$, both in percent per year. Here $r_b$ is the issue-month Kim–Wright (Kim and Wright, 2005) expected average short rate at the bond's rounded whole-year maturity, capped at ten years. Because $r_b$ is the fitted yield less its term premium, a term premium rising with maturity stays out of $\bar{X}_b$. Means of $\bar{X}_b$ in the five maturity bins are 33.9, 41.5, 50.5, 56.8 and 53.3 bp.

[29] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-11–13-12, for the allocation and the 20% disallowance.

The current-year benefit $X_t$ reflects only the issue year's financing cost. It falls from 104 bp in 2000 to an average of 9 bp in 2013–17, and premia do not track it, so ratios of premia to $X_t$ depend on the years pooled (Appendix C.7.5). I therefore divide each bond's yield by its own $\bar{X}_b$. The resulting premium estimates the capitalization share $\theta_k$ of (130), the share of the expected benefit reflected in yields.

### C.7.3 Identification

The design requires that, given the controls, crossing the cap changes offering yields only through designation. Differences between measured and statutory issuance make the design fuzzy.

**The fuzzy first stage.** Statutory issuance includes subordinate entities' issues and bank loans and leases, mostly absent from Mergent. It excludes refundings of designated bonds, which Section 265(b)(3)(D) deems designated, and private-activity bonds other than 501(c)(3) bonds. The statute also tests reasonably anticipated issuance at each sale, so a March designation stands even when an unanticipated October sale takes the year over the cap. As a result, 21% of bonds \$0.5–4 million above the cap are designated. In that window, 4% of issues designate in single-sale issuer-years, against 44% in issuer-years with multiple sales. Refundings are 46% of the bonds in that window, and measuring issuance net of refundings reduces designation in the corresponding window to 9%. Designation falls from 24% at \$0.5–1 million above the cap to 15–16% at \$4–8 million and 5% beyond \$10 million, consistent with measurement noise.

**Bunching at the cap.** The half-million donut excludes managed bunching, with 2,081 issuer-years within \$0.25 million below the cap against 572 above. It also excludes 1,034 issuer-years with 12,076 bonds exactly at the \$10 million cap (\$30 million in 2009–10). Such issuers are eligible (Section 6.2), so $E_{jt}$ uses a weak inequality. They designate 79.6% of their bonds, close to the below-cap rate of 79%, against 27% at \$1–50 thousand above the cap. A strict inequality would misclassify them, lowering the first stage without a donut from 0.39 to 0.30 but barely changing 2SLS, because dividing by the first stage offsets the attenuation.

**Balance at the cap.** Local linear regressions of each covariate in the baseline window, with maturity bucket × year and state × year effects, find callability and the number of maturities per issue balanced in every bin. The remaining differences concern controlled covariates. Ratings, observed for 62% of bonds, differ only at 1–2 years (0.17 notch, $t = 2.1$). Coupons, set at pricing and covered by coupon bin × maturity bucket fixed effects, are 0.09–0.14 percentage points lower below the cap in the four bins beyond two years. Log par, always controlled, is lower below the cap at 1–2 years ($-0.26$) and higher at 10–20 years ($+0.12$).

### C.7.4 Results

Table 29: Bank-qualified designation interacted with the expected annual benefit.

| Maturity (years) | 1–2 | 2–5 | 5–10 | 10–20 | > 20 |
|---|---|---|---|---|---|
| Yield reduction per benefit unit | $0.058^{**}$ | $0.096^{***}$ | $0.154^{***}$ | $0.175^{***}$ | $0.233^{***}$ |
| | (0.024) | (0.025) | (0.027) | (0.031) | (0.089) |
| 10–20 minus 1–2 | $0.117^{***}$ | | | | |
| | (0.032) | | | | |
| Fixed effects | | | Yes | | |
| Issuer cluster | | | Yes | | |
| Observations | | | 355,372 | | |
| First-stage $F$ (Kleibergen–Paap, joint) | | | 122.0 | | |

*Notes.* Entries are $-\beta_g$ in equation (132), the offering-yield reduction in basis points per basis point of expected annual benefit $\bar{X}_b$. Each designation–benefit interaction is instrumented by the below-cap indicator times the benefit in that bin. The regression adds the benefit by bin to the controls of Table 28 and uses the same 2000–2021 sample within \$0.5–4 million of each year's cap. Standard errors clustered by issuer (15,889 clusters) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Premia in basis points.** In Table 28, below-cap status raises designation by 0.42 to 0.45 in every bin (column (1)). In column (2), the 10–20 year premium exceeds the 1–2 year premium by 16.5 basis points (s.e. 2.7). Figure 24 shows the offering-yield discontinuity separately for each maturity bin.

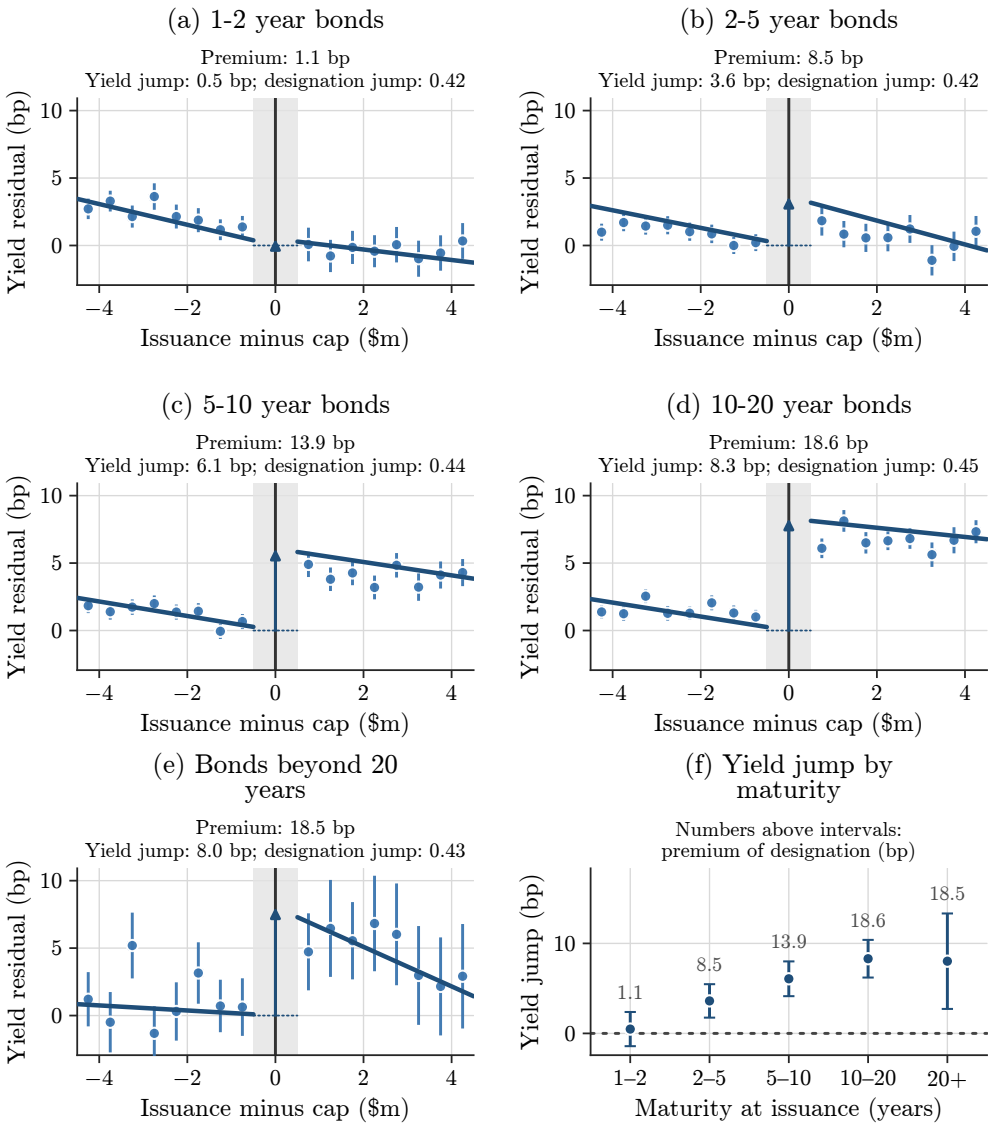



Figure 24: Offering-yield discontinuities by maturity. Panels (a)–(e) show $0.5 million issuance-bin means of offering-yield residuals after the fixed effects and log par of (28), relative to the below-cap fitted value at the cap, with local linear fits, donut shading and 95% intervals. Arrows mark the yield jumps, and headers divide each yield jump by the designation jump to obtain the premium. Panel (f) collects these yield jumps and premia. Its premia are the separate-bin counterparts of the joint 2SLS estimates in column (2) of Table 28. The common scale covers the central 96 percent of binned means.

**Premia in expected-benefit units.** Column (3) of Table 28 divides each offering yield by its own $\bar{X}_b$ before any pooling and then estimates (28) by 2SLS. The estimated capitalization shares are 0.17, 0.31, 0.41, 0.52 and 0.47, with a 10–20 minus 1–2 year contrast of 0.34 (s.e. 0.11). Section 6.2 relies on this specification for its statements that about half of the benefit is priced at 10–20 years and that the rising share rejects full capitalization.

**Interaction with the expected benefit.** A second specification checks that the rising profile does not come from dividing yields by the expected benefit. It keeps offering yields in basis points and interacts designation with the expected annual benefit:

$$y_b = \sum_{g=1}^{5} \beta_g B_b \bar{X}_b \mathbf{1}\{n_b \in g\} + \widetilde{Z}_b'\delta + u_b, \qquad (132)$$
$$B_b\bar{X}_b\mathbf{1}\{n_b \in g\} \quad \text{instrumented by } E_{jt}\bar{X}_b\mathbf{1}\{n_b \in g\}, \qquad g = 1, \ldots, 5.$$

Here $\widetilde{Z}_b$ adds $\bar{X}_b$ by bin to the controls $Z_b$ of (28), and $u_b$ is the regression error. Table 29 reports the yield reductions $-\beta_g$ and the joint first-stage $F$ statistic. Because designation and its instruments are multiplied by $\bar{X}_b$, high-benefit years carry most of the weight. Premia are the smallest fraction of the benefit in those years (Appendix C.7.5), so the estimates are lower than those of column (3) of Table 28. Both specifications give rising maturity profiles.

### C.7.5 Robustness and a placebo cutoff

**Donut, bandwidth and window.** Across donut half-widths from zero to \$1.5 million and bandwidths from \$2 million to \$8 million (Table 30), the 10–20 minus 1–2 year contrast is 13.5–18.4 bp and the 1–2 year premium is never significant. In the \$5–15 million window of Dagostino (2025), without a donut, the contrast is 15.4 (1.8) bp, one of the most precise estimates. The last row of Table 30 drops the region that her bunching estimation treats as contaminated, \$8.5–13.5 million at the \$10 million cap. Within $|R| \leq 8$, it compares issuers at least \$1.5 million below the cap with issuers at least \$3.5 million above it. First stages rise to 0.47–0.54, and with this longer extrapolation the contrast is 9.8 (4.2) bp.

Table 30: Bank-qualified discontinuity: sensitivity to the donut, the bandwidth and the window.

| | first stage | 1–2y | 2–5y | 5–10y | 10–20y | 20y+ | 10–20 minus 1–2 |
|---|---|---|---|---|---|---|---|
| *Donut half-width, bandwidth \$4 million* | | | | | | | |
| none | 0.39*** | 1.9 (1.8) | 7.5*** (1.8) | 13.9*** (1.8) | 17.7*** (1.8) | 17.7*** (4.5) | 15.7*** (2.0) |
| \$0.25 million | 0.40*** | 1.6 (2.0) | 7.6*** (2.0) | 13.4*** (2.0) | 16.7*** (2.2) | 16.3*** (5.5) | 15.1*** (2.4) |
| \$0.5 million (baseline) | 0.42*** | 1.8 (2.2) | 8.7*** (2.2) | 13.8*** (2.2) | 18.4*** (2.3) | 18.2*** (6.3) | 16.5*** (2.7) |
| \$1 million | 0.44*** | 1.2 (2.9) | 8.2*** (2.7) | 13.3*** (2.8) | 19.7*** (3.1) | 20.6*** (8.0) | 18.4*** (3.5) |
| \$1.5 million | 0.44*** | 2.0 (4.4) | 7.3* (3.8) | 11.8*** (4.0) | 15.5*** (4.2) | 19.5* (10.2) | 13.5*** (5.2) |
| *Bandwidth, donut \$0.5 million* | | | | | | | |
| \$2 million | 0.38*** | 0.8 (4.7) | 7.9 (4.9) | 13.3*** (4.8) | 16.9*** (5.0) | 21.0 (14.1) | 16.1*** (4.8) |
| \$3 million | 0.40*** | 2.7 (3.0) | 8.9*** (3.0) | 15.2*** (2.9) | 19.2*** (3.1) | 19.9*** (7.6) | 16.5*** (3.3) |
| \$5 million | 0.42*** | −0.9 (1.9) | 5.7*** (1.8) | 11.4*** (1.8) | 14.9*** (1.9) | 15.9*** (4.9) | 15.8*** (2.3) |
| \$6 million | 0.43*** | −0.5 (1.7) | 5.4*** (1.5) | 10.8*** (1.6) | 13.2*** (1.6) | 11.4*** (4.1) | 13.6*** (2.1) |
| \$8 million | 0.45*** | 0.0 (1.5) | 6.3*** (1.2) | 10.1*** (1.3) | 13.5*** (1.3) | 12.3*** (3.5) | 13.5*** (1.8) |
| *Other windows* | | | | | | | |
| \$5–15 million, no donut | 0.40*** | 0.1 (1.6) | 5.6*** (1.5) | 12.1*** (1.6) | 15.5*** (1.5) | 16.7*** (3.8) | 15.4*** (1.8) |
| \$8.5–13.5 million dropped, $\lvert R\rvert \leq 8$ | 0.47*** | 0.2 (3.5) | 6.0** (3.0) | 7.6** (3.1) | 10.0*** (3.4) | −1.8 (9.5) | 9.8** (4.2) |

*Notes.* Cells report the 2SLS designation premium in basis points by maturity bin, pooled over 2000–2021 at each year's cap, with the baseline fixed effects and issuer-clustered standard errors in parentheses. The bandwidth and the donut half-width are the largest and smallest $|R_{jt}|$ retained, and the first-stage column refers to the 1–2 year bin. The \$5–15 million window is the one used in the spread comparison of Dagostino (2025). The last row drops the region her bunching estimation excludes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Construction checks and the relation to Dagostino (2025).** Dagostino (2025) measures issuer-year issuance with Ipreo MuniIC data and documents bunching below the cap and a jump in designation at the \$30 million cap of 2009–10, as in my data. Her regressions on \$1 million issuance bins control for size, maturity, insurance, rating and use of proceeds, with year and issuer effects and standard errors clustered by year. In that study, tax-adjusted spreads over treasuries jump about 10 bp at the cap for general-obligation (GO) bonds issued before 2008. The tax-adjusted spread is $100\,[y/(1-\tau_{B,t}) - y^{\text{Tsy}}]$, with $y$ the offering yield and $y^{\text{Tsy}}$ the maturity-matched treasury par yield, both in percent. Within my design, I add bond insurance and enhancement by year and, with these controls, restrict the sample to general-obligation bonds or adopt her issuer fixed effects, tax-adjusted spread or year clusters. Using only the baseline controls, I net refunding par from the running variable, alone or with refunding bonds excluded. In every variant, the 10–20 year premium exceeds the 1–2 year premium at the one percent level, by 13.5 to 22.3 bp against 16.5 bp in the baseline, with

first stages of 0.40 to 0.51 in every bin. Issuer effects steepen the gradient, while netting refundings moves some issuer-years across the cap and widens confidence intervals.

I apply her $1 million issuance-bin estimator to her window, with maturity-bucket, state and rating effects by year and use-of-proceeds controls. Bin means are flat below the cap and show a step above it, matching her Figure 3. Pooled across maturities, the step is 4.2 bp in the offering yield and 5.6 bp in the tax-adjusted spread. The step rises with maturity, from 1.1 and 1.4 bp at 1–5 years to 7.8 and 10.2 bp at 10–20 years. In her pooled spread comparison for GO bonds issued before 2008, with her controls but with state fixed effects and clustering by issuer, the jump at the cap is 10.4 bp (s.e. 2.5), against her estimate of about 10. With her issuer fixed effects and clustering by year, which leaves eight clusters, the jump is 6.3 (3.8) bp, because within-issuer variation across the cap is thin before 2008.

**The benefit measure and benefit regimes.** Measuring the expected rate in $\bar{X}_b$ with the Adrian–Crump–Moench (Adrian et al., 2013) decomposition, which removes larger term premia than Kim–Wright at five and ten years, gives capitalization shares of 0.17, 0.35, 0.46, 0.55 and 0.48 (contrast 0.39). With years split into terciles of $X_t$, the 10–20 minus 1–2 year contrasts are 0.57 in low-benefit years (2013–18 and 2020–21), 0.37 in middle years (2003–04, 2009–12 and 2019) and 0.27 in high-benefit years (2000–02 and 2005–08). In the high-benefit years, expected benefits were nearly flat in maturity (62.8–71.2 bp) and yield curves often inverted, so the benefit's maturity structure does not drive the gradient. Excluding 2020–21, years of strong bank demand, gives 0.18, 0.25, 0.35, 0.39 and 0.36 (contrast 0.21). In the interaction specification of Table 29, excluding 2009–10 leaves the 10–20 minus 1–2 year contrast at 0.113 (s.e. 0.032).

**Placebo cutoff.** At a placebo cutoff of $5 million, first stages are 0.00–0.05 and all yield jumps at the cutoff are within two bp of zero. The 10–20 year jump exceeds the 1–2 year jump by 0.4 (1.2) bp, against 7.8 (1.2) at the actual cap.

**The 2009–10 cap.** The designation jump moves with the $30 million cap of the American Recovery and Reinvestment Act (ARRA). A $0.5–8 million window around it covers 16 thousand bonds issued in 2009–10. With the baseline fixed effects, the first stages are 0.19, 0.20, 0.19, 0.16 and 0.26, jointly significant at five percent. At the same cutoff, first stages are 0.01–0.04 in 2004–08 and −0.05 to 0.01 in 2011–15. At $10 million in 2009–10, they are 0.01–0.06 and not jointly significant. With designation jumps smaller than in other years, the reduced form (the yield jump) is the informative estimate for 2009–10. Its yield reductions at the $30 million cap are 0.5, 3.2, 1.6, 10.9 and 21.5 bp, and the average reduction beyond ten years exceeds the 1–2 year reduction by 15.7 (6.7), significant at five percent.

**By period.** Estimates by period test whether premia track the current-year benefit. Those for 2000–04, 2005–08, 2009–10, 2011–13, 2014–17 and 2018–19 use the baseline specification, with the ARRA cap and window in 2009–10. Before 2009, when $X_t$ is 65–66 bp, premia are at most about 0.3 of $X_t$ at every maturity, so the benefit is weakly capitalized when the financing-tax cost of nonqualified bonds approaches a percentage point. In 2011–17, when $X_t$ is 9–14 bp, premia are 2.4–2.5 bp at 1–2 years, below $X_t$, and 14–28 bp at 5–20 years, at or above $X_t$. Ratios of the 5–20 year premium to $X_t$ exceed two in some low-cost periods.

In 2018–19, after the cut in the bank tax rate, premia by bin are 6.9 (4.3), 6.4 (5.0), 7.7 (5.7), 14.9 (7.7) and 3.7 (24.2) bp. A difference-in-discontinuities estimate pools 2014–2019 and interacts designation with an indicator for 2018–19. It gives changes in 2018–19 of 3.1 (6.5), −9.1 (6.7), −19.2 (7.6), 3.6 (10.0) and −1.3 (28.7) bp, with unchanged first stages. The benefit measure uses the corporate rate because Subchapter S banks, whose income is taxed to their shareholders and whose tax rate the TCJA cut less, hold only 6–14% of banks' municipal holdings over 2000–2021.

Annual reduced-form yield reductions at the $10 million cap, pooled across maturities, are largest in 2016 and 2019–21 (11–18 bp), 6–8 bp in 2011–13 and within about 3 bp of zero in 2000–01, 2004–07 and 2018. First stages are 0.30–0.62 in every year except 2009–10, when the cap was $30 million. Bank holdings of municipal securities rose from $97 billion in 2000 to $370 billion in 2017, fell after the 2018 rate cut to $322 billion on average in 2018–19 and then averaged $396 billion in 2020–21. Premia do not track the current-year benefit, and the pattern is consistent with changing bank participation.

## C.8 Investor holdings by tax status

The municipal portfolio shares of insurers and pension funds cited in Section 3.4 vary with exemption benefits (Table 2). In 2000–2017, exempt municipal bonds account for 34–48% of the average P&C insurer's bond portfolio, compared with 6–9% for life insurers (Figure 25). In the public pension holdings panel, mean

portfolio shares never exceed 0.13% for exempt and 0.78% for taxable municipals (Figure 26). After the TCJA corporate-rate cut lowered P&C insurers' exemption benefit (Table 35), their exempt share fell from 38.0% in 2017 to 28.6% in 2019, while the share of other taxable bonds rose from 50.2% to 57.5%.

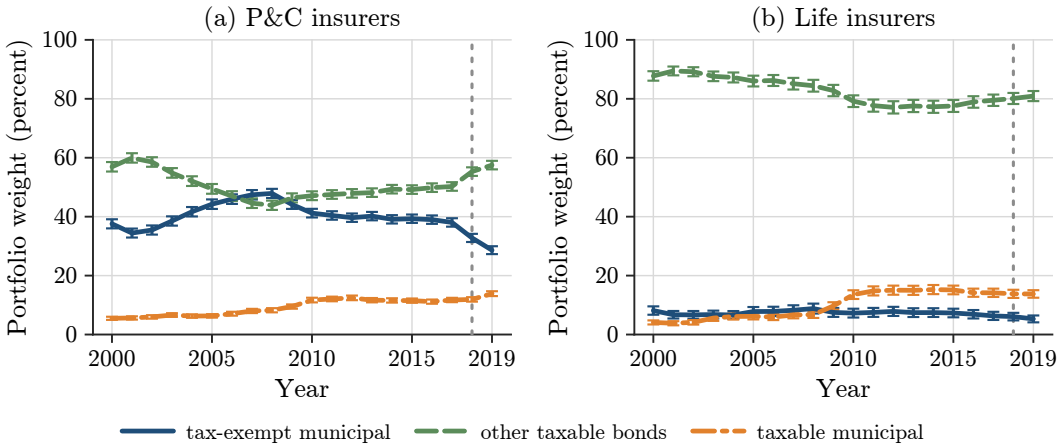



Figure 25: Tax-exempt municipals, taxable municipals and other taxable bonds as shares of insurers' total bond holdings: P&C in panel (a) and life in panel (b), 2000–2019, NAIC Schedule D. The dotted vertical line marks the TCJA corporate-rate cut. Bars are 95% intervals for yearly mean insurer portfolio shares, with standard errors clustered by insurer.

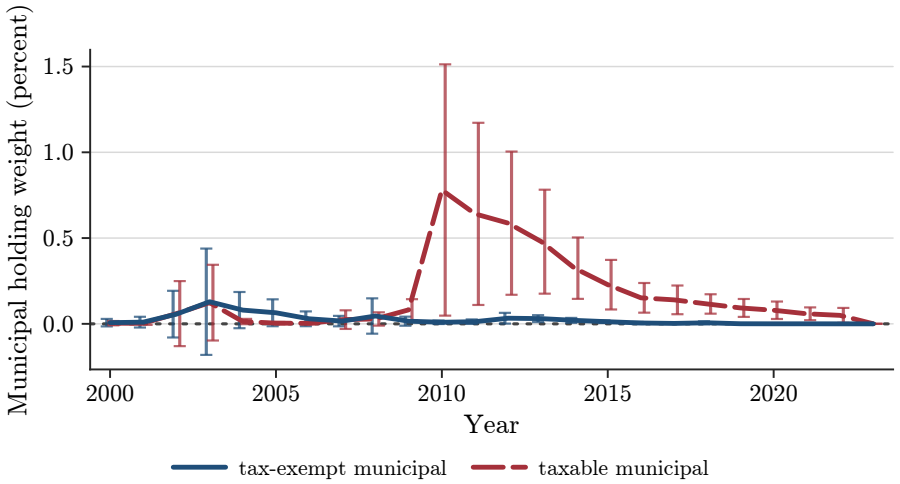



Figure 26: Municipal holdings by federal tax status as shares of total bond portfolios in the public pension holdings panel collected through Freedom of Information Act (FOIA) requests (Section 3.1), 2000–2023. All thirty plans are covered from 2015. Bars are 95% intervals for yearly mean plan portfolio shares, with standard errors clustered by plan. Other taxable bonds are omitted. Taxable municipals include Build America Bonds.

## C.9 P3: rate responses by maturity

P3 (Section 6.3) tests whether municipal yields respond more to a given movement in maturity-matched treasury yields as maturity rises. The comparison concerns this relative response, so longer bonds' greater duration alone does not explain the prediction. The test uses only the model's maturity ordering, because the empirical coefficients measure coupon-yield responses and so differ from the model's zero-coupon relative responses. The maturity gradient survives a treasury benchmark without roll-down, alternative filters, state clustering and a corporate reference curve.

**Design and estimating equation.** Beyond the screens in the notes to Table 31, the sample excludes trades less than 92 days after issue, when offering yields still anchor prices, and less than 365 days before maturity, where the fitted treasury curve is erratic. Two consecutive customer purchases, or two consecutive customer sales, of the same CUSIP at most 90 days apart form a pair. Its municipal yield change is $\Delta y^M_{b,t}$ and

Table 31: P3: municipal coupon-yield responses to treasury par yields by remaining maturity.

| | (1) Continuous $\Delta y^{M}_{b,t}$ | (2) Bin $\Delta y^{M}_{b,t}$ |
|---|---|---|
| $\Delta y^{\text{Tsy}}_{n_b,t}$ | $0.346^{***}$ | $0.351^{***}$ |
| | (0.016) [0.012] | (0.013) [0.006] |
| $\Delta y^{\text{Tsy}}_{n_b,t} \times n_b$ | $0.007^{***}$ | |
| | (0.002) [0.002] | |
| $\Delta y^{\text{Tsy}}_{n_b,t} \times Long_b$ | | $0.075^{***}$ |
| | | (0.012) [0.009] |
| $Long_b$ | | $0.016^{***}$ |
| | | (0.002) [0.002] |
| $n_b$ | $0.002^{***}$ | |
| | (0.000) [0.000] | |
| Trade-date cluster | Yes | Yes |
| Observations | 952,874 | 952,874 |
| Weights | Inverse pair count | Inverse pair count |

*Notes.* The sample is pairs of consecutive same-side MSRB customer trades within 90 days in federally tax-exempt, noncallable bonds rated AA and above (agency score $\leq 3$), January 2005 to August 2020. Between the two trades of each pair, the maturity-matched treasury yield moved at least 10 basis points in either direction. The treasury yield is interpolated on the daily par-yield curve of Gürkaynak et al. (2007) at the bond's exact remaining maturity on each transaction date. Observations are weighted by the inverse number of their CUSIP's pairs in this sample, counted before observations with missing or trimmed outcomes are dropped. The dependent variable is trimmed at its first and ninety-ninth percentiles. $Long_b = \mathbf{1}\{n_b \geq 5\}$. Parentheses give trade-date-clustered standard errors (3,864 clusters). Brackets give state-clustered standard errors (54 clusters: the 50 states, the District of Columbia, Puerto Rico, Guam and the U.S. Virgin Islands). Stars are identical under either clustering. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

its matched treasury change is $\Delta y^{\text{Tsy}}_{n_b,t} = y^{\text{Tsy}}_{n_b(t),t} - y^{\text{Tsy}}_{n_b(t-1),t-1}$, where $t-1$ denotes the trade preceding $t$, $n_b(t)$ is the bond's remaining maturity in years at trade $t$ and $y^{\text{Tsy}}$ is the daily treasury par-yield curve of Gürkaynak et al. (2007). Column (1) of Table 31 estimates (29), which includes the maturity level $n_b$ and no fixed effects, so the response at maturity $n_b$ is $\beta_0 + \beta_1 n_b$. Column (2) replaces maturity by $Long_b = \mathbf{1}\{n_b \geq 5\}$, so its short and long responses are $\beta_0$ and $\beta_0 + \beta_1$.

**Identification.** First differencing within a bond absorbs its time-invariant characteristics, and noncallability removes call-option repricing. Comparing purchases with purchases and sales with sales limits bid–ask and intraday noise.

**Results.** In the continuous specification, the coefficient on the treasury change is 0.346 and the maturity interaction is 0.007, both significant at one percent (Table 31, column (1)). In the binned specification, the response rises from 0.351 below five years to 0.426 from five years, both significant at one percent.

**Robustness: treasury roll-down.** Part of the matched treasury change is roll-down, which comes from the bond becoming shorter between trades. Holding the later date's curve fixed, roll-down is $y^{\text{Tsy}}_{n_b(t),t} - y^{\text{Tsy}}_{n_b(t-1),t}$. Fixing the benchmark at the first trade's maturity, $\Delta y^{\text{Tsy,fixed}}_{b,t} = y^{\text{Tsy}}_{n_b(t-1),t} - y^{\text{Tsy}}_{n_b(t-1),t-1}$, removes it. On upward-sloping curves, roll-down lowers the reference yield by an average of 2.6 bp below five years and 1.6 bp beyond, against absolute changes near 26 bp. Removing it lowers the short response more than the long response and so increases the gradient. The continuous maturity interaction rises from 0.0070 to 0.0084 and the binned interaction from 0.0749 to 0.0825. Municipal roll-down does not fully offset treasury roll-down because the two curves have different slopes. Controlling linearly for municipal roll-down, with elapsed time between trades and its interaction with maturity, raises both interactions further, to 0.0086 and 0.0850. All these estimates are significant at one percent. The headline specification keeps the benchmark matched at each trade, which gives the smallest gradients and is therefore conservative.

**Robustness: filter thresholds.** Across the treasury-move cutoffs and trade windows of Table 32, every maturity interaction is positive. Every binned cell is significant at one percent and every continuous cell at five percent, twenty-seven of the thirty at one percent.

Table 32: P3 sensitivity: maturity interaction by treasury-move cutoff and trade window (weighted by the inverse number of pairs per CUSIP).

| *Panel A. Continuous interaction* $\Delta y^{\text{Tsy}}_{n_b,t} \times n_b$ | | | | | |
|---|---|---|---|---|---|
| Cutoff \ window | 15d | 30d | 60d | 90d | 120d |
| $\geq 0.05$ | $0.0053^{***}$ | $0.0057^{***}$ | $0.0051^{**}$ | $0.0071^{***}$ | $0.0090^{***}$ |
| $\geq 0.075$ | $0.0061^{***}$ | $0.0057^{***}$ | $0.0047^{**}$ | $0.0068^{***}$ | $0.0087^{***}$ |
| $\geq 0.10$ | $0.0065^{***}$ | $0.0060^{***}$ | $0.0049^{**}$ | $0.0070^{***}$ | $0.0087^{***}$ |
| $\geq 0.15$ | $0.0080^{***}$ | $0.0071^{***}$ | $0.0059^{***}$ | $0.0077^{***}$ | $0.0091^{***}$ |
| $\geq 0.20$ | $0.0088^{***}$ | $0.0082^{***}$ | $0.0067^{***}$ | $0.0081^{***}$ | $0.0094^{***}$ |
| $\geq 0.25$ | $0.0104^{***}$ | $0.0083^{***}$ | $0.0074^{***}$ | $0.0087^{***}$ | $0.0097^{***}$ |
| *Panel B. Binned interaction* $\Delta y^{\text{Tsy}}_{n_b,t} \times Long_b$ | | | | | |
| Cutoff \ window | 15d | 30d | 60d | 90d | 120d |
| $\geq 0.05$ | $0.049^{***}$ | $0.061^{***}$ | $0.070^{***}$ | $0.074^{***}$ | $0.088^{***}$ |
| $\geq 0.075$ | $0.056^{***}$ | $0.063^{***}$ | $0.068^{***}$ | $0.074^{***}$ | $0.086^{***}$ |
| $\geq 0.10$ | $0.062^{***}$ | $0.066^{***}$ | $0.069^{***}$ | $0.075^{***}$ | $0.086^{***}$ |
| $\geq 0.15$ | $0.079^{***}$ | $0.077^{***}$ | $0.079^{***}$ | $0.083^{***}$ | $0.092^{***}$ |
| $\geq 0.20$ | $0.094^{***}$ | $0.090^{***}$ | $0.085^{***}$ | $0.088^{***}$ | $0.097^{***}$ |
| $\geq 0.25$ | $0.111^{***}$ | $0.100^{***}$ | $0.096^{***}$ | $0.096^{***}$ | $0.102^{***}$ |

*Notes.* Each cell re-estimates column (1) (Panel A) or (2) (Panel B) of Table 31 on pairs whose maturity-matched treasury move is at least the cutoff, in percentage points, and whose trades are at most the window apart. The headline sample uses 0.10 and 90 days. Standard errors are clustered by trade date as in Table 31. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Robustness: state clustering.** Brackets in Table 31 give standard errors clustered by issuing state, which allow correlation among a state's bonds across dates, while the headline trade-date clustering allows common shocks across states on the same date. Both headline interactions remain significant at one percent. In Table 32, the binned interactions keep their significance levels. Among the continuous interactions, the 0.05, 0.075 and 0.10 cutoffs give $p < 0.05$ at thirty days and $p < 0.10$ at sixty days, and the 0.15 cutoff gives $p < 0.05$ at sixty days. All other cells keep their trade-date significance levels, so twenty-seven of the thirty continuous cells remain significant at five percent.

**Robustness: corporate reference curve.** Section 4.6 derives P3 relative to the model's jointly priced taxable market. Against corporate zero-coupon yields, the maturity gradient is larger than against treasury par yields. Both reference curves here are monthly averages at the bond's remaining maturity in each trade month. On the 517,864 cross-month pairs whose absolute monthly treasury change is at least 10 bp, with the headline weights and trade-date clustering, the maturity interaction is 0.0132 (s.e. 0.0026) against corporate changes and 0.0072 (s.e. 0.0025) against treasury changes.

## C.10 Robustness to the rating cut

The yield pattern and the empirical tests use two rating cuts of the unrounded mean agency score on the twenty-two-notch scale, on which 1 = AAA, 3 = AA and 7 = A−. The curves in Sections 3 and 5 use scores of seven or better, the A-or-above sample, matching the corporate benchmark's credit band. The BAB figure in Section 3 and the P1 and P3 tests in Section 6 use scores of three or better, the AA-and-above sample.

The curve pattern survives the narrower cut, and the BAB, P1 and P3 tests retain their gradient signs on the wider cut. P1's pre-2018 state-tax pass-through declines across all five bins under OLS and IV (Table 33), with the short-to-long decline significant at five percent under OLS and one percent under IV. The wider sample gives larger P3 interactions, with similar BAB and P1 gradients.

### C.10.1 Curve construction at AA and above

On the AA-and-above sample, the implied tax also falls with maturity, and the model gap $\widetilde{\mathcal{G}}_T$ of (23) matches the residual before the model to within ten basis points at fourteen of fifteen maturities.

Figure 27 recomputes the AA-and-above curves with the three-stage averaging of Appendix C.1.1. At each maturity, the implied tax is one minus the ratio of these mean municipal to taxable yields. Net of calls and issuance premia, it falls from 40.1% at two years to 25.5% at thirty, a decline of 14.6 points against 17.1 on the A-or-above sample.

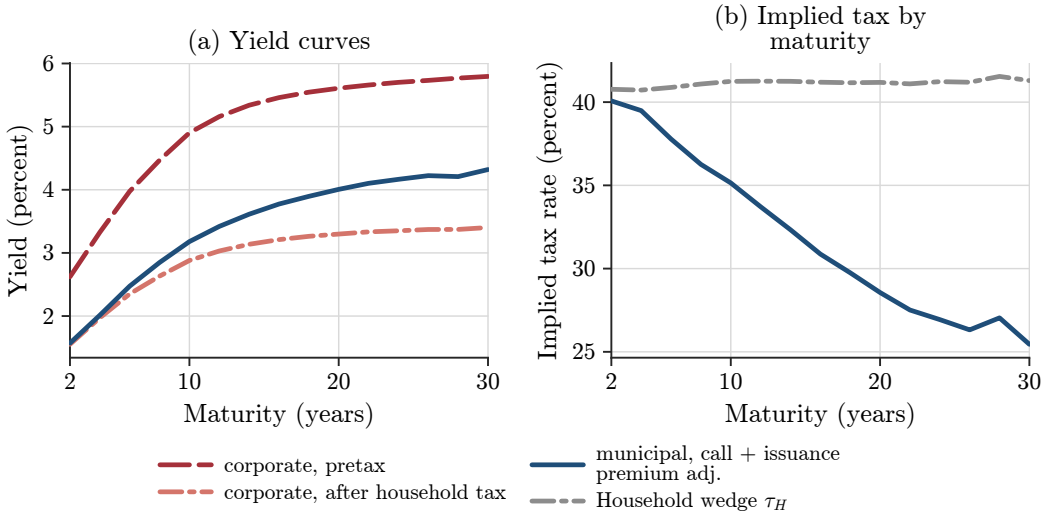



Figure 27: The AA-and-above municipal puzzle, 2000–2020, with this sample's own weights and household benchmark. Panel (a) shows pretax and household after-tax corporate zero-coupon yields and the municipal yield after removing callability and the issuance premium, and panel (b) shows the implied tax $1-y_T^M/y_T^C$ and the household wedge $1-\bar{\xi}_H(T)$, labeled $\tau_H$ in the legend. Call values come from the calibrated model of each bond's issue period (Appendix C.1.2), and the sample's all-size issuance premium is scaled by the A-or-above sample's ratio of institutional-size to all-size premia (see text).

The model comparison repeats the accounting of Section 5.3, applying the model's ratio of municipal to taxable mean yields, averaged with this sample's weights, to observed corporate yields (the corporate normalization of (23)). Call, liquidity and default estimates are sample-specific, and capital gains use the common method of Appendix C.1.4. To match the institutional-size baseline of Appendix C.1.3, the issuance-premium adjustment scales this sample's premium from all trade sizes by the A-or-above sample's ratio of institutional-size to all-size premia at each maturity, giving 11.5 bp at thirty years. The model gap exceeds the residual before the model at every maturity beyond two years, 96.2 against 89.9 basis points at thirty years, so the unexplained residual is negative at those maturities. It is −6.3 basis points at thirty years and 7.1 in root mean square across maturities.

### C.10.2 Empirical tests on the A-or-above sample

Table 21 (Appendix C.2) compares the two BAB rating cuts in columns (1) and (3), with call matching in columns (2) and (4). Table 33 repeats the P1 specification of Appendix C.5, so its AA-and-above columns reproduce the headline P1 estimates. Table 34 repeats the P3 specification of Table 31 but clusters standard errors by state, so its AA-and-above columns reproduce that table's headline coefficients and bracketed standard errors.

The wider sample also preserves the binned gradient across all thirty combinations of cutoff and window in Table 32. Coefficients range from 0.078 to 0.134, against 0.049 to 0.111 for AA and above. All are significant at one percent with state-clustered standard errors.

# D Institutional detail and data

This appendix states the tax rules behind the relative after-tax shares of Section 3 and shows how the model and the empirical adjustments implement them, separating statutory rules from model assumptions. Appendix D.4 converts these rules into tax receipts and financing costs under home-state repeal and equal

Table 33: P1 by rating cut, 2000–2017.

| | OLS | | IV | |
|---|---|---|---|---|
| | Score $\leq 3$ | Score $\leq 7$ | Score $\leq 3$ | Score $\leq 7$ |
| $\beta_1$: $[0, 5)$ years | $1.069^{***}$ | $0.986^{***}$ | $1.088^{***}$ | $1.080^{***}$ |
| | (0.195) | (0.255) | (0.225) | (0.233) |
| $\beta_2$: $[5, 10)$ years | $0.677^{***}$ | $0.607^{***}$ | $0.694^{***}$ | $0.665^{***}$ |
| | (0.069) | (0.100) | (0.086) | (0.117) |
| $\beta_3$: $[10, 15)$ years | $0.484^{***}$ | $0.380^{***}$ | $0.447^{***}$ | $0.384^{***}$ |
| | (0.096) | (0.083) | (0.118) | (0.098) |
| $\beta_4$: $[15, 20)$ years | $0.390^{***}$ | $0.285^{***}$ | $0.326^{***}$ | $0.264^{***}$ |
| | (0.102) | (0.086) | (0.116) | (0.095) |
| $\beta_5$: $[20, 30]$ years | $0.305^{***}$ | $0.240^{***}$ | $0.229^{***}$ | $0.214^{***}$ |
| | (0.075) | (0.077) | (0.081) | (0.079) |
| Decline $\beta_1 - \beta_5$ | $0.764^{***}$ | $0.746^{**}$ | $0.859^{***}$ | $0.866^{***}$ |
| $p$-value of decline | 0.002 | 0.011 | $< 0.001$ | $< 0.001$ |
| Issue month × maturity × call FE | Yes | Yes | Yes | Yes |
| State cluster | Yes | Yes | Yes | Yes |
| Observations | 681,838 | 1,397,768 | 681,838 | 1,397,768 |
| First-stage $F$ (Kleibergen–Paap) | — | — | 10.4 | 12.6 |

*Notes.* FE denotes fixed effects. Controls, fixed effects and state-clustered inference (47 states) follow Appendix C.5, 2000–2017, with call-adjusted zero-coupon yields. Score $\leq 3$ is the AA-and-above sample and score $\leq 7$ the A-or-above sample. IV uses bin-interacted 1990 rates. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 34: P3: municipal coupon-yield responses to treasury par yields, by rating cut.

| | (1) continuous | | (2) bin <5y vs ≥5y | |
|---|---|---|---|---|
| | score $\leq 3$ | score $\leq 7$ | score $\leq 3$ | score $\leq 7$ |
| $\Delta y^{\text{Tsy}}_{n_b,t}$ | $0.346^{***}$ | $0.281^{***}$ | | |
| | (0.012) | (0.011) | | |
| $\Delta y^{\text{Tsy}}_{n_b,t} \times n_b$ | $0.007^{***}$ | $0.013^{***}$ | | |
| | (0.002) | (0.002) | | |
| $\Delta y^{\text{Tsy}}_{n_b,t} \times \mathit{Long}_b$ | | | $0.075^{***}$ | $0.113^{***}$ |
| | | | (0.009) | (0.006) |
| State cluster | Yes | Yes | Yes | Yes |
| Observations | 952,874 | 2,023,637 | 952,874 | 2,023,637 |
| Weights | | Inverse pair count | | |

*Notes.* The estimates use Table 31's first-differenced specification for coupon yields and par yields, weighted by the inverse number of pairs per CUSIP and clustered by state, with $\mathit{Long}_b = \mathbf{1}\{n_b \geq 5\}$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

treatment, Appendix D.5 examines financing, short-sale and hedging restrictions, and Appendix D.6 lists the data sources.

## D.1 Federal tax treatment by investor group since 2000

The model prices bonds through each investor's relative after-tax share, so the federal rules that set this share determine which investors value the exemption most. Let $\xi_i^C = 1-\tau_i^C$ and $\xi_i^M = 1-\tau_i^M$ be the fractions of a dollar of taxable and municipal interest that investor $i$ keeps, before the normalization of Section 4.2. Pricing uses their ratio $\xi_i = \xi_i^C/\xi_i^M$, and the exemption benefit $1-\xi_i = (\tau_i^C - \tau_i^M)/(1-\tau_i^M)$ equals $\tau_i^C$ only when municipal interest is fully exempt. Households differ in marginal rates, insurers lose part of the benefit through proration, and banks can lose the deduction for financing costs. Lost financing deductions are a separate cost,

described for banks below and for other investors in Appendix D.5. Tables 2 and 35 give the pre-2018 and 2018–2020 values.

Table 35: Investor heterogeneity in the municipal market after the 2017 Act

| Investor type | Federal treatment | State treatment | Net exemption benefit |
|---|---|---|---|
| High-tax household | Interest exempt | Exempt on in-state bonds only | 0.35–0.41 |
| Middle-tax household | Interest exempt | Exempt on in-state bonds only | 0.22–0.32 |
| Low-tax household | Interest exempt | Exempt on in-state bonds only | 0.10–0.12 |
| P&C insurer | Exempt, less proration | Premium taxes, no distinction | 0.17 |
| Life insurer | Exempt, less proration | Premium taxes, no distinction | 0.16 |
| Bank | Exempt, financing cost disallowed | Varies by state | 0.16–0.18 |
| Other (pensions, etc.) | Varies | Varies | $< 0.05$ |

*Notes.* Household values are federal marginal tax rates in three groups (35% and above, 22% to 32%, below 22%), and the top value includes the 3.8% net investment income tax. The $10,000 cap on the state and local tax deduction makes the home-state exemption more valuable to households whose cap binds. Insurers and banks face the 21% corporate rate. P&C proration rises to 25% of exempt interest, giving $\xi = 0.79/0.9475$, and the life insurer's policyholders' share is set at 30%, giving $\xi = 0.79/0.937$. The bank value is the break-even implied tax defined under Table 2, evaluated at 2018–2020 taxable yields and that period's financing-tax cost of 0.117% a year.

**Households.** Municipal interest is generally federally exempt, and taxable bond interest is taxed as ordinary income. A household's federal after-tax shares are therefore $\xi_i^C = 1 - \tau_{oi}$ and $\xi_i^M = 1$, where $\tau_{oi}$ is its ordinary-income tax rate. These shares omit the net investment income tax and the tax on Social Security benefits that interest income can trigger. The calibration includes both when it computes federal and state rates for each SCF tax unit under each year's law (Appendix B.1).

The top ordinary rate in Table 36 fell under the 2001 tax cuts and again when they were accelerated in 2003, rose in 2013 and fell in 2018. The top long-term capital-gains rate fell to 15% in 2003 and rose in 2013. Since 2013, the 3.8% net investment income tax (NIIT) has applied to taxable interest and many gains, but not to exempt interest, widening the exemption benefit of the households it covers. In P1, I set the federal after-tax share of taxable interest to one minus the year's top ordinary rate, excluding NIIT.

Table 36: Top federal tax rates relevant to municipal pricing, 2000–2020

| Period | Top ordinary rate | + NIIT (on taxable interest) | Top corporate rate |
|---|---|---|---|
| 2000 | 39.6% | 39.6% | 35% |
| 2001 | 39.1% | 39.1% | 35% |
| 2002 | 38.6% | 38.6% | 35% |
| 2003–2012 | 35.0% | 35.0% | 35% |
| 2013–2017 | 39.6% | 43.4% | 35% |
| 2018–2020 | 37.0% | 40.8% | 21% |

**Banks.** A bank's municipal coupons are exempt, but interest expense that finances nonqualified exempt bonds is generally nondeductible. Let $c$ be the annual financing cost, the interest expense allocated to a dollar of holdings, and $\tau_B$ the bank's tax rate, as in Section 6.2. Without the deduction, financing a dollar of holdings costs $c$ after tax, against $c(1 - \tau_B)$ with it, a penalty of $\tau_B c$. This penalty is the financing-tax cost netted from the bank's benefit in Tables 2 and 35. Section 265(b)(3) excepts bank-qualified bonds, but Section 291 cuts their deduction by 20%, leaving 80% and an after-tax cost of $c(1 - 0.8\tau_B)$. Interest expense is allocated to exempt bonds pro rata, in proportion to adjusted tax basis.[30]

The Tax Equity and Fiscal Responsibility Act of 1982 disallowed 15% of banks' deduction for interest allocable to exempt bonds, a cut raised to 20% in 1984. The 1986 Act generally denied financial institutions' interest deductions allocable to exempt obligations acquired after August 7, 1986. The bank-qualified exception covers bonds that small issuers designate, subject to the $10 million annual issuance cap and other conditions. Temporary 2009–2010 rules expanded eligibility.

[30] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-9 and 13-11. Special rules apply to certain 2009–2010 issues and grandfathered obligations.

In the model, a bank's relative after-tax share is $\xi_B = 1 - \tau_B$, 0.65 before 2018 and 0.79 after, which gives the gross exemption benefit of Table 3. Each dollar of bank municipal holdings also bears the financing-tax cost $\tau_B c(1 - 0.8 q_{BQ})$, where $q_{BQ}$ is the bank-qualified share of those holdings. Because interest expense is allocated pro rata, $c$ is measured by the ratio of interest expense to assets. The quantitative model sets $q_{BQ} = 0$. Appendix C.7.1 derives the capitalization share, the fraction of the bank benefit reflected in the designation premium.

**Property and casualty insurers.** P&C insurers pay the corporate rate $\tau_c$ on taxable bond interest. Proration cuts their losses-incurred deduction by a percentage of exempt interest, so that percentage of municipal interest is in effect taxed at $\tau_c$. Before the 2017 Tax Cuts and Jobs Act, the percentage was 15%, giving an effective municipal rate of $0.15 \times 0.35 = 0.0525$. The Act replaced the percentage with 5.25% divided by the highest corporate rate, or 25% at $\tau_c = 21\%$. The municipal after-tax share thus stays at 0.9475 while the taxable share rises from 0.65 to 0.79. The relative after-tax share rises from 0.686 to 0.834, and the exemption benefit falls from 31.4% to 16.6%. Cutting the corporate rate while keeping 15% proration would not reproduce the 2018 law.

The calibration applies the top corporate rate to insurers and banks. Before 2018, marginal corporate rates were 15–34% below \$100,000 and 34–39% from \$100,000 to \$18,333,333, including bubble rates that recapture the lower brackets' benefit, and a flat 35% above. Since 2018 the rate is 21% at every taxable-income level.

**Life insurers and other institutions.** Section 812 divides a life insurer's net investment income into company and policyholder shares, and the policyholders' share of exempt interest reduces its reserve deduction. With a one-half policyholders' share and a 35% corporate rate, the pre-2018 calibration has an effective municipal rate of 17.5%, after-tax shares of 0.825 on municipal and 0.65 on taxable interest, and a relative after-tax share of 0.788. The post-2017 calibration's 30% policyholders' share and 21% corporate rate give an effective municipal rate of 6.3%, after-tax shares of 0.937 and 0.79, and a relative after-tax share of 0.843. In both calibrations the life insurer's exemption benefit is below the P&C insurer's, and the Act narrows the difference. Tax-exempt institutions and tax-deferred accounts already receive taxable interest free of current tax and gain no additional benefit from the exemption.

## D.2 State exemptions and their implementation in the model

State exemptions add a benefit that depends on both the issuer's state and the investor's residence. Home-state repeal (Section 7.2) removes the distinction between own-state and other-state interest.

**Resident households.** State income taxes generally exempt residents' interest on home-state municipal bonds and tax their interest on other states' bonds. The quantitative model applies a common home-state exemption in every income-tax market. The P1 sample keeps only jurisdictions where the exemption applies and excludes the four states with exceptional rules (Illinois, Iowa, Wisconsin and Oklahoma).

For a household resident in state $s$, let $\tau_s^{\text{st}}$ be its state tax rate and $\xi^{\text{fed}}$ its federal relative after-tax share. The baseline normalizes the home-state municipal share to one and sets the out-of-state municipal share to $1 - \tau_s^{\text{st}}$. It grants no federal deduction of state tax on out-of-state municipal interest, a convention varied in Appendix D.2.1. Through 2017, the share on taxable interest, formed tax unit by tax unit, is $\xi^{\text{fed}}(1 - \tau_s^{\text{st}})$, which includes the federal deduction of state tax on taxable interest. After 2017, the model treats the state and local tax deduction cap as binding, so the share is $\xi^{\text{fed}} - \tau_s^{\text{st}}$, as in (125).

**Funds and institutions.** Municipal funds generally pass the federal exemption through to shareholders as exempt-interest dividends. Their state treatment depends on the shareholder's residence and the share of interest from that state's bonds. A single-state fund preserves the home-state exemption for resident shareholders, while a national fund passes it through only on the eligible share, which weakens the household's state-tax incentive. Among institutions, tax-deferred accounts and many pensions owe no current state tax on interest. Insurers commonly pay premium taxes, which do not distinguish municipal from taxable interest, and banks' and other corporations' state taxes depend on the tax base and business apportionment. P1 therefore exploits a household exemption that depends jointly on issuer and holder residence (Section 6.1). For insurers and banks, the model applies only federal rules, including proration and financing deductions.

### D.2.1 Sensitivity to the federal deduction of state tax on out-of-state municipal interest

Granting every household the federal deduction of state tax on out-of-state municipal interest in the pre-2018 periods, which the baseline omits, lowers the level of state-tax pass-through but preserves its declining profile and the large long-maturity gap.

**The rule and the test.** Before 2018, itemizers could deduct state income taxes on federally exempt interest against ordinary federal income tax. Taking the standard deduction, paying the alternative minimum tax or choosing a larger sales-tax deduction could eliminate this saving.[31] I grant every tax unit the full deduction, whatever its eligibility, valued at the ordinary-income component of its baseline federal marginal rate, which excludes NIIT. With federal and state shares multiplied tax unit by tax unit, I re-solve the model in the six pre-2018 periods, holding all other inputs and estimated coefficients fixed. After 2017, the deduction cap binds as in the baseline.

**Why the deduction lowers pass-through.** For the 2015–2017 California high-tax group, the effective after-tax shares with the deduction are 0.498 on taxable interest and 0.919 on out-of-state municipal interest. At a riskless taxable yield of 5%, a municipal yield of 2.71% equates the two after-tax returns. Without the added deduction, this break-even out-of-state yield is 2.87%, the same as for the high-tax group in states without an income tax. The taxable share already includes the baseline deduction, so the added municipal deduction lowers the yield needed to match it. Residents of high-tax states can therefore accept lower out-of-state yields than residents of states without an income tax. Cross-state yield differences shrink, and state-tax pass-through falls even at the riskless maturity.

**Result.** With the 2000–2020 averaging and corporate normalization of Section 5.3, the thirty-year model gap falls by only 0.23 basis points, from 97.39 to 97.16. The model's P1 pass-through, computed as in Figure 9 and averaged over the six periods, falls from 1.000 to 0.904 at the two-year riskless maturity and from 0.460 to 0.416 at thirty years.

## D.3 Taxation over a bond's life: accrual, basis and resale

The model applies the investor's after-tax share of interest income to the whole return $\mathcal{R} = \mu + u$ in (33), without separately tracking coupons, accrual and resale gains.[32] The capital-gains adjustment of Appendix C.1.4 measures the effect of separating interest from resale gains, using the basis accounting below.

**Buying at issue and holding to maturity.** For individuals, federal law separates interest accrual from gains realized at disposition (Sections 171, 1272–1278 and 1288). Consider a bond with face value 100, annual coupon payment $\bar{c}$ in dollars and annual effective issue yield $y_0$. Under constant-yield accrual, its adjusted issue price (AIP) is the original buyer's tax basis after each payment. It starts at the original price $\text{AIP}_0$ and satisfies

$$\text{AIP}_t = (1 + y_0)\text{AIP}_{t-1} - \bar{c}, \qquad t = 1, \ldots, n, \qquad \text{AIP}_n = 100. \tag{133}$$

A discount, $\text{AIP}_0 < 100$, is original-issue discount (OID), and a premium, $\text{AIP}_0 > 100$, is original-issue premium (OIP). Annual interest $y_0\text{AIP}_{t-1} = \bar{c} + (\text{AIP}_t - \text{AIP}_{t-1})$, the coupon plus the noncash accrual, is taxed at the ordinary-income rate $\tau_{oi}$:

$$\text{tax}_t = \begin{cases} \tau_{oi} y_0 \text{AIP}_{t-1}, & \text{taxable bond,} \\ 0, & \text{tax-exempt bond.} \end{cases} \tag{134}$$

The basis change $\text{AIP}_t - \text{AIP}_{t-1}$ prevents taxing accrued interest again at redemption. OID raises basis, including for zero-coupon bonds, and OIP lowers it. Under this accrual convention, the original buyer's basis reaches face at maturity, leaving no residual gain for the capital-gains adjustment. Amortizing premium on a taxable bond requires a Section 171 election, which the capital-gains adjustment assumes, while amortization on an exempt bond gives no deduction. For taxable bonds, de minimis OID below a quarter point per full year

[31] IRS Publication 550 (2017), "State income taxes"; 2017 Instructions for Form 6251, line 3; and 2017 Instructions for Schedule A, line 5. 2016 Instructions for Form 8960, line 9b, restrict NIIT deductions to taxes attributable to net investment income, which excludes exempt interest.

[32] The after-tax share multiplies both the log of the expected gross return, $\log \mathbb{E}[1 + R]$, and the centered log-return innovation. Taxing a simple return $R$ gives the after-tax gross return $1+\xi R$, whose log is $\log(1+\xi R)$. For a deterministic two-year simple return of 5.4% and $\xi = 0.62$, this log differs from $\xi \log(1 + R)$ by 3.2 basis points per two-year period (1.6 a year), and by zero when $\xi = 1$. Appendix A.1.1 reports the effect of taxing simple returns exactly in the solved model. The empirical implied tax and the after-tax yields of Appendix C.1.4 use the model's return convention.

can be treated as zero, with the discount recognized as capital gain at maturity. This de minimis rule generally does not apply to tax-exempt obligations.[33]

**Selling before maturity.** A sale at price $P_h$ at date $h$ realizes the gain $G_h = P_h - \text{basis}_h$, measured from adjusted basis. A holding period of more than one year makes a capital gain long-term, taxed at rate $\tau_{cg}$.

**The next buyer's basis.** A secondary buyer starts with basis equal to the purchase price $P$. If $P$ exceeds face, the premium amortizes at the buyer's yield, mandatorily for exempt bonds and by election for taxable bonds. On an OID bond bought at date $s$ for $\text{AIP}_s < P \leq 100$, the acquisition premium $P - \text{AIP}_s$ reduces the remaining OID by the fraction $(P - \text{AIP}_s)/(100 - \text{AIP}_s)$, leaving basis 100 at maturity.

**Market discount and the tax at sale.** Market discount is the shortfall of the purchase price below $\text{AIP}_s$ for OID bonds or below face for par and premium bonds. It accrues in equal daily amounts, or at constant yield by election. Unless included in income annually, accrued market discount $D_h$ is taxed as ordinary income at sale or redemption, up to any positive gain. This rule also covers exempt bonds bought after April 30, 1993. Below the de minimis threshold (119), the discount is treated as zero and the gain is a capital gain. The sale tax therefore applies $\tau_{oi}$ to the positive gain up to accrued market discount and $\tau_{cg}$ to the residual gain or loss:

$$\text{tax at sale} = \tau_{oi} \min\{\max(G_h, 0), D_h\} + \tau_{cg}[G_h - \min\{\max(G_h, 0), D_h\}], \qquad G_h = P_h - \text{basis}_h. \tag{135}$$

A buyer at a market discount owes this tax, so the capital-gains adjustment builds it into the price that buyer pays (Appendix C.1.4).

**Losses, institutions and the adjustment.** Individual capital losses offset gains and up to $3,000 of ordinary income a year, with carryforwards (Sections 1211–1212). Corporations pay their ordinary rate on gains but offset capital losses only against gains, and banks treat gains and losses on debt as ordinary income (Section 582(c)). The capital-gains adjustment applies the household accrual and basis rules above with the federal rates of Appendix C.1.4 and assumes full loss offsets at the capital-gains rate, as in (120).

## D.4 Fiscal accounting and counterfactual evidence

This subsection supports the fiscal results of Section 7. On average, home-state repeal raises own-state financing costs by 0.68–0.88 per dollar of added state receipts across the seven states, and equal treatment raises all issuers' costs by 1.196 per dollar of added federal and state receipts.

### D.4.1 Measuring tax receipts and financing costs

Portfolio choice depends on relative after-tax shares, but tax receipts require absolute tax rates. For example, the pre-2018 P&C relative after-tax share of 0.6860 reflects taxes of 5.25% on municipal and 35% on taxable interest. Taxing only taxable interest, at $1 - 0.6860$ or 31.4%, would preserve the relative share but misstate receipts in both markets. Each reform therefore changes absolute tax rates, while preferences and liability hedges stay in each investor's baseline payoff units (Section 4.2). For households, I keep the dated federal marginal rates, approximate state taxes and approximate additional tax on Social Security benefits of Appendix B.1, and state receipts go to the state of residence. Before 2018, federal and state after-tax shares multiply, so the federal tax per dollar of taxable interest is the total tax less the state rate, $\tau_i^{\text{fed}}(1 - \tau_i^{\text{st}})$, where $\tau_i^{\text{fed}}$ and $\tau_i^{\text{st}}$ are investor $i$'s federal and state tax rates.

**Scope of the accounting.** The accounting covers interest on bonds and on taxable cash. Type $i$ holds $m_i(W_i - \mathbf{1}'x_i)$ dollars of cash, its funding less its bond positions, and cash interest faces the same absolute tax rates as taxable bond interest. Aggregate funding, bond supplies and the short rate are fixed, so aggregate cash interest is unchanged. When cash moves between investors with different tax rates, the change in its tax receipts is offset by the change in after-tax cash income, preserving identity (30). Changes in the bank financing-tax cost are recorded as federal receipts and deducted from banks' income, without counting interest expense or issuer payments twice. Other liability flows, issuer deductions, refinancing timing and fiscal feedback are outside the accounting.

**Financing costs and averaging.** Issuer costs are annual financing costs at the solved yields, computed as debt outstanding at each maturity times its yield, $y_T D_T$, summed across maturities at $z = 0$. The contractual payments of existing bonds stay unchanged. For both reforms, I weight calendar years equally and divide mean

[33] IRS Publication 550 (2017), "Original Issue Discount (OID)"; Publication 1212, "De minimis rule." The market-discount threshold in (119) applies separately.

dollar costs by mean dollar receipts. The dollar value of the puzzle in the introduction uses the same accounting. It sums the puzzle at each two-year maturity times municipal debt outstanding at that maturity and averages the six periods of 2000–2017 equally, which gives $13.0 billion a year, from $5.8 billion in 2000–2002 to $16.8 billion in 2015–2017.

### D.4.2 Portfolio reallocation under the two reforms

Receipts depend on who holds each bond after a reform, so Table 37 reports the portfolio changes behind the fiscal effects.

Table 37: Portfolio allocation under the two tax reforms

| Panel A. Home-state repeal: average resident allocations, 2000–2017 | | | |
|---|---|---|---|
| | Home-state municipal | All risky | Riskless |
| California | 20.33 → 4.35 | 30.02 → 31.56 | 69.98 → 68.44 |
| New York | 19.65 → 3.92 | 29.97 → 31.18 | 70.03 → 68.82 |
| New Jersey | 18.63 → 1.32 | 31.34 → 31.12 | 68.66 → 68.88 |
| Massachusetts | 18.96 → 1.55 | 29.95 → 30.99 | 70.05 → 69.01 |
| Ohio | 19.10 → 1.09 | 29.75 → 31.00 | 70.25 → 69.00 |
| Pennsylvania | 17.42 → 2.15 | 30.76 → 30.58 | 69.24 → 69.42 |
| Minnesota | 20.19 → 0.64 | 29.60 → 31.38 | 70.40 → 68.62 |
| **Panel B. Equal treatment: average investor allocations, 2000–2017** | | | |
| | Risky maturity (years) | Share of long bonds (%) | Riskless |
| High-tax households | 6.71 → 4.91 | 2.6 → 0.0 | 75.31 → 76.96 |
| Lower-tax households | 12.85 → 12.14 | 29.7 → 27.2 | 66.07 → 64.95 |
| Banks | 11.76 → 8.03 | 8.6 → 0.0 | 0.00 → 0.00 |
| P&C insurers | 9.93 → 6.83 | 5.3 → 1.7 | 21.70 → 0.65 |
| Life insurers | 11.81 → 12.07 | 14.6 → 14.0 | 0.38 → 0.68 |
| Tax-neutral investors | 7.73 → 8.53 | 39.2 → 57.1 | 27.90 → 29.56 |

*Notes.* Entries are baseline → reform values. Home-state, all-risky and riskless shares are percentages of each group's funding, which equals its bond holdings plus cash and excludes outside assets and liabilities. The share-of-long-bonds column reports each group's percentage of all municipal and taxable bonds maturing in more than ten years and sums to 100% across groups. Panel A aggregates resident households within each state and averages their portfolio shares across the six periods of 2000–2017. Each row is a separate repeal. Panel B averages each investor group's allocations across the same periods under 99.9% equal treatment. Both panels weight types by their mass $m_i$ and use rate state $z = 0$. Tax-neutral investors are the residual group of Table 3. Riskless holdings comprise cash and one-period municipal bonds (two years on the numerical grid). All longer bonds are risky. In Panel A, home-state municipal holdings are also counted in the last two columns, which sum to 100%. In Panel B, risky maturity is the value-weighted mean maturity of the group's risky municipal and taxable bonds. Under equal treatment, risky municipal and taxable bonds of the same maturity become close substitutes, so Panel B reports their combined maturity profile (Appendix B.7.4).

**Home-state repeal.** Panel A combines the two resident household groups, which respond differently. High-tax residents raise their riskless share by 0.6–2.7 percentage points across the seven states. Lower-tax residents raise their taxable-bond share by 5.1–14.6 points and lower their riskless share. For both groups together, the riskless share rises in New Jersey and Pennsylvania and falls in the other five states. The high-tax group's mean bond maturity rises by 0.4–0.9 years because it replaces riskless municipal bonds with cash, which that average excludes, while its risky bonds become slightly shorter. In California, its mean bond maturity rises from 5.99 to 6.44 years and its mean risky-bond maturity falls from 7.05 to 6.52, while the lower-tax group's mean risky-bond maturity falls from 14.47 to 11.84 years. The added receipts thus accompany substitution across markets and tax groups, without a shift toward longer risky bonds.

**Equal treatment.** In Panel B, the tax-neutral group's share of municipal and taxable bonds beyond ten years rises from 39.2% to 57.1%. The high-tax household and bank shares fall to almost zero, the P&C share falls from 5.3% to 1.7%, and the lower-tax household and life-insurer shares decline slightly.

### D.4.3 Home-state repeal: financing costs and receipts

Panel A of Table 38 reports each state experiment by period, and Panel B decomposes the six-period averages.

Table 38: Home-state repeal: financing ratios by period and the decomposition of costs and receipts

| Panel A. Own-state / all-issuer financing costs per added state receipt dollar | | | | | | | |
|---|---|---|---|---|---|---|---|
| State | 2000–2002 | 2003–2005 | 2006–2008 | 2009–2011 | 2012–2014 | 2015–2017 | Average |
| CA | 0.80 / 1.19 | 0.61 / 0.84 | 0.63 / 0.67 | 0.78 / 1.13 | 0.94 / 1.41 | 0.96 / 1.52 | 0.81 / 1.17 |
| NY | 0.74 / 1.15 | 0.64 / 0.89 | 0.59 / 0.49 | 0.78 / 1.11 | 0.94 / 1.38 | 0.92 / 1.41 | 0.78 / 1.08 |
| NJ | 0.82 / 1.35 | 0.79 / 1.23 | 0.80 / 1.11 | 0.83 / 1.18 | 1.00 / 1.49 | 0.98 / 1.46 | 0.88 / 1.30 |
| MA | 0.66 / 1.03 | 0.51 / 0.75 | 0.56 / 0.49 | 0.66 / 0.95 | 0.80 / 1.04 | 0.90 / 1.20 | 0.68 / 0.89 |
| OH | 0.73 / 1.17 | 0.57 / 0.83 | 0.60 / 0.57 | 0.70 / 1.04 | 0.88 / 1.19 | 0.99 / 1.39 | 0.73 / 0.99 |
| PA | 0.66 / 1.11 | 0.56 / 0.88 | 0.66 / 1.02 | 0.63 / 0.87 | 0.90 / 1.12 | 1.02 / 1.32 | 0.73 / 1.03 |
| MN | 0.69 / 1.07 | 0.52 / 0.74 | 0.53 / 0.50 | 0.68 / 0.95 | 0.86 / 1.22 | 0.83 / 1.16 | 0.68 / 0.93 |

| Panel B. Average annual changes ($million) | | | | | | | |
|---|---|---|---|---|---|---|---|
| State | Cost at fixed $y^0$ | Change in $y^0$ | Total cost | Tax at $y^0$ | Yield changes | Reallocation | Total receipts |
| CA | 798.7 | 17.2 | 815.9 | 902.5 | 1.0 | 100.8 | 1004.3 |
| NY | 455.4 | 6.6 | 462.0 | 526.0 | 0.1 | 65.2 | 591.4 |
| NJ | 138.4 | 0.2 | 138.6 | 141.8 | -0.2 | 15.2 | 156.9 |
| MA | 76.1 | 0.1 | 76.2 | 98.9 | -0.1 | 14.0 | 112.8 |
| OH | 75.1 | 0.1 | 75.1 | 90.1 | -0.1 | 12.7 | 102.6 |
| PA | 59.8 | -0.3 | 59.5 | 73.0 | -0.1 | 9.0 | 81.9 |
| MN | 59.5 | 0.1 | 59.6 | 75.8 | -0.1 | 11.3 | 87.0 |

*Notes.* Each state-period cell is a separate repeal from its dated baseline. All-issuer costs add other municipal and pooled taxable issuers to the state's own issuers, and receipts are the repealing state's added receipts. The average column divides six-period mean dollar costs by mean receipts. Where the all-issuer ratio is below the own-state ratio, as in several 2006–2008 cells, residents' reallocation lowers other issuers' yields and hence their financing costs. Panel B splits the average own-state costs and receipts around each period's comparison yield $y_T^0$ (Section 7.2), with the cost-at-fixed-$y^0$, change-in-$y^0$ and tax-at-$y^0$ columns defined in the text. Yield changes are the tax on residents' baseline home-state holdings at post-repeal yields less the tax at $y^0$, and reallocation is the net effect of residents' portfolio changes, including cash. Components sum to totals up to rounding. All dollar calculations use continuous yields and holdings at $z = 0$ and include residents' taxable cash interest.

**Costs and receipts around the comparison yield.** Panel B separates the fixed-yield comparison of Section 7.2 from yield changes and reallocation. Let $D_T$ be the state's debt at maturity $T$, $y_T^0$ the comparison yield (the pooled municipal yield of the states without an income tax) and $\tau_T^{\text{impl}} = 1 - y_T^M / y_T^0$ the implied state tax, with primes marking post-repeal values. The own-state cost increase is

$$\Delta C = \sum_T D_T \, y_T^0 \left(\tau_T^{\text{impl}} - \tau_T^{\text{impl}\prime}\right) + \sum_T D_T \left(1 - \tau_T^{\text{impl}\prime}\right)\left(y_T^{0\prime} - y_T^0\right).$$

The first term holds $y^0$ fixed, and the second comes from the change in $y^0$. Averaged over the six periods, they give Panel B's cost-at-fixed-$y^0$ and change-in-$y^0$ columns. The tax at $y^0$ is $\sum_i \tau_i^{\text{st}} \sum_T H_{iT} y_T^0$, where $H_{iT}$ is resident group $i$'s baseline home-state holding and $\tau_i^{\text{st}}$ its state rate.

At a fixed comparison yield, own-state cost per dollar of tax ranges from 0.770 in Massachusetts to 0.976 in New Jersey. Beyond the tax at the comparison yield, almost all remaining receipts come from residents' reallocation, which adds 10.8% to 14.9% of that tax, so costs stay below receipts in every state. By asset, California's added annual receipts of $1.004 billion comprise $0.227 billion from home-state bonds and net increases of $0.300 billion from other states' bonds, $0.368 billion from taxable bonds and $0.110 billion from cash. Interest on these last three assets was already taxable, so their increases come from changes in residents' holdings and the interest they earn.

California's $100.8 million reallocation term values purchases and sales at post-repeal yields. If every bond earned the post-repeal comparison yield $y_T^{0\prime}$ at its maturity and cash kept its taxable yield, the reallocation term would be $4.9 million. Taxable yields above this reference add $96.8 million and municipal yield differences −$0.9 million, so most of the increase comes from substitution into higher-yielding taxable bonds. All three terms value the same solved portfolio changes. The taxable-yield term is the largest part in the other six states as well, at 62% of the reallocation term in Pennsylvania, 64% in New Jersey and at least 90% elsewhere.

**Differences across states.** At longer maturities, lower state-tax pass-through reduces the cost per resident-held dollar, but debt held by nonresidents and institutions adds cost without adding tax. New Jersey's residents hold only 49% of its 12–20 year debt, on which the cost is 1.38 times the tax at the comparison yield, giving New Jersey the highest overall ratio. Minnesota's residents hold 80% of its 22–30 year debt, against 67% in New Jersey, and its lower-tax residents face a higher state rate. Minnesota's cost on those bonds is therefore 0.59 per dollar of tax, against 0.98 in New Jersey. Household composition, outside rate exposures and resident funding relative to local debt are common across states within each period, so these differences in resident ownership and cost come from tax schedules and market sizes.

### D.4.4 Equal treatment and an issuer subsidy

Table 39 reports the equal-treatment reform by period and a subsidy that returns all added federal and state receipts to municipal issuers. Per receipt dollar, the municipal cost after subsidy is the all-issuer cost less the taxable-issuer cost less one dollar. It averages 0.063, while taxable issuers still bear their 0.133 increase. In every period, all-issuer costs exceed the added receipts.

Table 39: Equal treatment and a revenue-financed issuer subsidy across the six pre-2018 periods

| | Thirty-year yield increase (bp) | | Costs per receipt dollar | | |
|---|---|---|---|---|---|
| Period | Municipal | Taxable | All issuers | Taxable issuers | Municipal after subsidy |
| 2000–2002 | 199.8 | 38.2 | 1.434 | 0.282 | 0.152 |
| 2003–2005 | 136.9 | 9.3 | 1.125 | 0.089 | 0.036 |
| 2006–2008 | 146.5 | 11.4 | 1.077 | 0.073 | 0.004 |
| 2009–2011 | 152.7 | 11.2 | 1.113 | 0.043 | 0.070 |
| 2012–2014 | 178.9 | 33.9 | 1.245 | 0.170 | 0.075 |
| 2015–2017 | 208.6 | 34.7 | 1.242 | 0.174 | 0.068 |
| Average 2000–2017 | 170.6 | 23.1 | 1.196 | 0.133 | 0.063 |

*Notes.* Each row removes 99.9% of the municipal–taxable tax distinction from its own calibrated baseline. Yield changes average each period's 36 monthly rate states. Fiscal ratios use continuous yields and holdings at $z = 0$, with bond and taxable cash interest in receipts. The last column subtracts a subsidy equal to all added federal and state receipts from municipal issuer costs. The other columns describe the reform before subsidy. The subsidy leaves gross yields and portfolios unchanged because bond supplies and investors' funding are fixed. The average row divides calendar-year-weighted mean costs by mean receipts.

## D.5 Financing, short sales and rate hedges

The model rules out short sales and borrowing (Sections 3.5 and 4.1). Financing a municipal purchase, hedging it and shorting a municipal bond are distinct trades with different tax and market costs. For municipal and corporate par coupons $c_M$ and $c_C$ of the same maturity and a marginal tax rate $\tau$, carry is net after-tax coupon income for equal principal in legs that need not match in duration, before fees, margin and collateral costs. The after-tax coupon gap $g = c_M - (1-\tau)c_C$ differs from the 118-basis-point thirty-year puzzle in zero-coupon yields (Section 5.3) and from the model's one-period return gap $g_T$. With equal short-seller and lender tax rates, financing a municipal bond with a corporate short and the reverse trade, cases (i) and (iii), both have nonpositive carry when $c_M$ lies between $(1-\tau)c_C$ and $c_C$ (139), as long-end municipal coupons do. Returns with adjusted gross income of $1 million or more, about 0.3% of all returns, reported 26–33% of tax-exempt interest in tax years 2008, 2016 and 2019, so the high-income households whose top-bracket rates define the household benchmark hold much of the market.[34]

### D.5.1 Financing and shorting

**(i) Financing municipals with debt or a corporate short.** Section 265(a)(2) denies deductions for interest on debt incurred or continued to purchase or carry tax-exempt obligations. With interest $c_F$ on a dollar borrowed per dollar of municipal principal, carry falls from $c_M - (1-\tau)c_F$ to

$$c_M - c_F = \{c_M - (1-\tau)c_F\} - \tau c_F. \tag{136}$$

[34] IRS Statistics of Income, Individual Income Tax Returns, Table 1.4, all returns by size of adjusted gross income, tax years 2008, 2016 and 2019. Tax-exempt interest includes exempt-interest dividends from funds.

At after-tax parity, $c_M = (1-\tau)c_F$, the denial turns zero carry into a loss of $\tau c_F$. A corporate short does not restore the deduction, because Section 265(a)(5) treats its short-sale payments $c_C$ as interest under Section 265(a)(2). Carry $c_M - c_C$ is then negative at every maturity on the sample-average curves, with thirty-year municipal and corporate par yields of 4.498% and 5.443% at annual compounding.[35] While the deduction was available, depository institutions held 51.8% of municipal bonds in 1972 and $235.1 billion (27.3%) in 1985. After the 1986 denial (Appendix D.1), their holdings fell to $120.4 billion in 1990 and $95.4 billion (7.5%) in 1995, while municipal debt outstanding grew from $862 billion in 1985 to $1.27 trillion in 1995.[36]

**(ii) Pairing corporate positions.** A corporate short whose proceeds buy a corporate bond is the hedge in (v), and one whose proceeds finance the municipal bond falls under (i). The case depends on the borrowing's purpose, which Revenue Procedure 72-18 tests through tracing of the proceeds, exempt collateral or the facts and circumstances, and its presumption that business borrowing is not incurred to carry exempt bonds lapses when the borrowing exceeds business needs. For an offsetting straddle, Section 263(g) also capitalizes interest and carrying charges, net of the positions' taxable income, into basis, deferring the deductions.[37]

**(iii) Shorting municipals.** A short seller pays the lender an amount equal to each coupon, called a substitute payment, which, unlike the exempt coupon, is taxable to the lender. A lender with tax rate $\tau_L$ who requires an after-tax payment of $c_M$ must therefore receive $L$ with

$$(1-\tau_L)L = c_M, \qquad L - c_M = \frac{\tau_L}{1-\tau_L} c_M. \tag{137}$$

The short also needs a willing lender of the specific issue and must meet the contract's collateral terms.[38] The reverse of (i) shorts a municipal bond and buys an equal-principal corporate bond. With short-seller tax rate $\tau_S$ and a currently deductible fraction $\delta \in [0,1]$ of the substitute payment, its coupon carry is

$$\pi_{M\to C} = (1-\tau_S)c_C - (1-\delta\tau_S)\frac{c_M}{1-\tau_L}. \tag{138}$$

Even with $\delta = 1$ and equal rates $\tau$, this carry $(1-\tau)c_C - c_M$ is nonpositive when $c_M \geq (1-\tau)c_C$, and the carry $c_M - c_C$ of (i) is nonpositive when $c_M \leq c_C$, so both are nonpositive throughout the band

$$(1-\tau)c_C \leq c_M \leq c_C, \tag{139}$$

which contains long-end municipal coupons. They exceed the household benchmark and, on the sample-average curves, fall below the corporate coupon. A municipal curve trade pays the gross-up (137) on its long-maturity short, and the payment is nondeductible as in (i) to the extent the proceeds buy municipal bonds.

**(iv) Floaters, de minimis holdings and other entities.** A tender-option-bond partnership splits a long municipal bond's cash flows between a floating-rate claim and a residual inverse floater, and exempt income passes through to both classes, so the structure needs no interest deduction. The inverse holder receives the municipal curve's slope and bears the bond's duration, so capturing $g$ still requires a hedge from (v)–(vii), and demand for the floaters limits the scale.[39] For insubstantial holdings, Revenue Procedure 72-18 ordinarily presumes that borrowing is not incurred to carry exempt bonds, absent tracing or exempt collateral. A corporation qualifies only up to 2% of average business assets, larger holdings face the facts-and-circumstances test, and Sections 163(d), 263(g), 1092 and 1211(b) still constrain deductions and losses.[40] Financial institutions outside the bank-qualified exception face the pro rata disallowance of Appendix D.1 without tracing, and Revenue

[35] 26 U.S.C. §265(a)(2), (5) and (b), 2018 U.S. Code. Section 265(a)(5)(B) excepts specified short-sale payments when the taxpayer posts cash collateral that earns no material return. That exception does not remove the opportunity cost of the collateral. Municipal fund disclosures tell investors that interest on debt allocable to exempt-interest dividends is nondeductible. See BlackRock Municipal Bond Fund, Inc. (2025), Part II, "Dividends and Taxes," p. II-131; iShares Trust (2026), "Taxes," p. 210; Nuveen Municipal Trust (2026), "Tax Matters," p. S-80. See also Hines and Horton (2024) on banks' financing-tax restrictions.

[36] Board of Governors of the Federal Reserve System, Financial Accounts of the United States (Z.1), release of September 8, 2023, Table L.212, year-end levels. Depository institutions comprise U.S.-chartered depository institutions, foreign banking offices in the U.S. and banks in U.S.-affiliated areas.

[37] Rev. Proc. 72-18, 1972-1 C.B. 740, §§3.01–3.04 and 6.01, as summarized in Revenue Ruling 2004-47 and in IRS Field Service Advice 200129009, which is nonprecedential. On straddles, see 26 U.S.C. §263(g)(1)–(2) and §1092(c)(1)–(2). Positions are offsetting when holding one substantially diminishes the risk of loss from holding the other.

[38] IRS Publication 550 (2016), "Reporting Substitute Payments"; Treasury Regulation §1.6045-2(a)(4).

[39] Revenue Procedure 2003-84, Section 2; 26 U.S.C. §702(b).

[40] Rev. Proc. 72-18, §3.05, as summarized in Revenue Ruling 2004-47 and Field Service Advice 200129009.

Procedure 72-18 applies a similar ratio to dealers' business borrowing that cannot be traced. Insurer proration reduces the exemption regardless of financing, through P&C insurers' loss deductions (Section 832(b)(5)(B)) and life insurers' reserve deductions (Sections 807 and 812). Partnerships pass the exempt character of municipal interest through to partners under Section 702(b), but their interest expense remains subject to Section 265(a)(2) under Section 703(a). Dealers and hedge funds hold few municipal bonds, dealers on average 0.85% and at most 1.94% over 2000–2020 and hedge funds, reported from 2012, at most 0.36%.[41]

### D.5.2 Rate hedges

**(v) Hedging duration with a corporate short.** An unlevered hedge of a municipal bond's duration shorts a long-maturity corporate bond with coupon $c_\ell$ and buys a short-maturity one with coupon $c_s$ using the proceeds, so that $g = c_M - (1-\tau)c_\ell$. With $\delta$ the currently deductible fraction of the short-sale payment, annual carry relative to holding the short corporate bond alone is

$$\Delta = \{c_M + (1-\tau)c_s - (1-\delta\tau)c_\ell\} - (1-\tau)c_s = g - (1-\delta)\tau c_\ell. \tag{140}$$

Denial under Section 265 gives $\delta = 0$ and $\Delta = c_M - c_\ell$, negative at every maturity and a loss of 95 basis points a year at thirty years on the observed sample-average par curves, larger still on the call-adjusted municipal curve. Where neither Section 265 nor Section 263(g) applies, Section 163(d) lets individuals deduct investment interest only if they itemize and only up to net investment income, which the short-maturity corporate coupons raise and wages and exempt coupons do not. Thus $\delta < 1$ once the coupon difference between the corporate legs exceeds the unused capacity.[42] Section 263(g) capitalization and Section 163(d) carryforwards defer deductions, so current carry approaches $c_M - c_\ell$ and later deductions bear a timing and liquidity cost. Even with $\delta = 1$, bond-specific risk (Appendix A.2.8) and credit, liquidity and call differences (Appendix C) remain, and shorting requires available bonds and incurs fees, recall risk and margin demands against an illiquid callable long position.[43] Fees for borrowing corporate bonds are 10–20 bp in a major lender's data used by Asquith et al. (2013).

**(vi) Hedging with swaps.** A pay-fixed, receive-floating swap hedges duration without borrowing or shorting a bond, but a person that is not an eligible contract participant can trade a swap only on, or subject to the rules of, a designated contract market. An individual qualifies only with more than $10 million invested on a discretionary basis, or more than $5 million when the swap manages the risk of an asset owned or a liability incurred.[44] Ordinary swap payments are not automatically interest on indebtedness, so Section 265(a)(2) alone does not deny their deduction, but Section 265(a)(1) disallows a household's Section 212 investment expenses attributable to exempt interest.[45] Whatever the classification, let $G$ be a hedge's net taxable gain and $d \in [0,1]$ the denied fraction of the tax saving on its losses, so that its after-tax value is

$$A_d(G) = (1-\tau)G - d\tau G^-, \qquad G^- := \max\{-G, 0\}. \tag{141}$$

Symmetric offsets give $d = 0$, and taxing gains while denying losses gives $d = 1$, which lowers expected proceeds by $\tau\mathbb{E}[G^-]$. Even then, netting gives the swap an after-tax advantage of $\tau$ times the smaller of matched periodic receipts and payments over a corporate short with a wholly nondeductible short-sale payment, and additional deduction capacity favors the corporate short. The swap avoids the corporate short's credit and borrowing exposures but carries its own collateral and counterparty exposure.

**(vii) Treasury futures.** Regulated futures avoid the eligibility rules but require cash margin and annual marking to market, with gains and losses treated as 60% long-term and 40% short-term. Individuals offset

[41] Rev. Proc. 72-18, §§5.03 and 7, as summarized in Revenue Ruling 2004-47; 26 U.S.C. §265(b)(1)–(2), §832(b)(5)(B)–(C), §807(a)–(b) and §812. Holdings: Financial Accounts of the United States (Z.1), release of September 8, 2023, Tables L.212 and B.101.f, quarterly levels.

[42] 26 U.S.C. §163(d)(1)–(3), which includes otherwise deductible short-sale expenses in investment interest; IRS Publication 550 (2016), Chapter 3, "Interest Expenses" and "Interest and carrying charges on straddles"; IRS Form 4952 and instructions (2016). Net capital gain and qualified dividends count as investment income only through an election that relinquishes their preferential rates to that extent.

[43] Financial Industry Regulatory Authority (FINRA) Rule 4210, paragraph (e)(2), sets margin requirements for bond shorts.

[44] Commodity Exchange Act §2(e) and §1a(18)(A)(xi), 7 U.S.C. §§2(e), 1a(18)(A)(xi). See also the Commodity Futures Trading Commission's 2012 final-rule summary.

[45] Treasury Regulations §1.446-3 and §1.265-1(a)(2), (c). IRS Field Service Advice 200037034 treats the ordinary swap payments before it as non-interest in the absence of a loan. The advice concerns a corporate taxpayer and is nonprecedential.

losses fully against capital gains, deduct at most $3,000 against ordinary income and carry the excess forward. Net Section 1256 losses can also be carried back three years against net Section 1256 gains, and Section 1092 defers losses to the extent of unrecognized gains on offsetting bonds. Gains can thus be taxed before losses are deducted, a timing and liquidity cost that acts in present value as a partial loss denial, $0 \le d < 1$ in (141).[46]

### D.5.3 The model constraint and risk after hedging

**The model constraint.** With a concave objective and linear shorting costs, a zero short position is optimal when the risk-adjusted benefit of the first dollar shorted does not exceed the tax, borrowing and financing costs. The quantitative model imposes this corner solution, with bond positions nonnegative and limited by measured funding, and banks bear the measured financing-tax cost.

**Residual risk and participation after hedging.** The mechanism can operate when investors also trade derivatives, because it depends on their cost of bearing additional rate risk after available hedges. Suppose the shorting constraint on cash is slack and bond-specific risk is zero. A hedge removes $q_i$ units of rate exposure at an expected charge $f_i q_i$ plus a differentiable convex cost $\psi_i(q_i)$, in normalized payoff units, and leaves residual exposure $\widehat{D}_i^{\text{hedge}} = \widehat{D}_i - q_i$, where $\widehat{D}_i$ is the net rate exposure (10). At an interior hedge, maximizing $-f_i q_i - \psi_i(q_i) - \gamma_i \sigma_r^2 (\widehat{D}_i^{\text{hedge}})^2/2$ gives the unit cost of rate risk after hedging,

$$\lambda_i := \gamma_i \sigma_r^2 \widehat{D}_i^{\text{hedge}} = f_i + \psi_i'(q_i). \tag{142}$$

A binding hedge limit turns this equality into an inequality. If the household holds the $T$-period municipal bond at an equilibrium with optimized hedges, the lower-tax investor's slack in its portfolio condition for that bond, its required return minus the offered return, is

$$\nu_{I,T}^M = (\xi_I - \xi_H) r - A_{T-1}^M (\lambda_H - \lambda_I), \tag{143}$$

and the lower-tax investor also holds the bond only if this slack is zero. This is the participation comparison (59) with unit costs of rate risk measured after hedging. Even when both investors hedge, differences in liabilities leave these costs unequal if hedge access differs, limits bind or hedging costs rise with the hedge. If unlimited hedging in either direction spans rate risk, with $\psi_i = 0$ and a common normalized charge $f$, then $\lambda_H = \lambda_I = f$ and, at a positive short rate, the slack reduces to the positive difference in exemption benefits $(\xi_I - \xi_H) r$, which excludes lower-tax municipal holdings. This spanning case requires a common after-tax hedging opportunity, which zero dealer fees alone do not provide, and observed insurer holdings (Figure 25) require a departure from it. Insurers already hedge, so their participation depends on the cost of rate risk after hedging. At year-end 2015, interest-rate swaps were 64% of the notional value of U.S. insurers' interest-rate hedges, and insurers using derivatives held 87% of life-insurer assets but 19% of P&C assets.[47]

### D.5.4 Household swaps: incentives and equilibrium

Only the wealthiest households can meet the thresholds of (vi). Giving every household access to a pay-fixed, receive-floating swap, which gains when rates rise, reduces the 2015–2017 thirty-year gap by only 9.1%.

**Swap quotes, pricing and taxes.** Swap quotes come from the 95 dates, all in 2001–2008 and outside the 2015–2017 period, on which the two-year zero rate bootstrapped from the swap curve lies within 10 bp of the model's mean rate, 3.0206% a year. Fixed rates at ten, twenty (interpolated) and thirty years are 4.8217%, 5.0976% and 5.3736%.[48] Floating payments are quarterly and fixed payments semiannual, with one curve for discounting and projection. Swaps are priced at inception and at liquidation after two years under a Gaussian rate state with quarter-by-quarter drifts fitted to these quotes, while physical rate dynamics preserve the calibrated two-year persistence and variance. Annual net gains and the second-year liquidation payoff are taxed as ordinary income, with losses denied in the baseline and offset symmetrically in a sensitivity check. Separately held swap cash earns taxable interest when positive and pays nondeductible interest when negative.

[46] 26 U.S.C. §1256(a)(1), (a)(3), (b) and (g)(1); §1211(b); §1212(b)–(c); §1092(a)(1)(A). Section 1256(a)(4) removes Sections 1092 and 263(g) only for straddles consisting entirely of Section 1256 contracts.

[47] NAIC, "Update on the Insurance Industry's Use of Derivatives and Exposure Trends," year-end 2015, Tables 1 and 8 and the accompanying discussion. Notional shares measure instrument use and are silent on how much liability risk is eliminated.

[48] Federal Reserve H.15 (Selected Interest Rates) series DSWP1–DSWP5, DSWP7, DSWP10 and DSWP30, numbered by tenor in years, are mid-market fixed rates against three-month London Interbank Offered Rate (LIBOR) and differ from household quotes.

Municipal funds likewise distribute swap gains as taxable ordinary income or capital gains and allocate net payments pro rata against taxable and exempt income.[49]

**Stage 1: the incentive at fixed bond portfolios.** Stage 1 holds bond portfolios and existing cash fixed for all 630 household types in the mean rate state. A swap with notional $q \geq 0$ and after-tax cash payoff $A_i$ per unit notional, unrelated to the rate loading $A_T^a$, adds $qA_i/\xi_i^M$ to investor $i$'s calibrated after-tax portfolio innovation $\widetilde{W}_i$ (35). Here $\xi_i^M$ is the after-tax share of home-state municipal interest (Section 4.1), and $\widetilde{W}_i$ is in normalized payoff units and includes the term $-h_i z'$ from outside rate exposure (Appendix B.1). The swap's annualized marginal benefit in basis points per unit notional is then

$$b_i = \frac{10^4}{2}\left[\mathbb{E}A_i - \gamma_i \operatorname{Cov}(\widetilde{W}_i, A_i)\right]. \tag{144}$$

A positive $b_i$ means that a small pay-fixed position improves the objective. The calculation uses the model's log-return approximation, and independent bond-specific shocks have zero covariance with the rate-only swap.

Table 40: Fixed-portfolio swap benefits with gains taxed and losses denied.

| Household group | Years | Mean payoff | Risk benefit | Total | Median | Positive |
|---|---|---|---|---|---|---|
| High tax | 10 | −408.6 | 852.8 | 444.3 | −14.4 | 135/315 |
| High tax | 20 | −473.5 | 996.0 | 522.6 | −13.1 | 135/315 |
| High tax | 30 | −473.9 | 988.2 | 514.2 | −17.2 | 135/315 |
| Lower tax | 10 | −387.9 | 675.6 | 287.7 | −28.8 | 135/315 |
| Lower tax | 20 | −450.0 | 787.6 | 337.7 | −31.5 | 135/315 |
| Lower tax | 30 | −449.5 | 783.3 | 333.9 | −33.3 | 135/315 |

*Notes.* Annualized marginal benefits $b_i$ of equation (144), in basis points per unit notional. Years is the swap tenor. Mean payoff and risk benefit are the two terms of $b_i$ and total is their sum, each averaged within the group with portfolio-asset weights. Positive counts types with positive $b_i$ out of all types in the group. The median and this count weight types equally. Entries are rounded separately.

Table 41: Household swap access and the equilibrium municipal gap.

| Maturity | Baseline | Swap access | Change |
|---|---|---|---|
| 2 years | 0.00 | 0.00 | 0.00 |
| 10 years | 23.68 | 22.24 | −1.44 |
| 20 years | 72.63 | 65.42 | −7.20 |
| 30 years | 104.27 | 94.79 | −9.48 |

*Notes.* Entries are the supply-weighted municipal yield minus the household benchmark, in basis points, averaged over the 36 monthly rate states of 2015–2017 with continuous yields. After-tax shares, parameters and dollar supplies are fixed, and both yields adjust. Changes are computed before rounding. Refining the simulation fourfold moves either yield curve or the gap by less than 0.003 bp, and price and optimality residuals are below $10^{-10}$.

**Stage 1 results.** At all three tenors, 270 types have positive benefits and 360 negative ones. These 270 types have the three largest of the seven outside rate exposures in each household tax group and home state. They hold 42.9% of household portfolio assets but only 9.8% of household municipal bonds, and their larger gains make the portfolio-asset-weighted mean positive despite the negative median (Table 40). With symmetric loss offsets, 585 of the 630 benefits are positive. Doubling pricing resolution and draws, or computing benefits from terminal dollar payoffs, preserves every sign, the doubling changes benefits by at most 2.57 bp, and a wider quote window gives between 275 and 302 positive benefits, some near zero. Additional costs per unit notional reduce hedge demand at given bond holdings, and margin and position limits restrict feasible hedges.

**Stage 2: equilibrium with household hedge access.** Stage 2 fixes the 2015–2017 calibration and lets every household choose bonds and a nonnegative thirty-year pay-fixed swap notional jointly. Outside dealers quote swaps with no fee, bid–ask spread, margin, collateral opportunity cost or notional limit and do not trade the modeled bonds. New swaps are struck at the current state's par rate, with the fitted drifts counted from their start date. Swap gains are taxed, losses are denied ($d = 1$), interim swap cash is financed separately,

[49] BlackRock Municipal Bond Fund, Inc. (2025), Part II, pp. II-20, II-126 and II-132; iShares Trust (2026), "Taxes," p. 213.

bonds and cash stay nonnegative and institutions' hedging is unchanged, while bond prices in both markets and all portfolios adjust. Swap access reduces the thirty-year gap by 9.48 bp, or 9.1%, on the 2015–2017 model curves before the full-sample corporate normalization (23) (Table 41). Thirty-year municipal yields fall 15.95 bp and taxable yields 12.04 bp. At $z = 0$, 270 types hedge, but over 99.99% of their municipal holdings remain riskless two-year bonds because these types mainly hedge outside rate exposure. Portfolio responses at other rate states still affect resale prices. The test substantially expands household access but keeps asymmetric taxes and nondeductible interest on negative swap cash. The largest notional at $z = 0$ is 6.88 times the type's funding.

## D.6 Data sources and analysis coverage

Table 42 lists what each source measures and its coverage, the span assembled for the paper. Each analysis states its own estimation window. Appendix B explains the calibration inputs, and Appendix C gives the matched samples for the adjustments and empirical tests.

Table 42: Data sources, variables and coverage.

| Source | Variables | Coverage |
|---|---|---|
| Municipal issues: Mergent Municipal Bond Database (FTSE Russell) | CUSIP, issue & maturity dates, coupon, rating, callability, tax status (federal/state), issuer state, purpose, amounts outstanding | 1986–2023 |
| Corporate issues: Mergent FISD (FTSE Russell) | CUSIP, issue & maturity dates, coupon, callability, rating, issuer industry, seniority, amounts outstanding | 1986–2023 |
| Insurers: NAIC Schedule D (annual, Q1–Q3 supplements) | CUSIP-level par, book and market values for U.S. P&C and life insurers | 1991–2023 |
| Banks: FDIC BankFind (Call Report income statement RI and securities schedule RC-B) | Annual interest expense, average assets and municipal securities holdings by bank | 2000–2021 |
| Public pensions: FOIA requests and annual financial reports | CUSIP-level municipal/taxable par and market values, 30 state/local plans (all 30 covered from 2015) | 2000–2023 |
| Sector holdings and assets: Financial Accounts (Z.1) | Quarterly municipal holdings by sector (L.212, B.101.f), sector debt securities and cash, treasury and corporate totals | 1972–2022 |
| Households: SCF, Federal Reserve | Tax-unit income and filing status, financial assets and debt | 2001–2019 waves |
| Household taxes: IRS Statistics of Income (SOI) | Filer counts and taxable and tax-exempt interest by AGI bracket | 2000–2022 |
| Municipal trades: MSRB Real-Time Transaction Reporting System (RTRS) | Trade-level price, par volume, customer-side flag, trade-condition codes | 2005–2023 |
| Corporate trades: FINRA TRACE Enhanced (Academic) | Trade-level price, par volume, customer-side flag, condition codes | 2002–2023 |
| Treasury curve: Gürkaynak–Sack–Wright (GSW), Federal Reserve | Daily fitted par yields, 1–30y, used in P3 | 2000–2023 |
| Short rates: TB3MS, FRED | Monthly three-month treasury bill rates for the rate process | 1952–2024 |
| Treasury supply: Monthly Statement of the Public Debt | Security-level amounts and maturities for the treasury maturity profile | 2001–2020 |
| Corporate zeros: A-or-above corporate zero curve (HQM, FRED) | Monthly A-or-above zero curve, 1–30y | 2000–2023 |
| Credit risk: Markit/IHS | State-government and AAA, AA and A corporate CDS at 1–30y tenors, for the matched adjustment sample | 2004–2020 |
| Expected rates: Kim–Wright, Federal Reserve; Adrian–Crump–Moench | Monthly fitted yields and term premia, expected-average-short-rate component capped at 10y | 2000–2021 |
| Swaps: Federal Reserve H.15 (FRED) | Par interest-rate swap rates, 1–30y, for household swap pricing | 2000–2016 |
| State taxes: Tax Foundation | Top marginal rate, bracket thresholds, treatment of out-of-state municipal interest | 2000–2022 |
| State income: Bureau of Economic Analysis (BEA) | Quarterly state-level personal income | 2000–2022 |
| Historical state rates: Institute on Taxation and Economic Policy (ITEP) | Top statutory rate by state, the P1 instruments | 1945–2000 vintages |